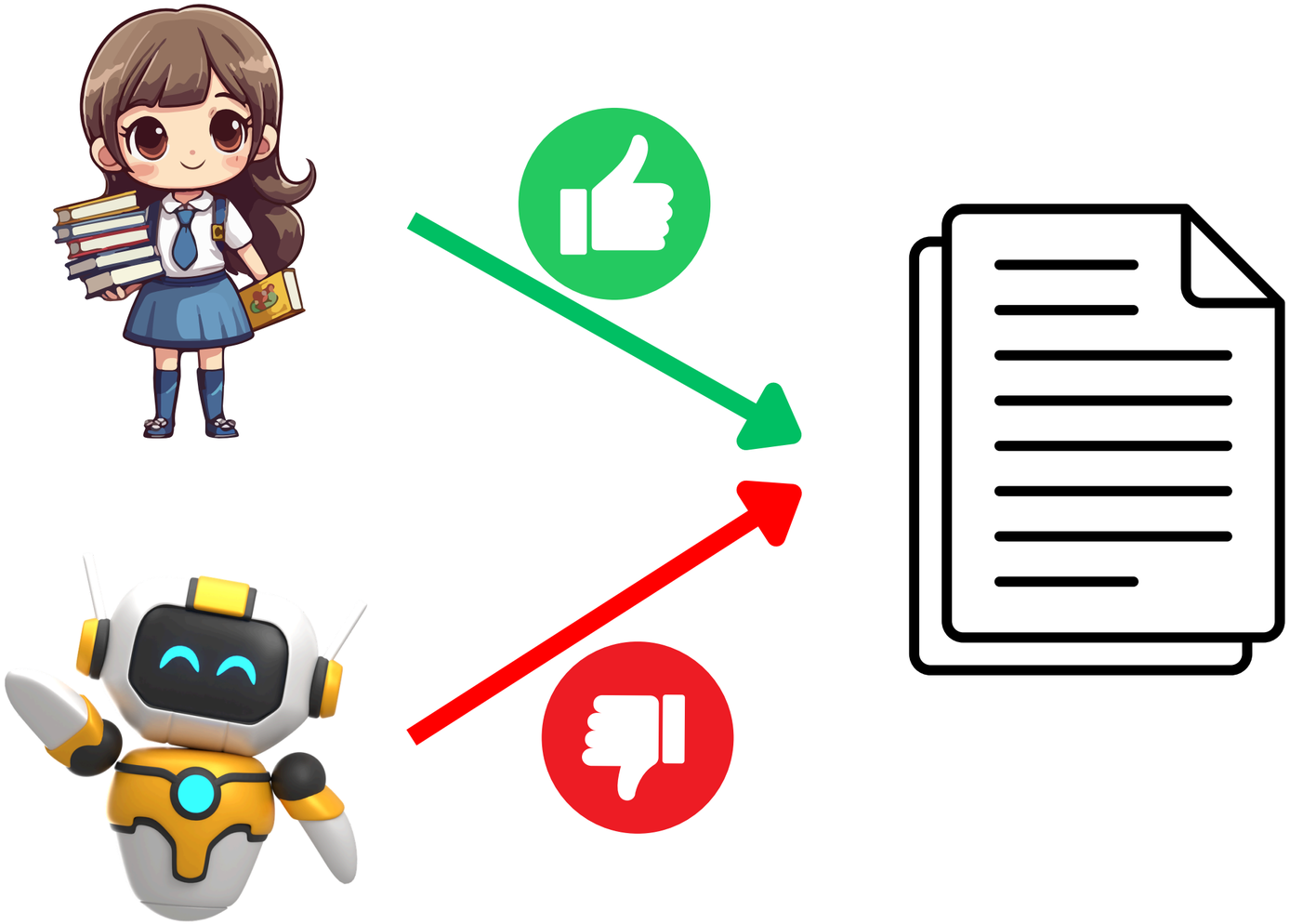
LLM, RAG & AI agent course

Machines do not understand words



Word Embedding

learn [3, 1, 2, ..., 1, 4]

One-hot encoding

Color
Red
Green
Blue

d1	d2	d3
1	0	0
0	1	0
0	0	1

Count-based methods

Bag of words

This movie is fun and funny. The dialogue is funny, the story is sweet, and the adventure is fun

adventure: 1
and: 2
dialogue: 1
fun: 2
funny: 2
is: 4
movie: 1
sweet: 1
story: 1
the: 3
this: 1

Bag of words

fun movie with great music
great movie great acting
fun game with music and friends
acting and music are great

make vocabulary →

acting, and, are, friends, fun, game, great, movie, music, with

score assignment →

acting	and	are	friends	fun	game	great	movie	music	with
0	0	0	0	1	0	1	1	1	1
1	0	0	0	0	0	2	1	0	0
0	1	0	1	1	1	0	0	1	1
1	1	1	0	0	0	1	0	1	0

Problem with BoW

This is a good job. I will not ignore it

=

This is not a good job. I will ignore it

N-grams

This is not a good job

Uni-gram	This | is | not | a | good | job
Bi-gram	This is | is not | not a | a good | good job
Tri-gram	This is not | is not a | not a good | a good job

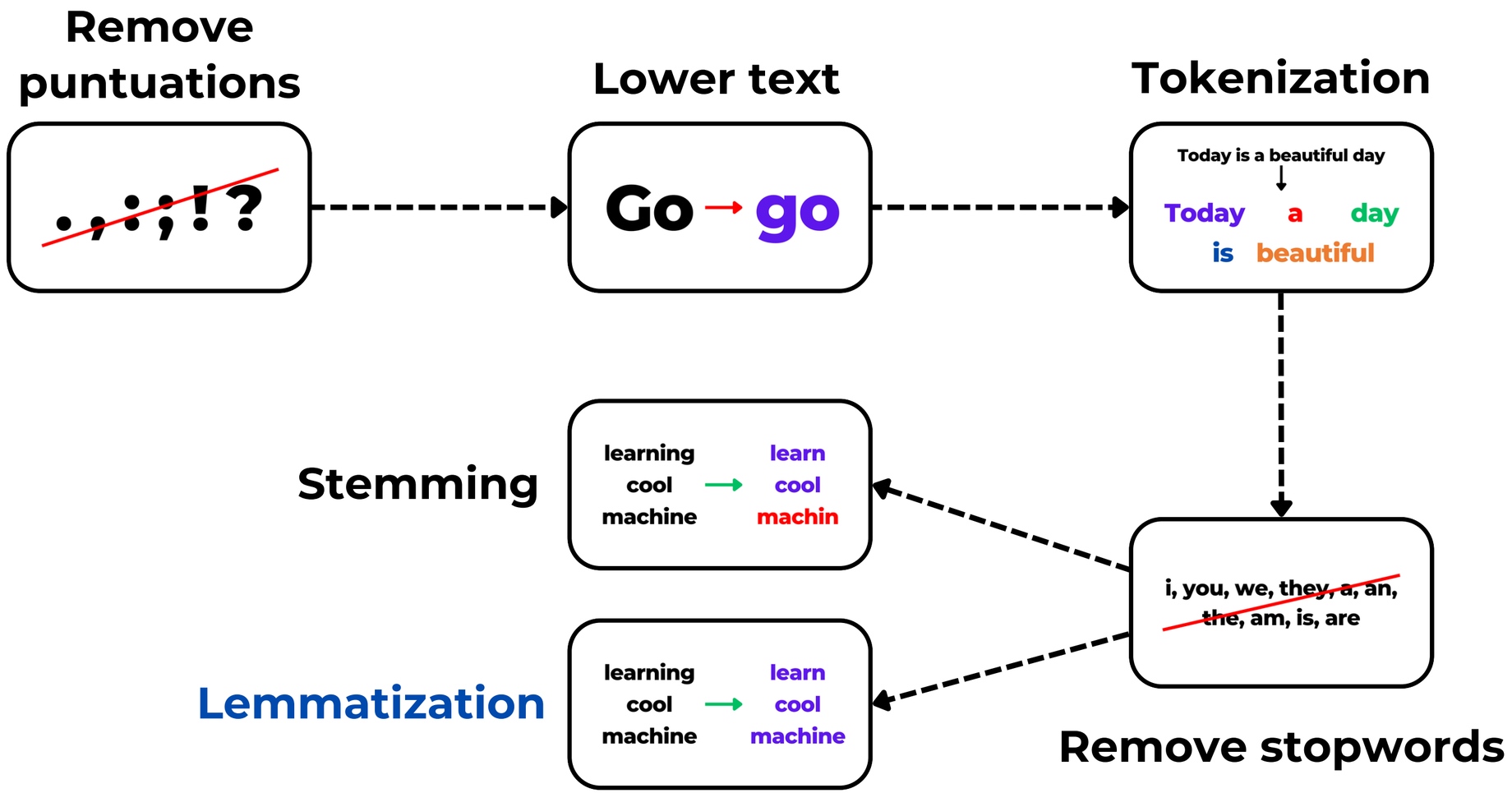
Term Frequency - Inverse Document Frequency

$$TF = \frac{\text{Số lần từ X xuất hiện trong document hiện tại}}{\text{Tổng số từ trong document hiện tại}}$$

$$IDF = \log\left(\frac{\text{Tổng số lượng document}}{\text{Tổng số lượng document có chứa từ X}}\right)$$

Term Frequency	Inverse Document Frequency
Tần suất xuất hiện của 1 từ trong 1 document cụ thể	Mức độ phổ biến/hiếm của 1 từ trong toàn bộ các document

Text preprocessing



Predictive methods

Giới thiệu về Word2vec

Word2vec là 1 mô hình giúp tạo Embedding vector cho các từ

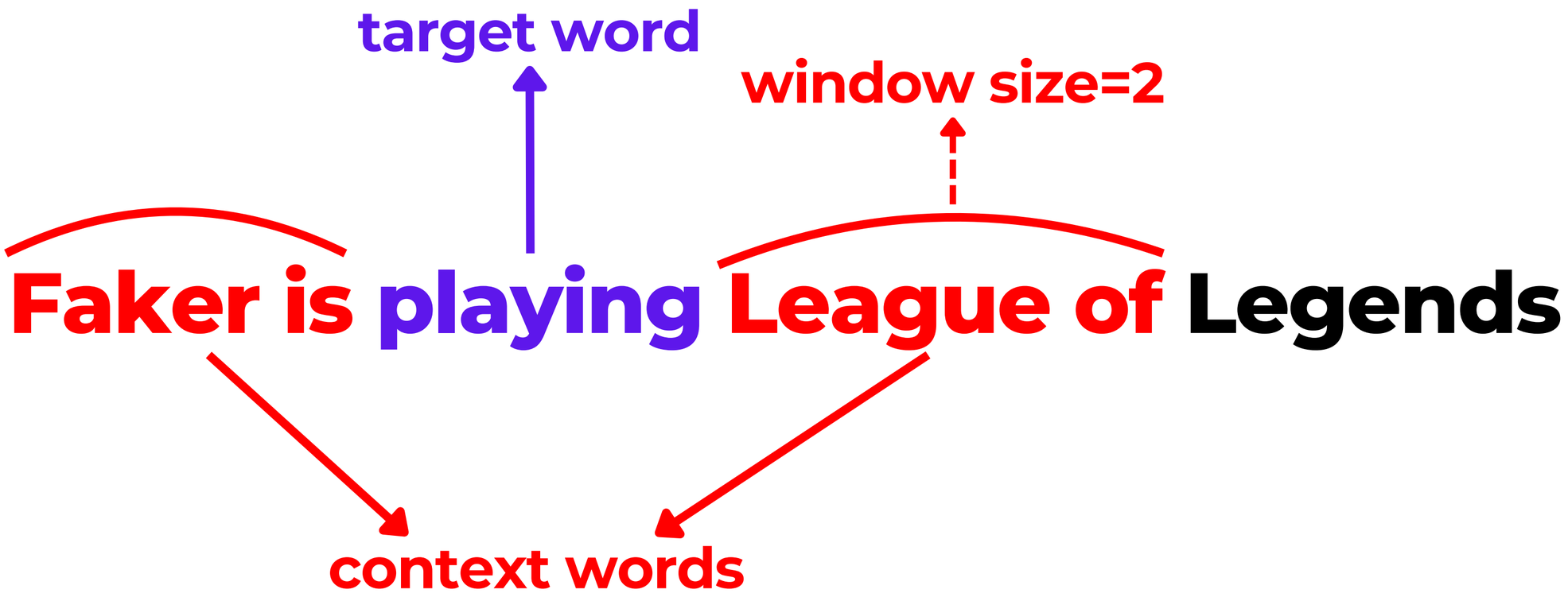
- **Ý tưởng 1: Các từ thường xuất hiện trong các văn cảnh giống nhau thường sẽ có ý nghĩa gần với nhau → embedding vector của chúng cũng sẽ gần nhau (e.g. Hanoi vs capital)**
- **Ý tưởng 2: Ta có thể dự đoán được 1 từ nếu biết các từ xung quanh từ đó**

Word2vec hoạt động dựa trên ngữ cảnh. Embedding được học bằng cách quan sát các từ ở gần nhau

Training data

Faker is playing League of Legends

Training data

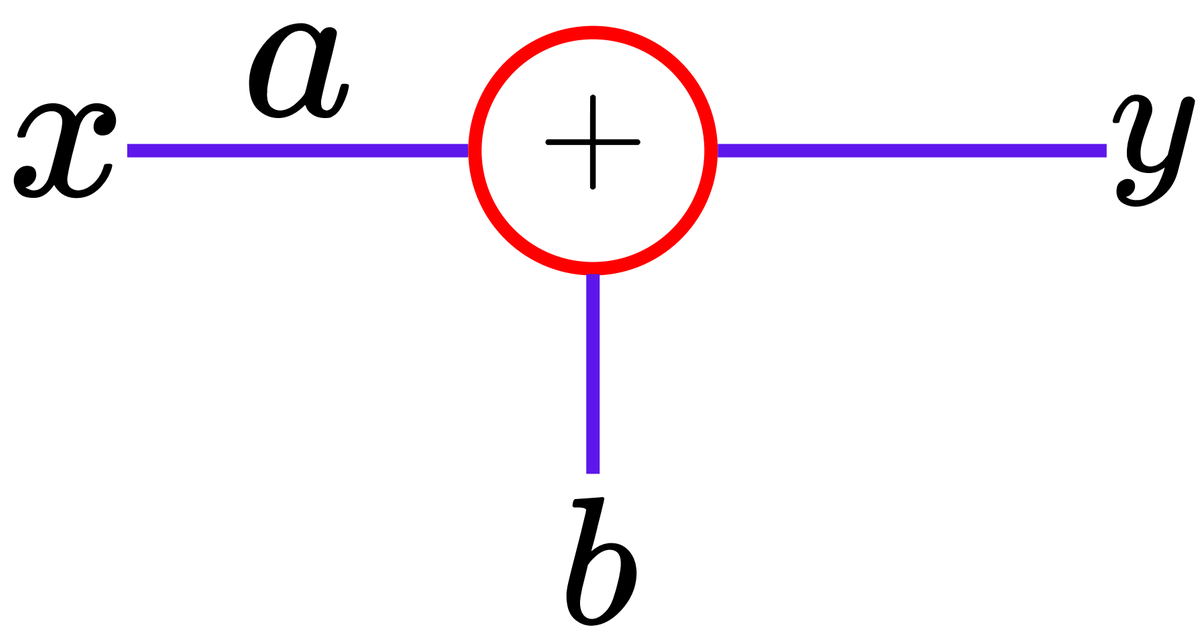


Training data

Text	Training samples
Faker is playing League of Legends	Faker, is Faker, playing
Faker is playing League of Legends	is, Faker is, playing is, League
Faker is playing League of Legends	playing, Faker playing, is playing, League playing, of

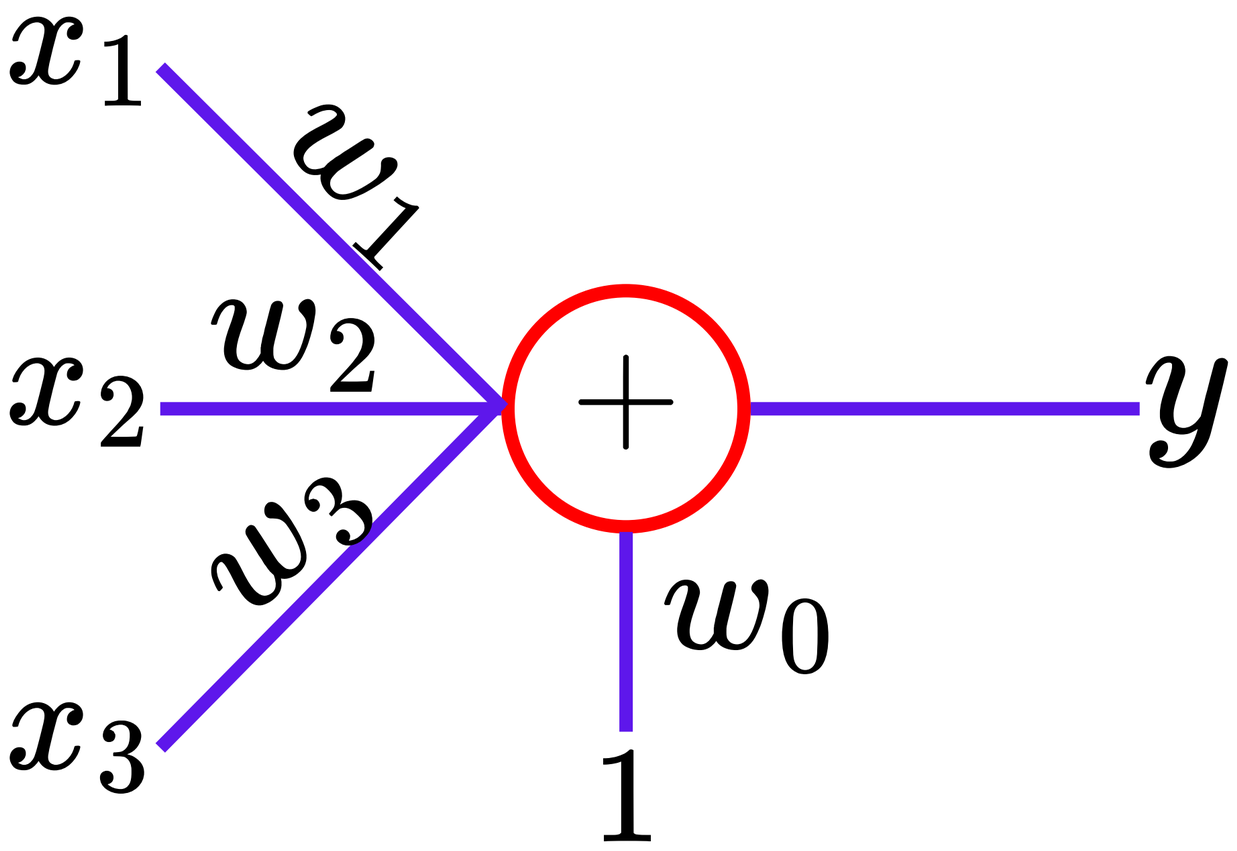
Linear function

$$y = ax + b$$



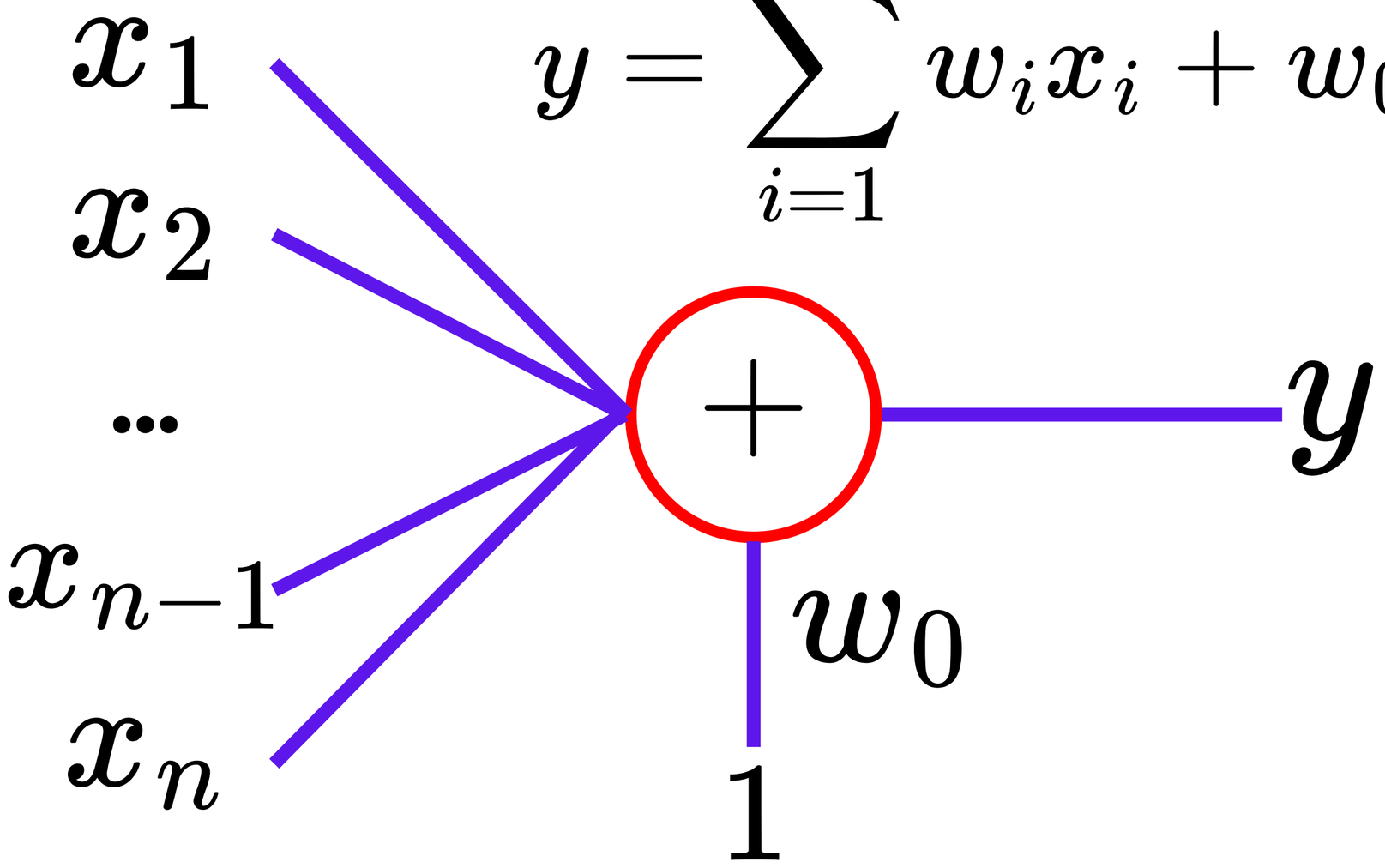
Linear function

$$y = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0$$



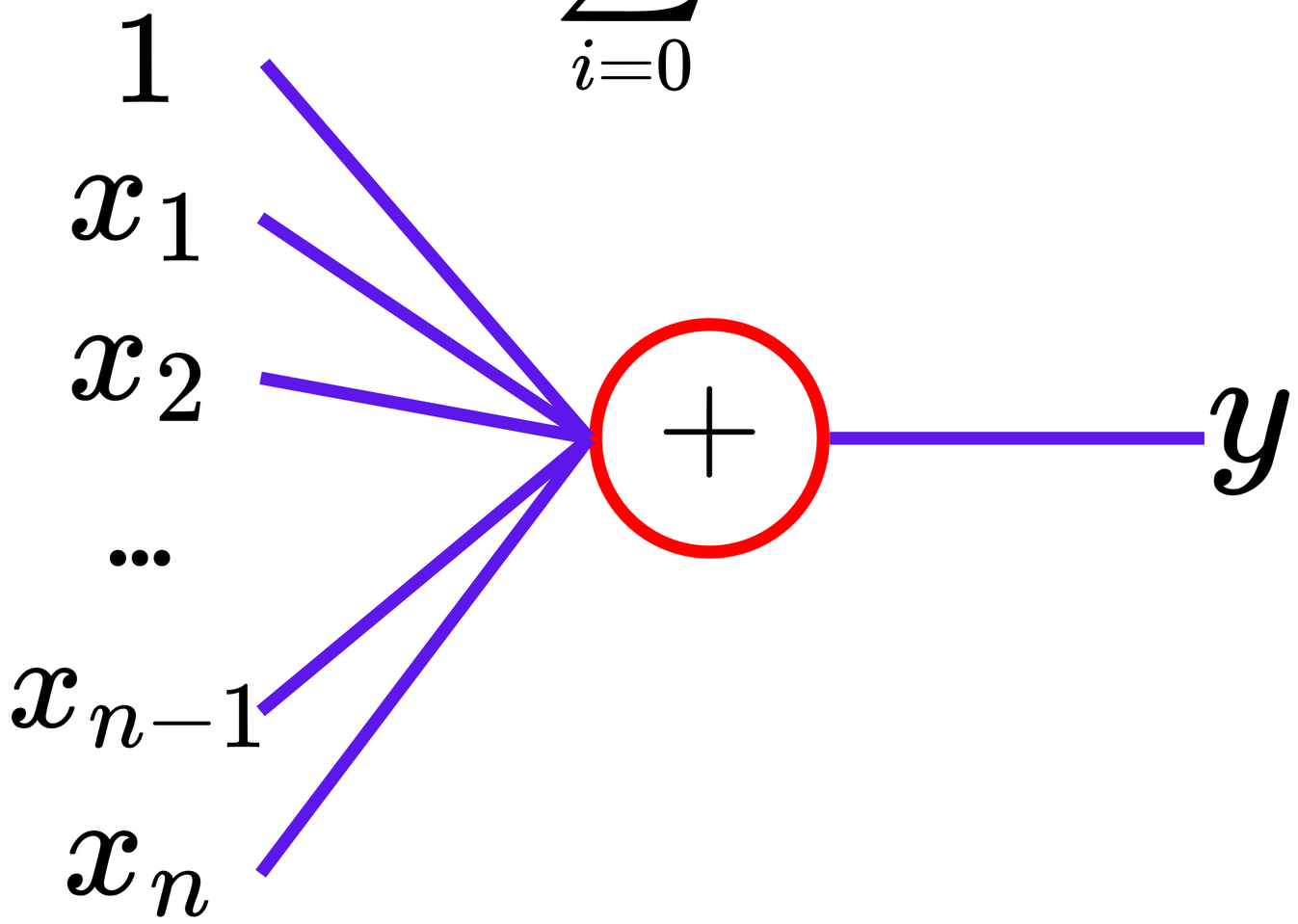
Linear function

$$y = \sum_{i=1}^{n} w_i x_i + w_0$$

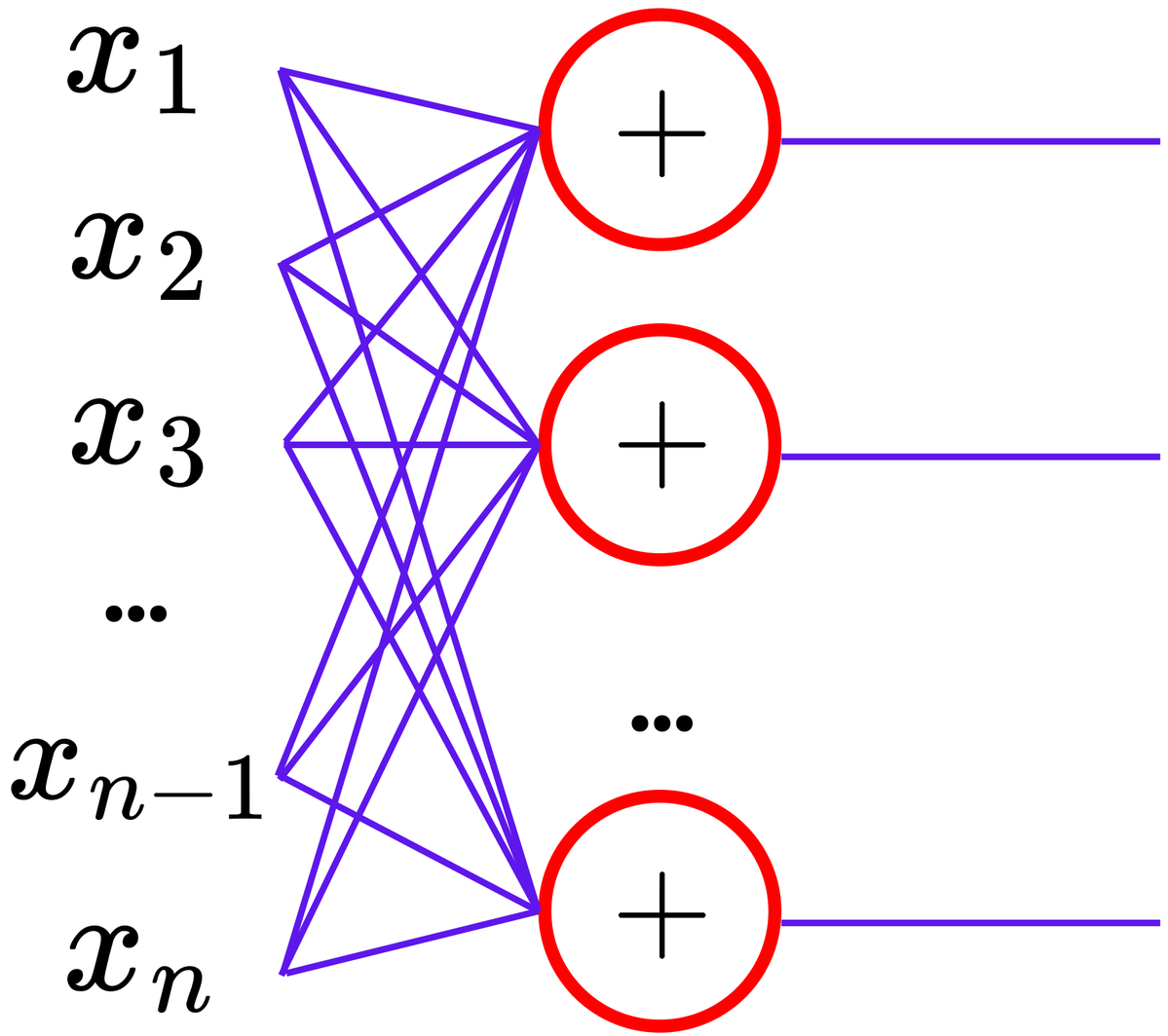


Linear function

$$y = \sum_{i=0}^{n} w_i x_i \qquad x_0 = 1$$



Linear functions

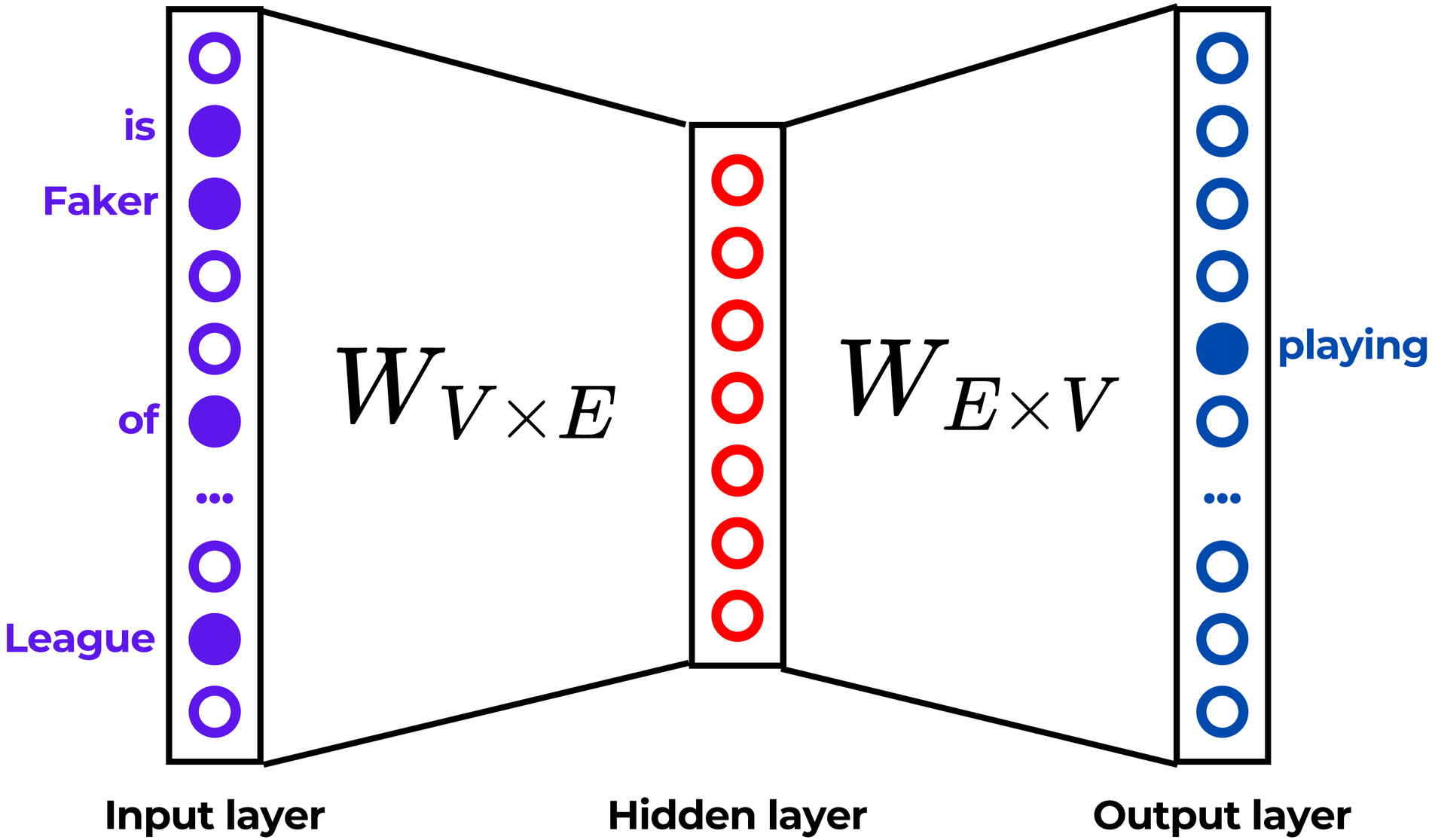


$$h_1 = \sum_{i=1}^{n} w_{1i} x_i$$

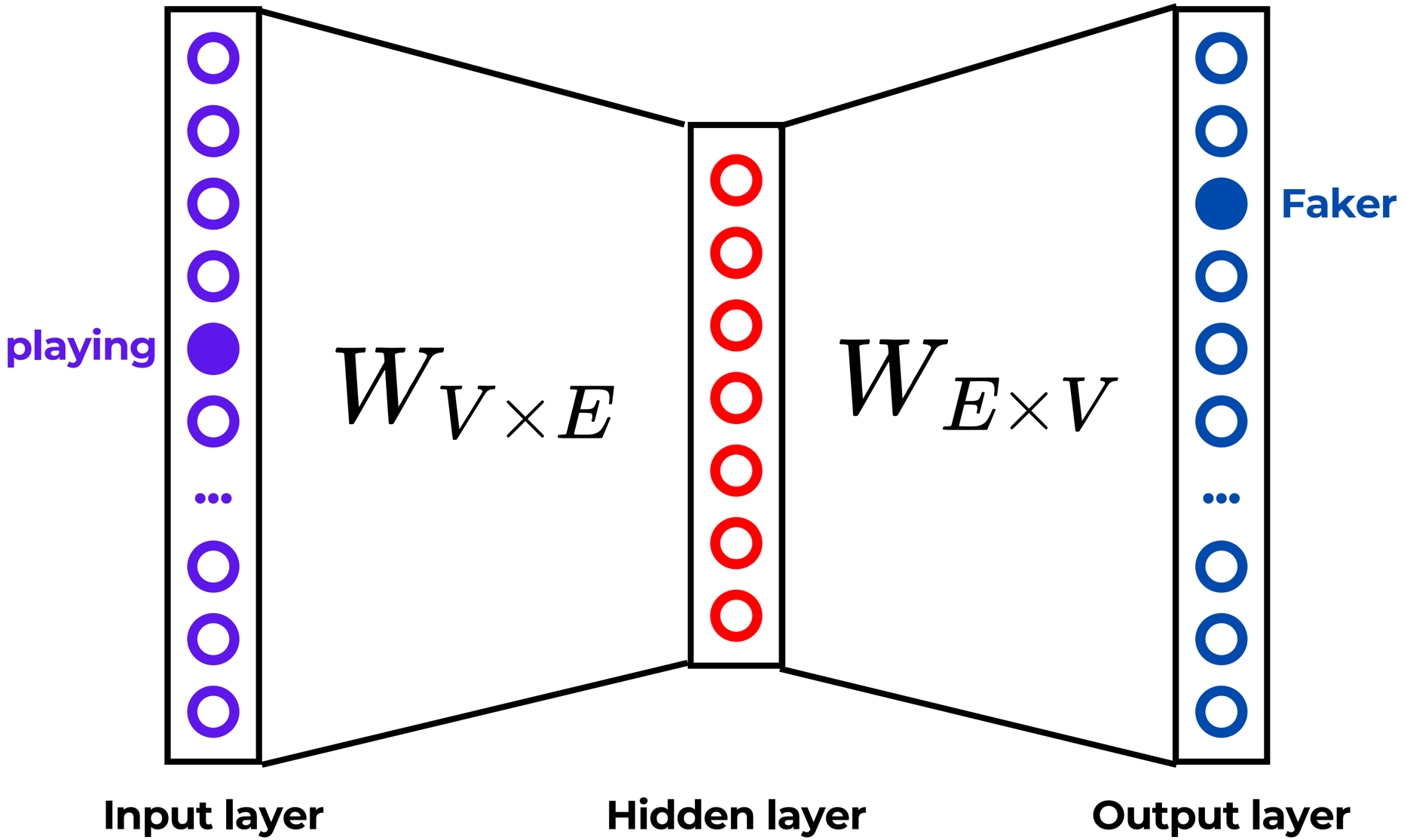
$$h_2 = \sum_{i=1}^{n} w_{2i} x_i$$

$$h_k = \sum_{i=1}^{n} w_{ki} x_i$$

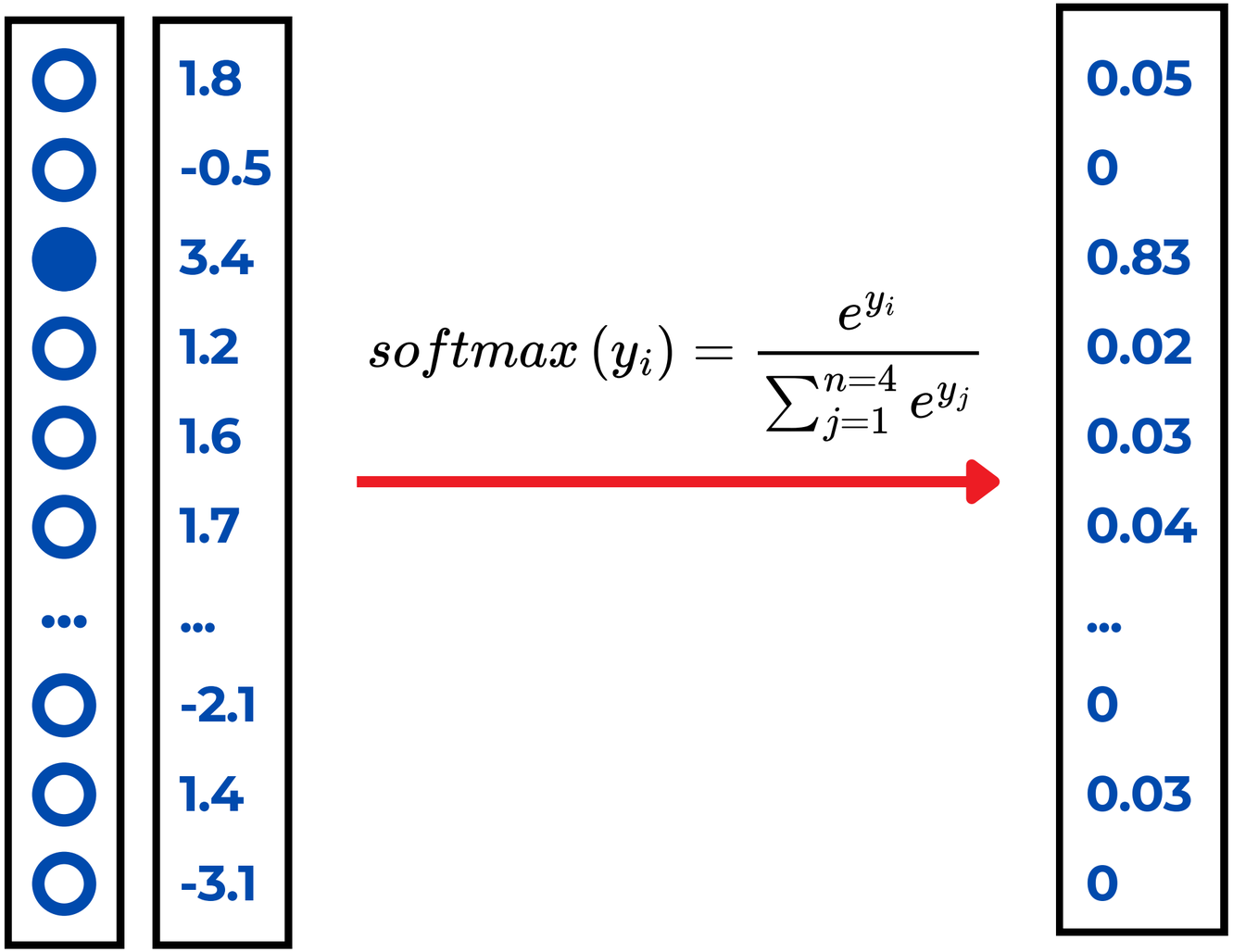
CBOW


is
Faker
of
...
League
$W_{V \times E}$
$W_{E \times V}$
playing
Input layer
Hidden layer
Output layer

Skip-gram


playing
$W_{V \times E}$
$W_{E \times V}$
Faker
Input layer
Hidden layer
Output layer

Softmax


1.8
-0.5
3.4
1.2
1.6
1.7
...
-2.1
1.4
-3.1
$softmax\left(y_i\right) = \frac{e^{y_i}}{\sum_{j=1}^{n=4} e^{y_j}}$
0.05
0
0.83
0.02
0.03
0.04
...
0
0.03
0

Negative sampling

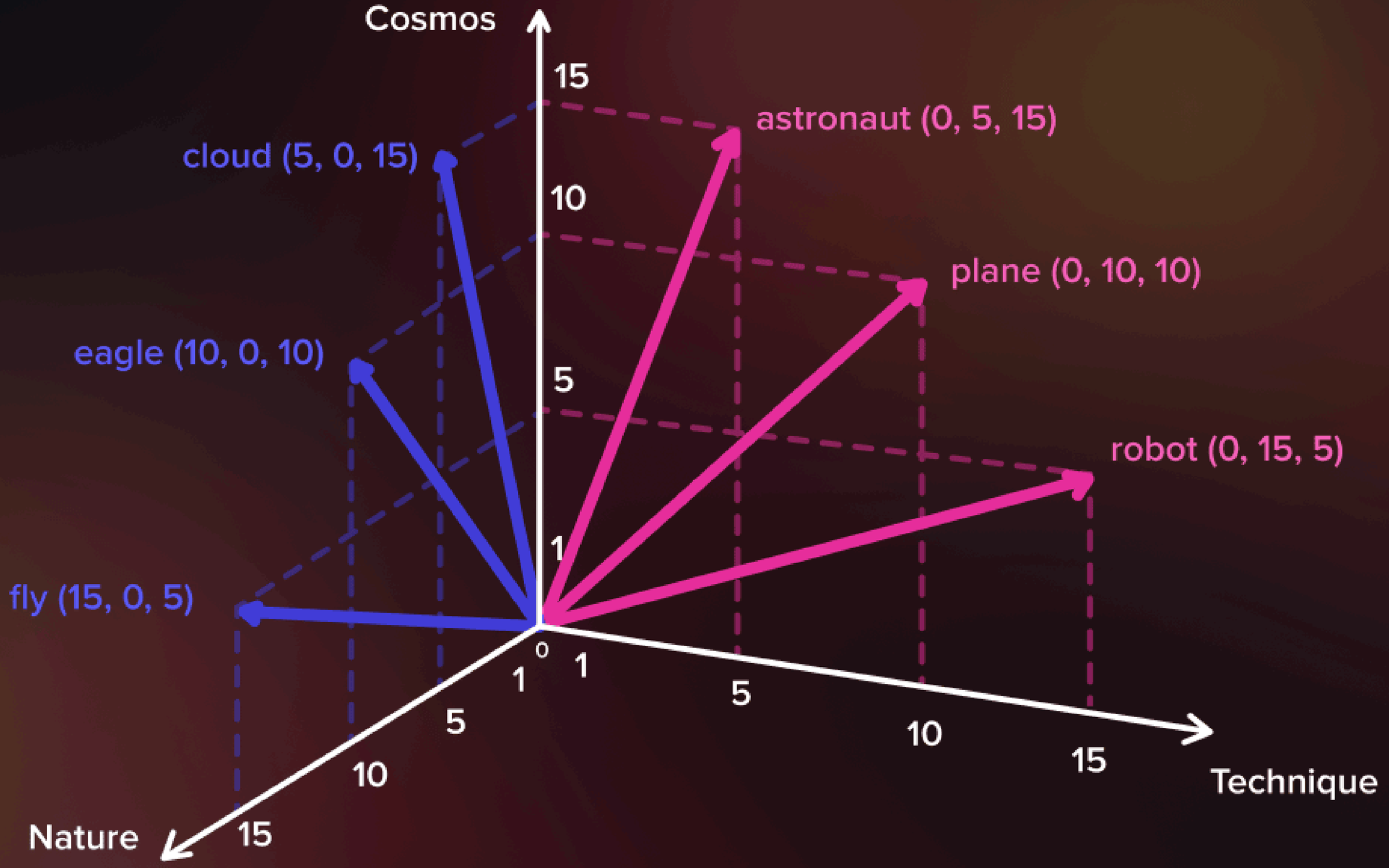
Word embedding

$$W_{V \times E}$$

E

V

0.45	-1.24	0.68	-2.12	1.33	...	-1.72	-1.12	0.34
-1.22	1.1	0.27	0.29	1.12	...	1.19	1.31	0.5
2.11	1.44	-0.23	-0.98	-1.11	...	0.14	-1.17	1.62
1.14	1.21	-0.29	0.21	2.12	...	1.61	-0.15	1.52
...	...	...	...	...	...	...	...	...
0.31	1.72	1.83	-1.81	2.15	...	-1.44	-1.51	1.42


Cosmos
15
10
5
1
astronaut (0, 5, 15)
cloud (5, 0, 15)
plane (0, 10, 10)
eagle (10, 0, 10)
robot (0, 15, 5)
fly (15, 0, 5)
0
1
1
5
10
15
5
10
15
Technique
Nature

CBOW

- Phù hợp với các bộ dữ liệu lớn
- Huấn luyện nhanh hơn
- Có khả năng biểu diễn tốt các từ thường xuyên xuất hiện
- Khi window size quá lớn dễ ảnh hưởng đến khả năng học

Skip-gram

- Phù hợp với các bộ dữ liệu bé
- Huấn luyện lâu hơn
- Có khả năng biểu diễn tốt các từ ít xuất hiện

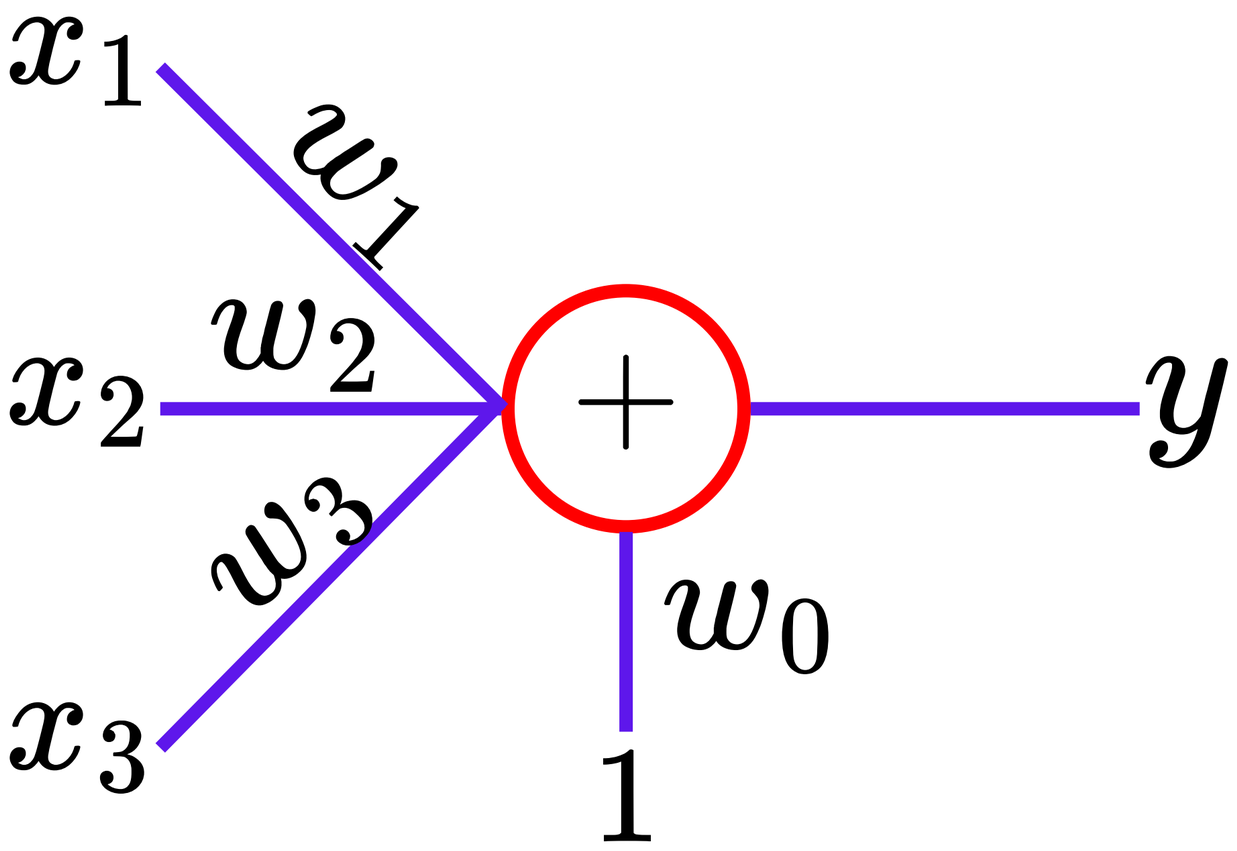
Hạn chế của word embedding truyền thống

- **Context-independent**: Luôn thể hiện 1 từ dưới dạng 1 vector cố định, bất kể từ ấy có nhiều nghĩa trong nhiều ngữ cảnh khác nhau
- **Out-of-Vocabulary (OOV)**: Không thể hiện được vector cho những từ mới, không xuất hiện trong quá trình huấn luyện

Neural Network

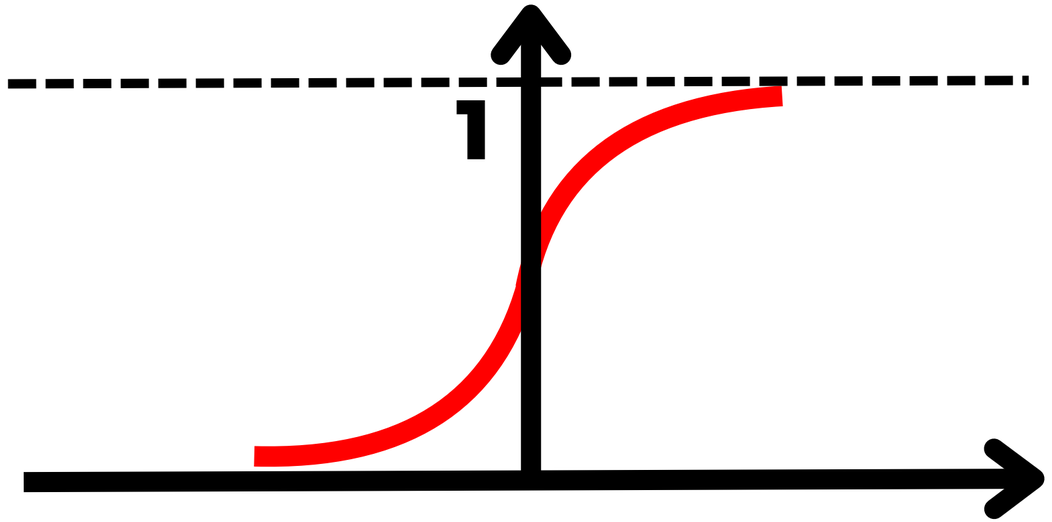
Linear function

$$y = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0$$



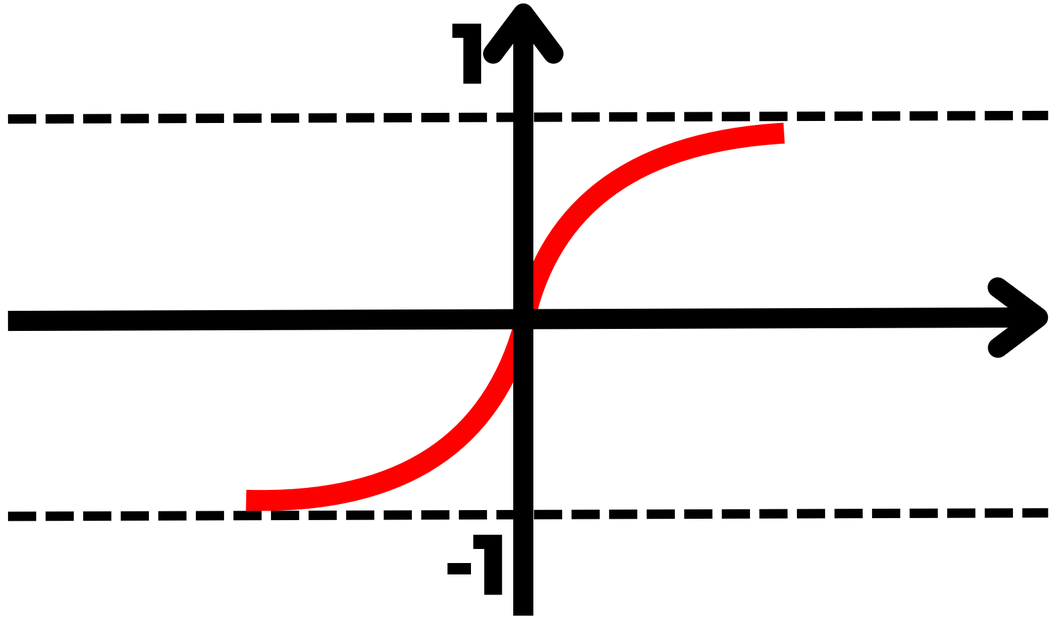
Activation function

sigmoid



$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

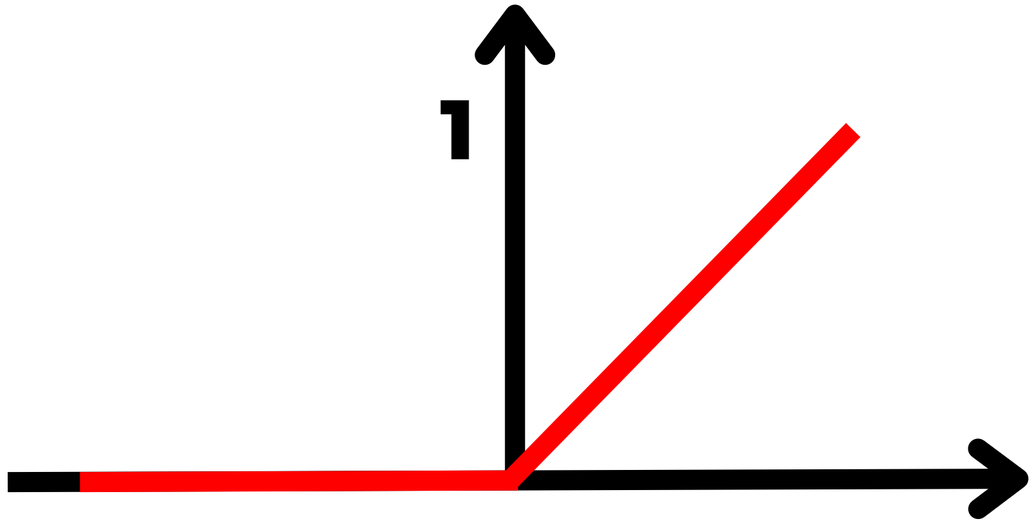
tanh



$$tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

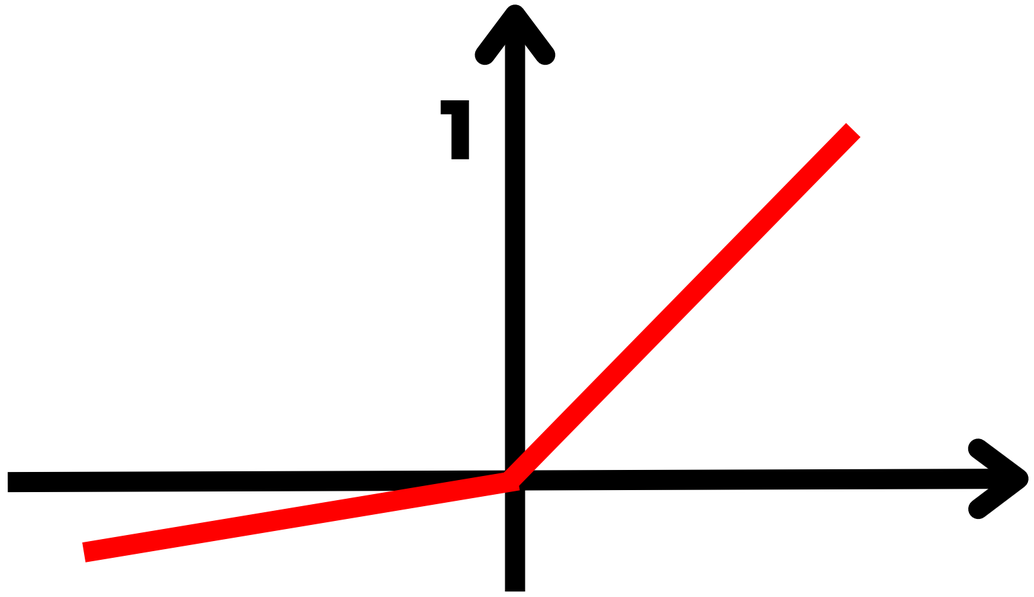
Activation function

ReLU



$$ReLU\left(x\right) = max\left(0, x\right)$$

Leaky ReLU

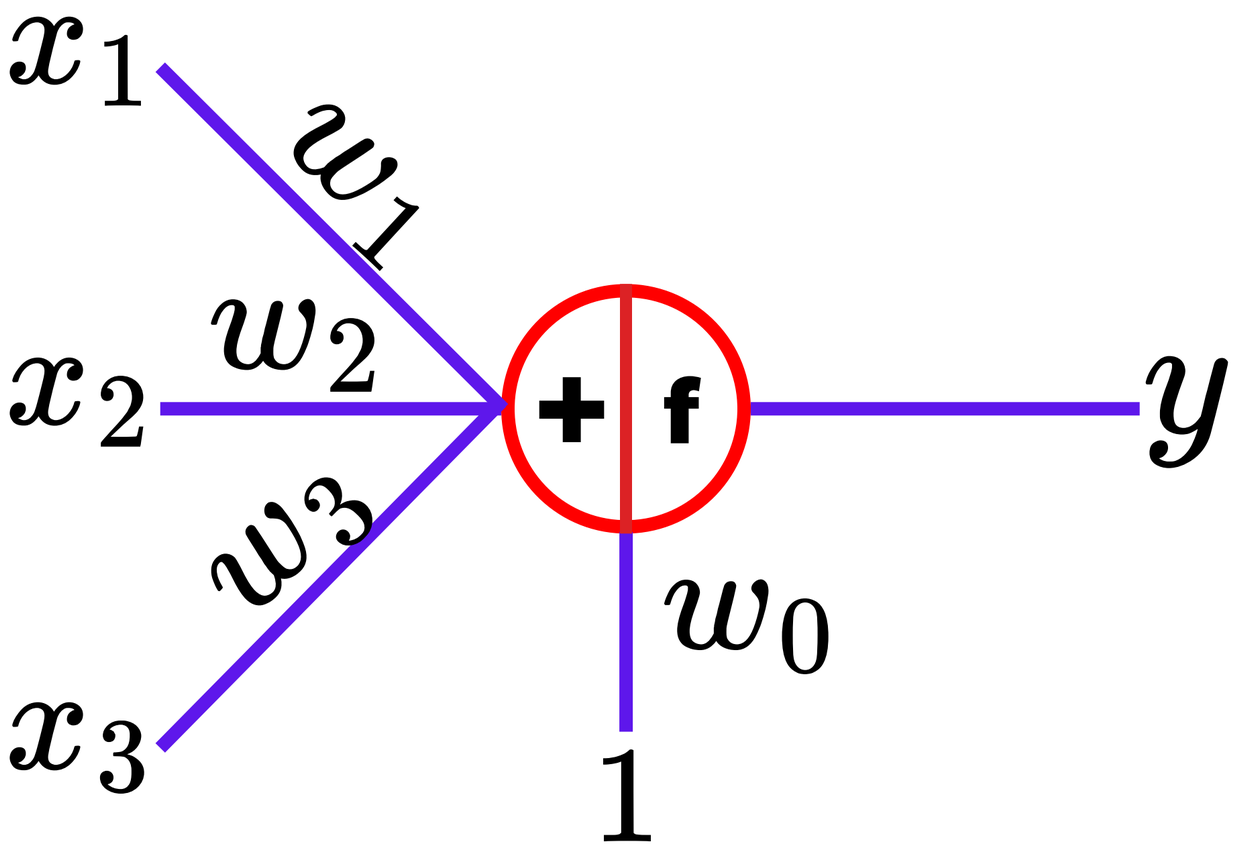


$$leakyReLU\left(x\right) = max\left(\varepsilon x, x\right)$$

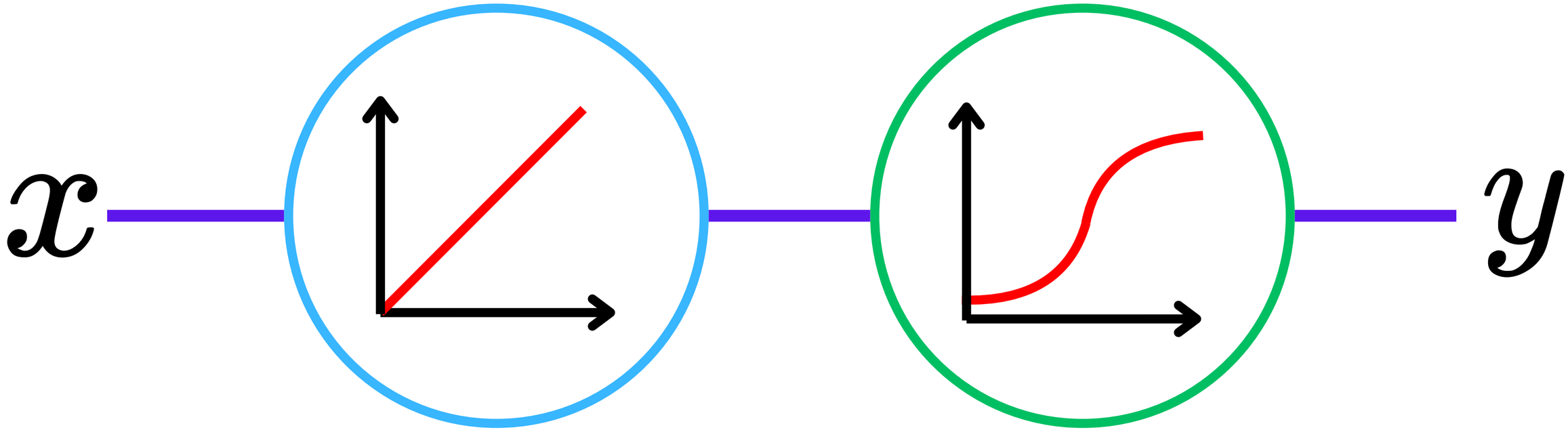
$$\varepsilon << 1$$

Neuron

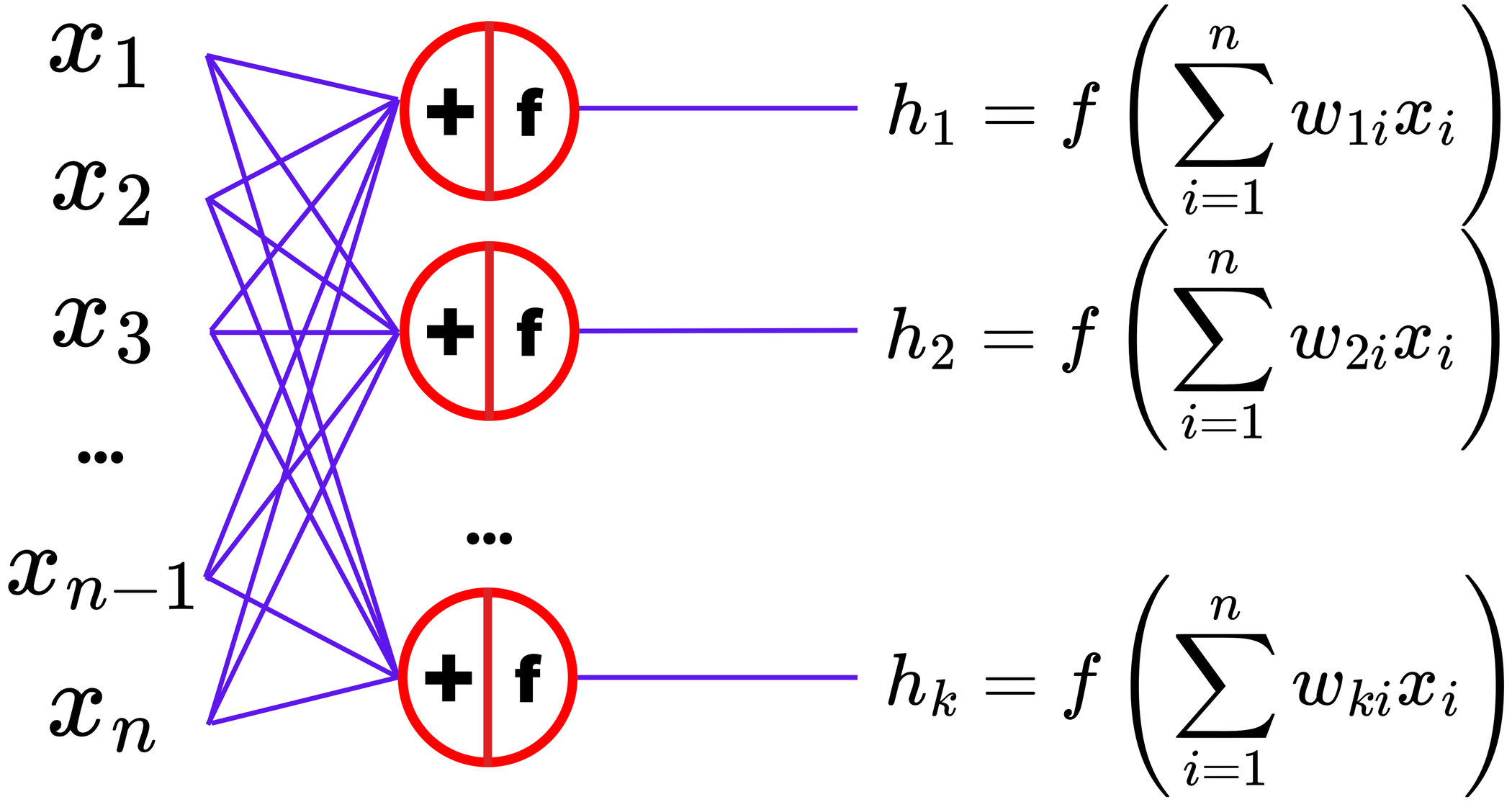
$$y = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0$$



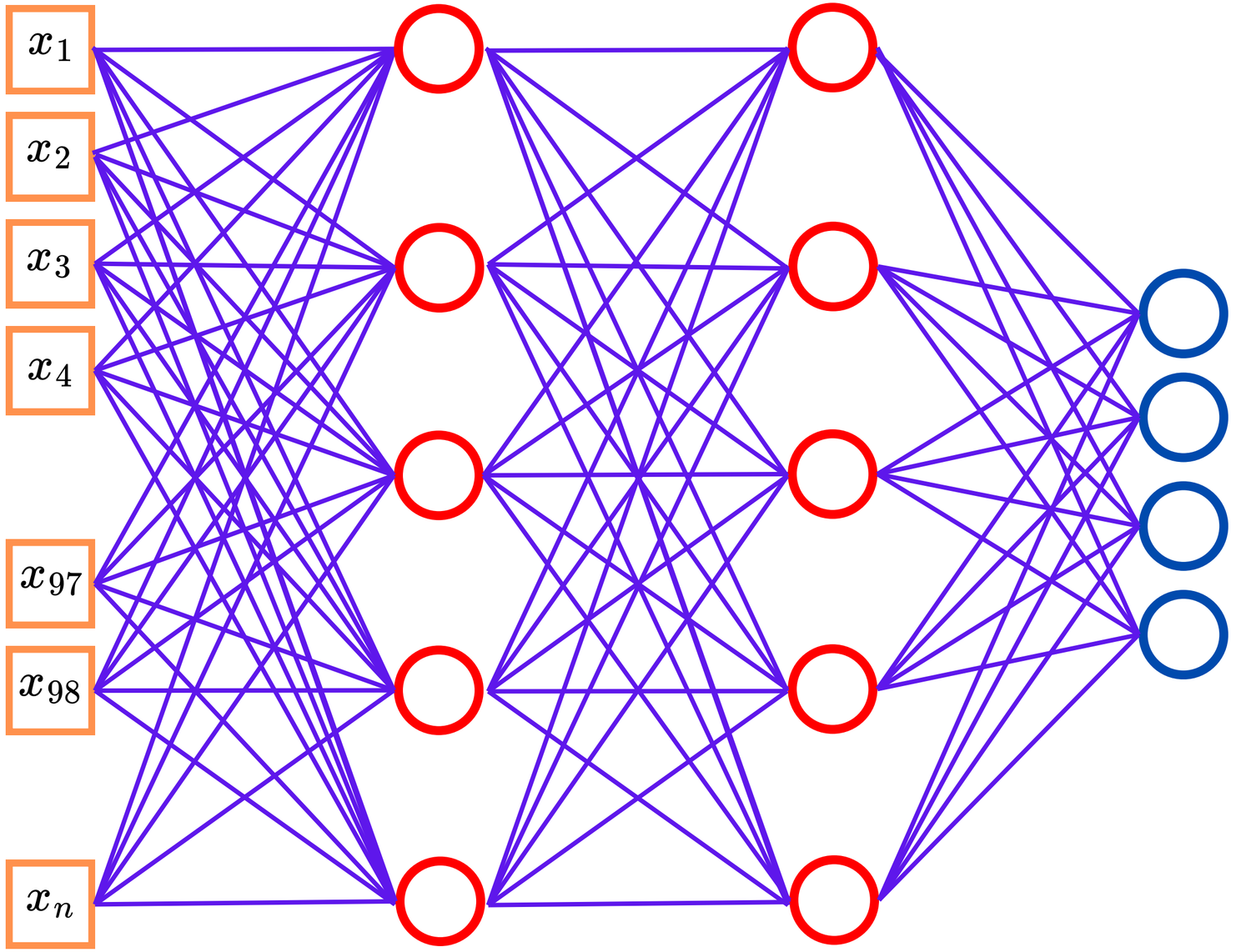
Neuron



Fully connected layer



Neural Network

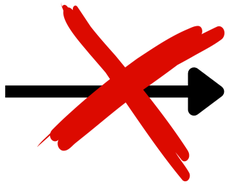


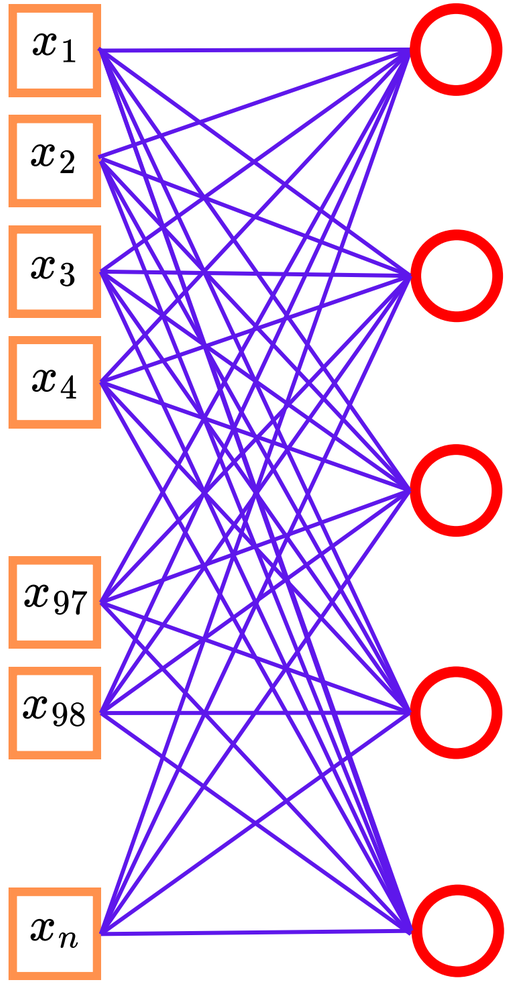
Vấn đề với Neural network dùng FCN

Embedding size

Text length

Faker	-1.24	0.68	-2.12	1.33	...	-1.72	-1.12	0.34
is	1.1	0.27	0.29	1.12	...	1.19	1.31	0.5
playing	1.44	-0.23	-0.98	-1.11	...	0.14	-1.17	1.62
League	1.21	-0.29	0.21	2.12	...	1.61	-0.15	1.52
...	...	...	...	...	...	...	...	...
match	1.72	1.83	-1.81	2.15	...	-1.44	-1.51	1.42





Flatten data

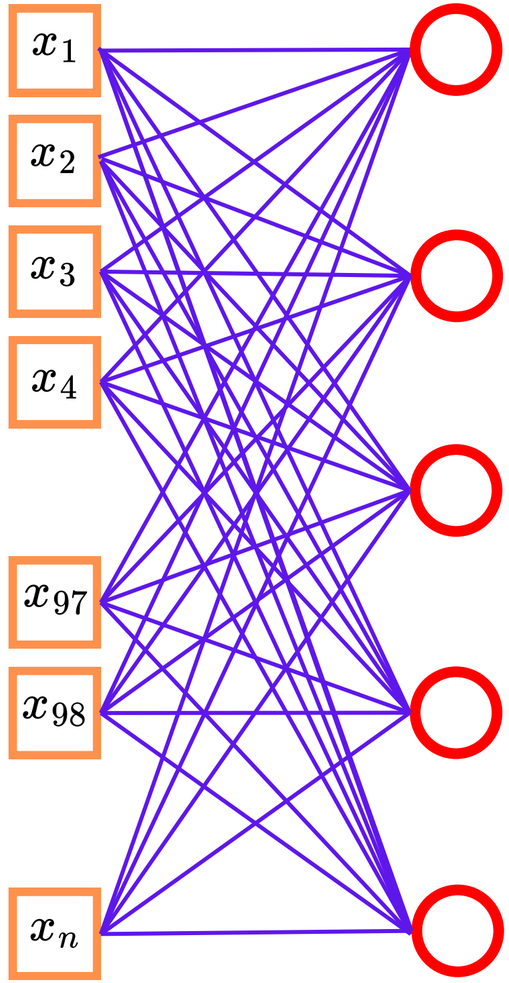
Embedding size

Faker	-1.24	0.68	-2.12	1.33	...	-1.72	-1.12	0.34
is	1.1	0.27	0.29	1.12	...	1.19	1.31	0.5
playing	1.44	-0.23	-0.98	-1.11	...	0.14	-1.17	1.62
League	1.21	-0.29	0.21	2.12	...	1.61	-0.15	1.52
...	...	...	...	...	...	...	...	...
match	1.72	1.83	-1.81	2.15	...	-1.44	-1.51	1.42

Text length

n = Text length x Embedding size

-1.24	0.68	...	...	...	...	-1.51	1.42



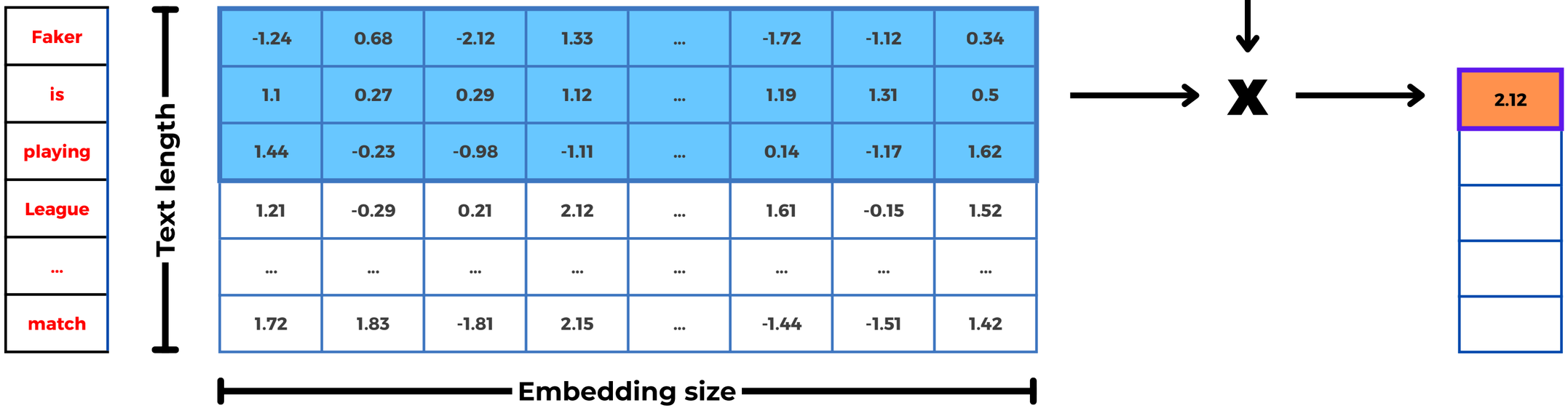
Hạn chế của Fully connected layer

- **Không hiểu cấu trúc text**: Khi dữ liệu bị làm phẳng, mô hình vẫn học được một số quan hệ vị trí giữa các từ, nhưng rất kém hiệu quả và khó tổng quát hóa.
- **Tốn tài nguyên**: Fully connected layer tốn rất nhiều tham số
- **Bị giới hạn bởi kích thước cố định**: Mô hình chỉ là việc với input có kích thước cố định, nên sẽ phải padding hoặc cắt ngắn text đầu vào

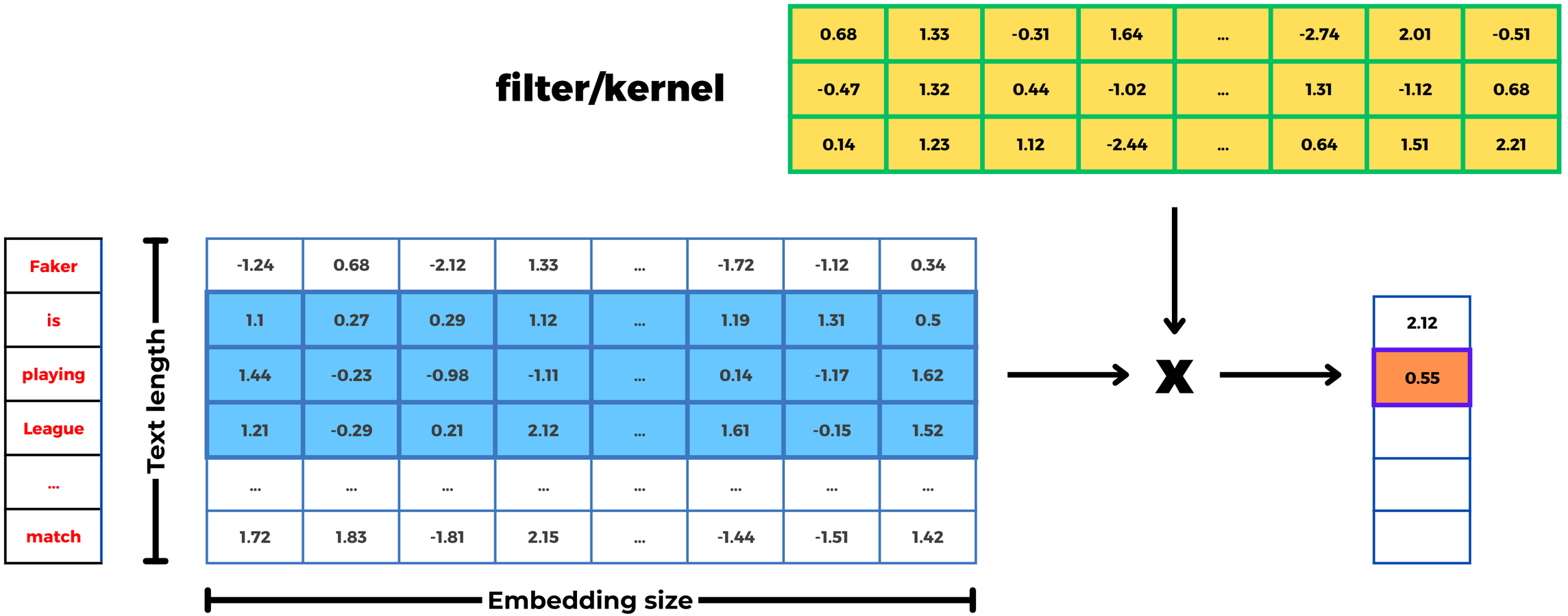
Convolutional layer

filter/kernel

0.68	1.33	-0.31	1.64	...	-2.74	2.01	-0.51
-0.47	1.32	0.44	-1.02	...	1.31	-1.12	0.68
0.14	1.23	1.12	-2.44	...	0.64	1.51	2.21

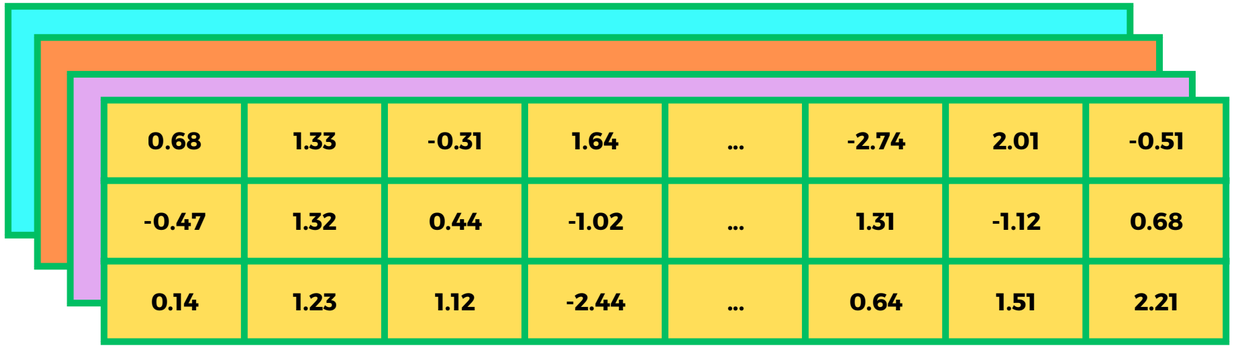


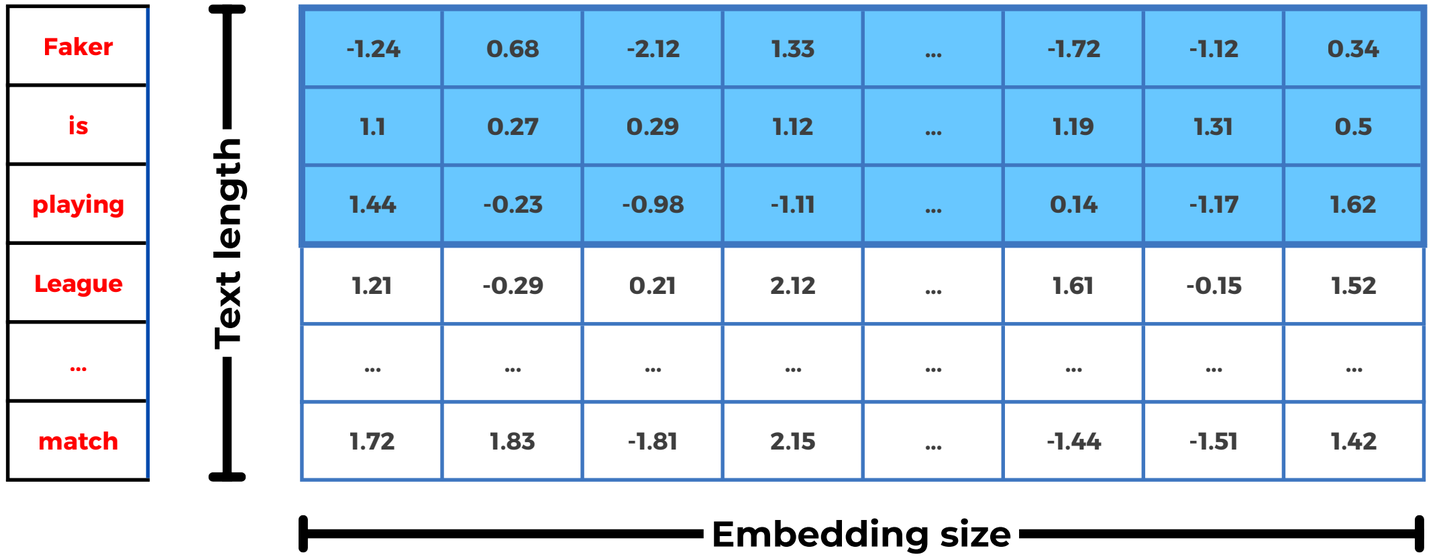
Convolutional layer

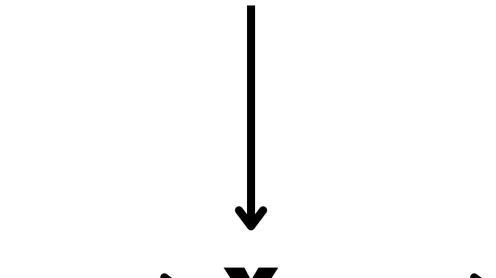


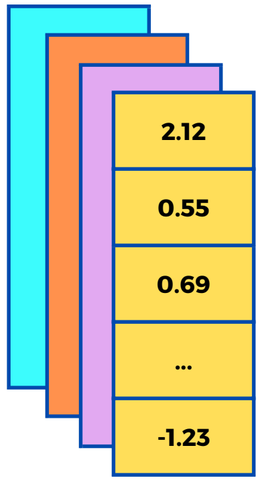
Convolutional layer

filter/kernel

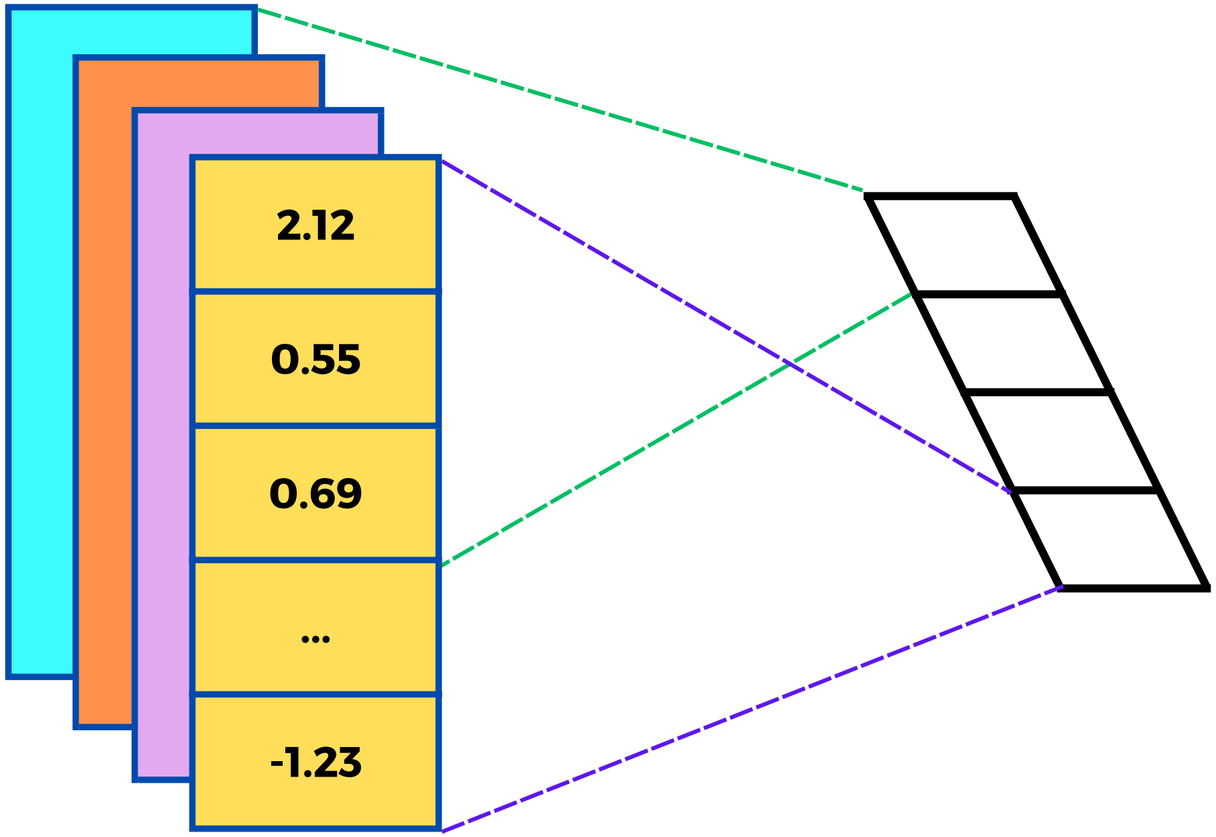




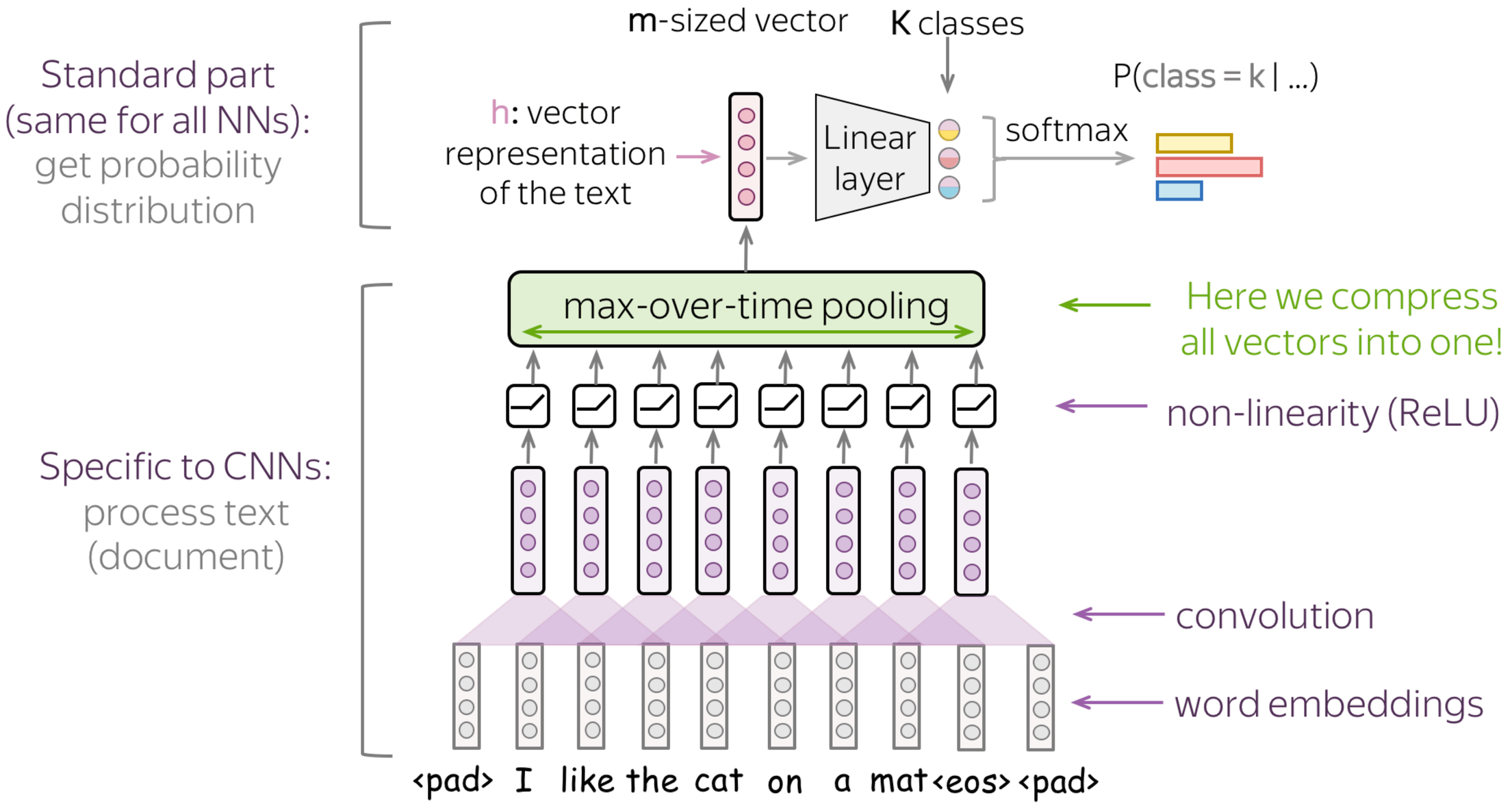




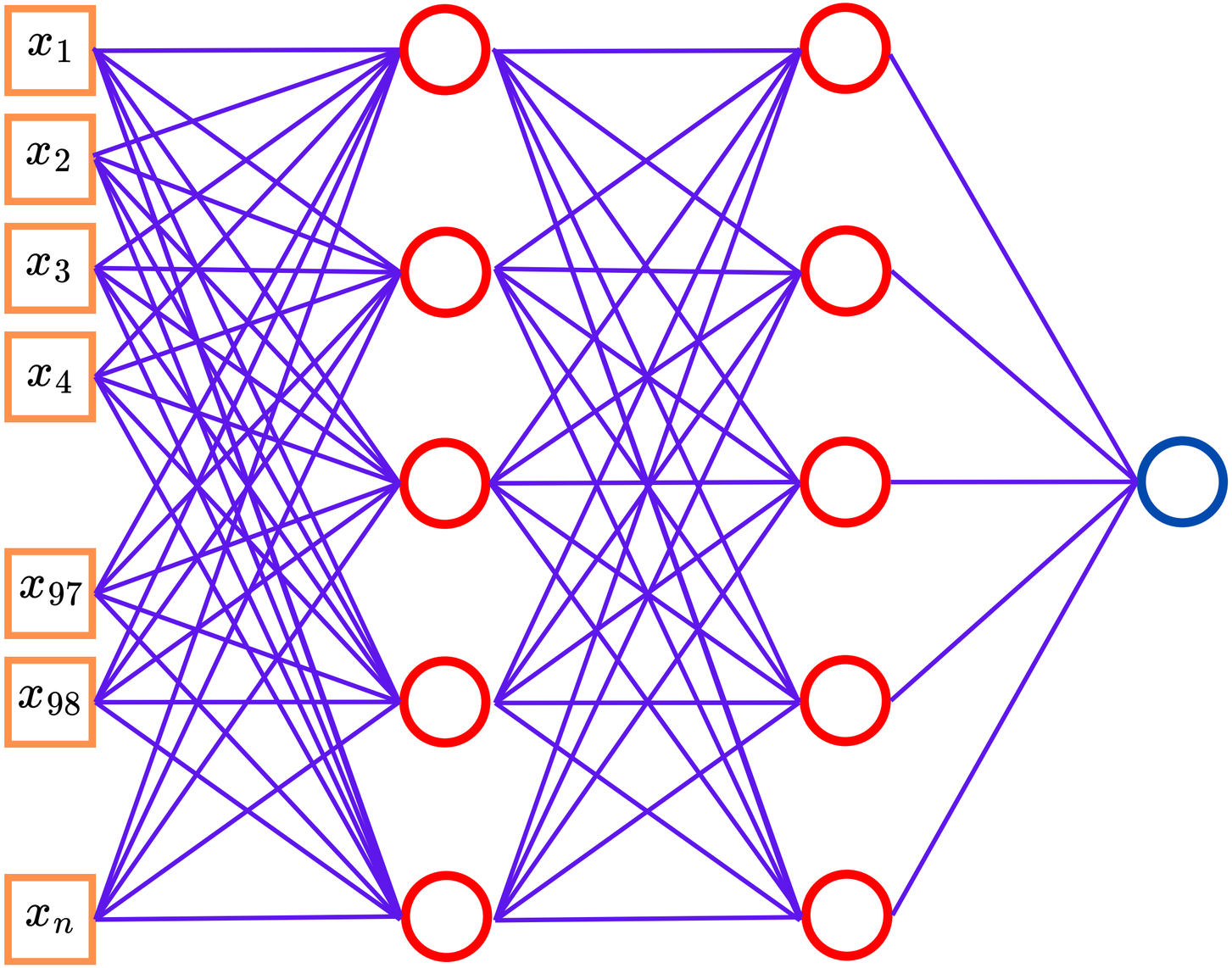
Max-over-time pooling



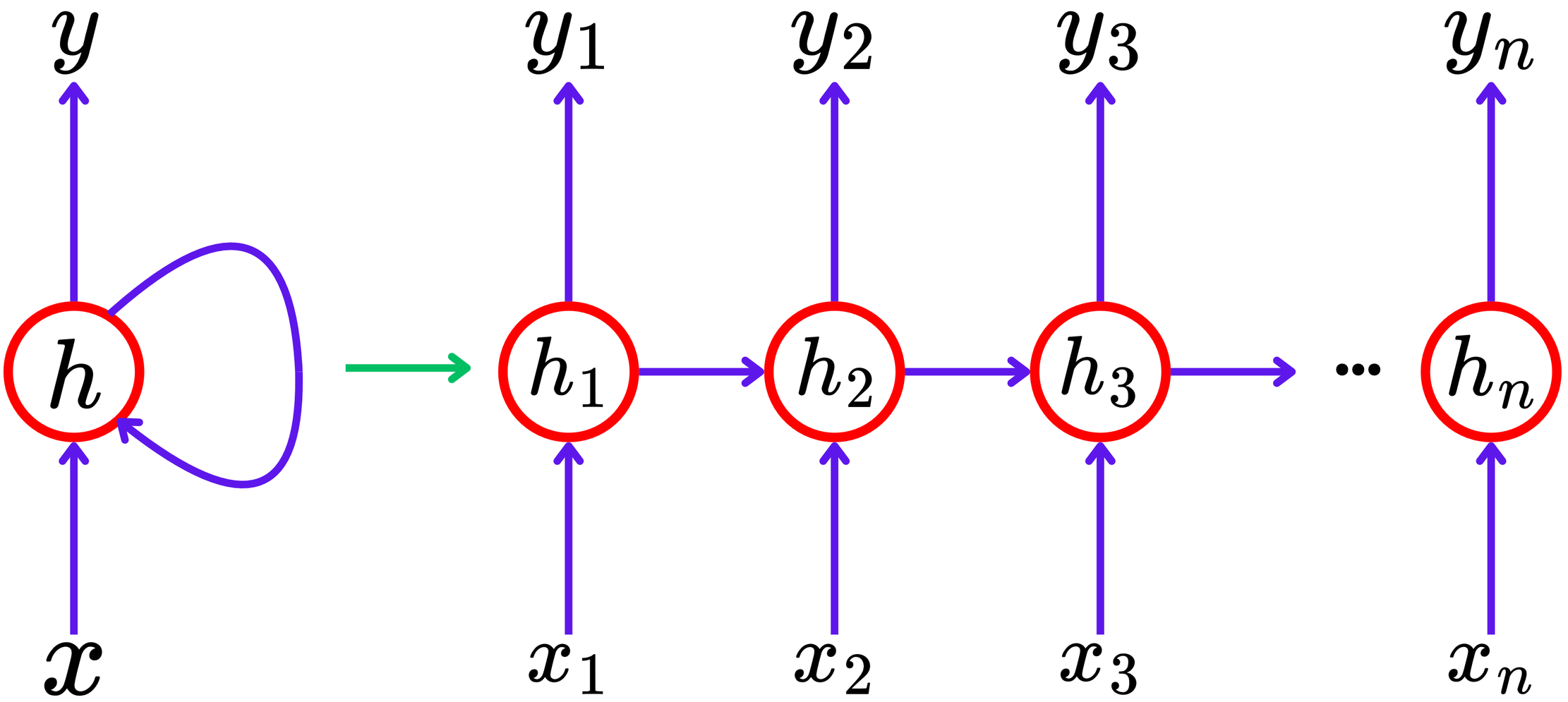
Convolutional Neural Network



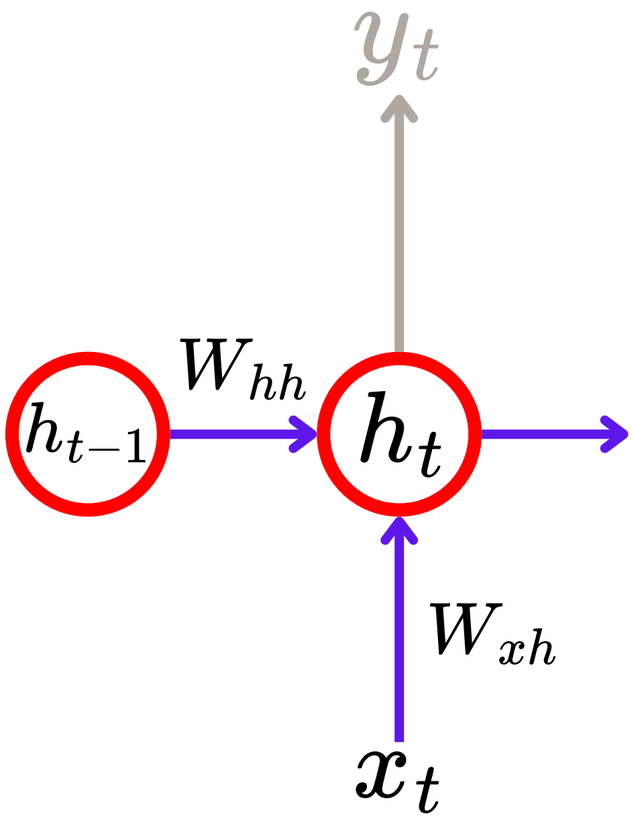
Neural Network



Recurrent Neural Network

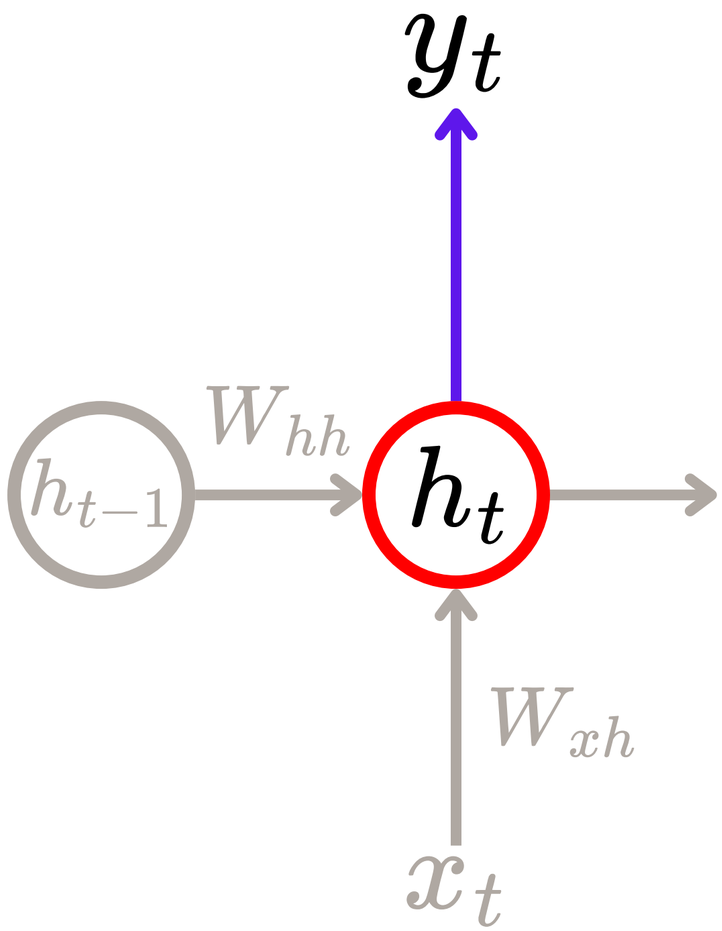


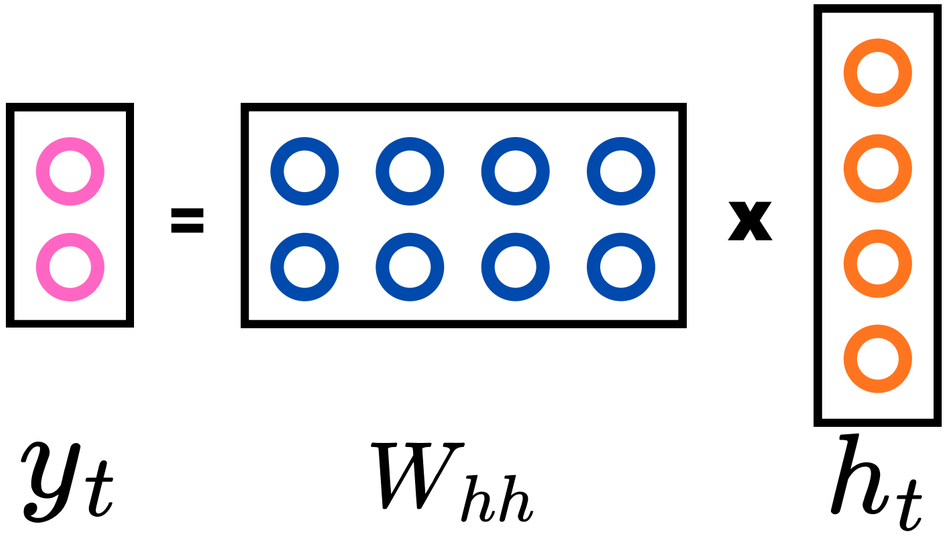
Recurrent Neural Network



h_t = tanh(W_{hh} x h_{t-1} + W_{xh} x x_t)

Recurrent Neural Network



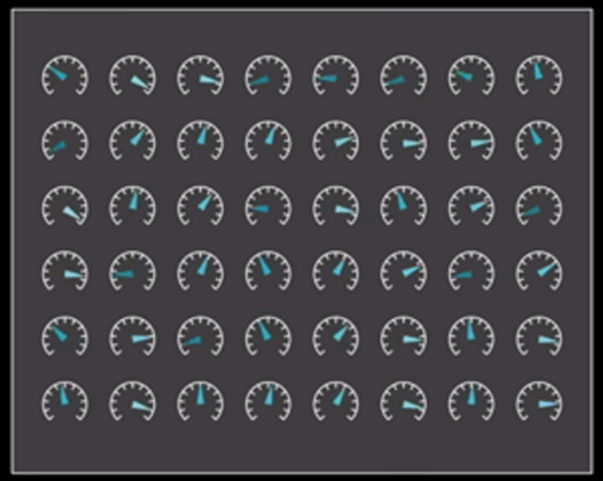


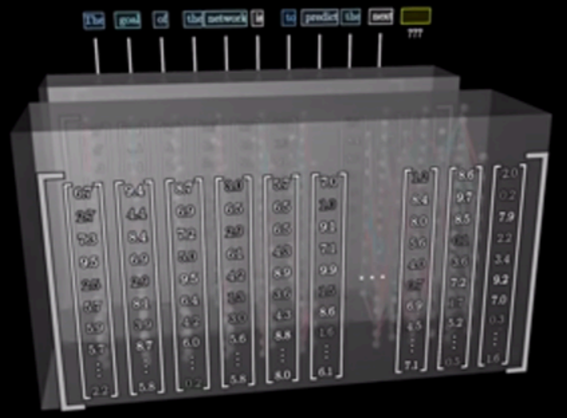
Introduction to Transformer

Generative Pre-trained Transformer

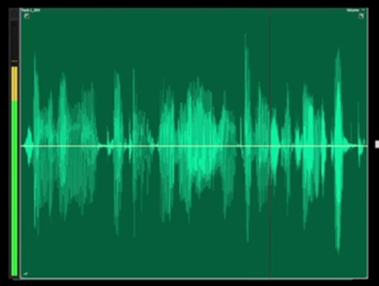
The most effective way to learn computer science is to actively engage with the material, practice regularly, and seek help

the moss and leaves, like the more fortunate hares who could find a living there. Chateau and hut, stone face and dangling figure, the red stain on the stone floor, and the pure water in the village well--thousands of acres of land--a whole province of France--all France itself--lay under the





voice-to-text



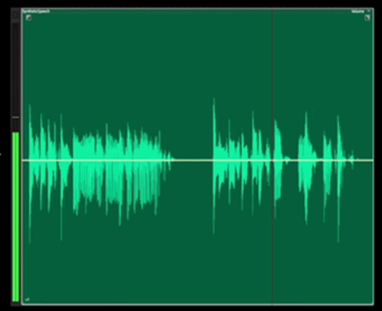
Transformer

Some models take in audio and produce a transcript

text-to-voice

This sentence comes from a model going the other way around, producing synthetic speech just from text.

Transformer



text-to-image

1960s photograph of a cute fluffy blue wild pi creature, a creature whose body is shaped like the symbol π, who is foraging in its native territory, staring back at the camera with an exotic scene in the background.

Transformer

Generated by DALL-E 3

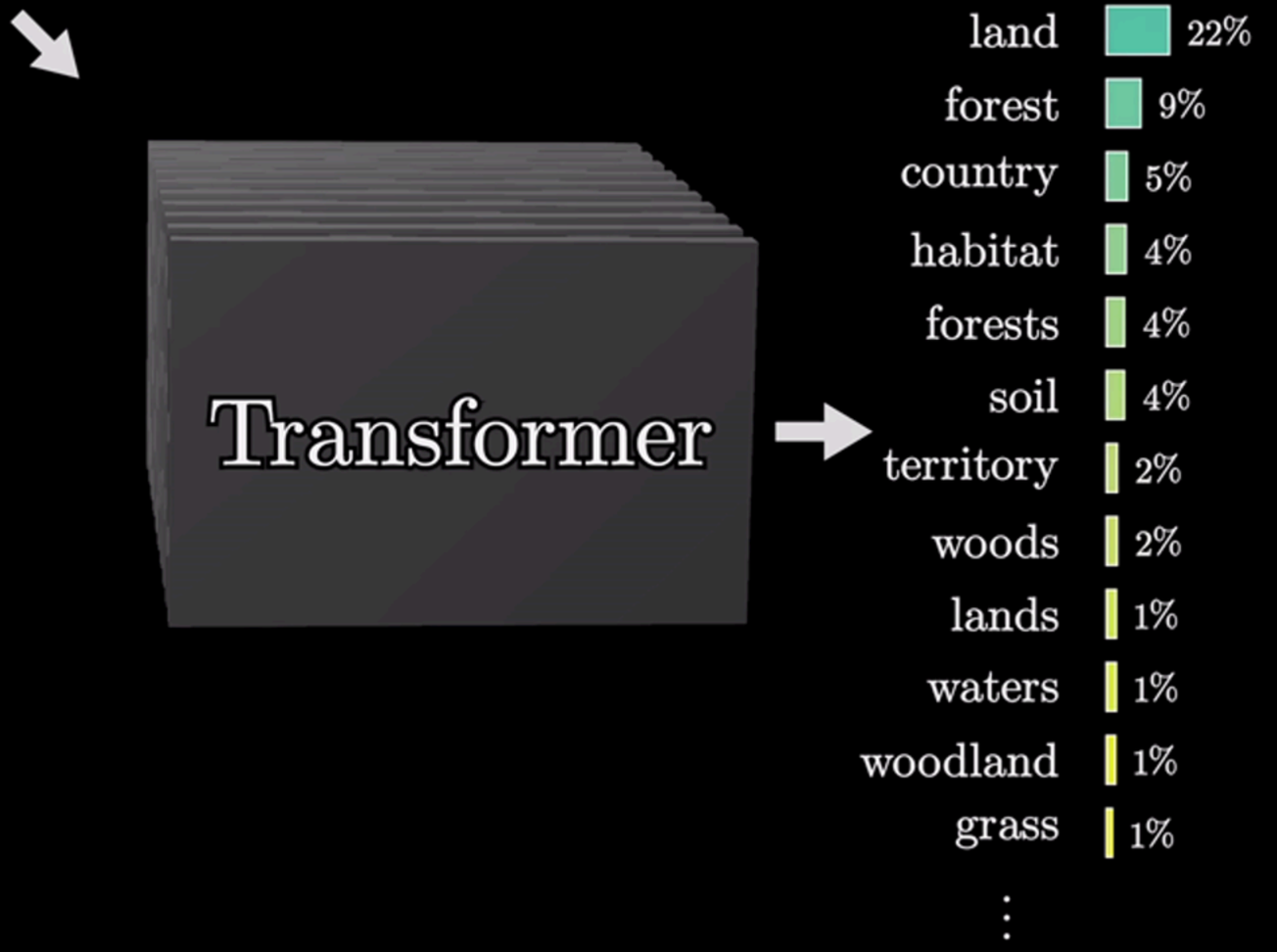
machine translation

Attention is all you need

Transformer

注意力就是你所需要的一切

Behold, a wild pi creature,
foraging in its native _____



The Self-Attention

a big blue cat chases the small mouse

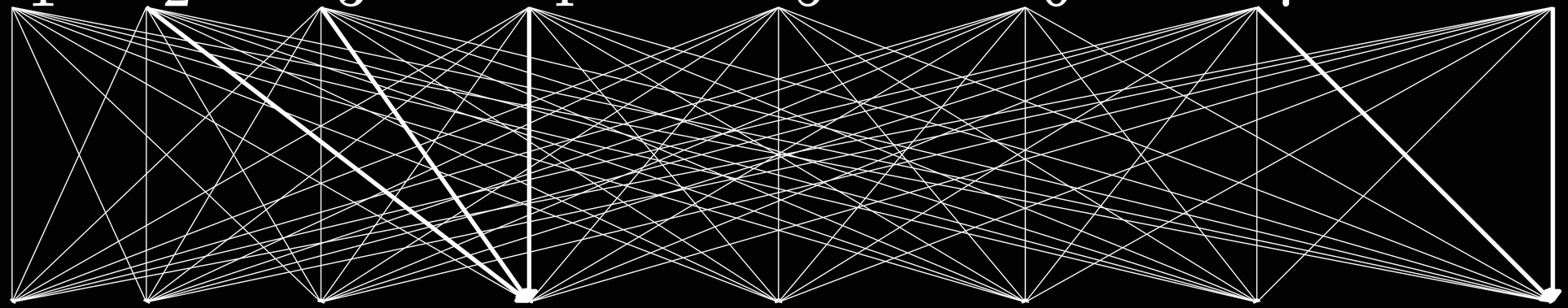
Position

1	2	3	4	5	6	7	8
↓	↓	↓	↓	↓	↓	↓	↓
5.4	3.1	9.2	8.2	3.6	5.6	5.7	8.7
7.1	4.1	3.2	0.0	4.3	4.3	2.2	5.8
6.0	0.6	6.6	6.7	6.9	9.8	9.4	8.7
5.4	6.9	1.3	2.7	0.6	1.0	4.4	6.9
4.2	5.6	7.1	7.3	6.6	2.1	8.4	7.2
6.4	2.6	2.9	9.5	6.6	1.6	6.9	5.0
4.3	5.2	1.8	2.5	2.1	6.5	2.9	9.5
8.8	0.9	5.8	5.7	1.3	2.5	8.1	6.4
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
3.8	5.7	0.2	5.9	3.6	2.4	3.9	4.2
→	→	→	→	→	→	→	→
E_1	E_2	E_3	E_4	E_5	E_6	E_7	E_8

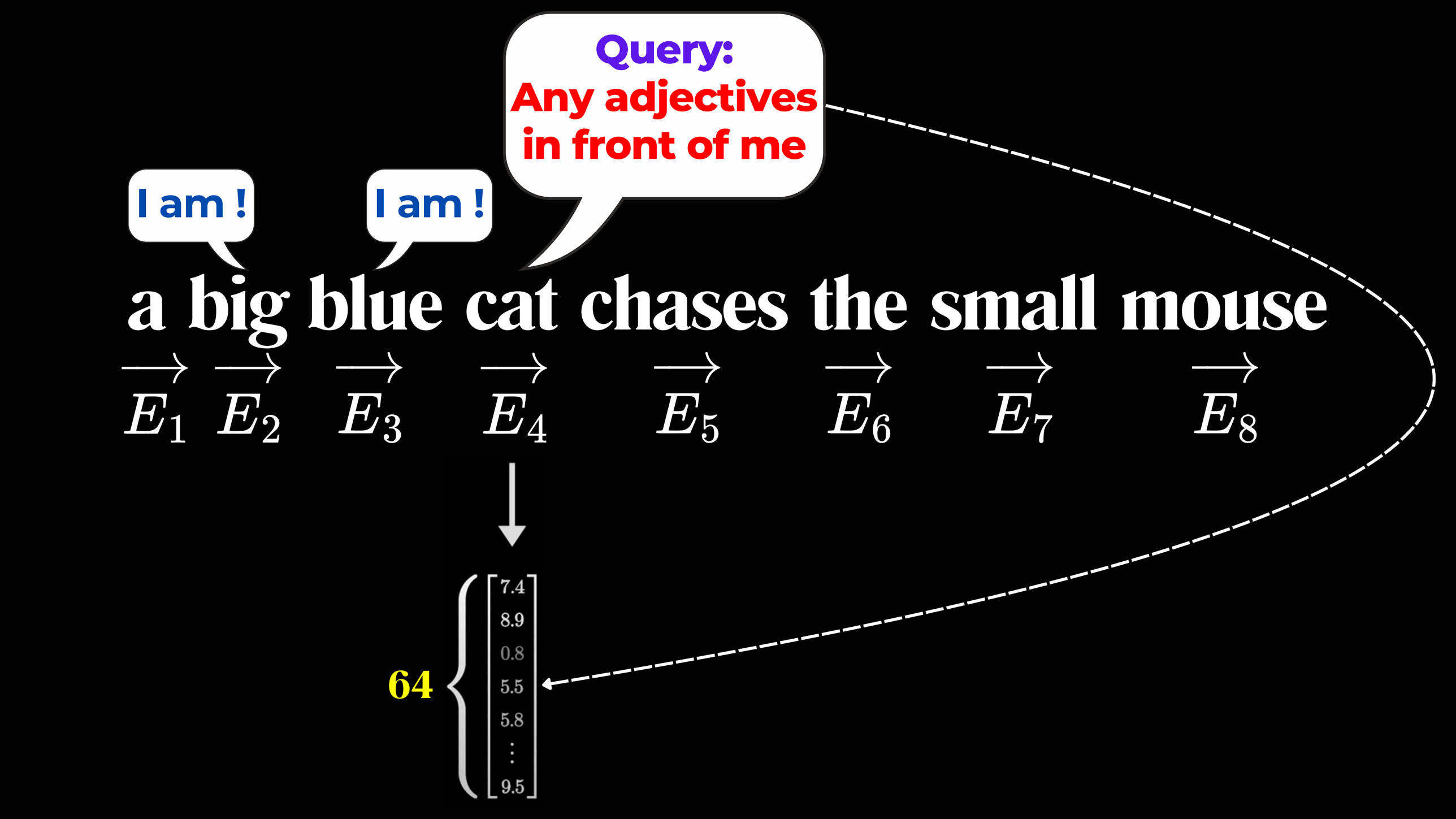
512

a big blue cat chases the small mouse

$\overrightarrow{E_1}$ $\overrightarrow{E_2}$ $\overrightarrow{E_3}$ $\overrightarrow{E_4}$ $\overrightarrow{E_5}$ $\overrightarrow{E_6}$ $\overrightarrow{E_7}$ $\overrightarrow{E_8}$



$\overrightarrow{E_1}$ $\overrightarrow{E_2}$ $\overrightarrow{E_3}$ $\overrightarrow{E_4}$ $\overrightarrow{E_5}$ $\overrightarrow{E_6}$ $\overrightarrow{E_7}$ $\overrightarrow{E_8}$


Query:
Any adjectives in front of me
I am !
I am !
a big blue cat chases the small mouse
$\overrightarrow{E_1}$ $\overrightarrow{E_2}$ $\overrightarrow{E_3}$ $\overrightarrow{E_4}$ $\overrightarrow{E_5}$ $\overrightarrow{E_6}$ $\overrightarrow{E_7}$ $\overrightarrow{E_8}$
64
7.4
8.9
0.8
5.5
5.8
9.5

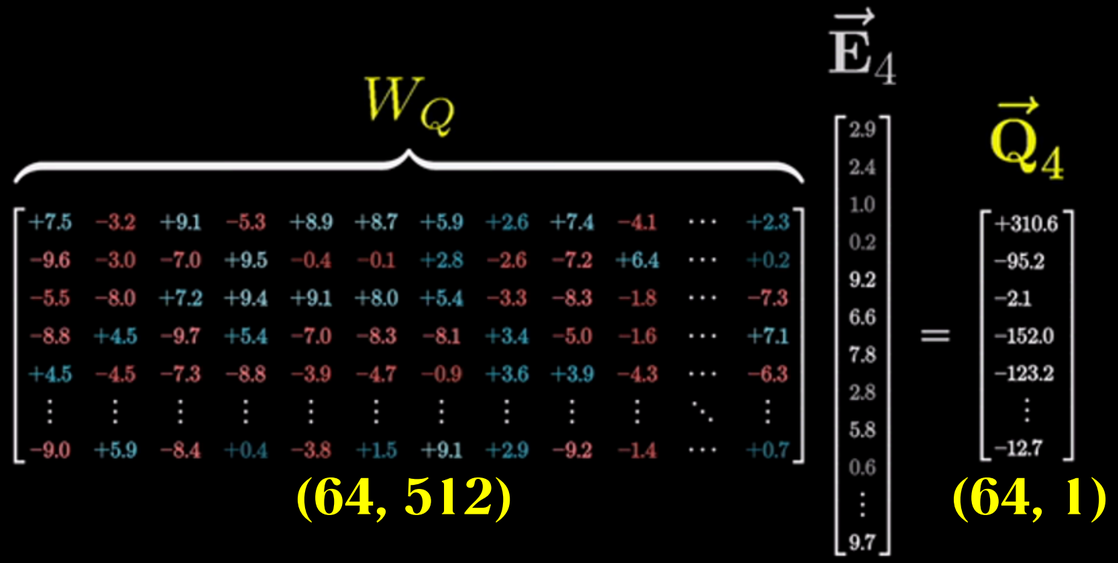
a big blue cat chases the small mouse

$$\overrightarrow{E_1} \quad \overrightarrow{E_2} \quad \overrightarrow{E_3} \quad \overrightarrow{E_4} \quad \overrightarrow{E_5} \quad \overrightarrow{E_6} \quad \overrightarrow{E_7} \quad \overrightarrow{E_8}$$

W_Q

$\vec{Q}_4$

Any adjectives
in front of me?



a big blue cat chases the small mouse

$\overrightarrow{E_1}$ $\overrightarrow{E_2}$ $\overrightarrow{E_3}$ $\overrightarrow{E_4}$ $\overrightarrow{E_5}$ $\overrightarrow{E_6}$ $\overrightarrow{E_7}$ $\overrightarrow{E_8}$

W_Q

$\vec{Q}_1$ $\vec{Q}_2$ $\vec{Q}_3$ $\vec{Q}_4$ $\vec{Q}_5$ $\vec{Q}_6$ $\vec{Q}_7$ $\vec{Q}_8$

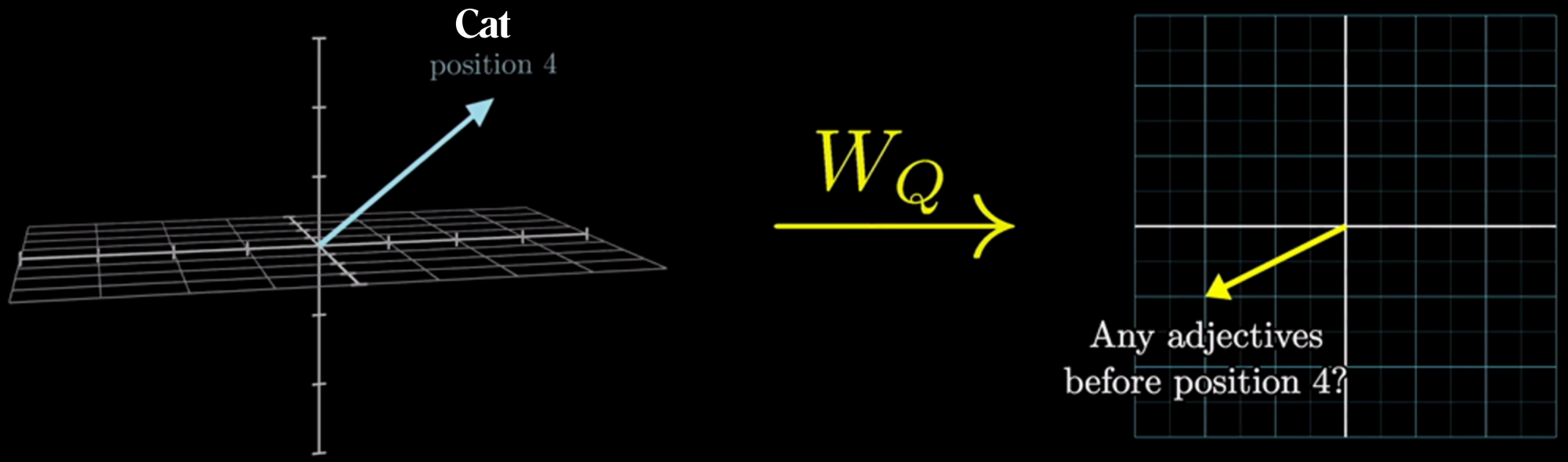
Any adjectives
in front of me?

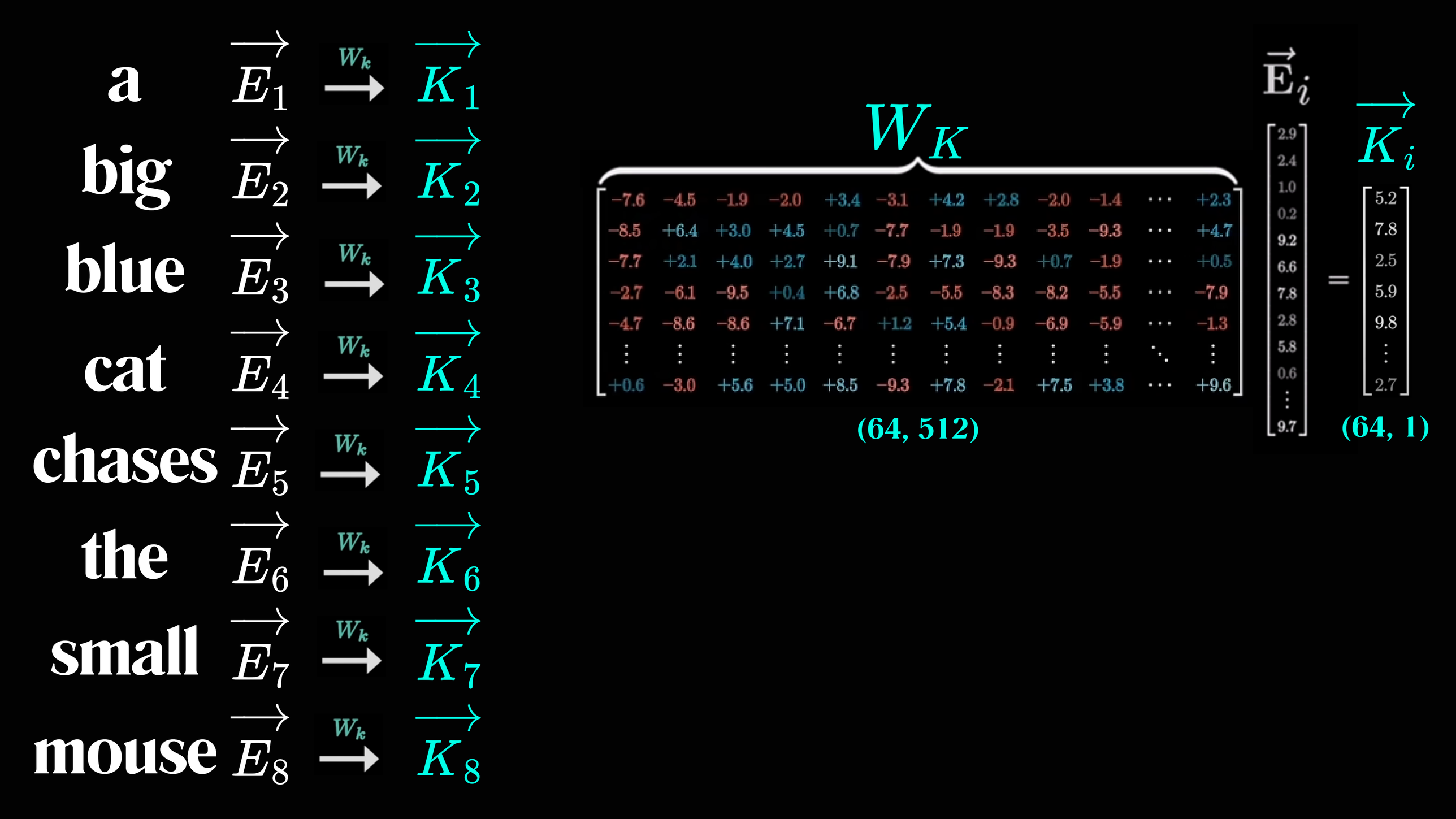
$$W_Q \begin{bmatrix} +0.9 & -0.9 & +7.6 & -0.8 & +4.4 & -2.0 & +8.0 & +3.8 & +4.0 & -3.4 & \cdots & +5.1 \\ +2.7 & -5.1 & -6.7 & +5.9 & +9.1 & -0.8 & +1.8 & +7.1 & -0.8 & +8.9 & \cdots & +1.5 \\ +6.4 & +8.1 & +6.2 & -6.7 & +2.6 & -2.0 & -8.7 & -1.5 & -4.8 & +6.9 & \cdots & -9.2 \\ +9.1 & -2.9 & -2.8 & -9.6 & -6.2 & -2.0 & +8.5 & -7.9 & +8.8 & +7.3 & \cdots & -0.9 \\ -3.4 & -5.3 & +2.3 & -9.2 & -9.6 & -1.4 & -8.6 & -4.9 & -5.5 & -4.9 & \cdots & -7.3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ -9.7 & -7.6 & +2.3 & +9.4 & +9.7 & -1.8 & -6.7 & +2.7 & -0.2 & +9.7 & \cdots & -8.6 \end{bmatrix} \vec{\mathbf{E}}_i \begin{bmatrix} 2.9 \\ 2.4 \\ 1.0 \\ 0.2 \\ 9.2 \\ 6.6 \\ 7.8 \\ 2.8 \\ 5.8 \\ 0.6 \\ \vdots \\ 9.7 \end{bmatrix} = \vec{\mathbf{Q}}_i \begin{bmatrix} +310.6 \\ -95.2 \\ -2.1 \\ -152.0 \\ -123.2 \\ \vdots \\ -12.7 \end{bmatrix}$$

$$W_Q \begin{bmatrix} +7.5 & -3.2 & +9.1 & -5.3 & +8.9 & +8.7 & +5.9 & +2.6 & +7.4 & -4.1 & \cdots & +2.3 \\ -9.6 & -3.0 & -7.0 & +9.5 & -0.4 & -0.1 & +2.8 & -2.6 & -7.2 & +6.4 & \cdots & +0.2 \\ -5.5 & -8.0 & +7.2 & +9.4 & +9.1 & +8.0 & +5.4 & -3.3 & -8.3 & -1.8 & \cdots & -7.3 \\ -8.8 & +4.5 & -9.7 & +5.4 & -7.0 & -8.3 & -8.1 & +3.4 & -5.0 & -1.6 & \cdots & +7.1 \\ +4.5 & -4.5 & -7.3 & -8.8 & -3.9 & -4.7 & -0.9 & +3.6 & +3.9 & -4.3 & \cdots & -6.3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ -9.0 & +5.9 & -8.4 & +0.4 & -3.8 & +1.5 & +9.1 & +2.9 & -9.2 & -1.4 & \cdots & +0.7 \end{bmatrix} \vec{E}_4 \begin{bmatrix} 2.9 \\ 2.4 \\ 1.0 \\ 0.2 \\ 9.2 \\ 6.6 \\ 7.8 \\ 2.8 \\ 5.8 \\ 0.6 \\ \vdots \\ 9.7 \end{bmatrix} = \vec{Q}_4 \begin{bmatrix} +310.6 \\ -95.2 \\ -2.1 \\ -152.0 \\ -123.2 \\ \vdots \\ -12.7 \end{bmatrix}$$

Embedding space (512)

Query space (64)

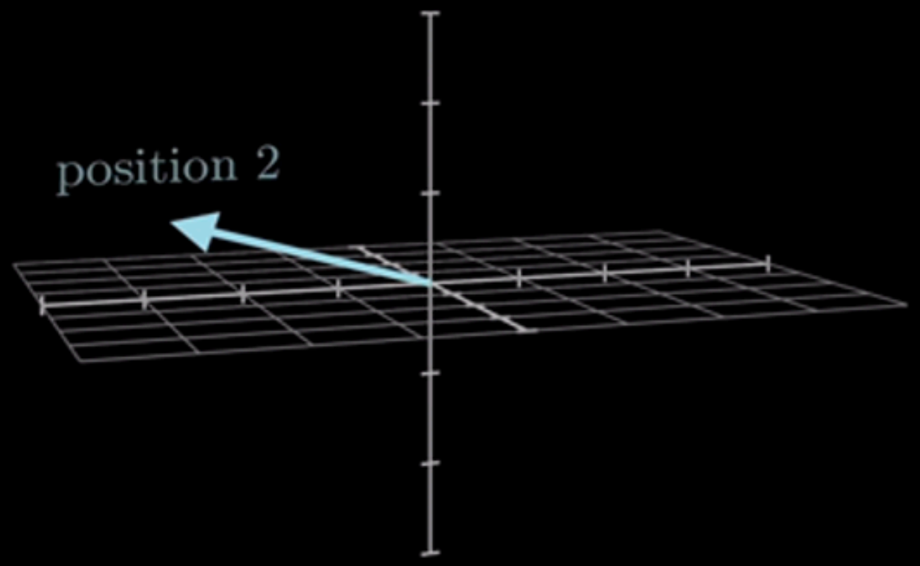



a $\overrightarrow{E_1}$ W_k $\overrightarrow{K_1}$
big $\overrightarrow{E_2}$ W_k $\overrightarrow{K_2}$
blue $\overrightarrow{E_3}$ W_k $\overrightarrow{K_3}$
cat $\overrightarrow{E_4}$ W_k $\overrightarrow{K_4}$
chases $\overrightarrow{E_5}$ W_k $\overrightarrow{K_5}$
the $\overrightarrow{E_6}$ W_k $\overrightarrow{K_6}$
small $\overrightarrow{E_7}$ W_k $\overrightarrow{K_7}$
mouse $\overrightarrow{E_8}$ W_k $\overrightarrow{K_8}$
W_K
−7.6 −4.5 −1.9 −2.0 +3.4 −3.1 +4.2 +2.8 −2.0 −1.4 ⋯ +2.3
−8.5 +6.4 +3.0 +4.5 +0.7 −7.7 −1.9 −1.9 −3.5 −9.3 ⋯ +4.7
−7.7 +2.1 +4.0 +2.7 +9.1 −7.9 +7.3 −9.3 +0.7 −1.9 ⋯ +0.5
−2.7 −6.1 −9.5 +0.4 +6.8 −2.5 −5.5 −8.3 −8.2 −5.5 ⋯ −7.9
−4.7 −8.6 −8.6 +7.1 −6.7 +1.2 +5.4 −0.9 −6.9 −5.9 ⋯ −1.3
⋮ ⋮ ⋮ ⋮ ⋮ ⋮ ⋮ ⋮ ⋮ ⋮ ⋱ ⋮
+0.6 −3.0 +5.6 +5.0 +8.5 −9.3 +7.8 −2.1 +7.5 +3.8 ⋯ +9.6
(64, 512)
$\vec{E}_i$
2.9
2.4
1.0
0.2
9.2
6.6
7.8
2.8
5.8
0.6
⋮
9.7
=
$\overrightarrow{K_i}$
5.2
7.8
2.5
5.9
9.8
⋮
2.7
(64, 1)

$$W_K \begin{bmatrix} -7.6 & -4.5 & -1.9 & -2.0 & +3.4 & -3.1 & +4.2 & +2.8 & -2.0 & -1.4 & \cdots & +2.3 \\ -8.5 & +6.4 & +3.0 & +4.5 & +0.7 & -7.7 & -1.9 & -1.9 & -3.5 & -9.3 & \cdots & +4.7 \\ -7.7 & +2.1 & +4.0 & +2.7 & +9.1 & -7.9 & +7.3 & -9.3 & +0.7 & -1.9 & \cdots & +0.5 \\ -2.7 & -6.1 & -9.5 & +0.4 & +6.8 & -2.5 & -5.5 & -8.3 & -8.2 & -5.5 & \cdots & -7.9 \\ -4.7 & -8.6 & -8.6 & +7.1 & -6.7 & +1.2 & +5.4 & -0.9 & -6.9 & -5.9 & \cdots & -1.3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ +0.6 & -3.0 & +5.6 & +5.0 & +8.5 & -9.3 & +7.8 & -2.1 & +7.5 & +3.8 & \cdots & +9.6 \end{bmatrix} \begin{bmatrix} 2.9 \\ 2.4 \\ 1.0 \\ 0.2 \\ 9.2 \\ 6.6 \\ 7.8 \\ 2.8 \\ 5.8 \\ 0.6 \\ \vdots \\ 9.7 \end{bmatrix} = \begin{bmatrix} 5.2 \\ 7.8 \\ 2.5 \\ 5.9 \\ 9.8 \\ \vdots \\ 2.7 \end{bmatrix}$$

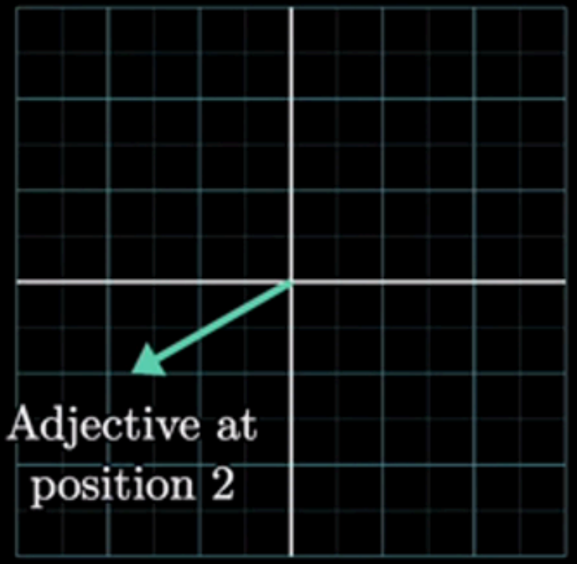
$\vec{\mathbf{E}}_i$ $\qquad$ $\overrightarrow{K_i}$

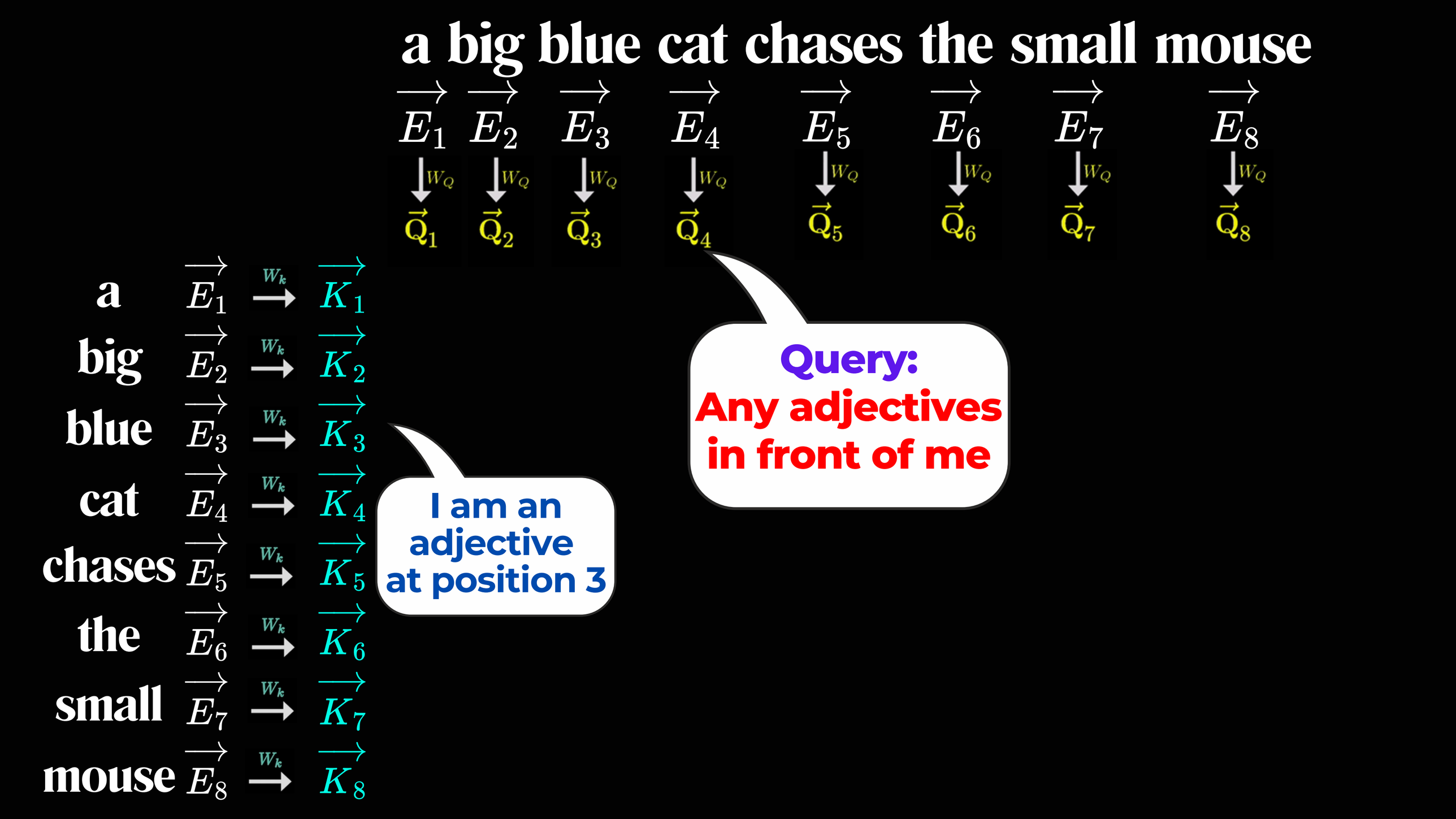
Embedding space (512)

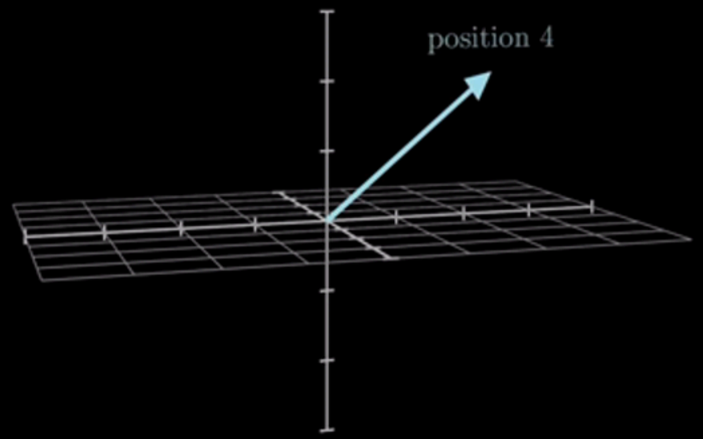


W_K

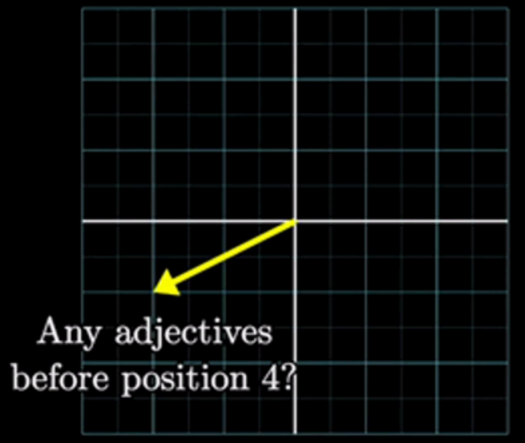
Key space (64)



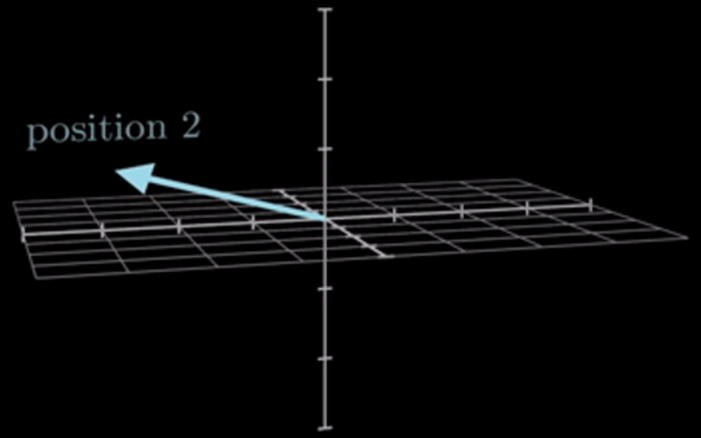

a big blue cat chases the small mouse
$\vec{E}_1$ $\vec{E}_2$ $\vec{E}_3$ $\vec{E}_4$ $\vec{E}_5$ $\vec{E}_6$ $\vec{E}_7$ $\vec{E}_8$
W_Q
$\vec{Q}_1$ $\vec{Q}_2$ $\vec{Q}_3$ $\vec{Q}_4$ $\vec{Q}_5$ $\vec{Q}_6$ $\vec{Q}_7$ $\vec{Q}_8$
Query:
Any adjectives
in front of me
a $\vec{E}_1$ W_k $\vec{K}_1$
big $\vec{E}_2$ W_k $\vec{K}_2$
blue $\vec{E}_3$ W_k $\vec{K}_3$
cat $\vec{E}_4$ W_k $\vec{K}_4$
chases $\vec{E}_5$ W_k $\vec{K}_5$
the $\vec{E}_6$ W_k $\vec{K}_6$
small $\vec{E}_7$ W_k $\vec{K}_7$
mouse $\vec{E}_8$ W_k $\vec{K}_8$
I am an
adjective
at position 3


position 4

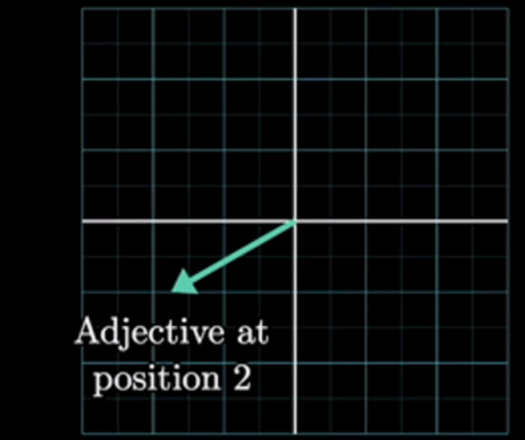
W_Q


Any adjectives
before position 4?

Adjective at
position 2


position 2

W_K

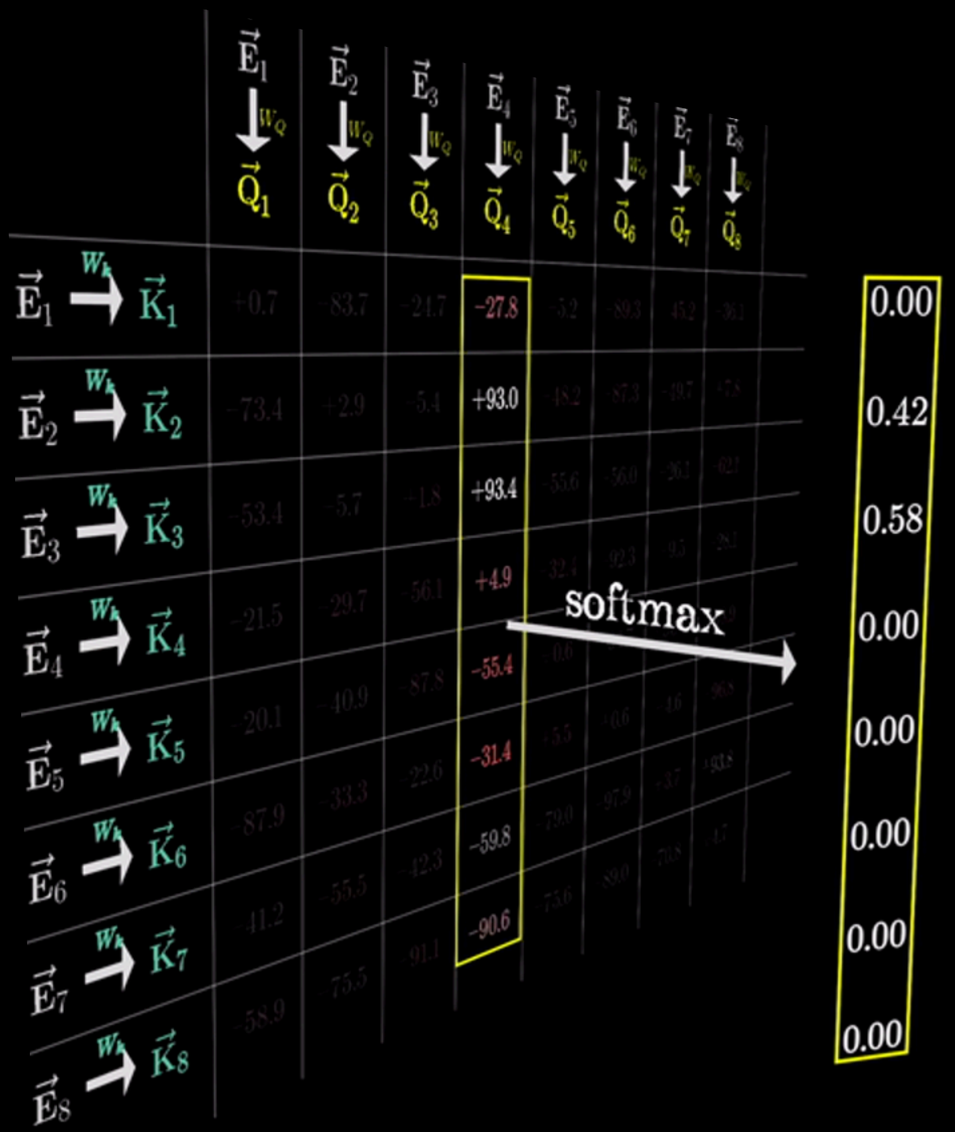

Adjective at
position 2

	a	big	blue	cat	chases	the	small	mouse
	$\vec{E}_1 \xrightarrow{W_Q} \vec{Q}_1$	$\vec{E}_2 \xrightarrow{W_Q} \vec{Q}_2$	$\vec{E}_3 \xrightarrow{W_Q} \vec{Q}_3$	$\vec{E}_4 \xrightarrow{W_Q} \vec{Q}_4$	$\vec{E}_5 \xrightarrow{W_Q} \vec{Q}_5$	$\vec{E}_6 \xrightarrow{W_Q} \vec{Q}_6$	$\vec{E}_7 \xrightarrow{W_Q} \vec{Q}_7$	$\vec{E}_8 \xrightarrow{W_Q} \vec{Q}_8$
a $\rightarrow \vec{E}_1 \xrightarrow{W_K} \vec{K}_1$	$\vec{K}_1 \cdot \vec{Q}_1$	$\vec{K}_1 \cdot \vec{Q}_2$	$\vec{K}_1 \cdot \vec{Q}_3$	$\vec{K}_1 \cdot \vec{Q}_4$	$\vec{K}_1 \cdot \vec{Q}_5$	$\vec{K}_1 \cdot \vec{Q}_6$	$\vec{K}_1 \cdot \vec{Q}_7$	$\vec{K}_1 \cdot \vec{Q}_8$
big $\rightarrow \vec{E}_2 \xrightarrow{W_K} \vec{K}_2$	$\vec{K}_2 \cdot \vec{Q}_1$	$\vec{K}_2 \cdot \vec{Q}_2$	$\vec{K}_2 \cdot \vec{Q}_3$	$\vec{K}_2 \cdot \vec{Q}_4$	$\vec{K}_2 \cdot \vec{Q}_5$	$\vec{K}_2 \cdot \vec{Q}_6$	$\vec{K}_2 \cdot \vec{Q}_7$	$\vec{K}_2 \cdot \vec{Q}_8$
blue $\rightarrow \vec{E}_3 \xrightarrow{W_K} \vec{K}_3$	$\vec{K}_3 \cdot \vec{Q}_1$	$\vec{K}_3 \cdot \vec{Q}_2$	$\vec{K}_3 \cdot \vec{Q}_3$	$\vec{K}_3 \cdot \vec{Q}_4$	$\vec{K}_3 \cdot \vec{Q}_5$	$\vec{K}_3 \cdot \vec{Q}_6$	$\vec{K}_3 \cdot \vec{Q}_7$	$\vec{K}_3 \cdot \vec{Q}_8$
cat $\rightarrow \vec{E}_4 \xrightarrow{W_K} \vec{K}_4$	$\vec{K}_4 \cdot \vec{Q}_1$	$\vec{K}_4 \cdot \vec{Q}_2$	$\vec{K}_4 \cdot \vec{Q}_3$	$\vec{K}_4 \cdot \vec{Q}_4$	$\vec{K}_4 \cdot \vec{Q}_5$	$\vec{K}_4 \cdot \vec{Q}_6$	$\vec{K}_4 \cdot \vec{Q}_7$	$\vec{K}_4 \cdot \vec{Q}_8$
chases $\rightarrow \vec{E}_5 \xrightarrow{W_K} \vec{K}_5$	$\vec{K}_5 \cdot \vec{Q}_1$	$\vec{K}_5 \cdot \vec{Q}_2$	$\vec{K}_5 \cdot \vec{Q}_3$	$\vec{K}_5 \cdot \vec{Q}_4$	$\vec{K}_5 \cdot \vec{Q}_5$	$\vec{K}_5 \cdot \vec{Q}_6$	$\vec{K}_5 \cdot \vec{Q}_7$	$\vec{K}_5 \cdot \vec{Q}_8$
the $\rightarrow \vec{E}_6 \xrightarrow{W_K} \vec{K}_6$	$\vec{K}_6 \cdot \vec{Q}_1$	$\vec{K}_6 \cdot \vec{Q}_2$	$\vec{K}_6 \cdot \vec{Q}_3$	$\vec{K}_6 \cdot \vec{Q}_4$	$\vec{K}_6 \cdot \vec{Q}_5$	$\vec{K}_6 \cdot \vec{Q}_6$	$\vec{K}_6 \cdot \vec{Q}_7$	$\vec{K}_6 \cdot \vec{Q}_8$
small $\rightarrow \vec{E}_7 \xrightarrow{W_K} \vec{K}_7$	$\vec{K}_7 \cdot \vec{Q}_1$	$\vec{K}_7 \cdot \vec{Q}_2$	$\vec{K}_7 \cdot \vec{Q}_3$	$\vec{K}_7 \cdot \vec{Q}_4$	$\vec{K}_7 \cdot \vec{Q}_5$	$\vec{K}_7 \cdot \vec{Q}_6$	$\vec{K}_7 \cdot \vec{Q}_7$	$\vec{K}_7 \cdot \vec{Q}_8$
mouse $\rightarrow \vec{E}_8 \xrightarrow{W_K} \vec{K}_8$	$\vec{K}_8 \cdot \vec{Q}_1$	$\vec{K}_8 \cdot \vec{Q}_2$	$\vec{K}_8 \cdot \vec{Q}_3$	$\vec{K}_8 \cdot \vec{Q}_4$	$\vec{K}_8 \cdot \vec{Q}_5$	$\vec{K}_8 \cdot \vec{Q}_6$	$\vec{K}_8 \cdot \vec{Q}_7$	$\vec{K}_8 \cdot \vec{Q}_8$

	a	big	blue	cat	chases	the	small	mouse
	$\vec{E}_1 \downarrow W_Q \vec{Q}_1$	$\vec{E}_2 \downarrow W_Q \vec{Q}_2$	$\vec{E}_3 \downarrow W_Q \vec{Q}_3$	$\vec{E}_4 \downarrow W_Q \vec{Q}_4$	$\vec{E}_5 \downarrow W_Q \vec{Q}_5$	$\vec{E}_6 \downarrow W_Q \vec{Q}_6$	$\vec{E}_7 \downarrow W_Q \vec{Q}_7$	$\vec{E}_8 \downarrow W_Q \vec{Q}_8$
a $\rightarrow \vec{E}_1 \xrightarrow{W_k} \vec{K}_1$	$\vec{K}_1 \bullet \vec{Q}_1$	$\vec{K}_1 \bullet \vec{Q}_2$	$\vec{K}_1 \bullet \vec{Q}_3$	$\vec{K}_1 \bullet \vec{Q}_4$	$\vec{K}_1 \bullet \vec{Q}_5$	$\vec{K}_1 \bullet \vec{Q}_6$	$\vec{K}_1 \bullet \vec{Q}_7$	$\vec{K}_1 \bullet \vec{Q}_8$
big $\rightarrow \vec{E}_2 \xrightarrow{W_k} \vec{K}_2$	$\vec{K}_2 \bullet \vec{Q}_1$	$\vec{K}_2 \bullet \vec{Q}_2$	$\vec{K}_2 \bullet \vec{Q}_3$	$\vec{K}_2 \bullet \vec{Q}_4$	$\vec{K}_2 \bullet \vec{Q}_5$	$\vec{K}_2 \bullet \vec{Q}_6$	$\vec{K}_2 \bullet \vec{Q}_7$	$\vec{K}_2 \bullet \vec{Q}_8$
blue $\rightarrow \vec{E}_3 \xrightarrow{W_k} \vec{K}_3$	$\vec{K}_3 \bullet \vec{Q}_1$	$\vec{K}_3 \bullet \vec{Q}_2$	$\vec{K}_3 \bullet \vec{Q}_3$	$\vec{K}_3 \bullet \vec{Q}_4$	$\vec{K}_3 \bullet \vec{Q}_5$	$\vec{K}_3 \bullet \vec{Q}_6$	$\vec{K}_3 \bullet \vec{Q}_7$	$\vec{K}_3 \bullet \vec{Q}_8$
cat $\rightarrow \vec{E}_4 \xrightarrow{W_k} \vec{K}_4$	$\vec{K}_4 \bullet \vec{Q}_1$	$\vec{K}_4 \bullet \vec{Q}_2$	$\vec{K}_4 \bullet \vec{Q}_3$	$\vec{K}_4 \bullet \vec{Q}_4$	$\vec{K}_4 \bullet \vec{Q}_5$	$\vec{K}_4 \bullet \vec{Q}_6$	$\vec{K}_4 \bullet \vec{Q}_7$	$\vec{K}_4 \bullet \vec{Q}_8$
chases $\rightarrow \vec{E}_5 \xrightarrow{W_k} \vec{K}_5$	$\vec{K}_5 \bullet \vec{Q}_1$	$\vec{K}_5 \bullet \vec{Q}_2$	$\vec{K}_5 \bullet \vec{Q}_3$	$\vec{K}_5 \bullet \vec{Q}_4$	$\vec{K}_5 \bullet \vec{Q}_5$	$\vec{K}_5 \bullet \vec{Q}_6$	$\vec{K}_5 \bullet \vec{Q}_7$	$\vec{K}_5 \bullet \vec{Q}_8$
the $\rightarrow \vec{E}_6 \xrightarrow{W_k} \vec{K}_6$	$\vec{K}_6 \bullet \vec{Q}_1$	$\vec{K}_6 \bullet \vec{Q}_2$	$\vec{K}_6 \bullet \vec{Q}_3$	$\vec{K}_6 \bullet \vec{Q}_4$	$\vec{K}_6 \bullet \vec{Q}_5$	$\vec{K}_6 \bullet \vec{Q}_6$	$\vec{K}_6 \bullet \vec{Q}_7$	$\vec{K}_6 \bullet \vec{Q}_8$
small $\rightarrow \vec{E}_7 \xrightarrow{W_k} \vec{K}_7$	$\vec{K}_7 \bullet \vec{Q}_1$	$\vec{K}_7 \bullet \vec{Q}_2$	$\vec{K}_7 \bullet \vec{Q}_3$	$\vec{K}_7 \bullet \vec{Q}_4$	$\vec{K}_7 \bullet \vec{Q}_5$	$\vec{K}_7 \bullet \vec{Q}_6$	$\vec{K}_7 \bullet \vec{Q}_7$	$\vec{K}_7 \bullet \vec{Q}_8$
mouse $\rightarrow \vec{E}_8 \xrightarrow{W_k} \vec{K}_8$	$\vec{K}_8 \bullet \vec{Q}_1$	$\vec{K}_8 \bullet \vec{Q}_2$	$\vec{K}_8 \bullet \vec{Q}_3$	$\vec{K}_8 \bullet \vec{Q}_4$	$\vec{K}_8 \bullet \vec{Q}_5$	$\vec{K}_8 \bullet \vec{Q}_6$	$\vec{K}_8 \bullet \vec{Q}_7$	$\vec{K}_8 \bullet \vec{Q}_8$

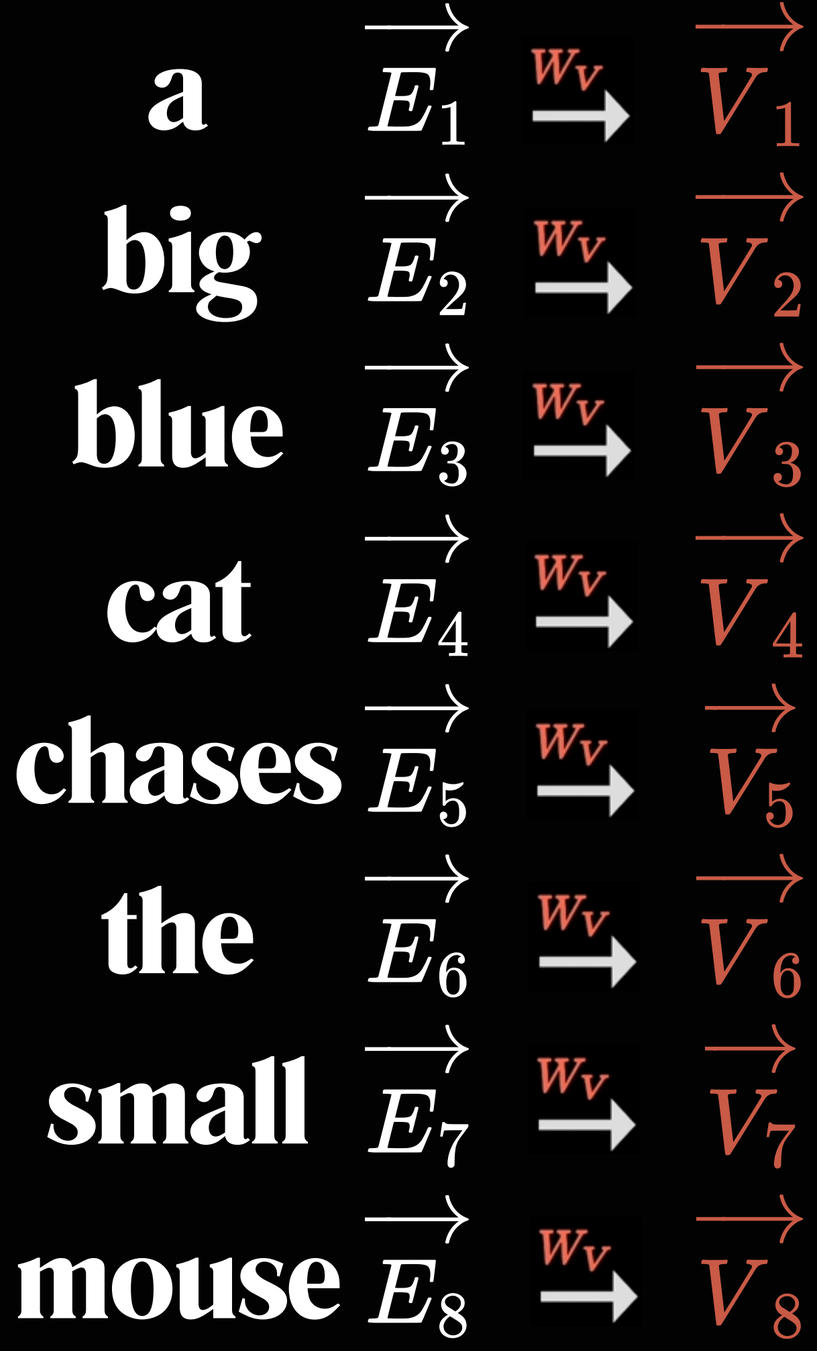
	a	big	blue	cat	chases	the	small	mouse
	$\vec{E}_1 \xrightarrow{W_Q} \vec{Q}_1$	$\vec{E}_2 \xrightarrow{W_Q} \vec{Q}_2$	$\vec{E}_3 \xrightarrow{W_Q} \vec{Q}_3$	$\vec{E}_4 \xrightarrow{W_Q} \vec{Q}_4$	$\vec{E}_5 \xrightarrow{W_Q} \vec{Q}_5$	$\vec{E}_6 \xrightarrow{W_Q} \vec{Q}_6$	$\vec{E}_7 \xrightarrow{W_Q} \vec{Q}_7$	$\vec{E}_8 \xrightarrow{W_Q} \vec{Q}_8$
a $\rightarrow \vec{E}_1 \xrightarrow{W_k} \vec{K}_1$	$\frac{\vec{K}_1 \cdot \vec{Q}_1}{\sqrt{d_k}}$	$\frac{\vec{K}_1 \cdot \vec{Q}_2}{\sqrt{d_k}}$	$\frac{\vec{K}_1 \cdot \vec{Q}_3}{\sqrt{d_k}}$	$\frac{\vec{K}_1 \cdot \vec{Q}_4}{\sqrt{d_k}}$	$\frac{\vec{K}_1 \cdot \vec{Q}_5}{\sqrt{d_k}}$	$\frac{\vec{K}_1 \cdot \vec{Q}_6}{\sqrt{d_k}}$	$\frac{\vec{K}_1 \cdot \vec{Q}_7}{\sqrt{d_k}}$	$\frac{\vec{K}_1 \cdot \vec{Q}_8}{\sqrt{d_k}}$
big $\rightarrow \vec{E}_2 \xrightarrow{W_k} \vec{K}_2$	$\frac{\vec{K}_2 \cdot \vec{Q}_1}{\sqrt{d_k}}$	$\frac{\vec{K}_2 \cdot \vec{Q}_2}{\sqrt{d_k}}$	$\frac{\vec{K}_2 \cdot \vec{Q}_3}{\sqrt{d_k}}$	$\frac{\vec{K}_2 \cdot \vec{Q}_4}{\sqrt{d_k}}$	$\frac{\vec{K}_2 \cdot \vec{Q}_5}{\sqrt{d_k}}$	$\frac{\vec{K}_2 \cdot \vec{Q}_6}{\sqrt{d_k}}$	$\frac{\vec{K}_2 \cdot \vec{Q}_7}{\sqrt{d_k}}$	$\frac{\vec{K}_2 \cdot \vec{Q}_8}{\sqrt{d_k}}$
blue $\rightarrow \vec{E}_3 \xrightarrow{W_k} \vec{K}_3$	$\frac{\vec{K}_3 \cdot \vec{Q}_1}{\sqrt{d_k}}$	$\frac{\vec{K}_3 \cdot \vec{Q}_2}{\sqrt{d_k}}$	$\frac{\vec{K}_3 \cdot \vec{Q}_3}{\sqrt{d_k}}$	$\frac{\vec{K}_3 \cdot \vec{Q}_4}{\sqrt{d_k}}$	$\frac{\vec{K}_3 \cdot \vec{Q}_5}{\sqrt{d_k}}$	$\frac{\vec{K}_3 \cdot \vec{Q}_6}{\sqrt{d_k}}$	$\frac{\vec{K}_3 \cdot \vec{Q}_7}{\sqrt{d_k}}$	$\frac{\vec{K}_3 \cdot \vec{Q}_8}{\sqrt{d_k}}$
cat $\rightarrow \vec{E}_4 \xrightarrow{W_k} \vec{K}_4$	$\frac{\vec{K}_4 \cdot \vec{Q}_1}{\sqrt{d_k}}$	$\frac{\vec{K}_4 \cdot \vec{Q}_2}{\sqrt{d_k}}$	$\frac{\vec{K}_4 \cdot \vec{Q}_3}{\sqrt{d_k}}$	$\frac{\vec{K}_4 \cdot \vec{Q}_4}{\sqrt{d_k}}$	$\frac{\vec{K}_4 \cdot \vec{Q}_5}{\sqrt{d_k}}$	$\frac{\vec{K}_4 \cdot \vec{Q}_6}{\sqrt{d_k}}$	$\frac{\vec{K}_4 \cdot \vec{Q}_7}{\sqrt{d_k}}$	$\frac{\vec{K}_4 \cdot \vec{Q}_8}{\sqrt{d_k}}$
chases $\rightarrow \vec{E}_5 \xrightarrow{W_k} \vec{K}_5$	$\frac{\vec{K}_5 \cdot \vec{Q}_1}{\sqrt{d_k}}$	$\frac{\vec{K}_5 \cdot \vec{Q}_2}{\sqrt{d_k}}$	$\frac{\vec{K}_5 \cdot \vec{Q}_3}{\sqrt{d_k}}$	$\frac{\vec{K}_5 \cdot \vec{Q}_4}{\sqrt{d_k}}$	$\frac{\vec{K}_5 \cdot \vec{Q}_5}{\sqrt{d_k}}$	$\frac{\vec{K}_5 \cdot \vec{Q}_6}{\sqrt{d_k}}$	$\frac{\vec{K}_5 \cdot \vec{Q}_7}{\sqrt{d_k}}$	$\frac{\vec{K}_5 \cdot \vec{Q}_8}{\sqrt{d_k}}$
the $\rightarrow \vec{E}_6 \xrightarrow{W_k} \vec{K}_6$	$\frac{\vec{K}_6 \cdot \vec{Q}_1}{\sqrt{d_k}}$	$\frac{\vec{K}_6 \cdot \vec{Q}_2}{\sqrt{d_k}}$	$\frac{\vec{K}_6 \cdot \vec{Q}_3}{\sqrt{d_k}}$	$\frac{\vec{K}_6 \cdot \vec{Q}_4}{\sqrt{d_k}}$	$\frac{\vec{K}_6 \cdot \vec{Q}_5}{\sqrt{d_k}}$	$\frac{\vec{K}_6 \cdot \vec{Q}_6}{\sqrt{d_k}}$	$\frac{\vec{K}_6 \cdot \vec{Q}_7}{\sqrt{d_k}}$	$\frac{\vec{K}_6 \cdot \vec{Q}_8}{\sqrt{d_k}}$
small $\rightarrow \vec{E}_7 \xrightarrow{W_k} \vec{K}_7$	$\frac{\vec{K}_7 \cdot \vec{Q}_1}{\sqrt{d_k}}$	$\frac{\vec{K}_7 \cdot \vec{Q}_2}{\sqrt{d_k}}$	$\frac{\vec{K}_7 \cdot \vec{Q}_3}{\sqrt{d_k}}$	$\frac{\vec{K}_7 \cdot \vec{Q}_4}{\sqrt{d_k}}$	$\frac{\vec{K}_7 \cdot \vec{Q}_5}{\sqrt{d_k}}$	$\frac{\vec{K}_7 \cdot \vec{Q}_6}{\sqrt{d_k}}$	$\frac{\vec{K}_7 \cdot \vec{Q}_7}{\sqrt{d_k}}$	$\frac{\vec{K}_7 \cdot \vec{Q}_8}{\sqrt{d_k}}$
mouse $\rightarrow \vec{E}_8 \xrightarrow{W_k} \vec{K}_8$	$\frac{\vec{K}_8 \cdot \vec{Q}_1}{\sqrt{d_k}}$	$\frac{\vec{K}_8 \cdot \vec{Q}_2}{\sqrt{d_k}}$	$\frac{\vec{K}_8 \cdot \vec{Q}_3}{\sqrt{d_k}}$	$\frac{\vec{K}_8 \cdot \vec{Q}_4}{\sqrt{d_k}}$	$\frac{\vec{K}_8 \cdot \vec{Q}_5}{\sqrt{d_k}}$	$\frac{\vec{K}_8 \cdot \vec{Q}_6}{\sqrt{d_k}}$	$\frac{\vec{K}_8 \cdot \vec{Q}_7}{\sqrt{d_k}}$	$\frac{\vec{K}_8 \cdot \vec{Q}_8}{\sqrt{d_k}}$

	a $\downarrow \vec{E}_1 \downarrow W_Q \vec{Q}_1$	big $\downarrow \vec{E}_2 \downarrow W_Q \vec{Q}_2$	blue $\downarrow \vec{E}_3 \downarrow W_Q \vec{Q}_3$	cat $\downarrow \vec{E}_4 \downarrow W_Q \vec{Q}_4$	chases $\downarrow \vec{E}_5 \downarrow W_Q \vec{Q}_5$	the $\downarrow \vec{E}_6 \downarrow W_Q \vec{Q}_6$	small $\downarrow \vec{E}_7 \downarrow W_Q \vec{Q}_7$	mouse $\downarrow \vec{E}_8 \downarrow W_Q \vec{Q}_8$
a $\vec{E}_1 \xrightarrow{W_K} \vec{K}_1$	+0.7	−83.7	−24.7	−27.8	−5.2	−89.3	−45.2	−36.1
big $\vec{E}_2 \xrightarrow{W_K} \vec{K}_2$	−73.4	+2.9	−5.4	+93.0	−48.2	−87.3	−49.7	+7.8
blue $\vec{E}_3 \xrightarrow{W_K} \vec{K}_3$	−53.4	−5.7	+1.8	+93.4	−55.6	−56.0	−26.1	−62.1
cat $\vec{E}_4 \xrightarrow{W_K} \vec{K}_4$	−21.5	−29.7	−56.1	+4.9	−32.4	−92.3	−9.5	−28.1
chases $\vec{E}_5 \xrightarrow{W_K} \vec{K}_5$	−20.1	−40.9	−87.8	−55.4	+0.6	−64.7	−96.7	−18.9
the $\vec{E}_6 \xrightarrow{W_K} \vec{K}_6$	−87.9	−33.3	−22.6	−31.4	+5.5	+0.6	−4.6	−96.8
small $\vec{E}_7 \xrightarrow{W_K} \vec{K}_7$	−41.2	−55.5	−42.3	−59.8	−79.0	−97.9	+3.7	+93.8
mouse $\vec{E}_8 \xrightarrow{W_K} \vec{K}_8$	−58.9	−75.5	−91.1	−90.6	−75.6	−89.0	−70.8	+4.7


$\vec{E}_1 \xrightarrow{W_Q} \vec{Q}_1$
$\vec{E}_2 \xrightarrow{W_Q} \vec{Q}_2$
$\vec{E}_3 \xrightarrow{W_Q} \vec{Q}_3$
$\vec{E}_4 \xrightarrow{W_Q} \vec{Q}_4$
$\vec{E}_5 \xrightarrow{W_Q} \vec{Q}_5$
$\vec{E}_6 \xrightarrow{W_Q} \vec{Q}_6$
$\vec{E}_7 \xrightarrow{W_Q} \vec{Q}_7$
$\vec{E}_8 \xrightarrow{W_Q} \vec{Q}_8$
$\vec{E}_1 \xrightarrow{W_K} \vec{K}_1$
$\vec{E}_2 \xrightarrow{W_K} \vec{K}_2$
$\vec{E}_3 \xrightarrow{W_K} \vec{K}_3$
$\vec{E}_4 \xrightarrow{W_K} \vec{K}_4$
$\vec{E}_5 \xrightarrow{W_K} \vec{K}_5$
$\vec{E}_6 \xrightarrow{W_K} \vec{K}_6$
$\vec{E}_7 \xrightarrow{W_K} \vec{K}_7$
$\vec{E}_8 \xrightarrow{W_K} \vec{K}_8$
−0.7
−83.7
−21.7
−27.8
−73.4
+2.9
+93.0
−53.4
−5.7
−1.8
+93.4
−21.5
29.7
−56.1
+4.9
−20.1
−40.9
−87.8
−55.4
−87.9
−33.3
−22.6
−31.4
41.2
−55.5
−42.3
−59.8
−58.9
75.5
91.1
−90.6
softmax
0.00
0.42
0.58
0.00
0.00
0.00
0.00
0.00

Attention pattern

	a $\vec{Q}_1$	big $\vec{Q}_2$	blue $\vec{Q}_3$	cat $\vec{Q}_4$	chases $\vec{Q}_5$	the $\vec{Q}_6$	small $\vec{Q}_7$	mouse $\vec{Q}_8$
a $\vec{E}_1 \xrightarrow{W_K} \vec{K}_1$	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
big $\vec{E}_2 \xrightarrow{W_K} \vec{K}_2$	0.00	1.00	0.00	0.42	0.00	0.00	0.00	0.00
blue $\vec{E}_3 \xrightarrow{W_K} \vec{K}_3$	0.00	0.00	1.00	0.58	0.00	0.00	0.00	0.00
cat $\vec{E}_4 \xrightarrow{W_K} \vec{K}_4$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
chases $\vec{E}_5 \xrightarrow{W_K} \vec{K}_5$	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00
the $\vec{E}_6 \xrightarrow{W_K} \vec{K}_6$	0.00	0.00	0.00	0.00	0.99	1.00	0.00	0.00
small $\vec{E}_7 \xrightarrow{W_K} \vec{K}_7$	0.00	0.00	0.00	0.00	0.00	0.00	1.00	1.00
mouse $\vec{E}_8 \xrightarrow{W_K} \vec{K}_8$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00



$$W_V \quad \vec{\mathbf{E}}_i \quad \overrightarrow{V_i}$$

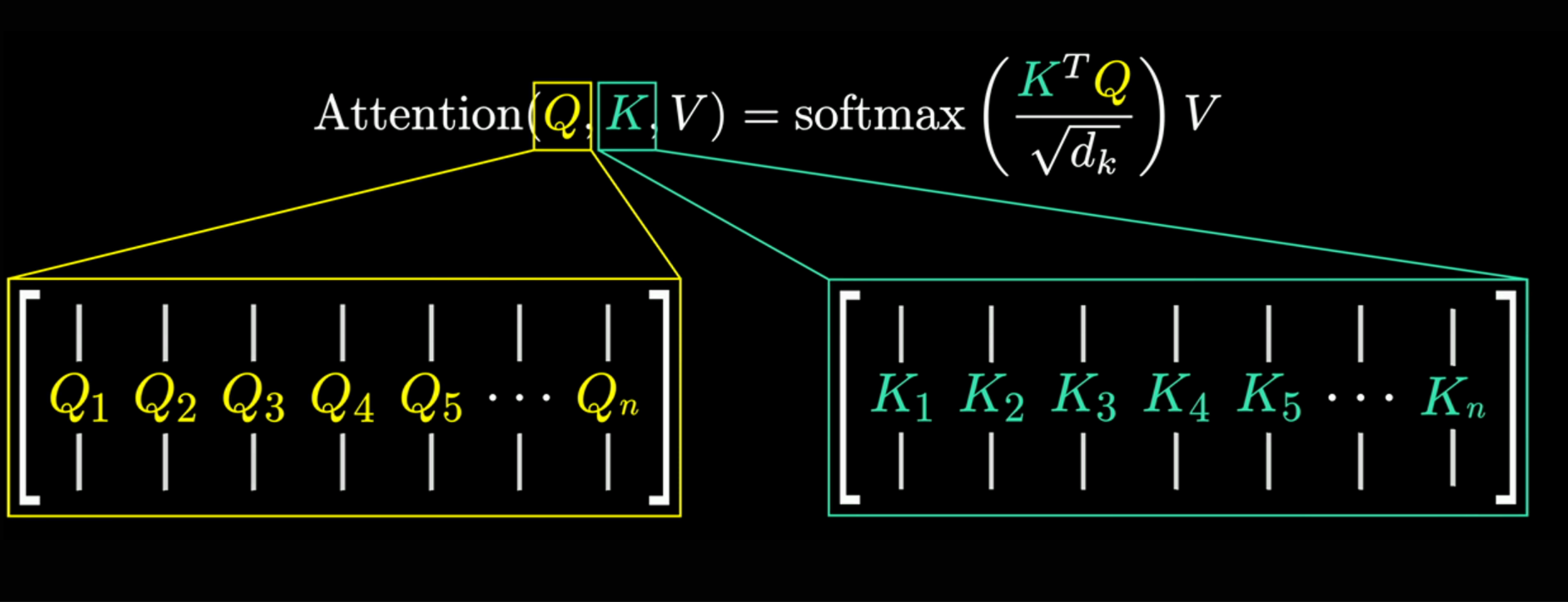
$$\begin{bmatrix} +7.3 & -3.2 & +9.1 & -5.2 & +8.8 & +8.5 & +6.0 & +2.6 & +7.3 & -4.1 & \cdots & +2.4 \\ -9.4 & -3.1 & -7.0 & +9.5 & -0.2 & -0.1 & +2.7 & -2.4 & -7.1 & +6.4 & \cdots & +0.2 \\ -5.2 & -7.7 & +7.2 & +9.1 & +9.0 & +7.9 & +5.2 & -3.3 & -8.2 & -1.7 & \cdots & -7.3 \\ -8.5 & +4.3 & -9.5 & +5.1 & -7.0 & -8.2 & -7.8 & +3.2 & -4.8 & -1.4 & \cdots & +7.0 \\ +4.3 & -4.6 & -7.1 & -8.8 & -4.0 & -4.6 & -1.0 & +3.5 & +3.7 & -4.3 & \cdots & -6.3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ -9.0 & +5.7 & -8.2 & +0.5 & -3.6 & +1.5 & +8.8 & +2.9 & -9.0 & -1.2 & \cdots & +0.5 \end{bmatrix} \begin{bmatrix} 2.9 \\ 2.4 \\ 1.0 \\ 0.2 \\ 9.2 \\ 6.6 \\ 7.8 \\ 2.8 \\ 5.8 \\ 0.6 \\ \vdots \\ 9.7 \end{bmatrix} = \begin{bmatrix} +310.6 \\ -95.2 \\ -2.1 \\ -152.0 \\ -123.2 \\ \vdots \\ -12.7 \end{bmatrix}$$

(64, 512) (64, 1)

In reality... (64, 512) (64, 1)

... but, let's assume that (512, 512) (512, 1)

	a	big	blue	cat	chases	the	small	mouse
	$\frac{\vec{K}\cdot\vec{Q}_1}{\sqrt{d_k}}$	$\frac{\vec{K}\cdot\vec{Q}_2}{\sqrt{d_k}}$	$\frac{\vec{K}\cdot\vec{Q}_3}{\sqrt{d_k}}$	$\frac{\vec{K}\cdot\vec{Q}_4}{\sqrt{d_k}}$	$\frac{\vec{K}\cdot\vec{Q}_5}{\sqrt{d_k}}$	$\frac{\vec{K}\cdot\vec{Q}_6}{\sqrt{d_k}}$	$\frac{\vec{K}\cdot\vec{Q}_7}{\sqrt{d_k}}$	$\frac{\vec{K}\cdot\vec{Q}_8}{\sqrt{d_k}}$
a $\rightarrow \vec{E}_1 \xrightarrow{W_V}$				0.00 $\vec{V}_1$				
big $\rightarrow \vec{E}_2 \xrightarrow{W_V}$				0.42 $\vec{V}_2$				
blue $\rightarrow \vec{E}_3 \xrightarrow{W_V}$				0.58 $\vec{V}_3$				
cat $\rightarrow \vec{E}_4 \xrightarrow{W_V}$				0.00 $\vec{V}_4$				
chases $\rightarrow \vec{E}_5 \xrightarrow{W_V}$				0.00 $\vec{V}_5$				
the $\rightarrow \vec{E}_6 \xrightarrow{W_V}$				0.00 $\vec{V}_6$				
small $\rightarrow \vec{E}_7 \xrightarrow{W_V}$				0.00 $\vec{V}_7$				
mouse $\rightarrow \vec{E}_8 \xrightarrow{W_V}$				0.00 $\vec{V}_8$				



In practice, we compute the attention function on a set of queries simultaneously, packed together into a matrix Q. The keys and values are also packed together into matrices K and V. We compute the matrix of outputs as:

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V \tag{1}$$

cat

$\downarrow$

$\vec{E}_4$

$+$

$\Delta\vec{E}_4$

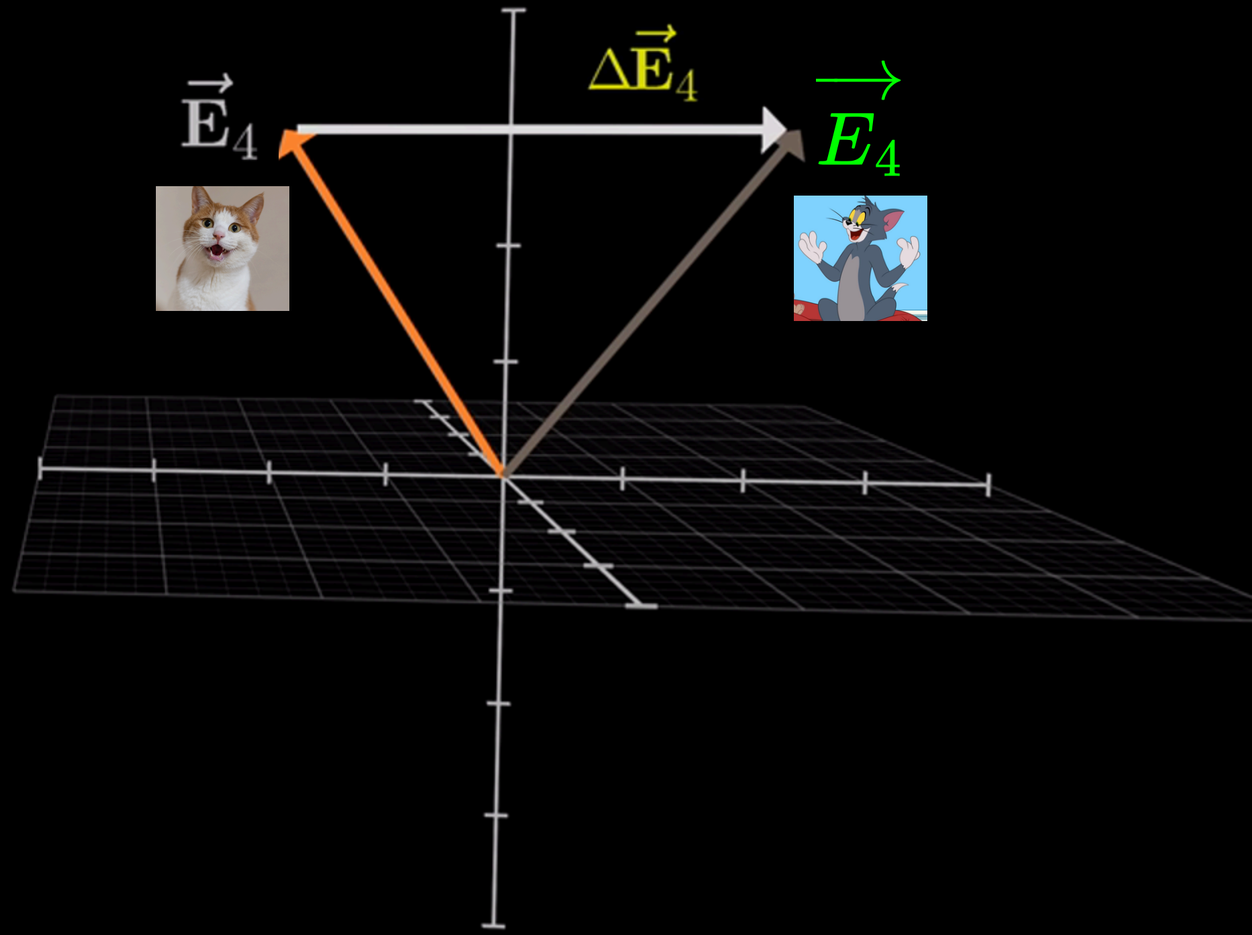
$||$

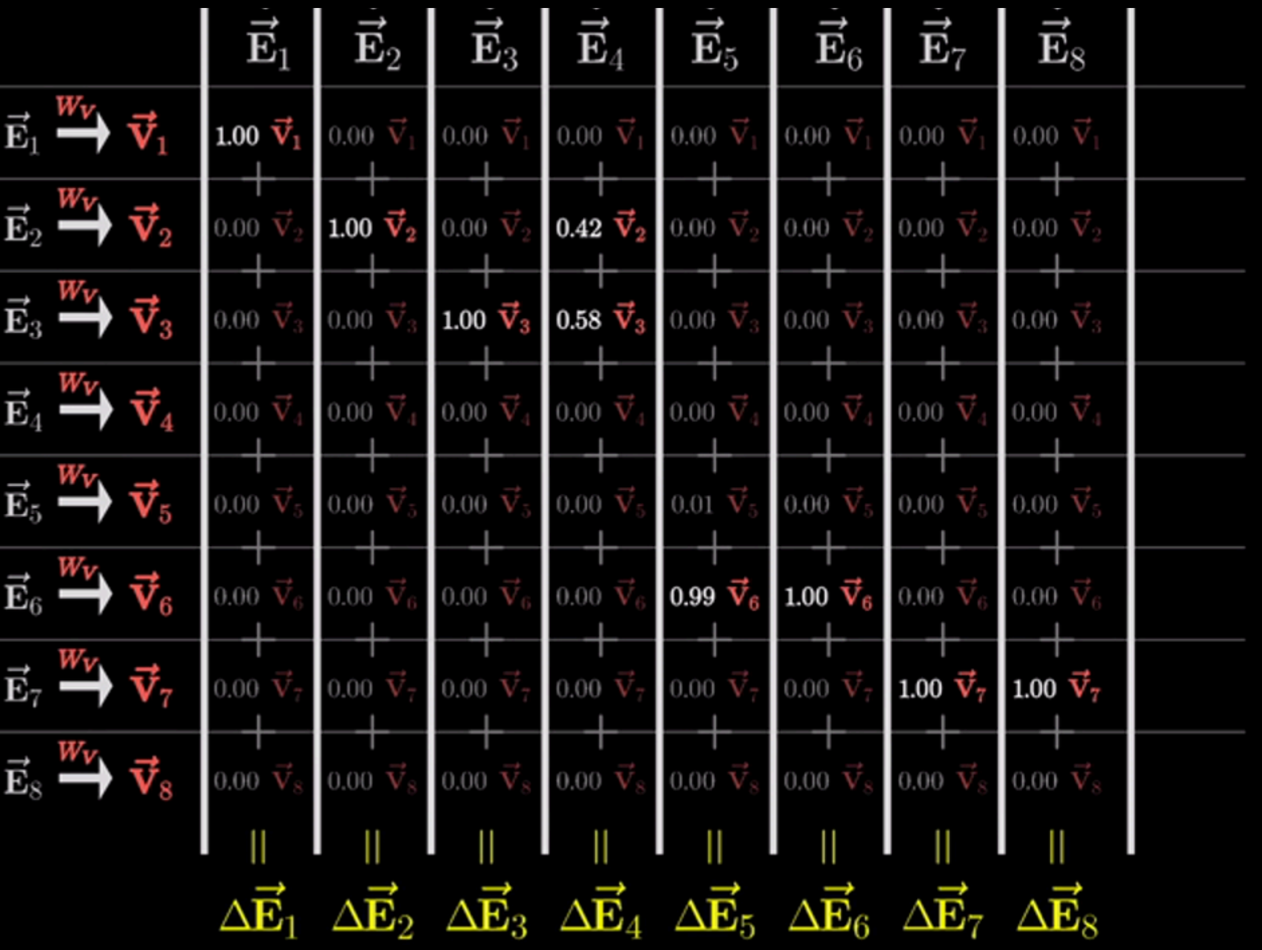
$\overrightarrow{E_4}$

$$\vec{\mathbf{E}}_4 \qquad \Delta\vec{\mathbf{E}}_4$$

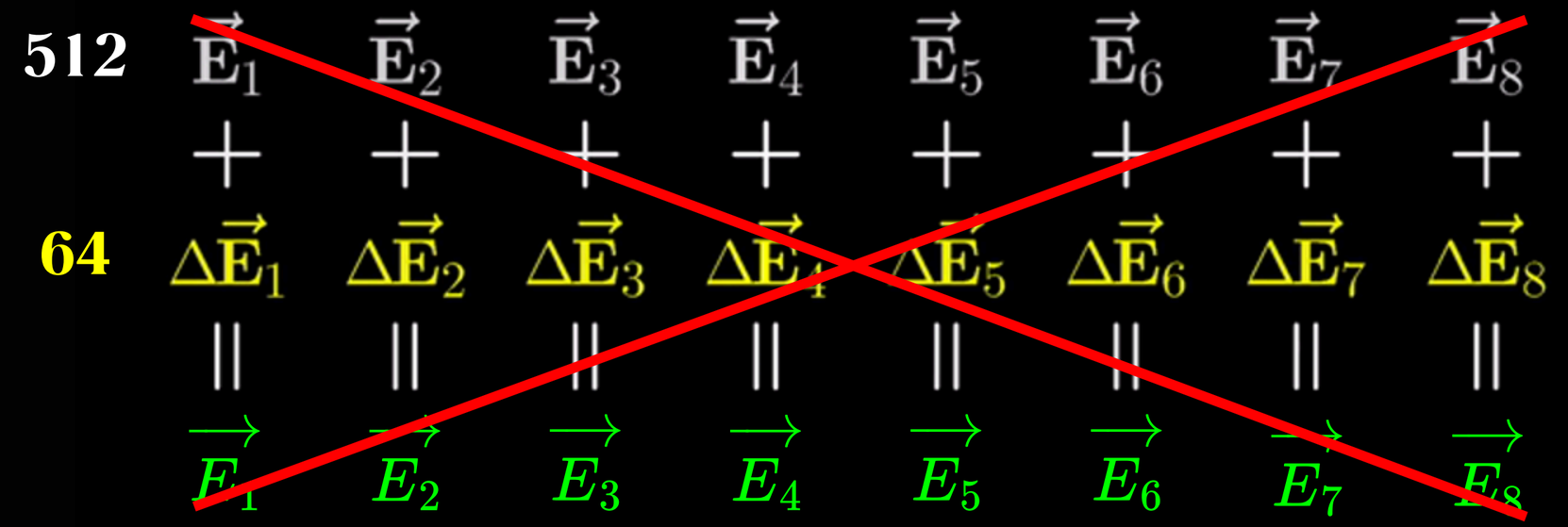
$$\begin{bmatrix} +9.2 \\ -2.3 \\ +5.8 \\ +0.6 \\ +1.3 \\ +8.4 \\ \vdots \\ -8.2 \end{bmatrix} \qquad \begin{bmatrix} -7.6 \\ +2.8 \\ -7.1 \\ +8.8 \\ +0.4 \\ -1.7 \\ \vdots \\ -4.7 \end{bmatrix}$$

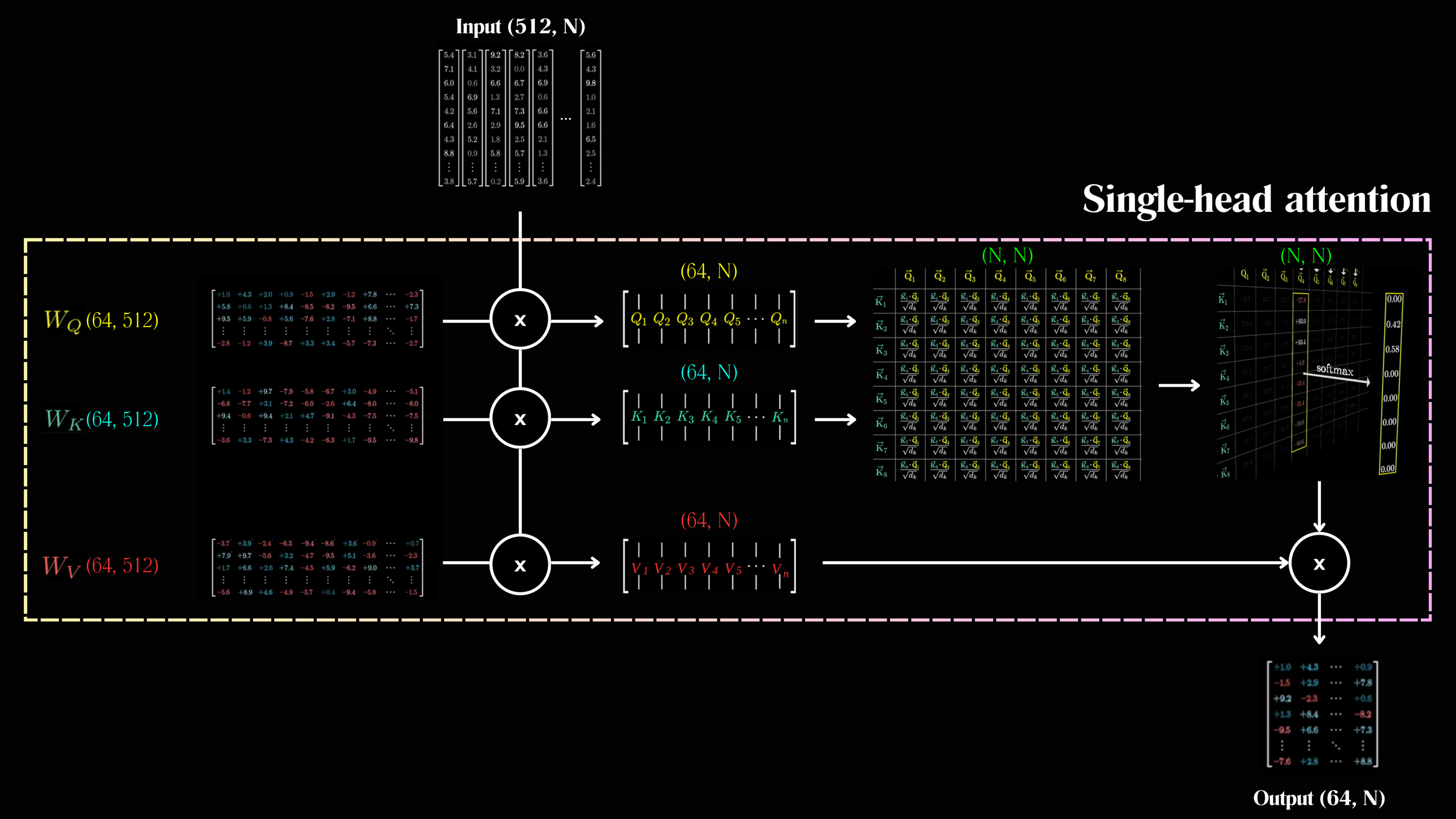
512-dimensional space

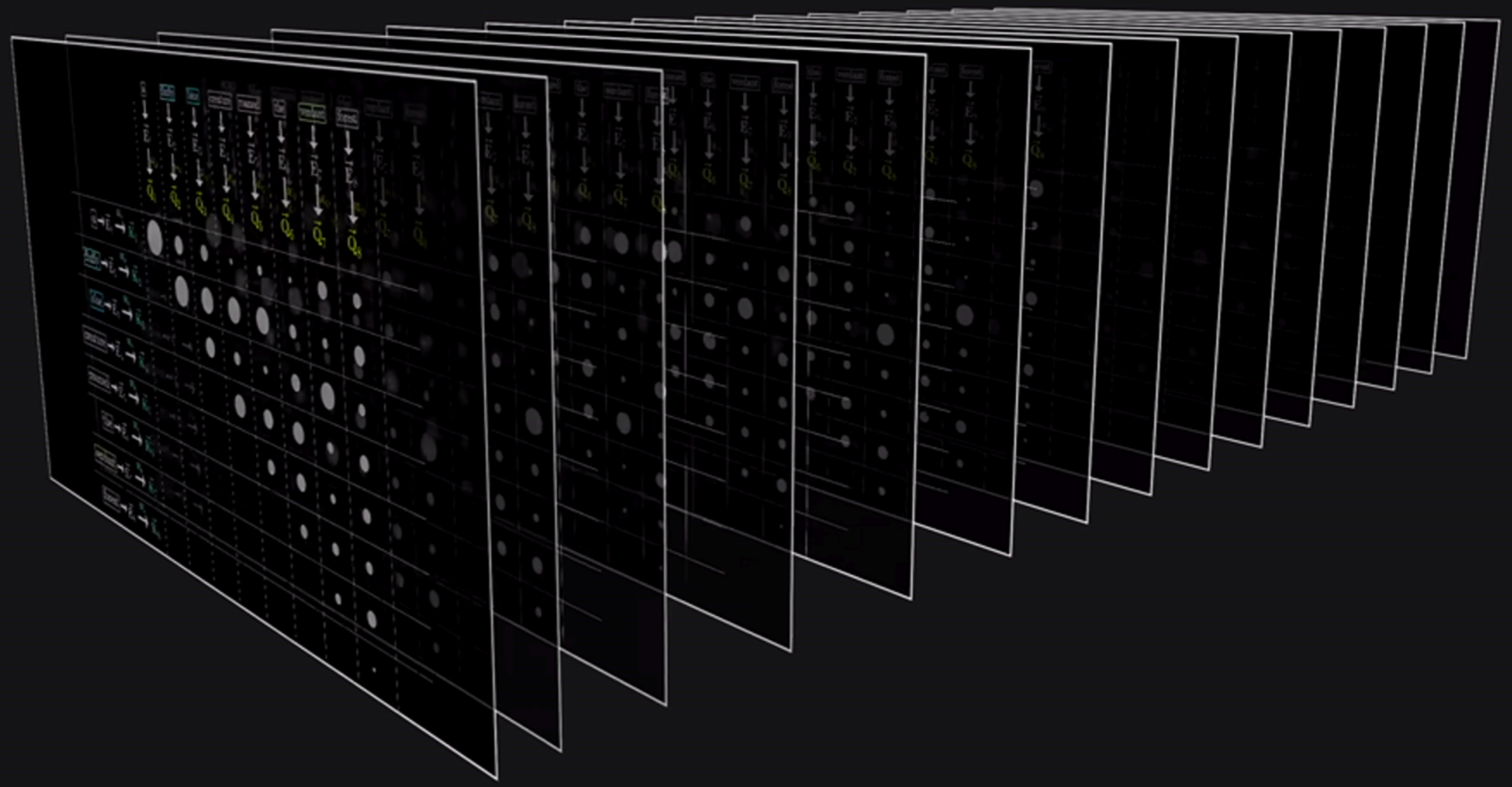


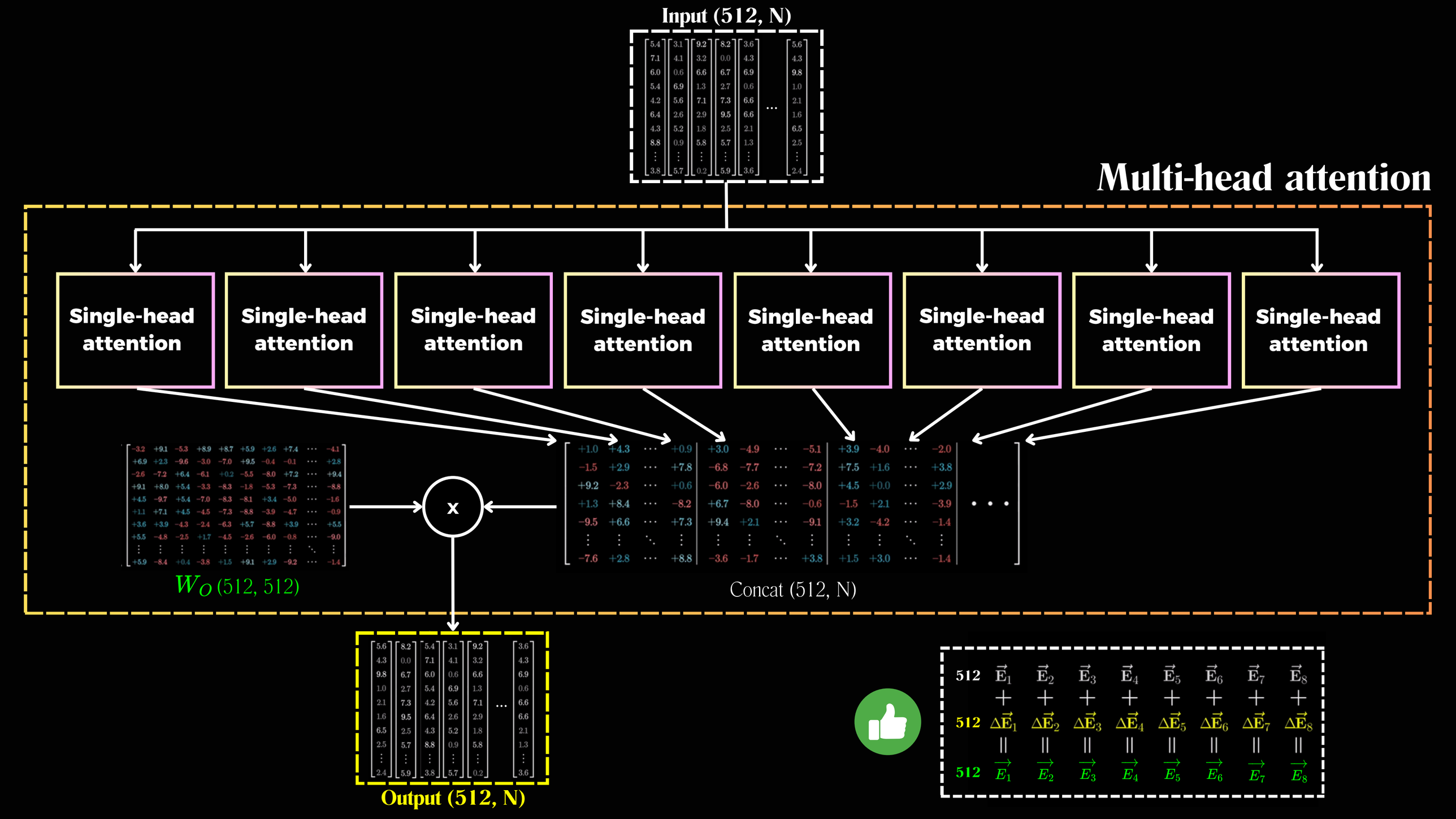


$$
\begin{array}{cccccccc}
\vec{E}_1 & \vec{E}_2 & \vec{E}_3 & \vec{E}_4 & \vec{E}_5 & \vec{E}_6 & \vec{E}_7 & \vec{E}_8 \\
+ & + & + & + & + & + & + & + \\
\Delta\vec{E}_1 & \Delta\vec{E}_2 & \Delta\vec{E}_3 & \Delta\vec{E}_4 & \Delta\vec{E}_5 & \Delta\vec{E}_6 & \Delta\vec{E}_7 & \Delta\vec{E}_8 \\
|| & || & || & || & || & || & || & || \\
\overrightarrow{E_1} & \overrightarrow{E_2} & \overrightarrow{E_3} & \overrightarrow{E_4} & \overrightarrow{E_5} & \overrightarrow{E_6} & \overrightarrow{E_7} & \overrightarrow{E_8}
\end{array}
$$


512
64
$\vec{E}_1$ $\vec{E}_2$ $\vec{E}_3$ $\vec{E}_4$ $\vec{E}_5$ $\vec{E}_6$ $\vec{E}_7$ $\vec{E}_8$
+ + + + + + + +
$\Delta\vec{E}_1$ $\Delta\vec{E}_2$ $\Delta\vec{E}_3$ $\Delta\vec{E}_4$ $\Delta\vec{E}_5$ $\Delta\vec{E}_6$ $\Delta\vec{E}_7$ $\Delta\vec{E}_8$
|| || || || || || || ||
$\overrightarrow{E_1}$ $\overrightarrow{E_2}$ $\overrightarrow{E_3}$ $\overrightarrow{E_4}$ $\overrightarrow{E_5}$ $\overrightarrow{E_6}$ $\overrightarrow{E_7}$ $\overrightarrow{E_8}$


Input (512, N)
Single-head attention
W_Q (64, 512)
W_K (64, 512)
W_V (64, 512)
x
x
x
(64, N)
(64, N)
(64, N)
$Q_1\ Q_2\ Q_3\ Q_4\ Q_5 \cdots Q_n$
$K_1\ K_2\ K_3\ K_4\ K_5 \cdots K_n$
$V_1\ V_2\ V_3\ V_4\ V_5 \cdots V_n$
(N, N)
(N, N)
softmax
x
Output (64, N)




Input (512, N)
Multi-head attention
Single-head attention
Single-head attention
Single-head attention
Single-head attention
Single-head attention
Single-head attention
Single-head attention
Single-head attention
W_O (512, 512)
x
Concat (512, N)
Output (512, N)
512
512
512

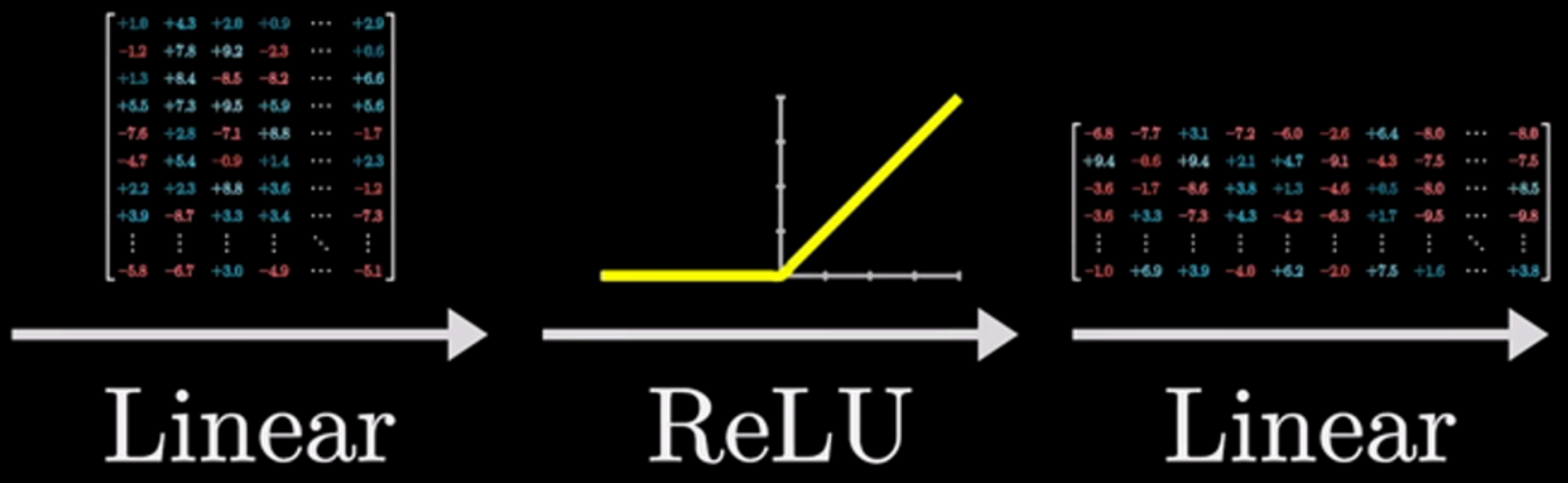
W_Q W_K W_V

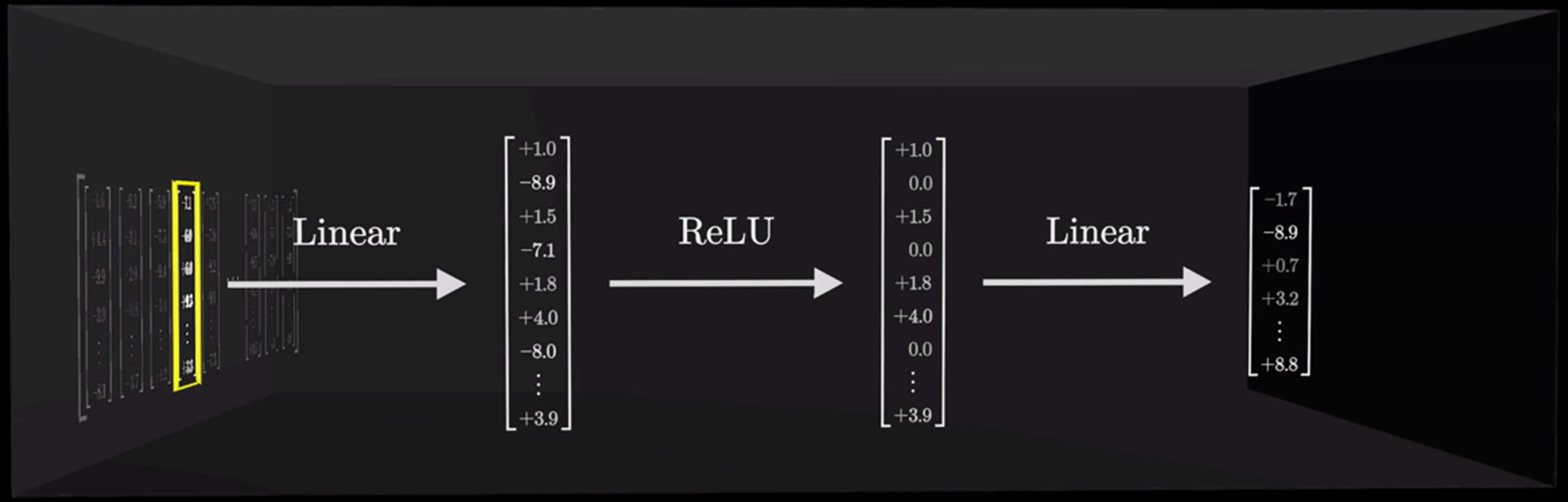
$$\begin{bmatrix} +3.2 & -4.7 & -9.5 & +5.1 & -3.6 & -2.3 & \cdots & +1.7 \\ +6.6 & +2.6 & +7.4 & -4.5 & +5.9 & -6.2 & \cdots & +9.0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ +3.7 & -5.6 & +8.9 & +4.6 & -4.9 & -5.7 & \cdots & +0.4 \end{bmatrix}$$

$$\begin{bmatrix} -9.4 & -5.8 & -1.5 & -2.5 & -0.7 & -4.4 & \cdots & +1.7 \\ +7.2 & -7.6 & +0.3 & -7.3 & +4.3 & -2.1 & \cdots & +1.3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ -6.3 & -7.0 & -0.2 & -2.9 & +8.7 & +5.3 & \cdots & +4.9 \end{bmatrix}$$

$$\begin{bmatrix} +8.9 & +8.7 & +5.9 & +2.6 & +7.4 & -4.1 & \cdots & +6.9 \\ +2.3 & -9.6 & -3.0 & -7.0 & +9.5 & -0.4 & \cdots & -0.1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ +2.8 & -2.6 & -7.2 & +6.4 & -6.1 & +0.2 & \cdots & -5.5 \end{bmatrix}$$

Multi-layer Perceptron


Linear
ReLU
Linear

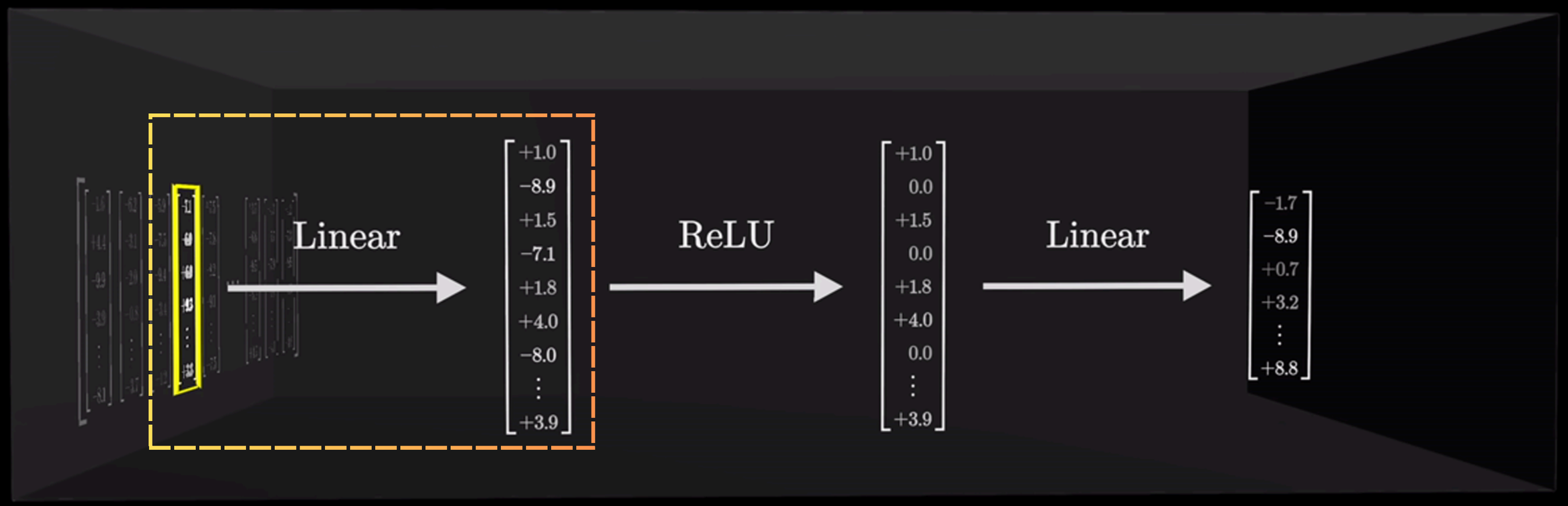

Linear
+1.0
−8.9
+1.5
−7.1
+1.8
+4.0
−8.0
+3.9
ReLU
+1.0
0.0
+1.5
0.0
+1.8
+4.0
0.0
+3.9
Linear
−1.7
−8.9
+0.7
+3.2
+8.8

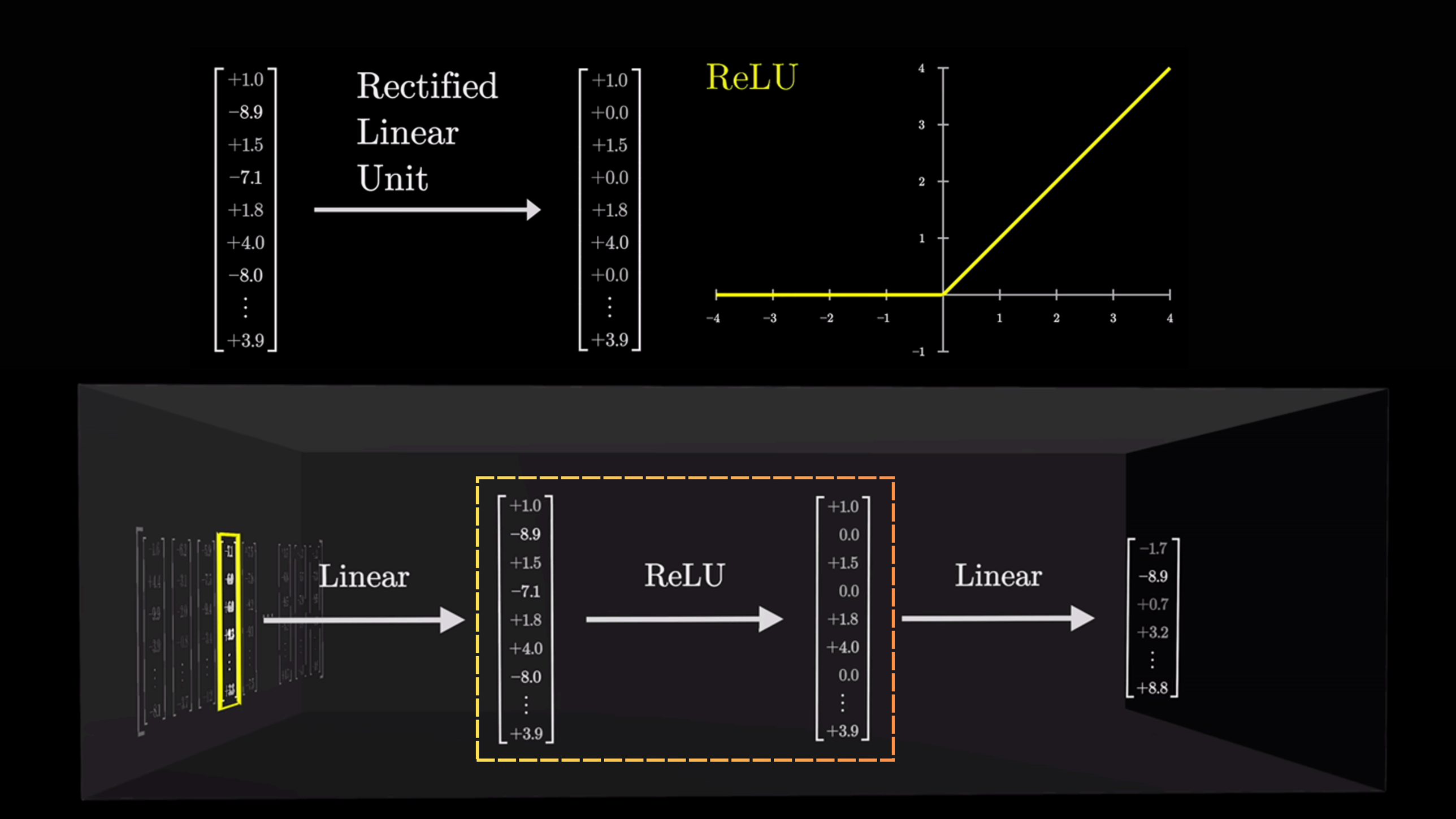
512

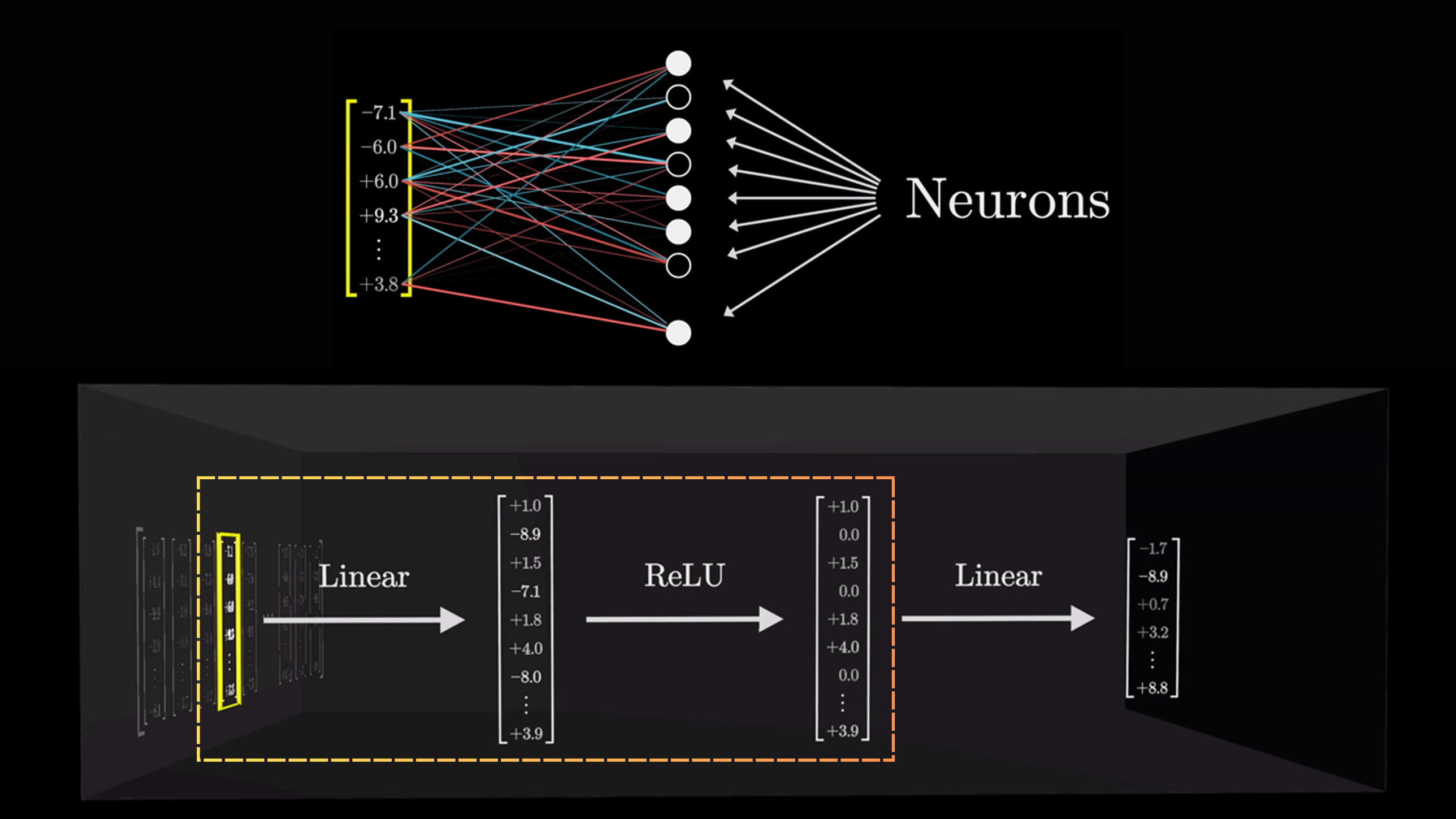
2048

$$\begin{bmatrix} -5.0 & +7.1 & +0.8 & +1.0 & \cdots & +6.8 \\ -7.4 & -4.4 & +1.7 & +9.3 & \cdots & +1.2 \\ -9.5 & +6.0 & -5.3 & +6.1 & \cdots & -2.2 \\ +7.2 & +4.9 & +1.1 & -7.2 & \cdots & -8.7 \\ -7.5 & -9.0 & -7.8 & -5.4 & \cdots & +4.2 \\ +1.2 & -9.7 & -8.5 & +9.3 & \cdots & +1.3 \\ -5.9 & -4.9 & +4.8 & -6.0 & \cdots & +1.6 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ +9.3 & +6.9 & -5.2 & -0.1 & \cdots & +2.4 \end{bmatrix} \begin{bmatrix} -7.1 \\ -6.0 \\ +6.0 \\ +9.3 \\ \vdots \\ +3.8 \end{bmatrix} + \begin{bmatrix} -1.0 \\ +1.6 \\ -2.4 \\ +1.0 \\ +4.9 \\ +3.4 \\ -4.7 \\ \vdots \\ -2.6 \end{bmatrix} = \begin{bmatrix} +1.0 \\ -8.9 \\ +1.5 \\ -7.1 \\ +1.8 \\ +4.0 \\ -8.0 \\ \vdots \\ +3.9 \end{bmatrix}$$

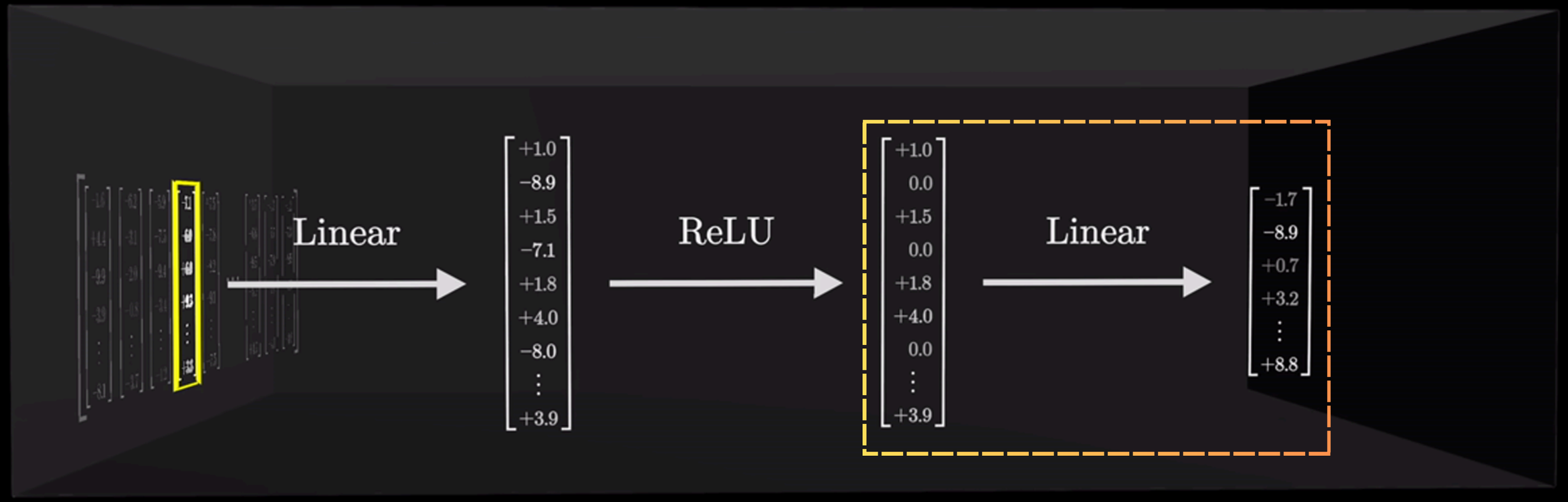
2048

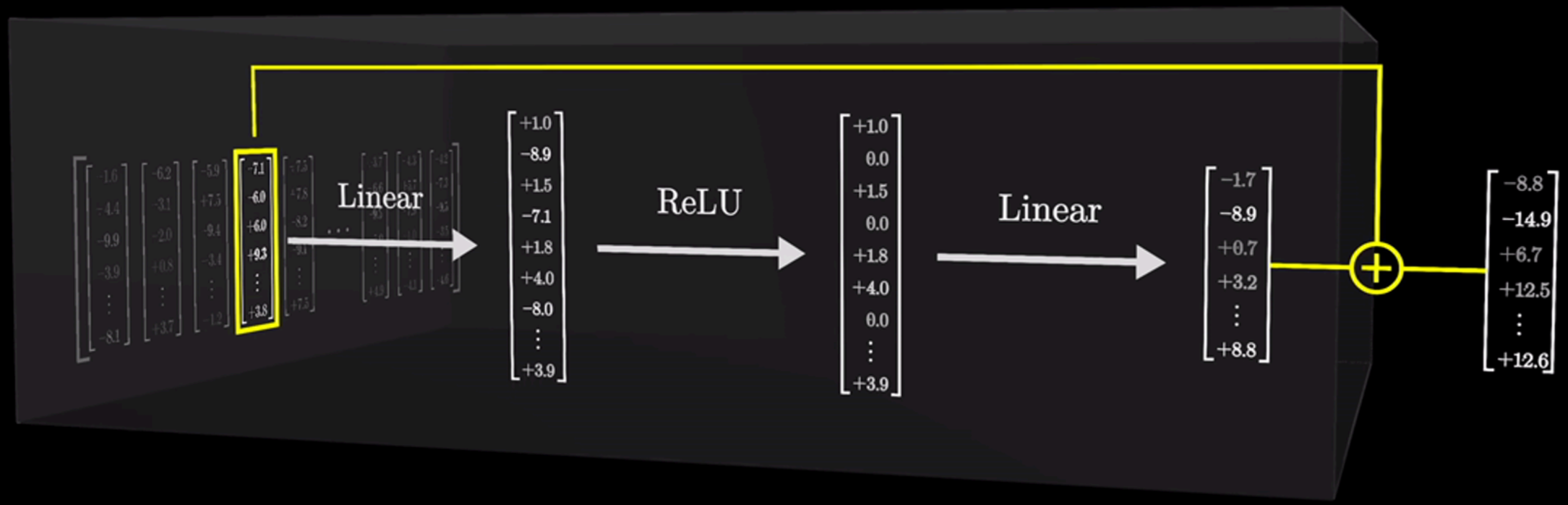


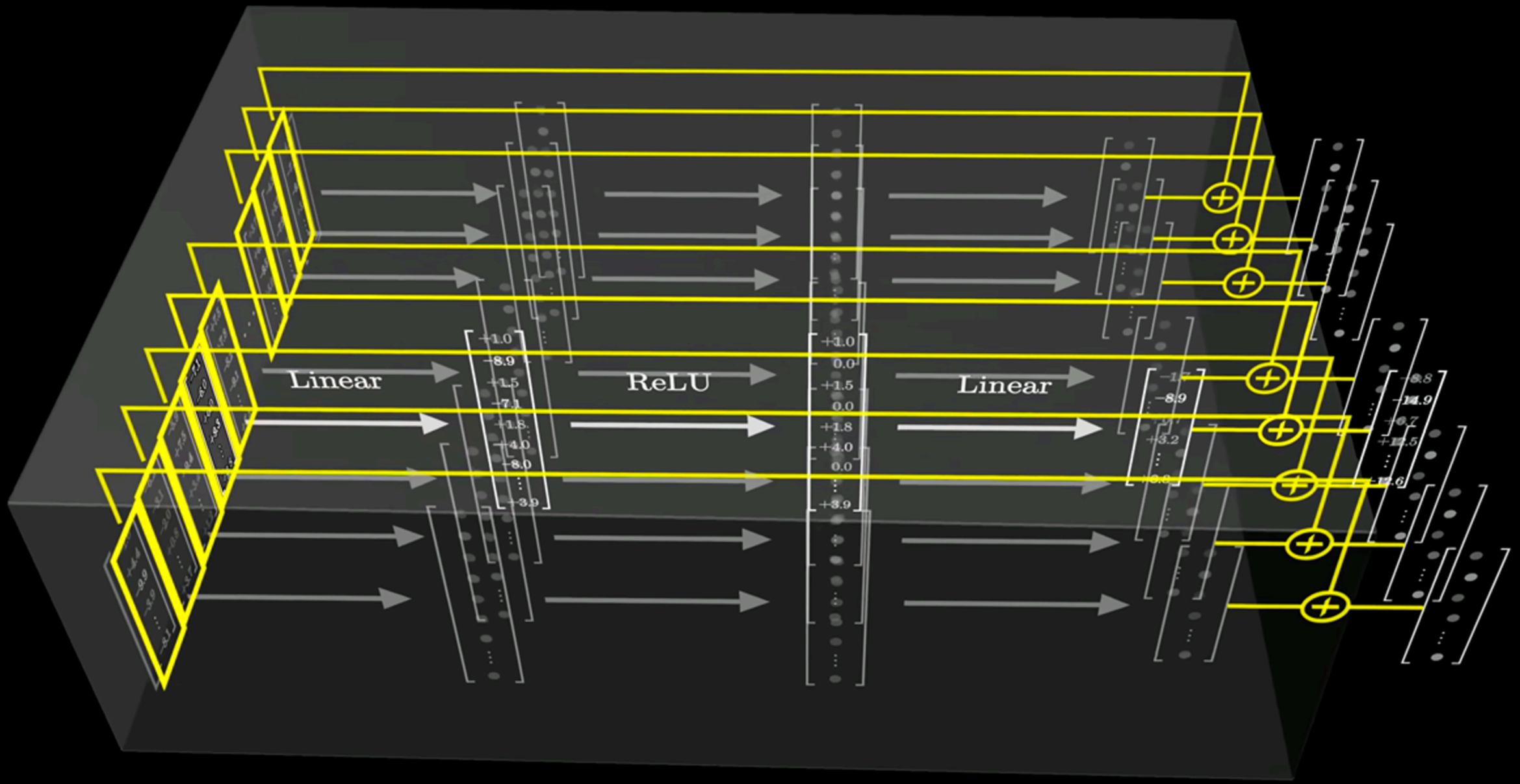

+1.0
−8.9
+1.5
−7.1
+1.8
+4.0
−8.0
+3.9
Rectified
Linear
Unit
+1.0
+0.0
+1.5
+0.0
+1.8
+4.0
+0.0
+3.9
ReLU
4
3
2
1
−1
−4
−3
−2
−1
1
2
3
4
Linear
+1.0
−8.9
+1.5
−7.1
+1.8
+4.0
−8.0
+3.9
ReLU
+1.0
0.0
+1.5
0.0
+1.8
+4.0
0.0
+3.9
Linear
−1.7
−8.9
+0.7
+3.2
+8.8


−7.1
−6.0
+6.0
+9.3
+3.8
Neurons
Linear
+1.0
−8.9
+1.5
−7.1
+1.8
+4.0
−8.0
+3.9
ReLU
+1.0
0.0
+1.5
0.0
+1.8
+4.0
0.0
+3.9
Linear
−1.7
−8.9
+0.7
+3.2
+8.8

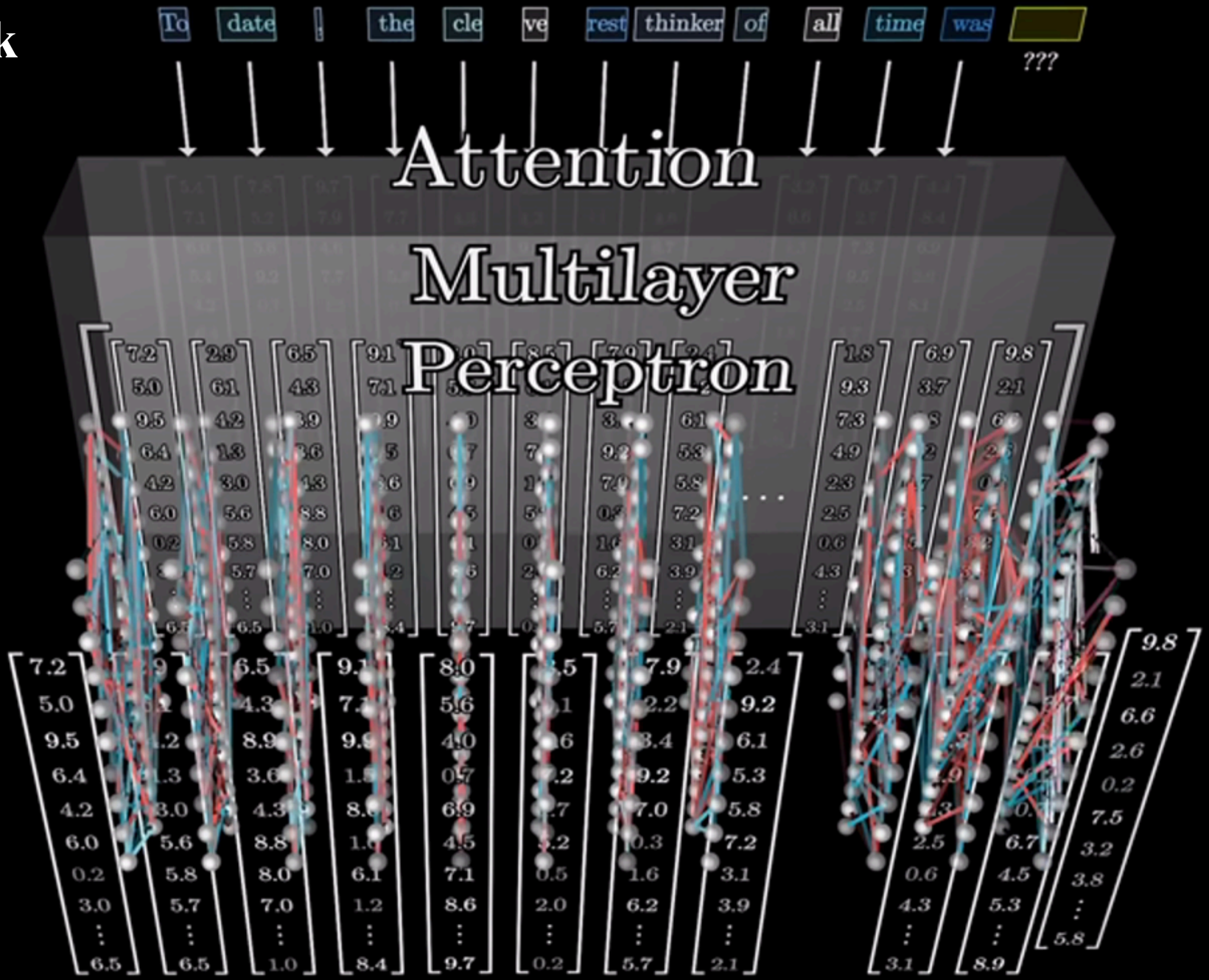
$$
\underset{512}{\begin{bmatrix}
+0.5 & +8.4 & -4.7 & -8.6 & +4.7 & +5.4 & +8.1 & \cdots & -9.6 \\
-5.3 & +2.3 & +8.9 & +8.9 & +1.1 & +8.2 & +2.8 & \cdots & -0.3 \\
+2.1 & +1.0 & +8.4 & +8.3 & -2.1 & +9.2 & -6.5 & \cdots & -7.2 \\
+0.1 & -9.5 & +8.9 & +6.5 & -9.6 & -6.4 & -3.3 & \cdots & +6.1 \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\
-4.2 & -0.2 & +2.0 & -9.6 & +1.9 & -1.3 & +6.1 & \cdots & +7.8
\end{bmatrix}}_{2048}
\begin{bmatrix} +1.0 \\ +0.0 \\ +1.5 \\ +0.0 \\ +1.8 \\ +4.0 \\ +0.0 \\ \vdots \\ +3.9 \end{bmatrix}
+
\begin{bmatrix} +1.5 \\ -6.3 \\ +5.7 \\ +2.2 \\ \vdots \\ -1.6 \end{bmatrix}
=
\underset{512}{\begin{bmatrix} -1.7 \\ -8.9 \\ +0.7 \\ +3.2 \\ \vdots \\ +8.8 \end{bmatrix}}
$$



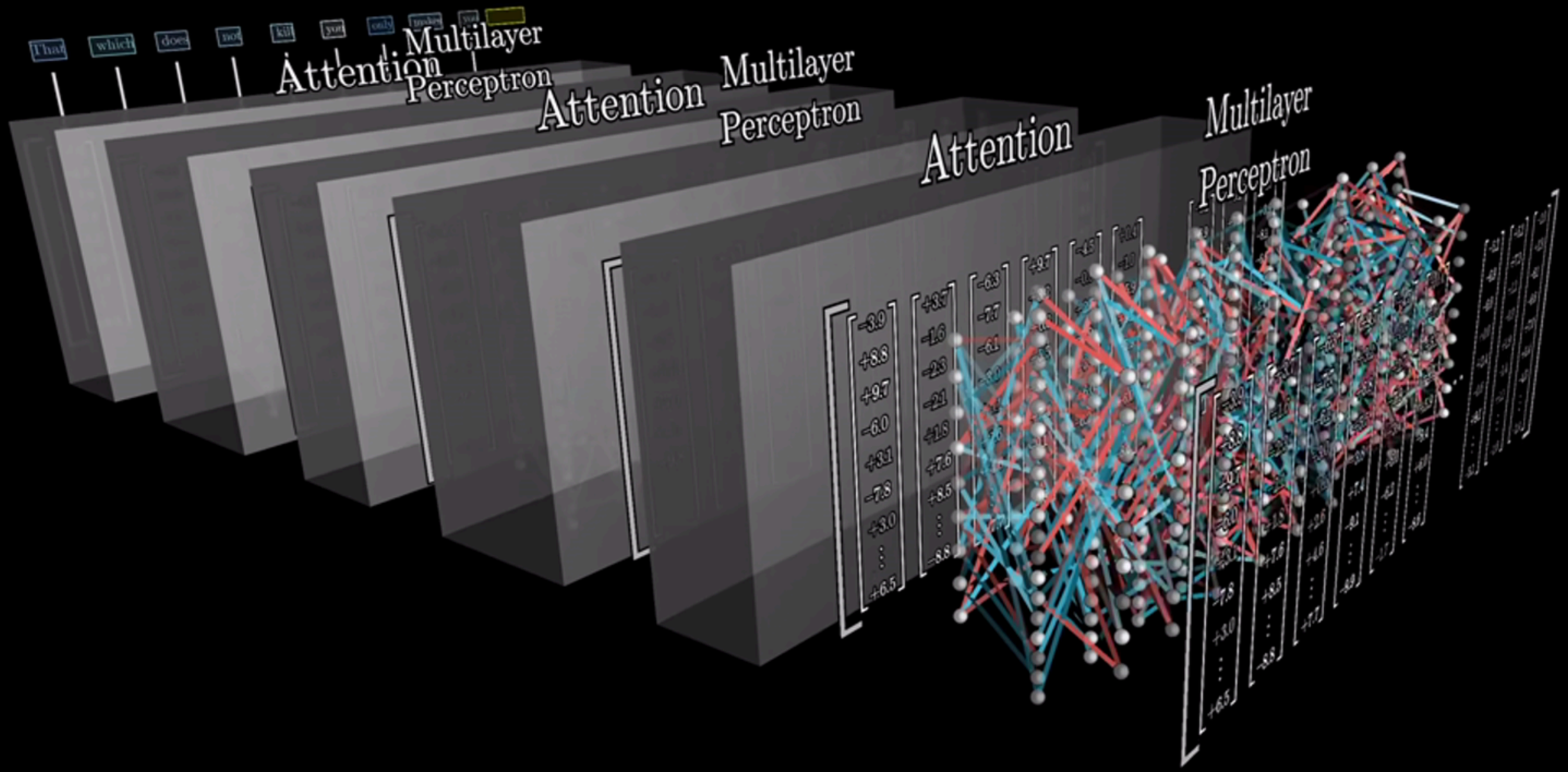

Linear
+1.0
−8.9
+1.5
−7.1
+1.8
+4.0
−8.0
+3.9
ReLU
+1.0
0.0
+1.5
0.0
+1.8
+4.0
0.0
+3.9
Linear
−1.7
−8.9
+0.7
+3.2
+8.8
−8.8
−14.9
+6.7
+12.5
+12.6


Linear
ReLU
Linear

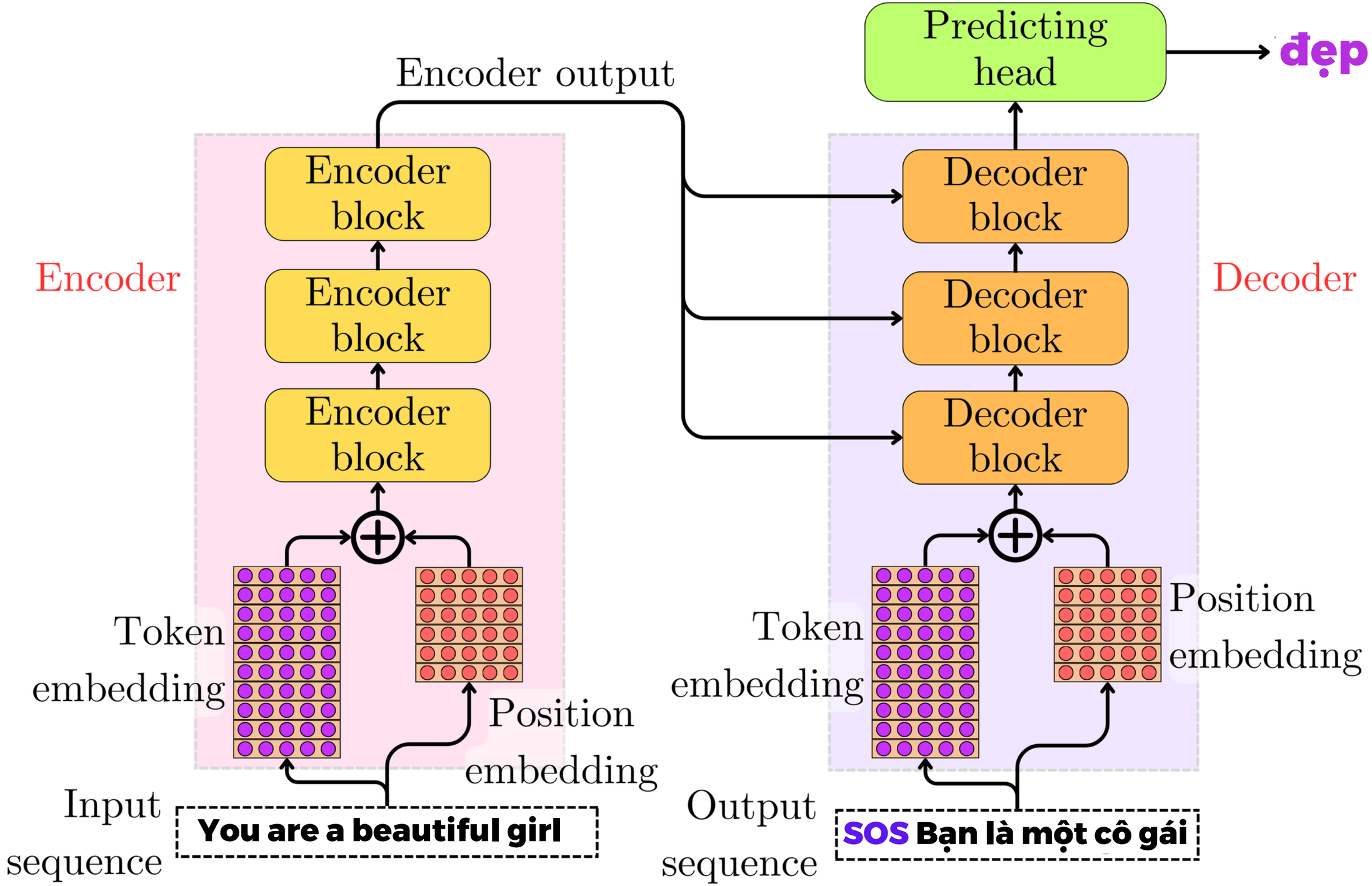
Encoder Block

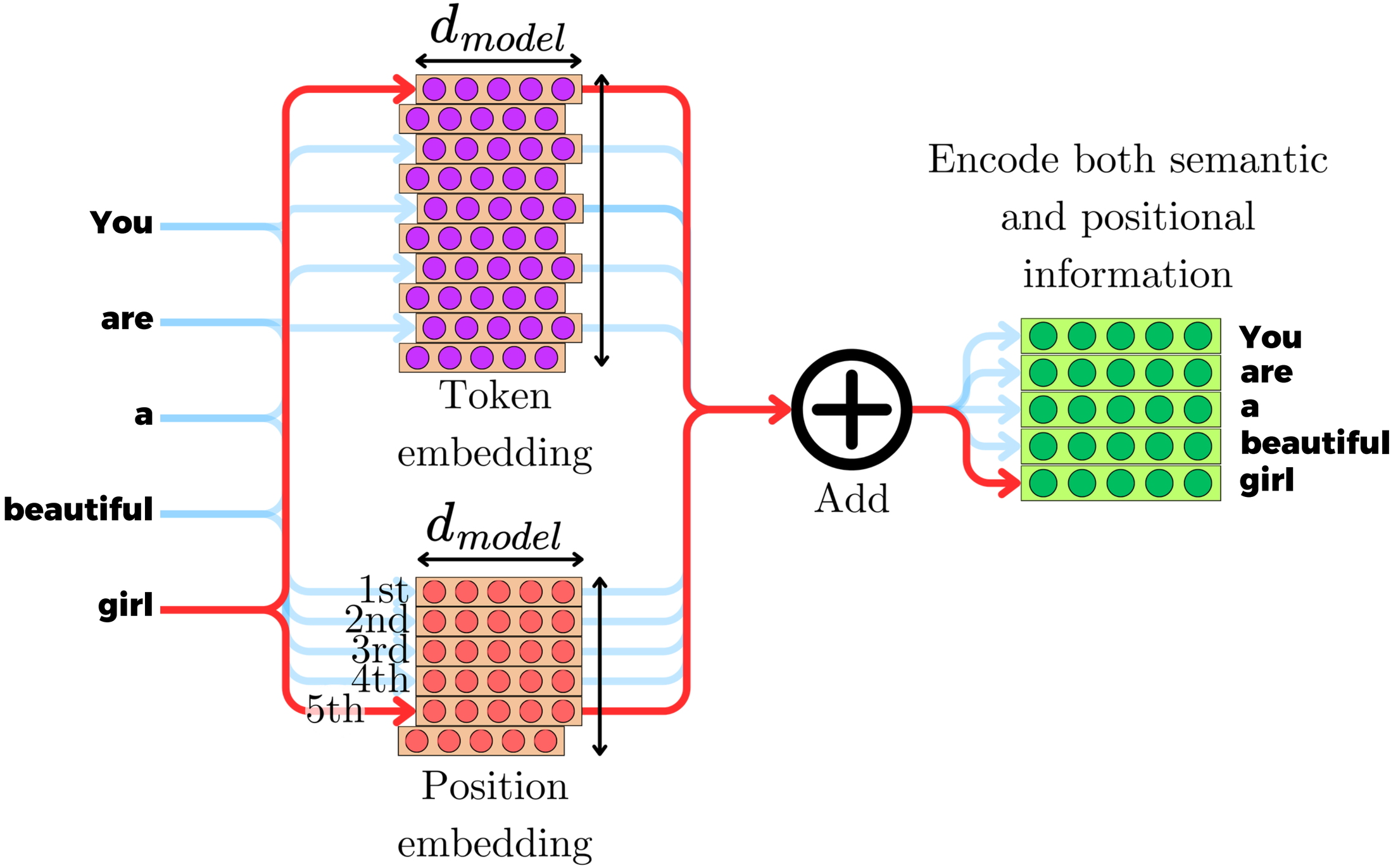


Encoder = 6 Encoder Blocks

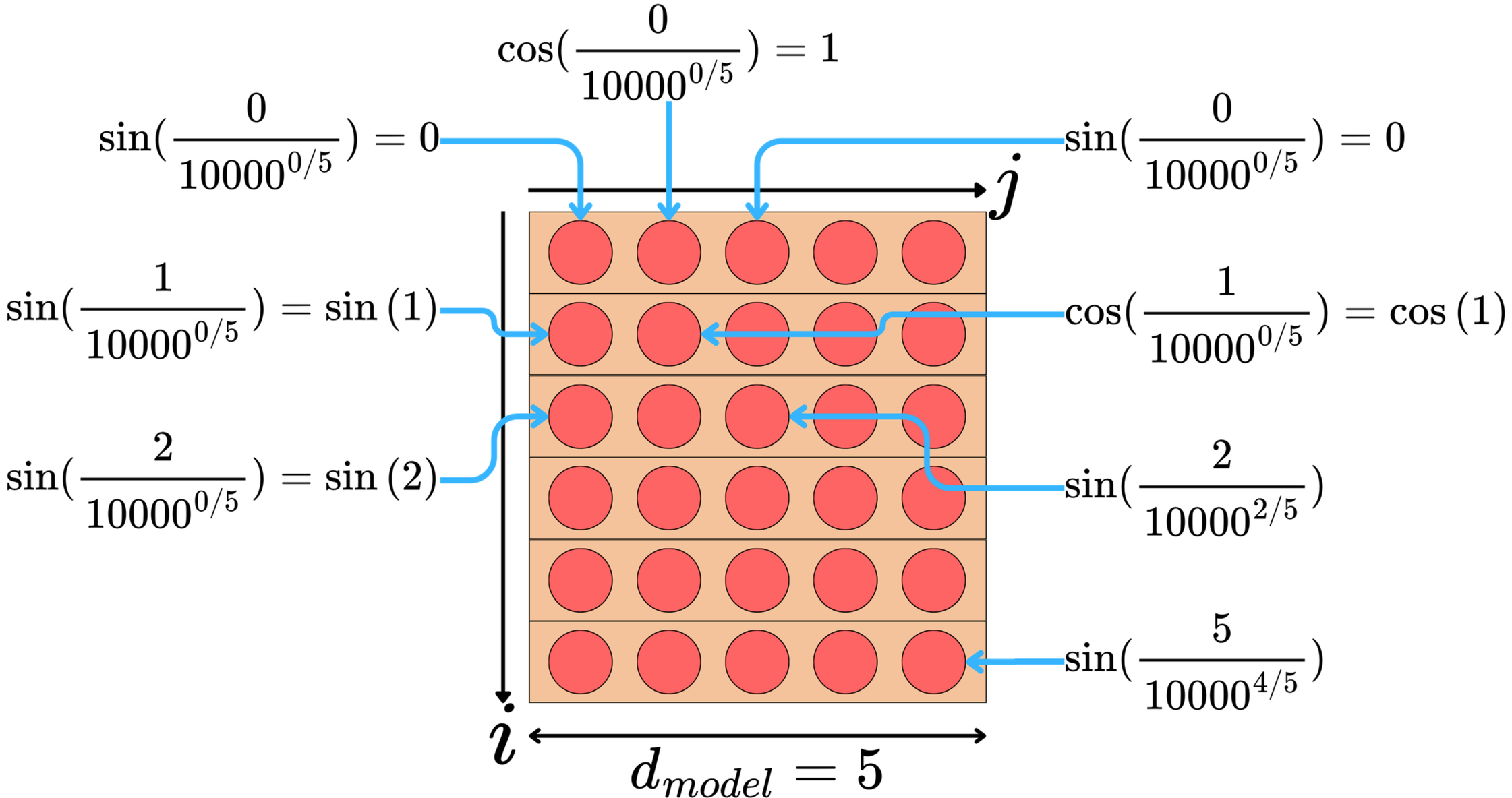


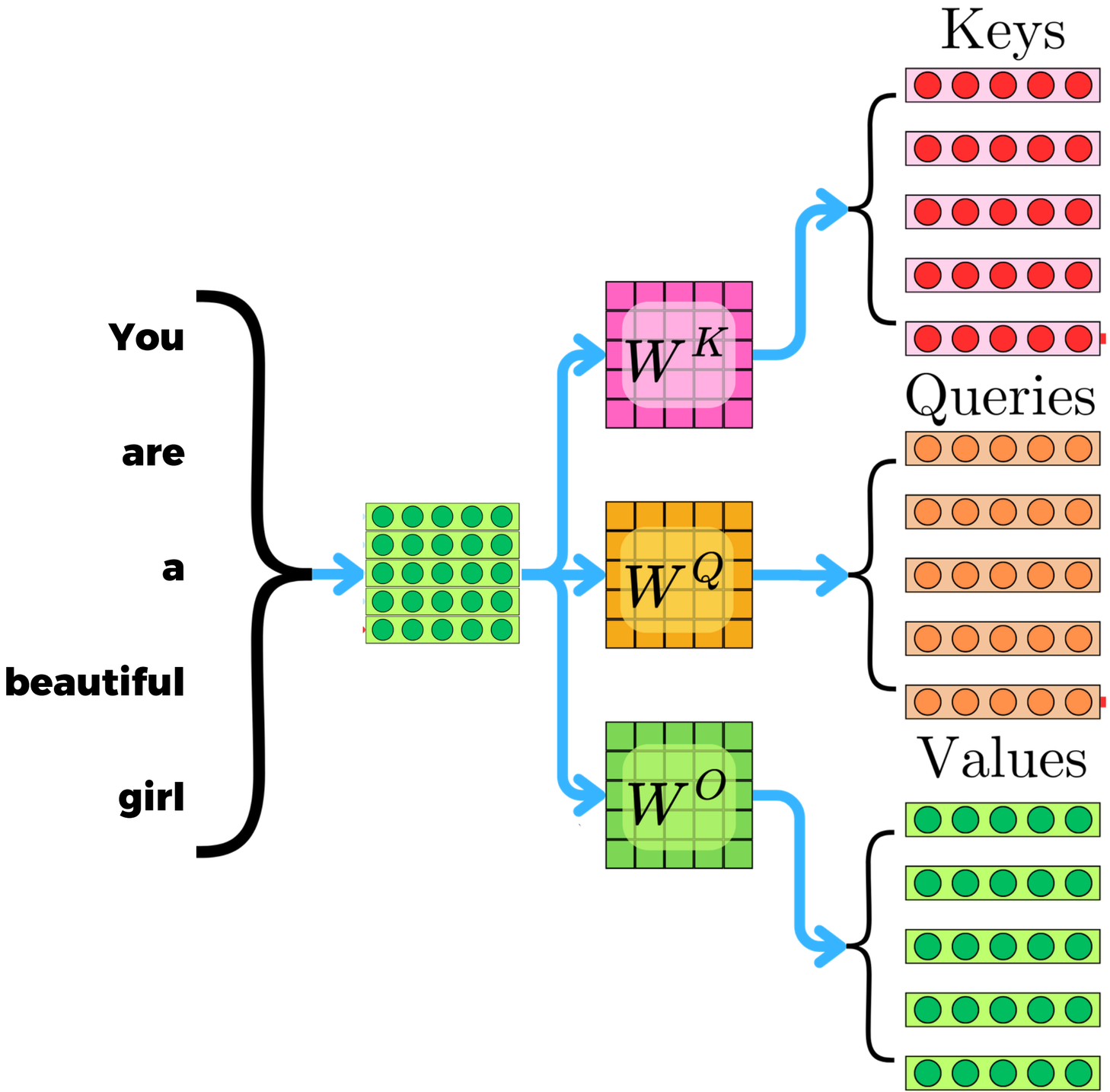
Transformer’s architecture

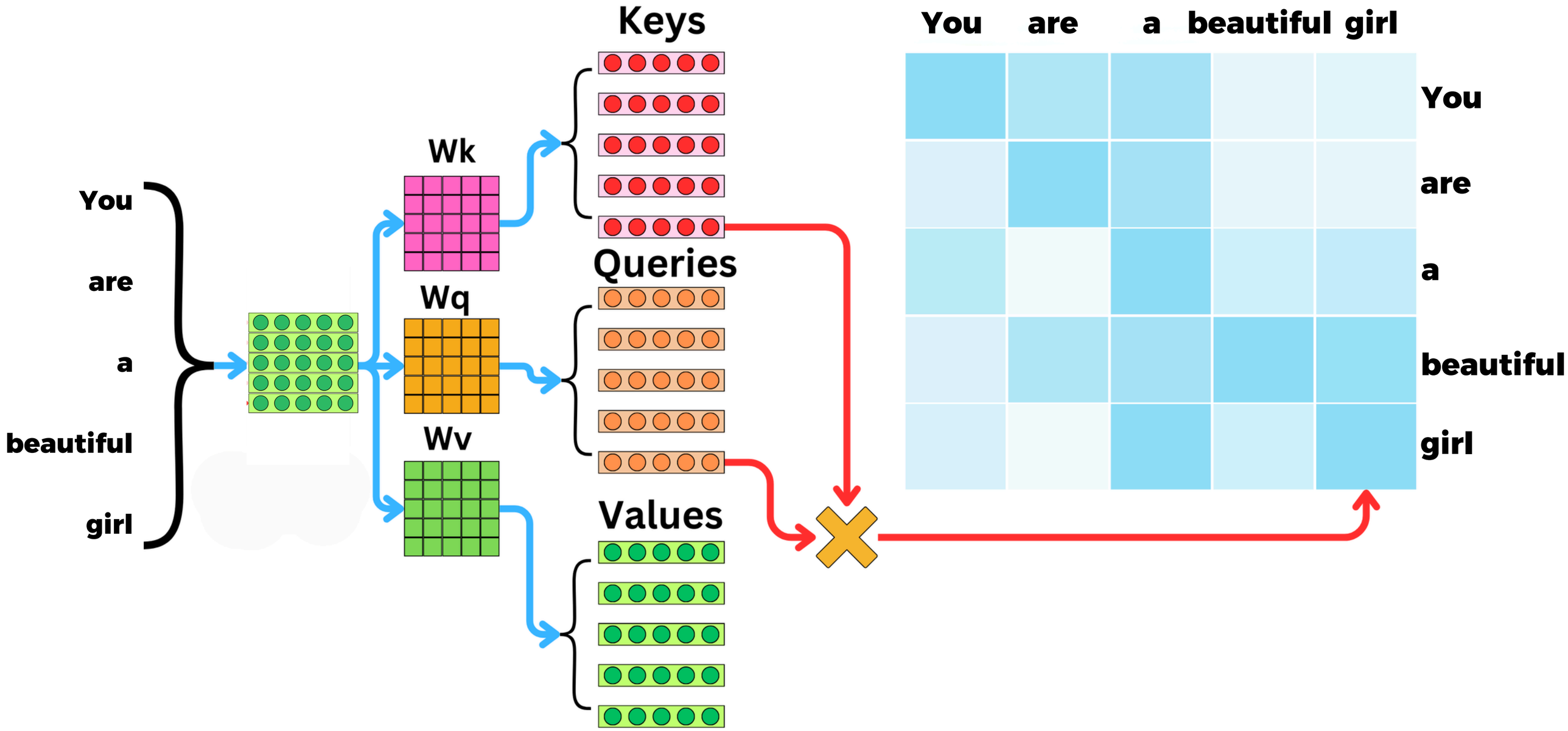

Predicting head
đẹp
Encoder output
Encoder block
Encoder block
Encoder block
Encoder
Decoder block
Decoder block
Decoder block
Decoder
Token embedding
Position embedding
Token embedding
Position embedding
Input sequence
You are a beautiful girl
Output sequence
SOS Bạn là một cô gái


d_{model}
You
are
a
beautiful
girl
Token
embedding
d_{model}
1st
2nd
3rd
4th
5th
Position
embedding
Add
Encode both semantic
and positional
information
You
are
a
beautiful
girl

$$\mathrm{PE}(i,j) = \begin{cases} \sin\left(\frac{i}{1000^{j/d_{\text{model}}}}\right) & \text{if } j \text{ is even,} \\ \cos\left(\frac{i}{1000^{(j-1)/d_{\text{model}}}}\right) & \text{if } j \text{ is odd,} \end{cases}.$$

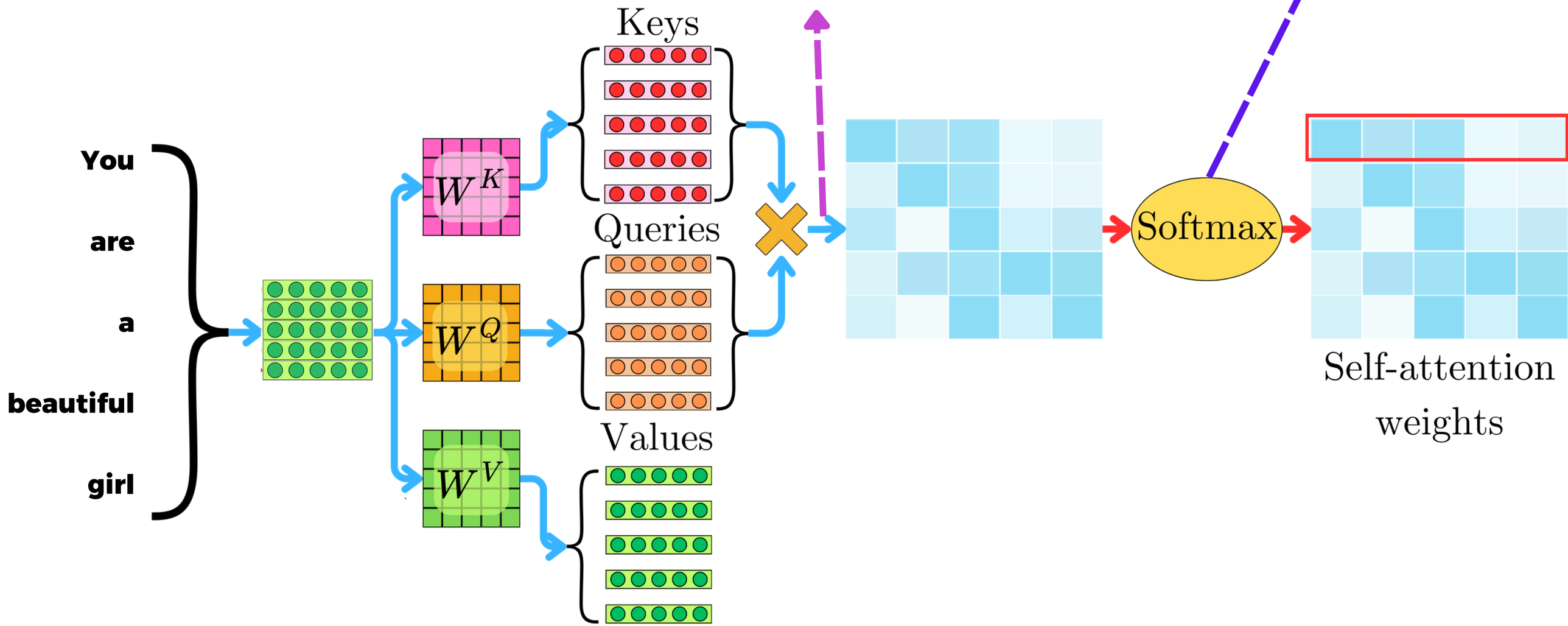


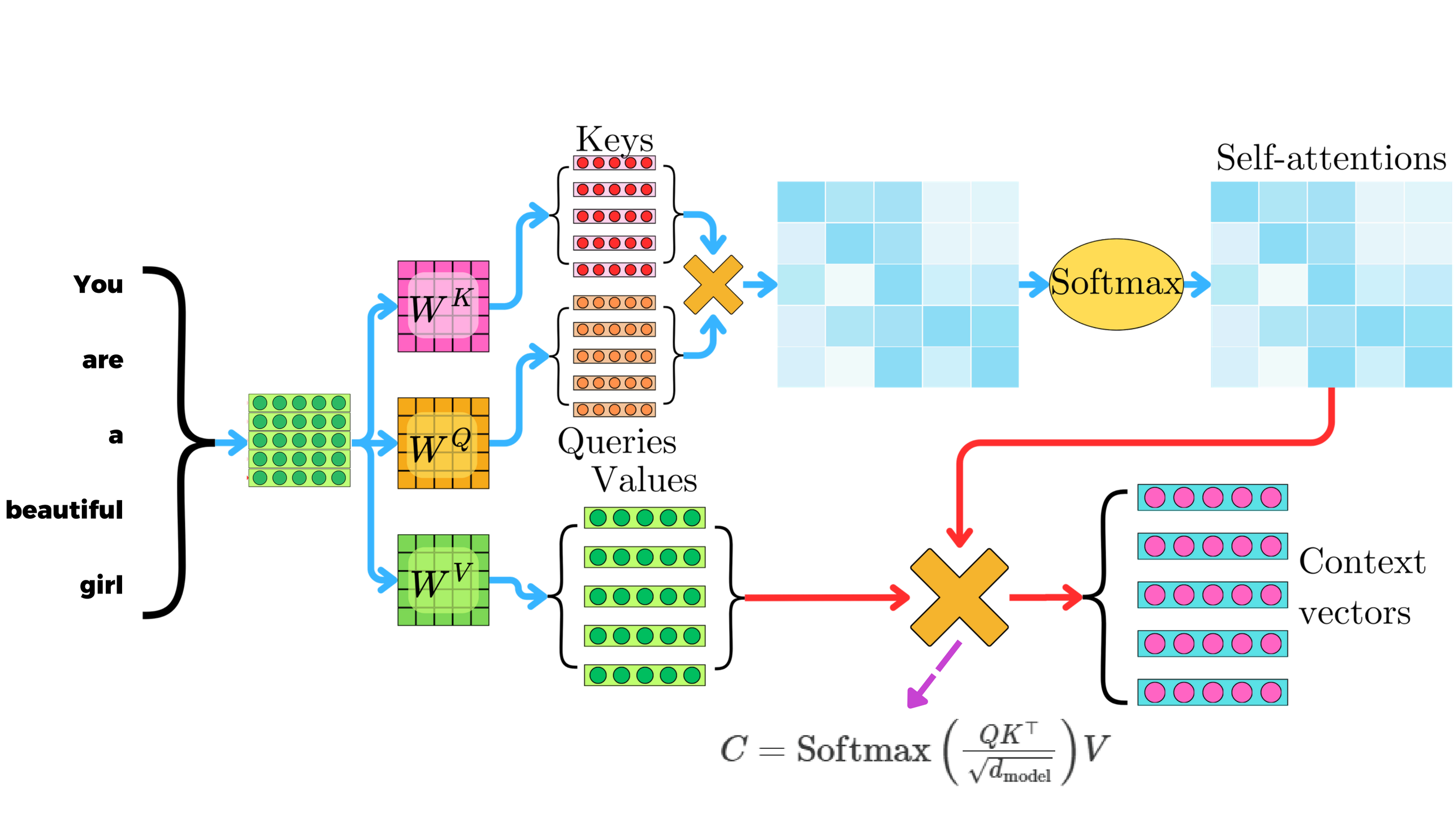

Keys
Queries
Values
You
are
a
beautiful
girl
W^K
W^Q
W^O

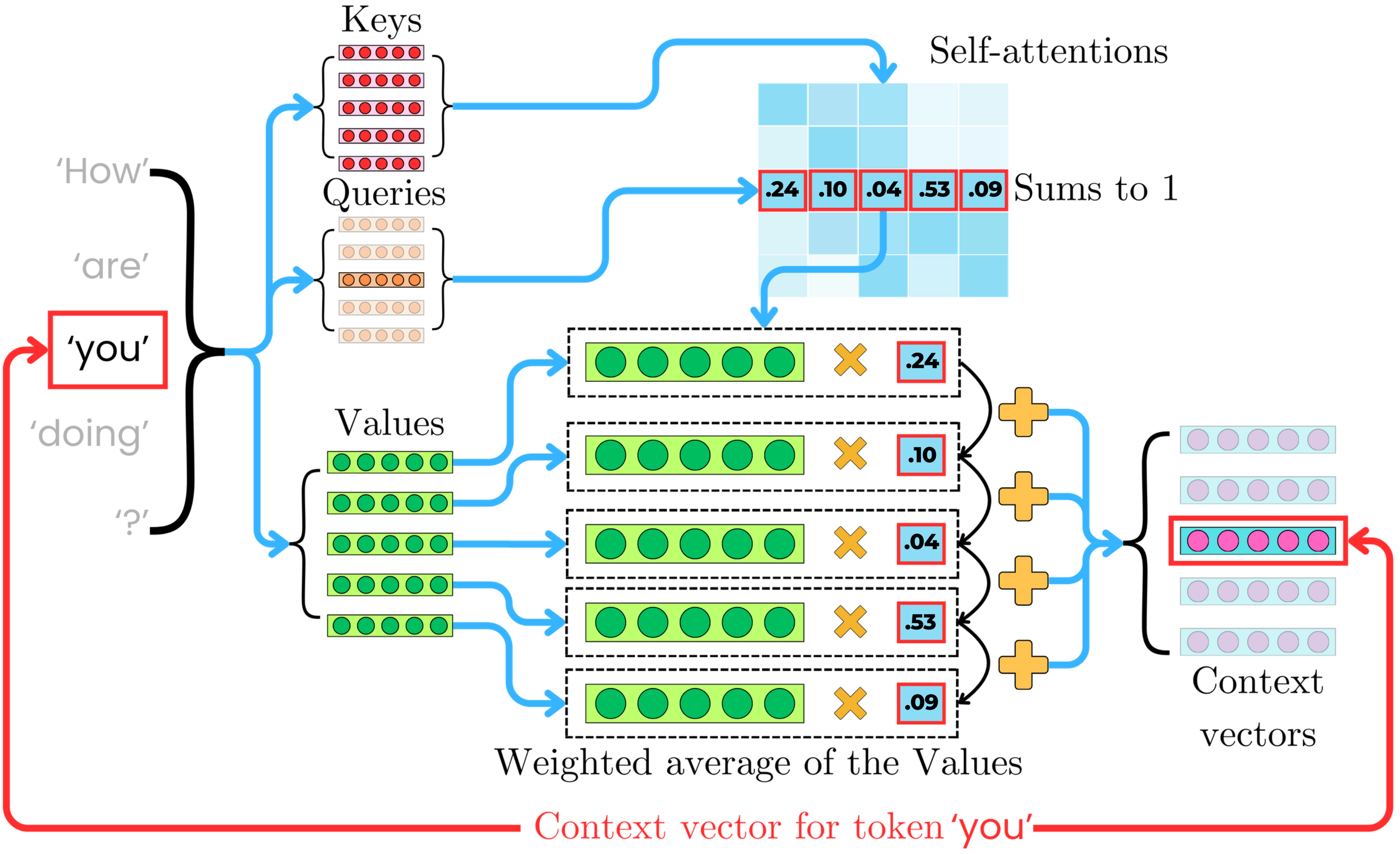

Keys
Wk
You
are
a
beautiful
girl
Queries
Wq
Wv
Values
You
are
a
beautiful
girl
You
are
a
beautiful
girl

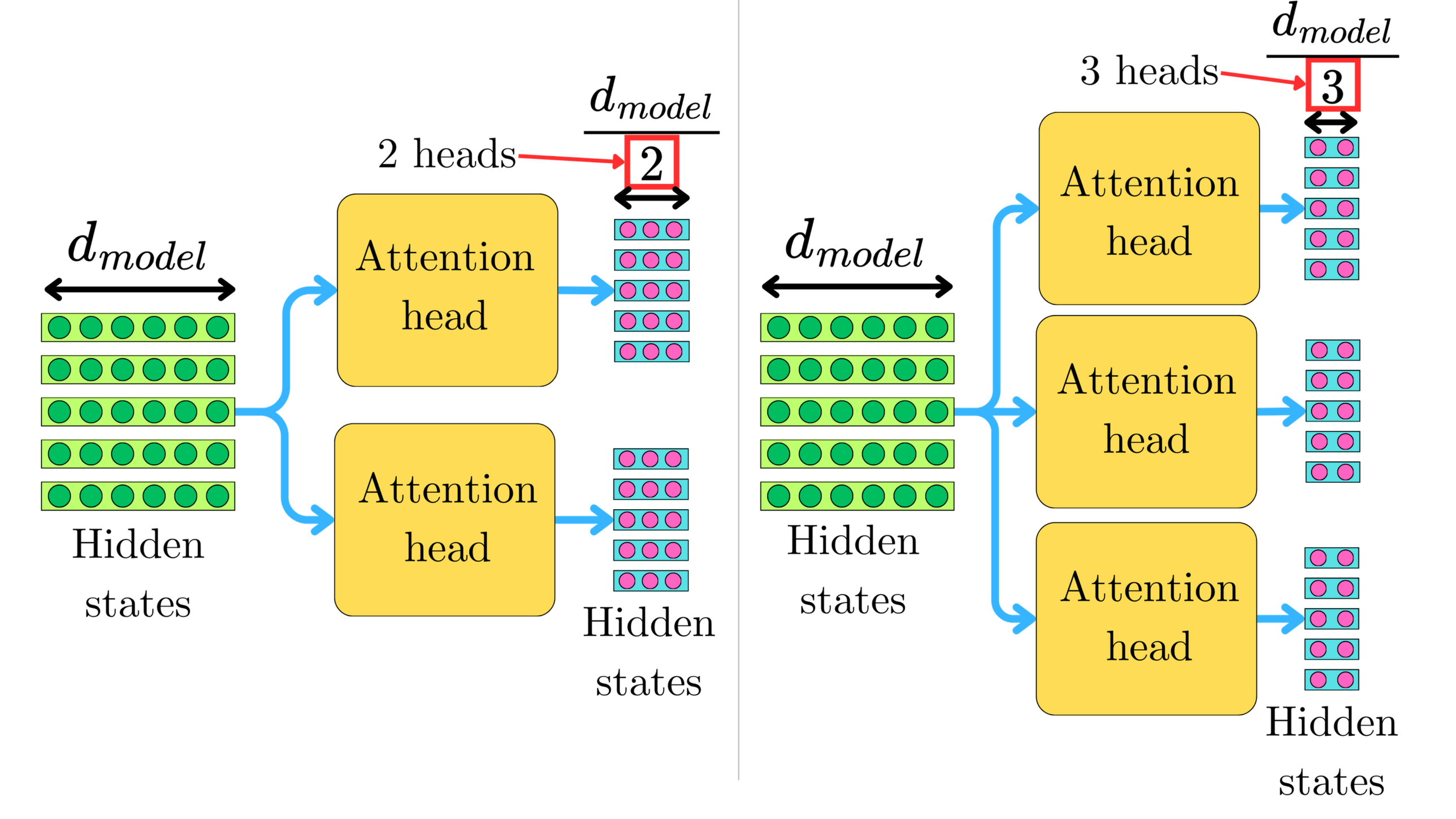
$$E = \frac{1}{\sqrt{d_{\text{model}}}} \begin{bmatrix} \mathbf{q}_1^\top \mathbf{k}_1 & \mathbf{q}_1^\top \mathbf{k}_2 & \cdots & \mathbf{q}_1^\top \mathbf{k}_N \\ \mathbf{q}_2^\top \mathbf{k}_1 & \mathbf{q}_2^\top \mathbf{k}_2 & \cdots & \mathbf{q}_2^\top \mathbf{k}_N \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{q}_N^\top \mathbf{k}_1 & \mathbf{q}_N^\top \mathbf{k}_2 & \cdots & \mathbf{q}_N^\top \mathbf{k}_N \end{bmatrix}$$

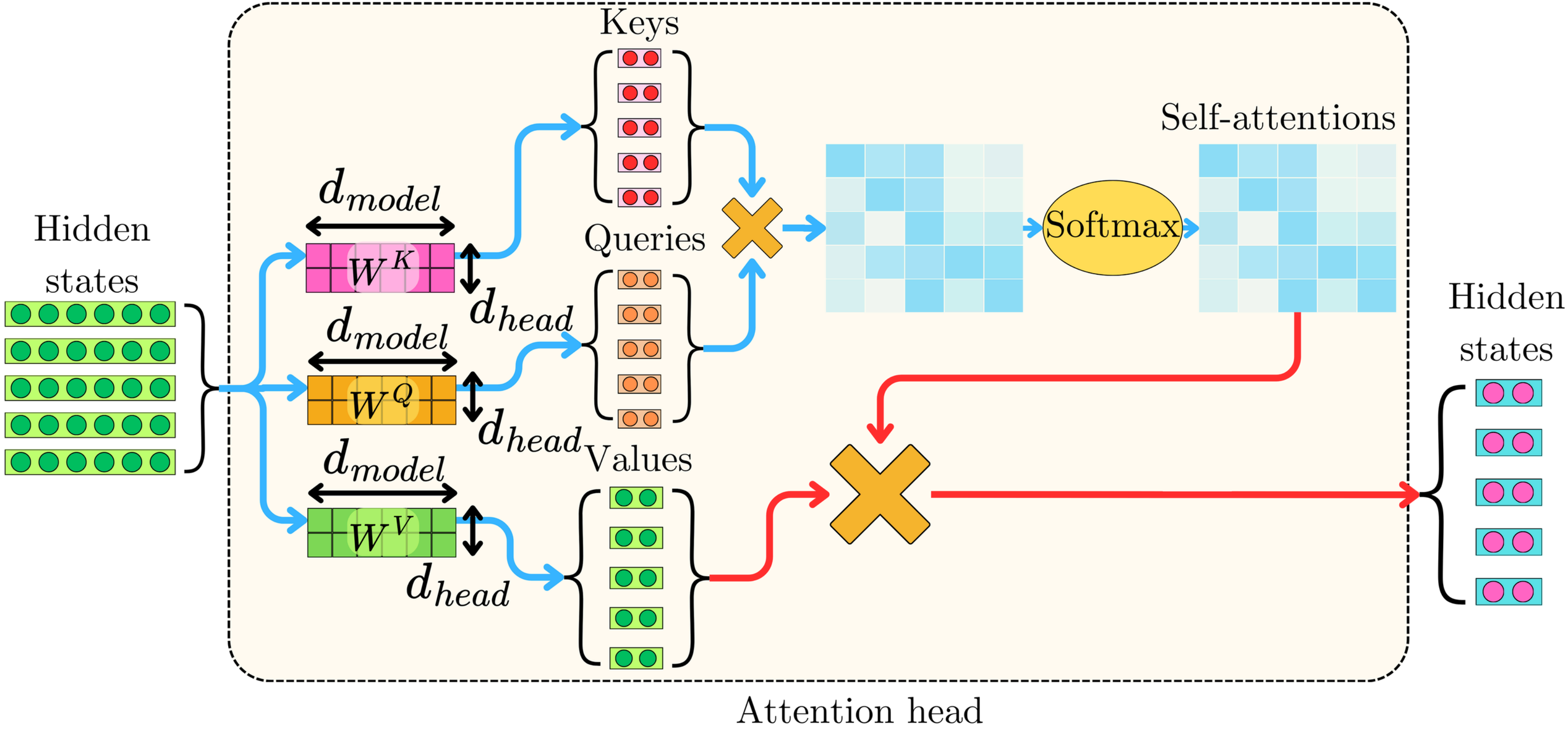
$$a_{ij} = \text{Softmax}(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{j=1}^{N} \exp(e_{ij})}$$

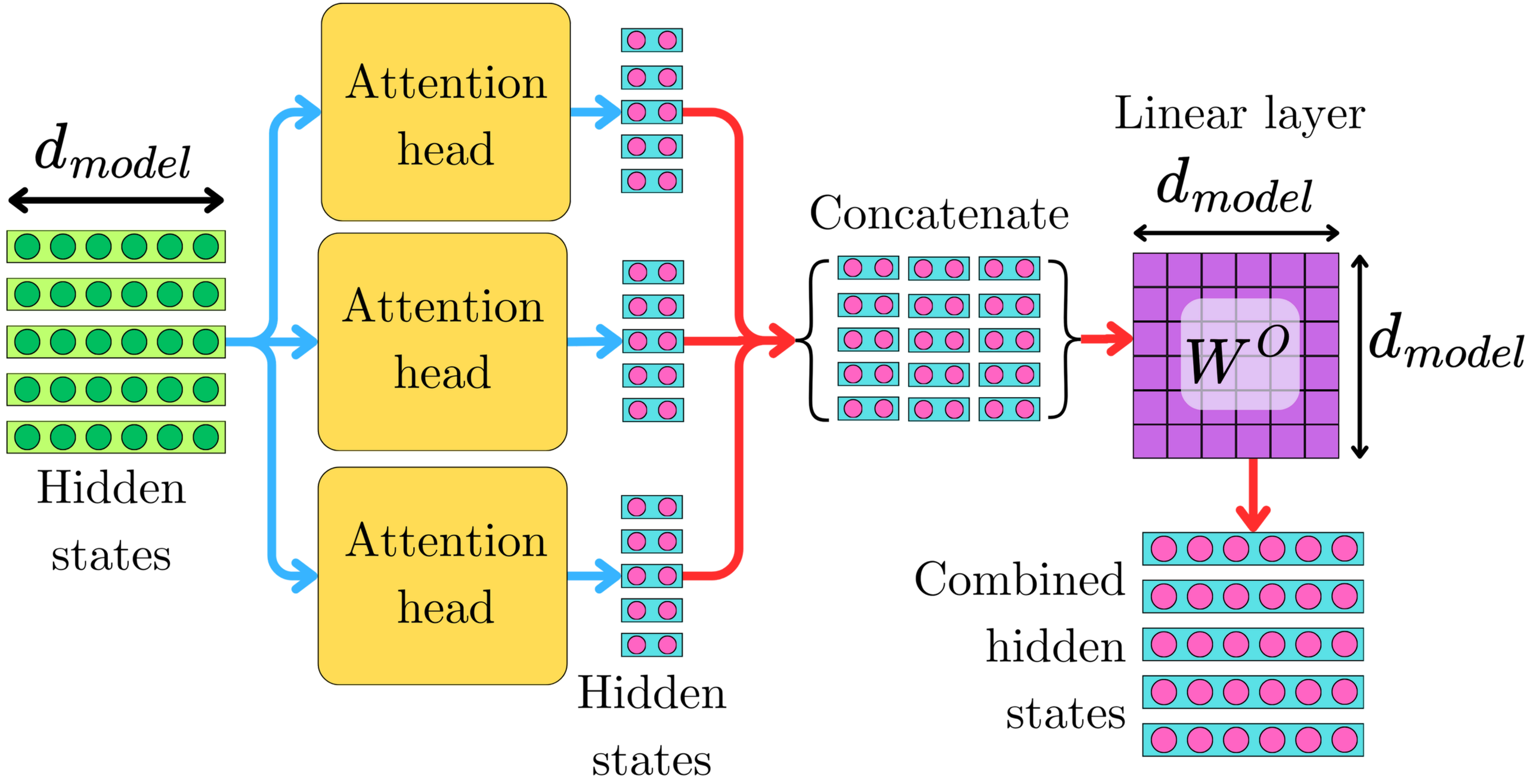


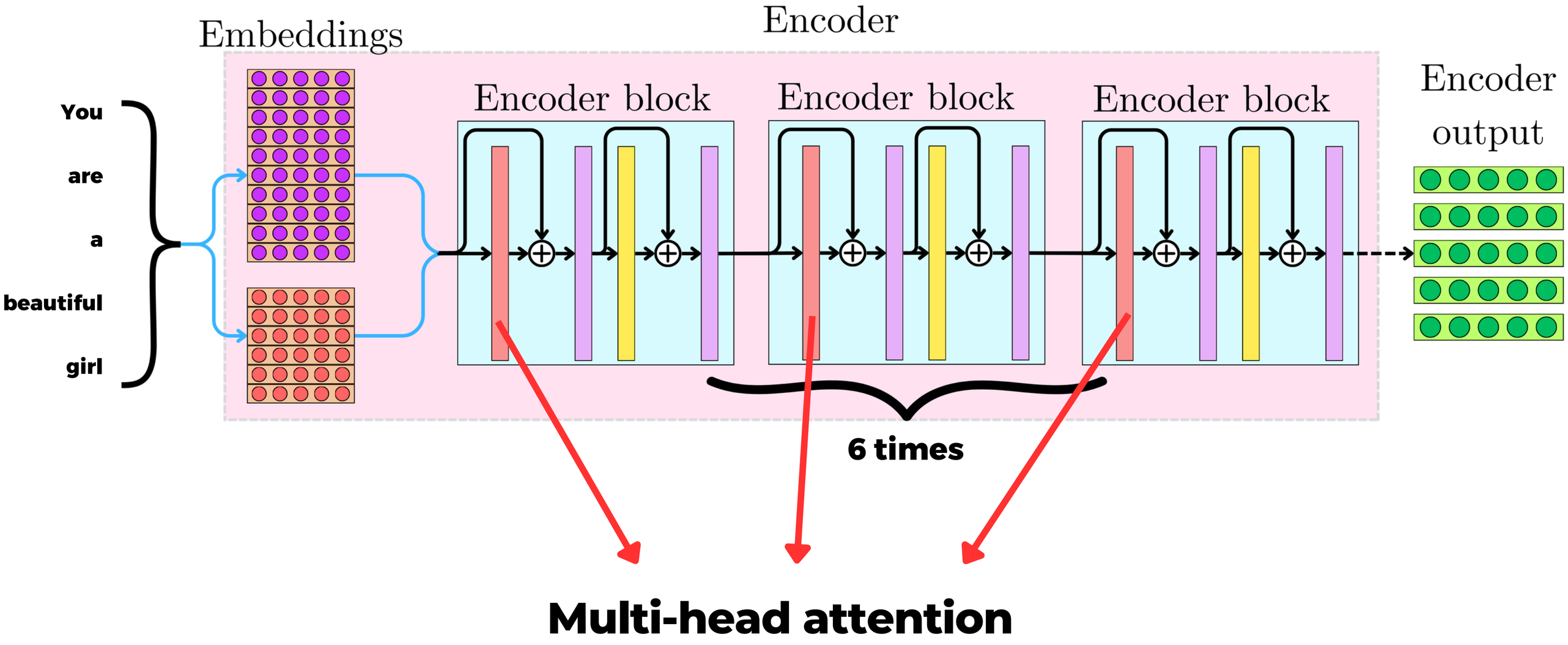

You
are
a
beautiful
girl
W^K
W^Q
W^V
Keys
Queries
Values
Softmax
Self-attentions
Context
vectors
$C = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_{\text{model}}}}\right)V$

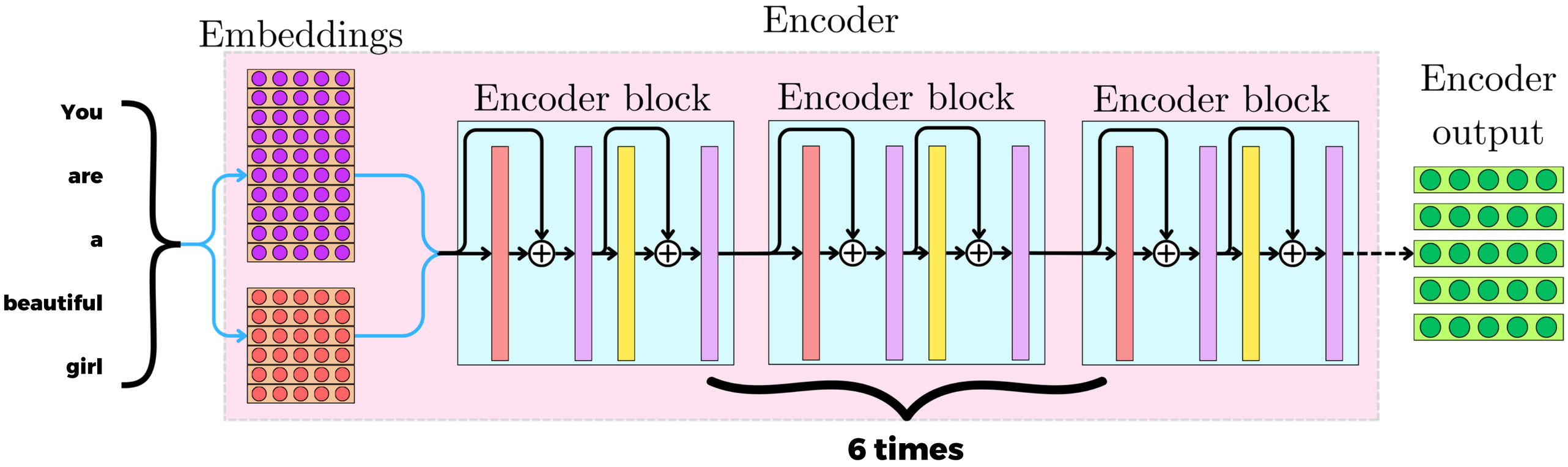

Keys
Queries
Values
Self-attentions
.24
.10
.04
.53
.09
Sums to 1
'How'
'are'
'you'
'doing'
'?'
Weighted average of the Values
Context vectors
Context vector for token 'you'

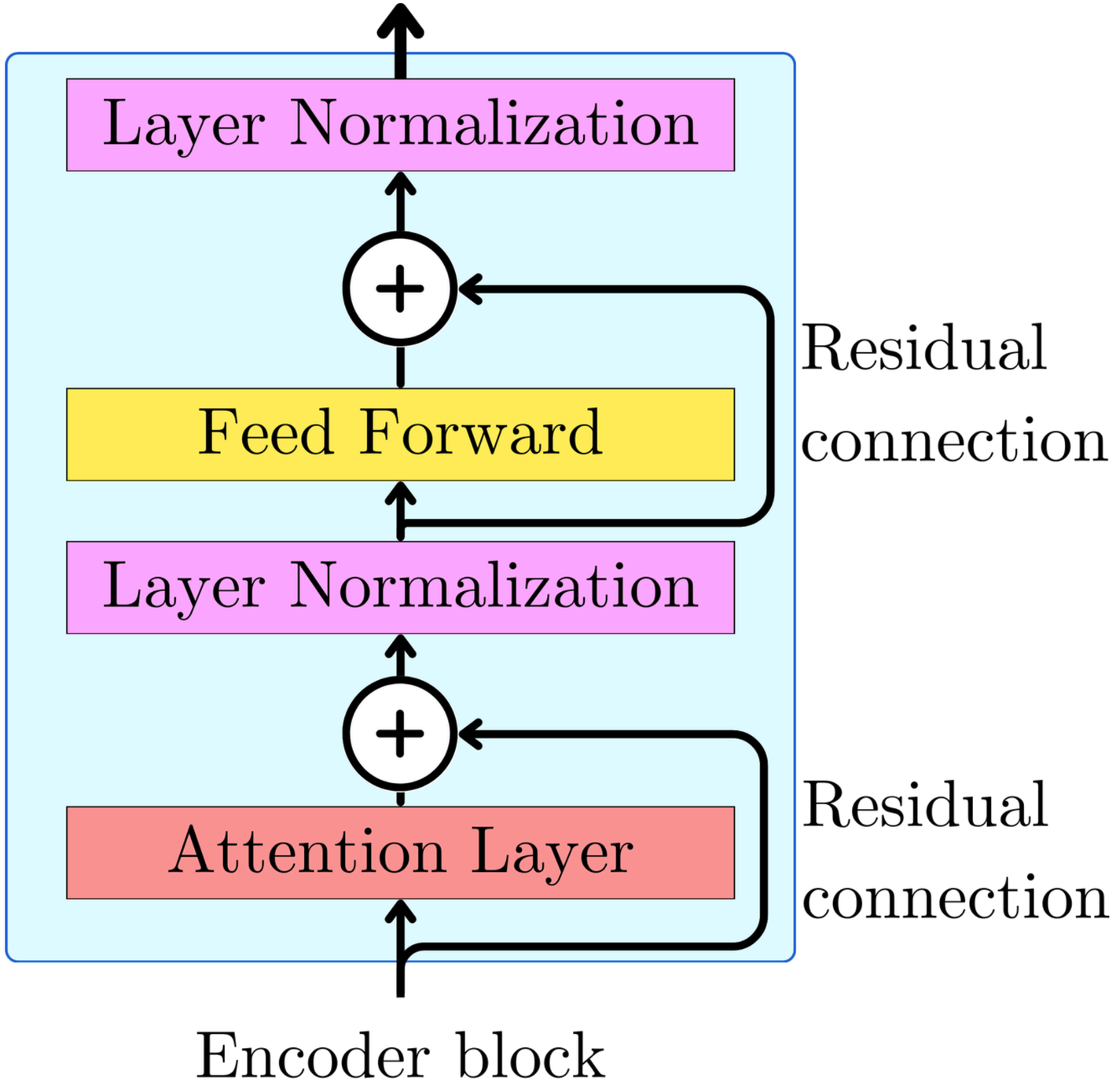

d_{model}
2 heads
2
Attention head
Attention head
Hidden states
d_{model}
Hidden states
d_{model}
3 heads
3
Attention head
Attention head
Attention head
d_{model}
Hidden states

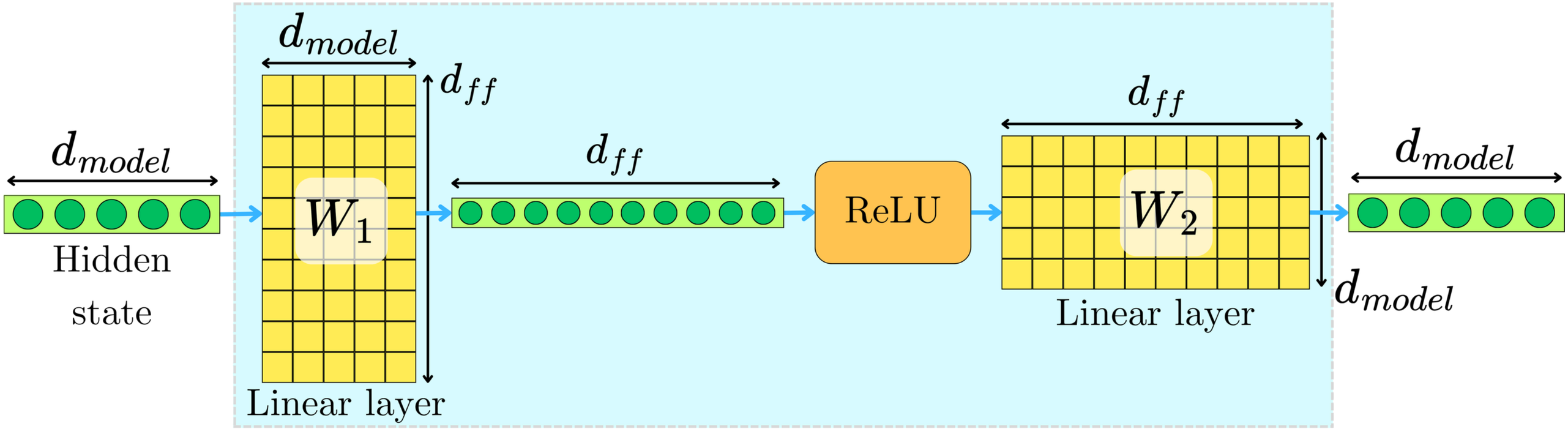

Hidden states
d_{model}
W^K
d_{head}
Keys
d_{model}
W^Q
d_{head}
Queries
d_{model}
W^V
d_{head}
Values
Softmax
Self-attentions
Hidden states
Attention head


d_{model}
Hidden states
Attention head
Attention head
Attention head
Hidden states
Concatenate
Linear layer
d_{model}
W^O
d_{model}
Combined hidden states

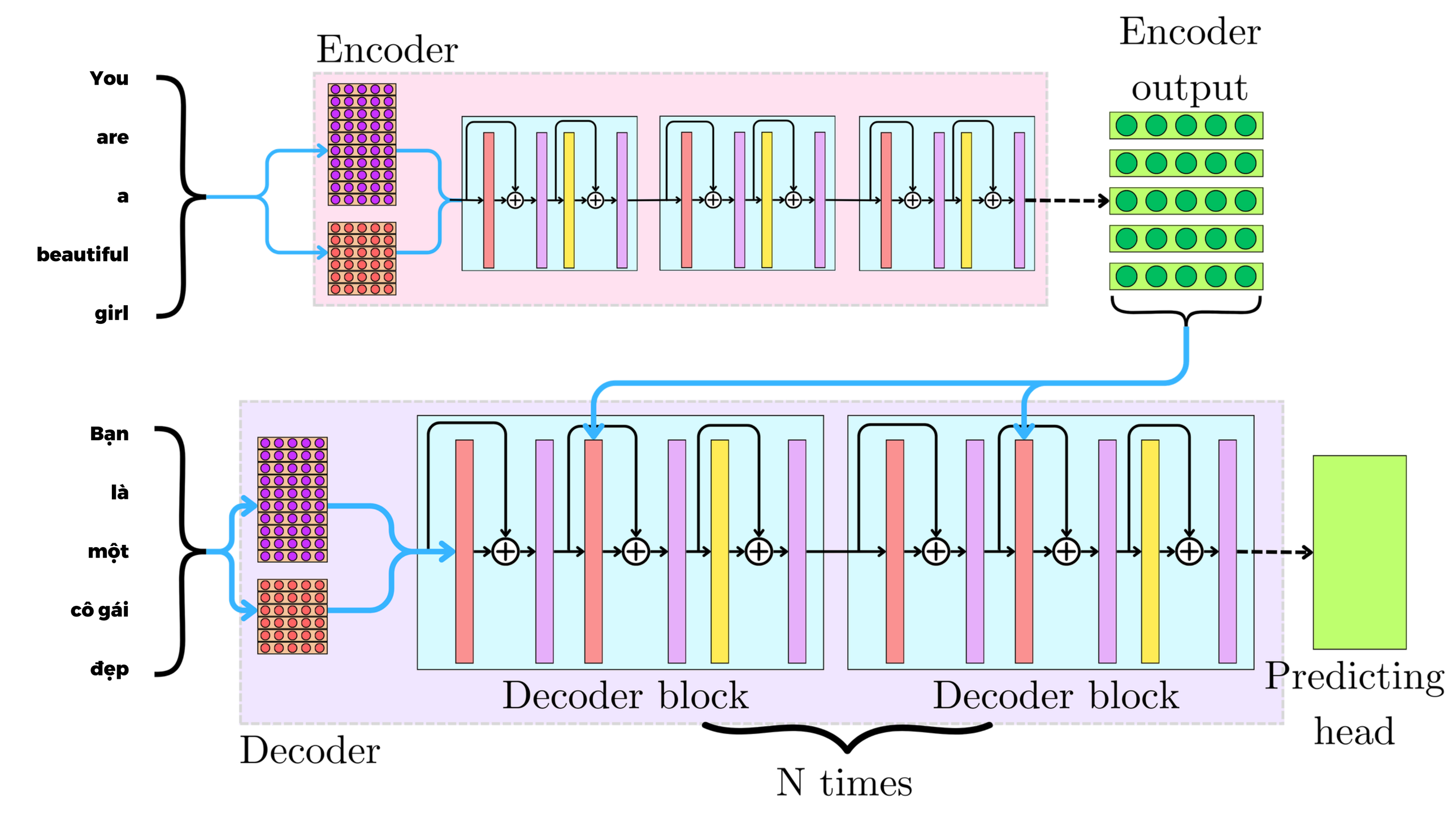

Embeddings
Encoder
You
are
a
beautiful
girl
Encoder block
Encoder block
Encoder block
Encoder output
6 times
Multi-head attention

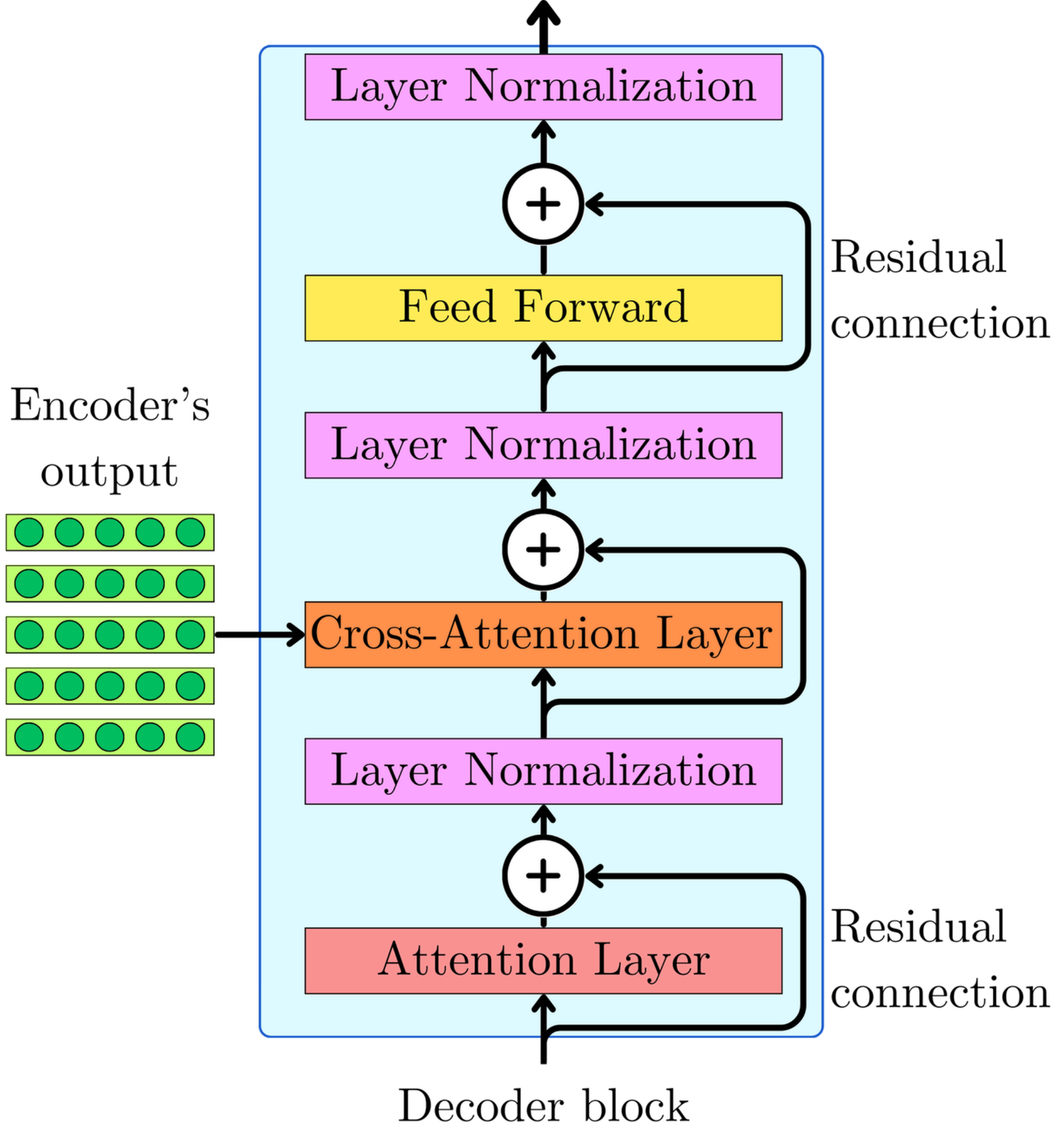

Embeddings
Encoder
You
are
a
beautiful
girl
Encoder block
Encoder block
Encoder block
Encoder output
6 times

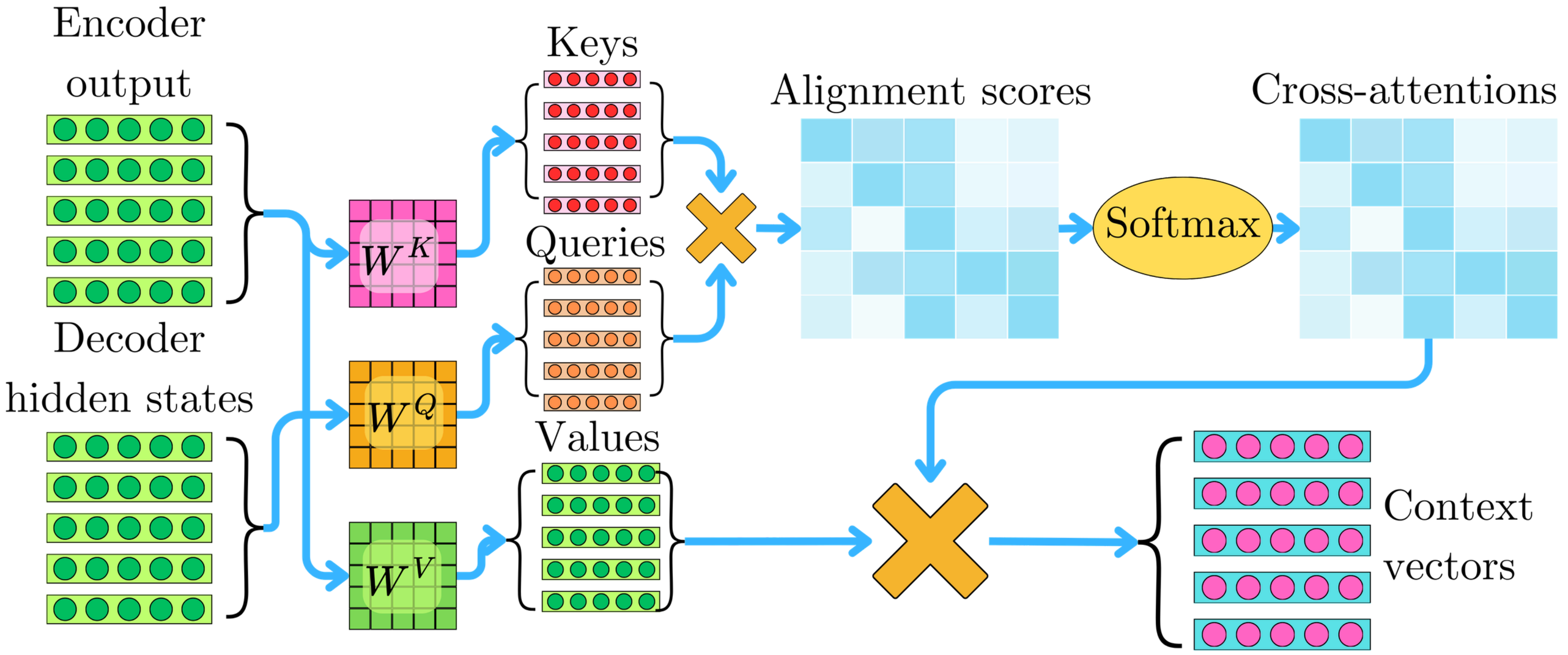

Layer Normalization
Feed Forward
Layer Normalization
Attention Layer
Residual connection
Residual connection
Encoder block



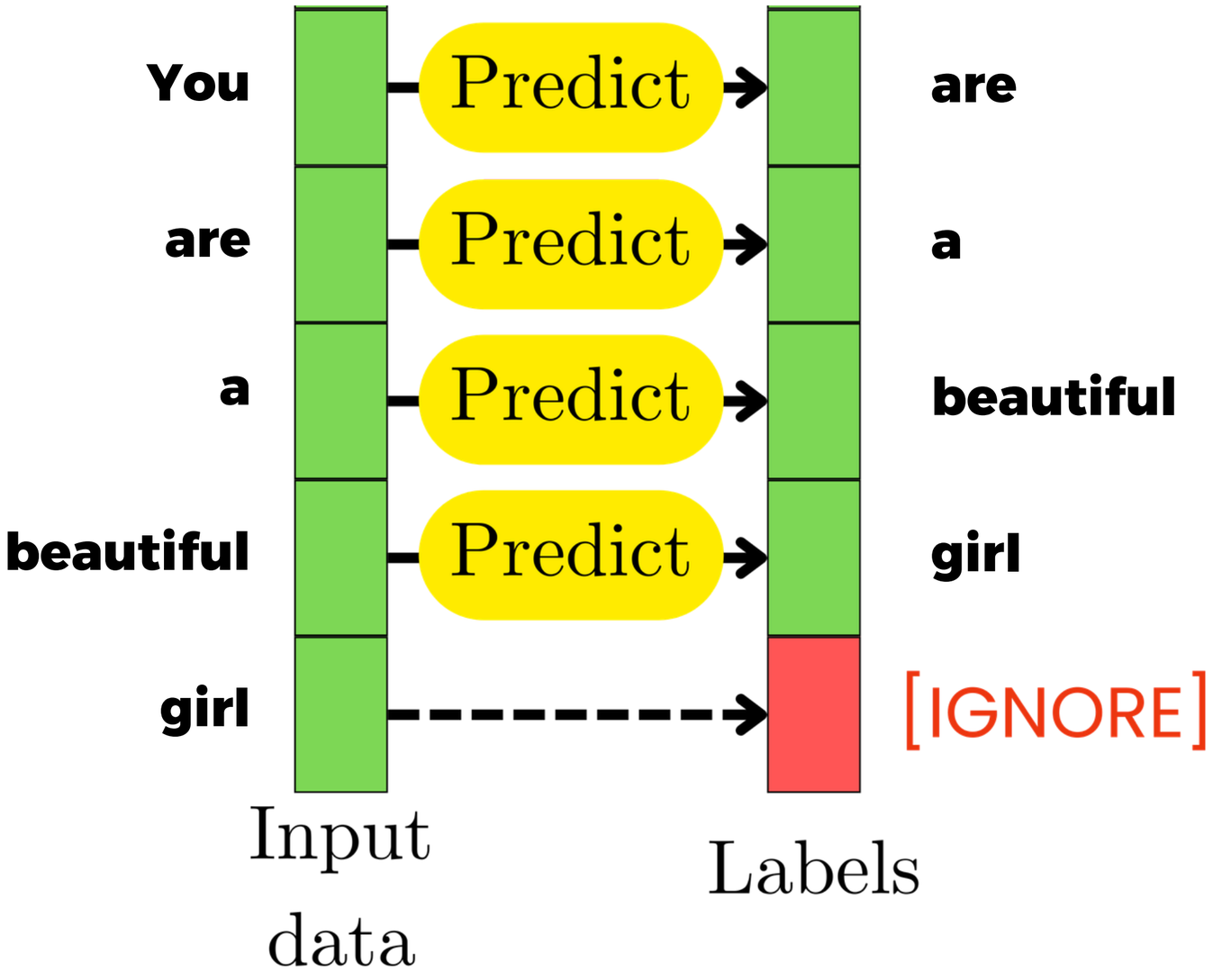
The Position-wise Feed-forward Network

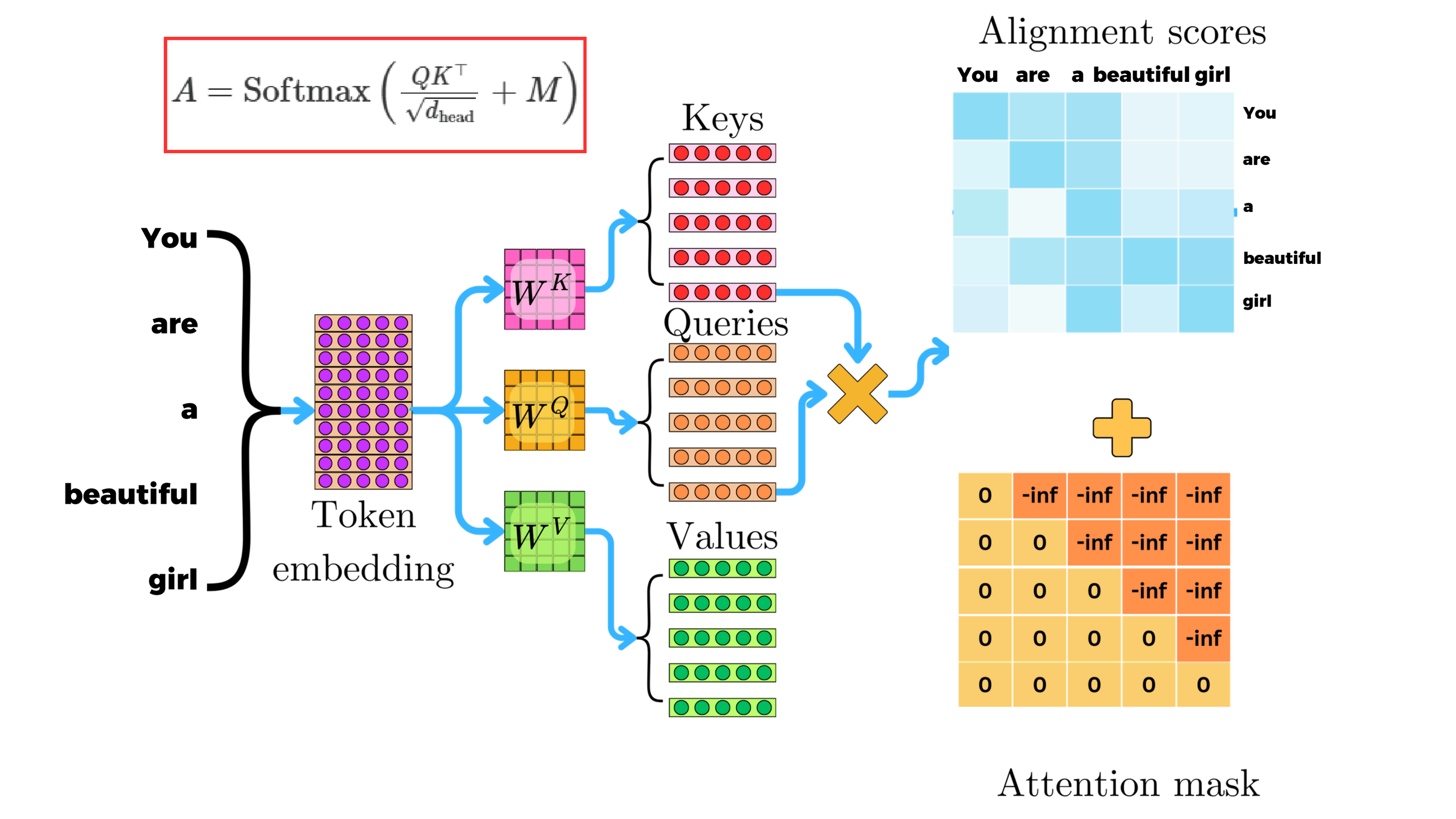

Encoder
Encoder output
You
are
a
beautiful
girl
Bạn
là
một
cô gái
đẹp
Decoder block
Decoder block
Decoder
N times
Predicting head

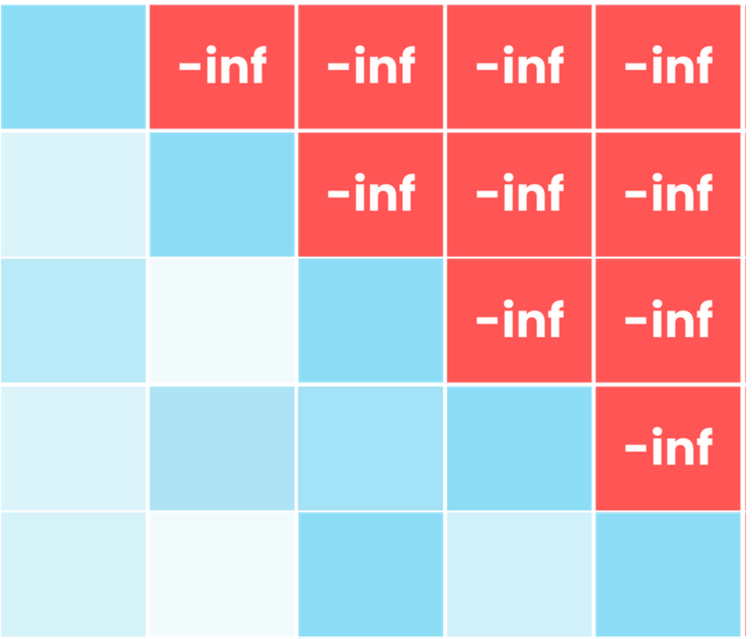

Layer Normalization
Feed Forward
Residual connection
Layer Normalization
Encoder's output
Cross-Attention Layer
Layer Normalization
Attention Layer
Residual connection
Decoder block


Encoder
output
Decoder
hidden states
W^K
W^Q
W^V
Keys
Queries
Values
Alignment scores
Softmax
Cross-attentions
Context
vectors

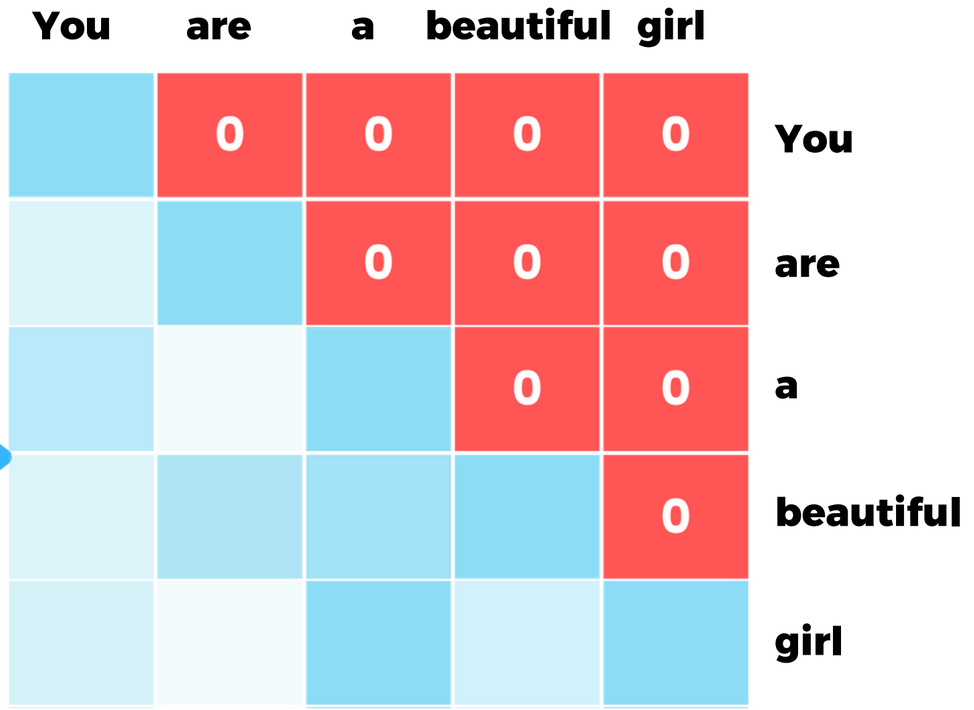
Causal Language Modeling




$A = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_{\text{head}}}} + M\right)$
Alignment scores
You are a beautiful girl
You
are
a
beautiful
girl
Keys
Queries
Values
W^K
W^Q
W^V
You
are
a
beautiful
girl
Token embedding
0 -inf -inf -inf -inf
0 0 -inf -inf -inf
0 0 0 -inf -inf
0 0 0 0 -inf
0 0 0 0 0
Attention mask

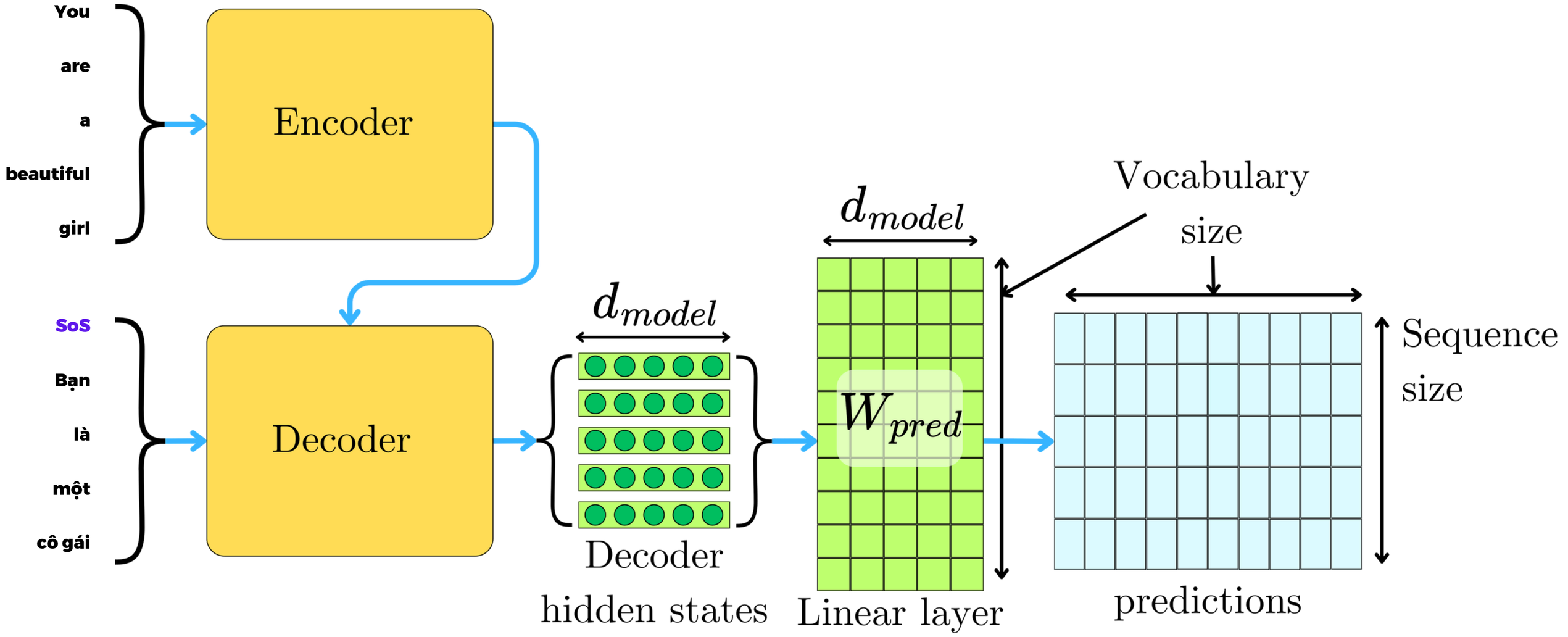


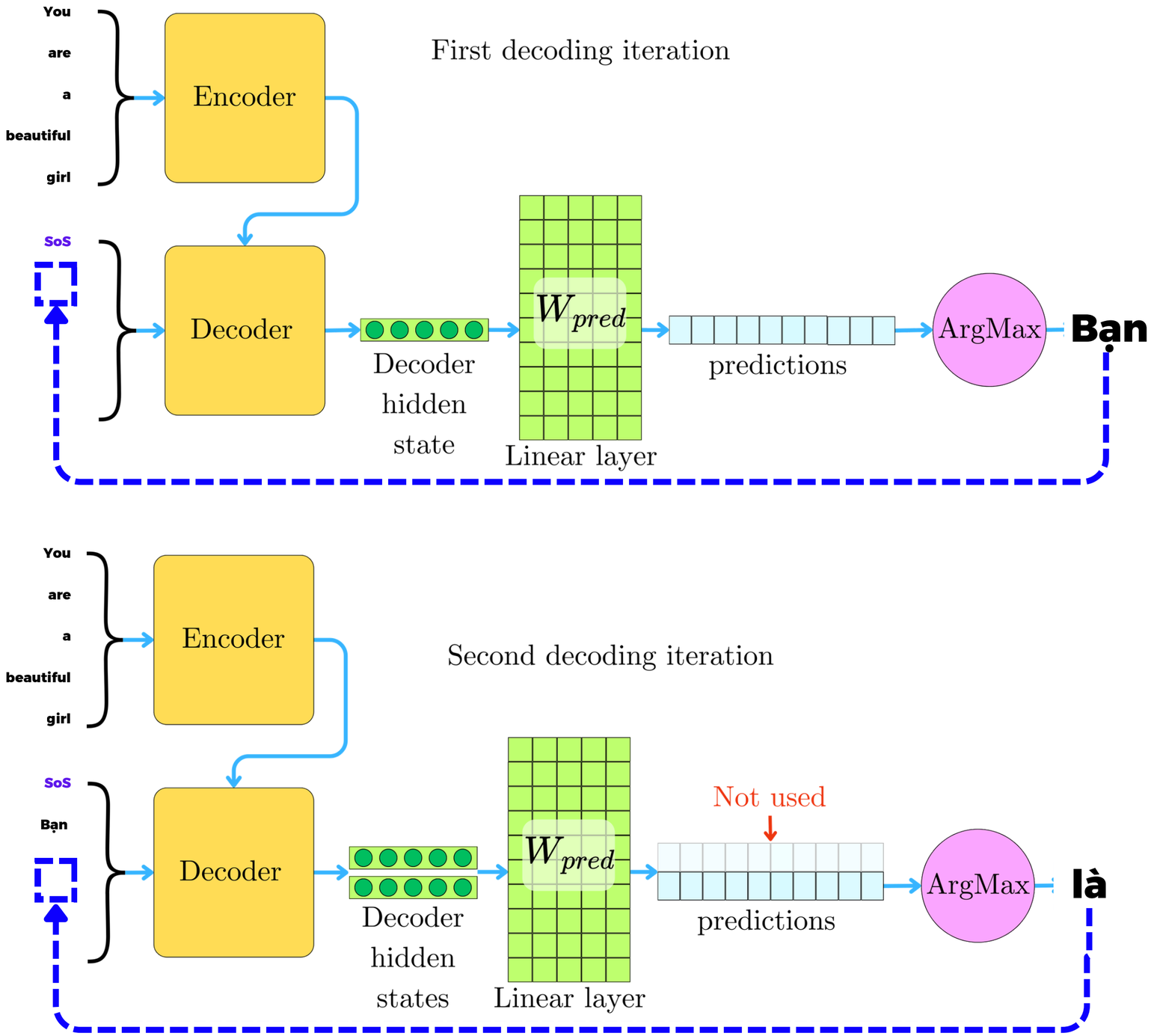
Softmax

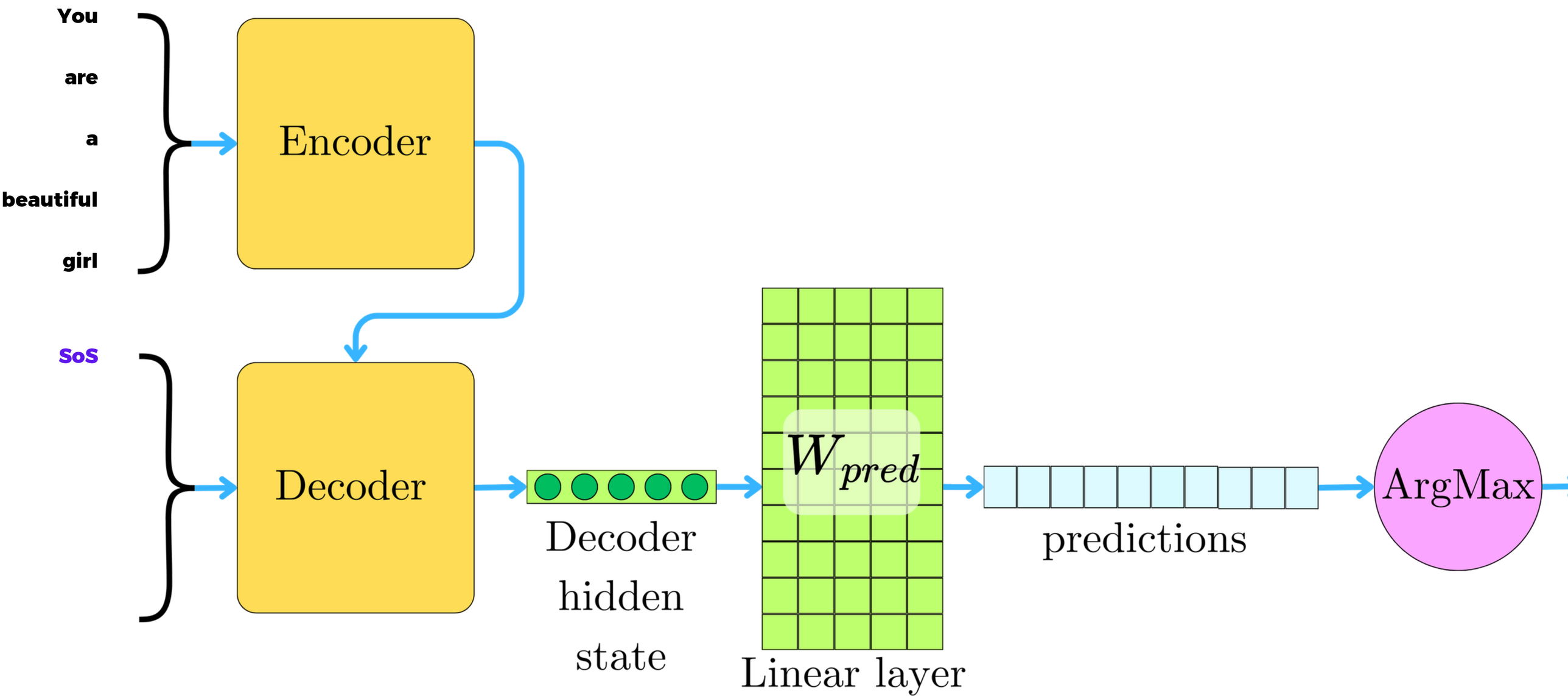


Alignment scores

Self-attentions

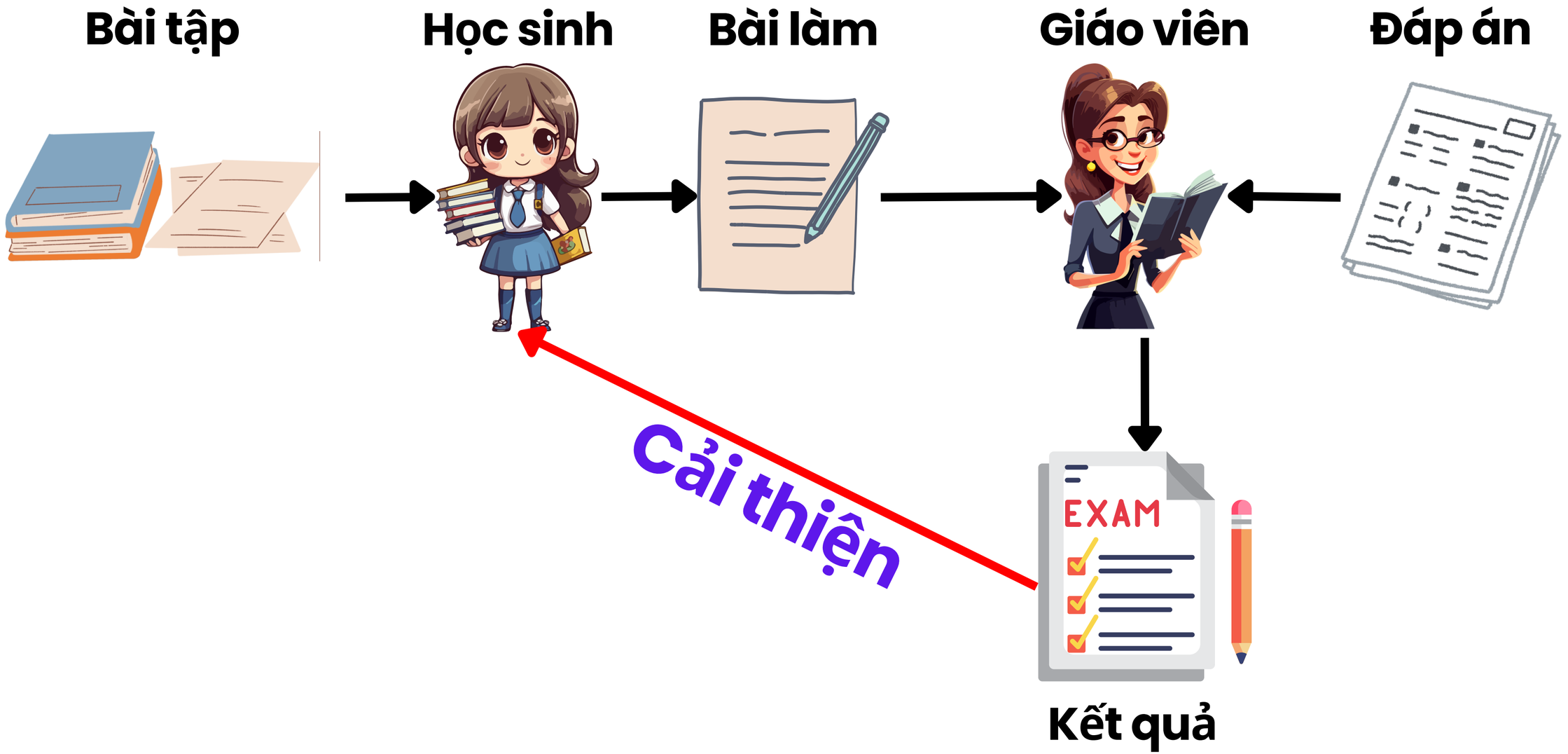

You
are
a
beautiful
girl
Encoder
SoS
Bạn
là
một
cô gái
Decoder
d_{model}
Decoder
hidden states
d_{model}
W_{pred}
Linear layer
Vocabulary
size
Sequence
size
predictions


You
are
a
beautiful
girl
Encoder
First decoding iteration
SoS
Decoder
Decoder
hidden
state
W_{pred}
Linear layer
predictions
ArgMax
Bạn
You
are
a
beautiful
girl
Encoder
Second decoding iteration
SoS
Bạn
Decoder
Decoder
hidden
states
W_{pred}
Linear layer
Not used
predictions
ArgMax
là


You
are
a
beautiful
girl
Encoder
SoS
Decoder
Decoder
hidden
state
W_{pred}
Linear layer
predictions
ArgMax

How AI models are trained ?

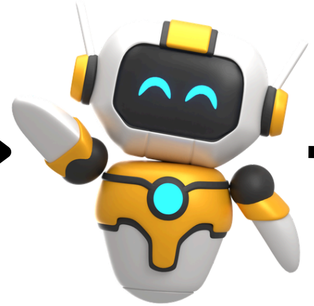
Cách con người học



Cách máy học

Dữ liệu

Mô hình

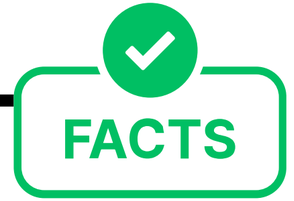


Dự đoán

Hàm mất mát

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(\hat{Y}_i - Y_i)^2$$

Thực tế



Cập nhật

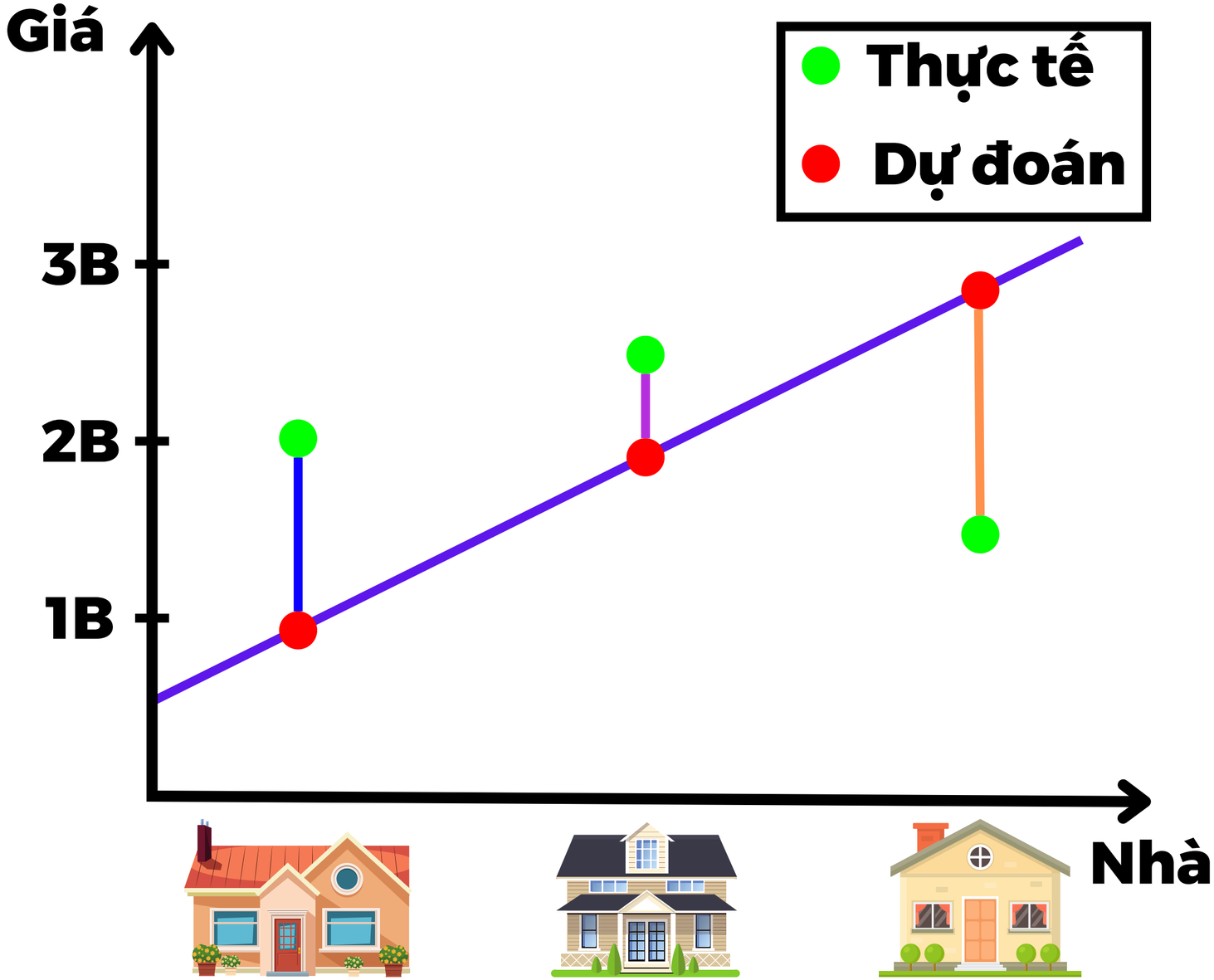
Giá trị mất mát

Tối ưu hóa mô hình

=

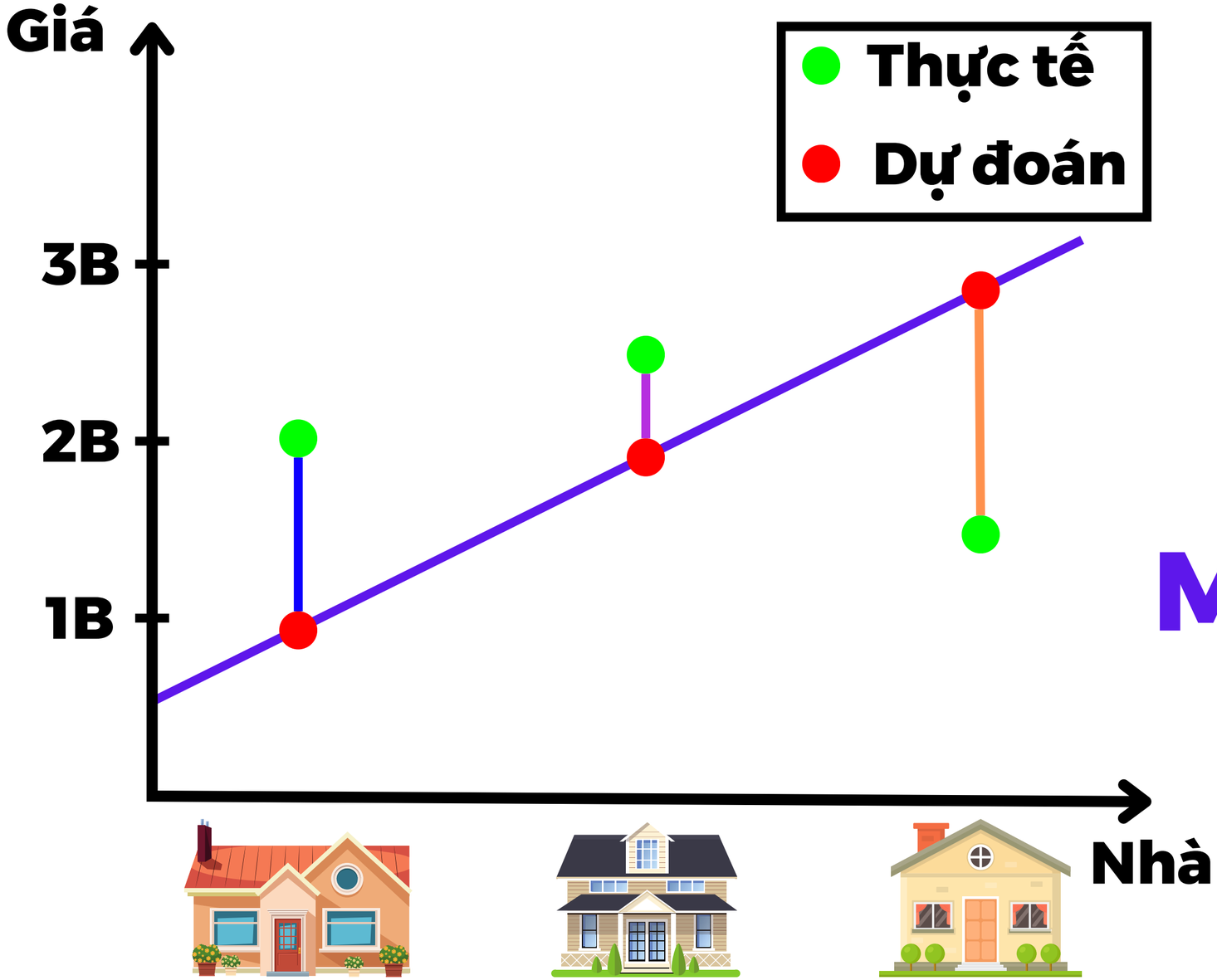
Tối thiểu hóa giá trị mất mát

Hàm mất mát



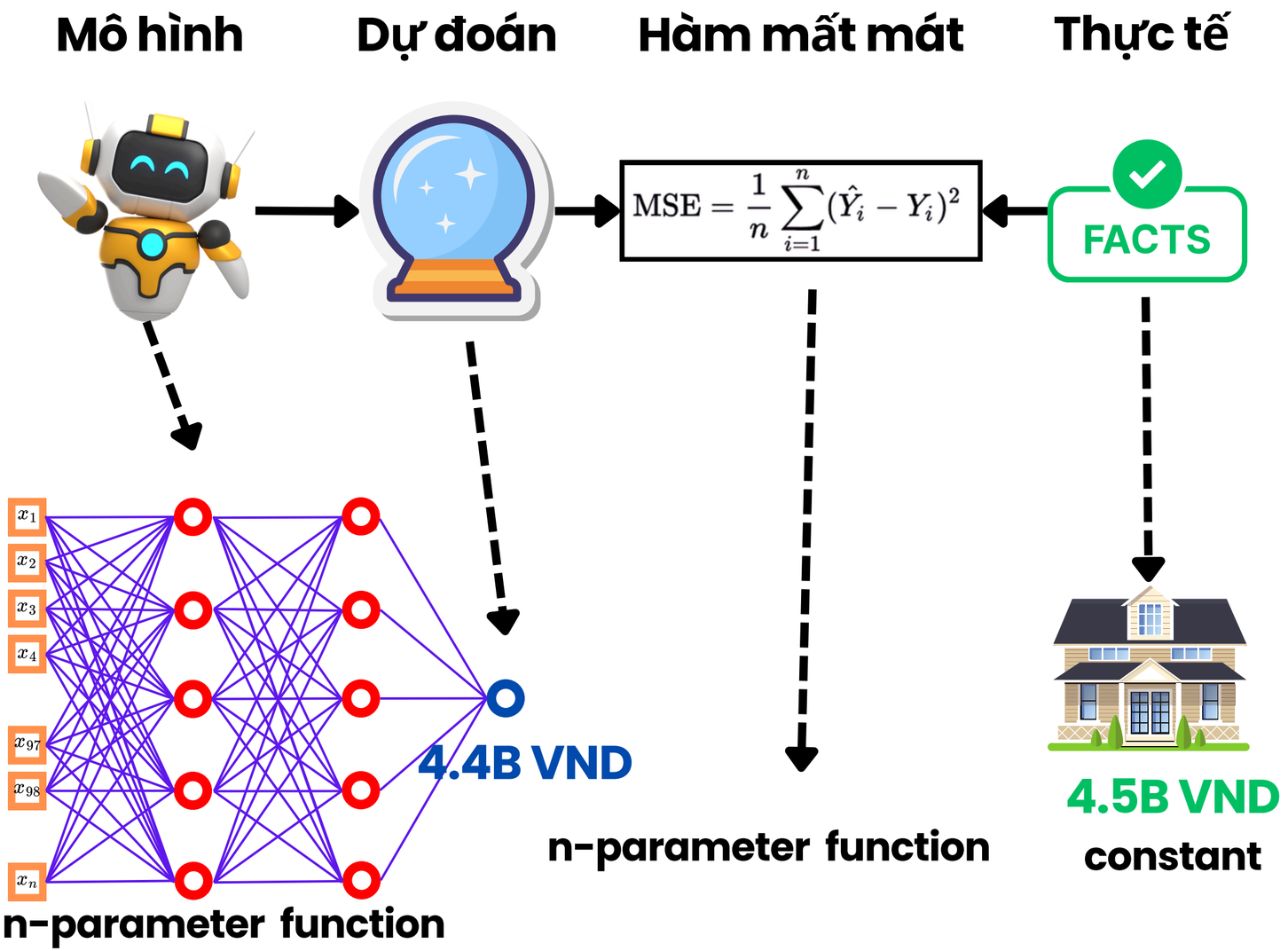
$$MAE = \frac{|\ +\ |\ +\ |}{3}$$

Hàm mất mát



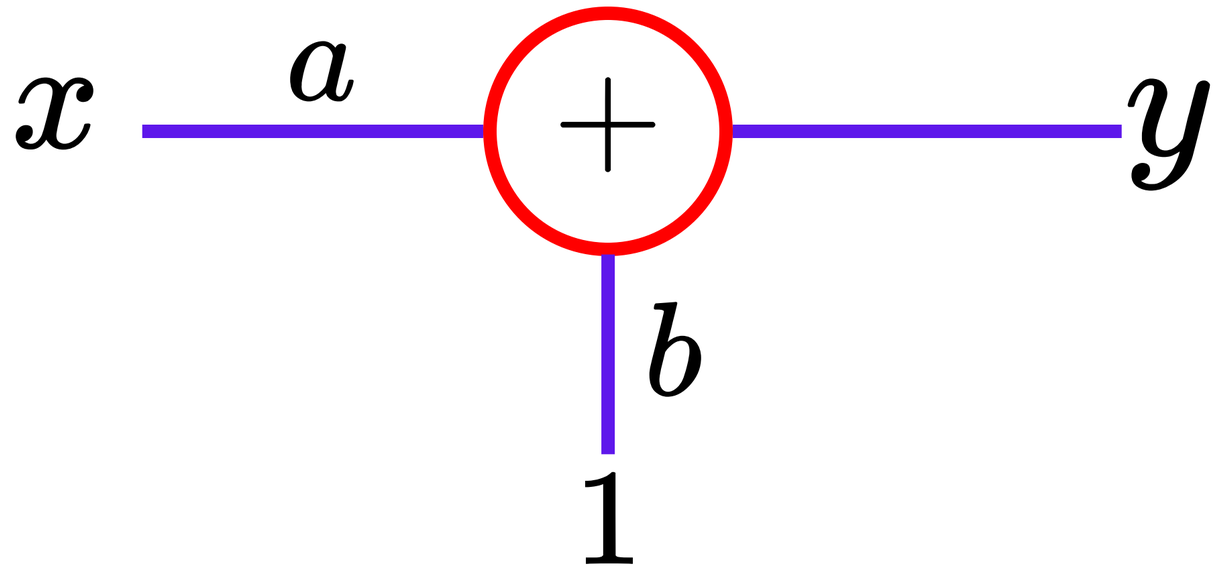
$$MSE = \frac{\blacksquare + \blacksquare + \blacksquare}{3}$$

Cách máy học


Mô hình
Dự đoán
Hàm mất mát
Thực tế
$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(\hat{Y}_i - Y_i)^2$
FACTS
x_1
x_2
x_3
x_4
x_{97}
x_{98}
x_n
4.4B VND
n-parameter function
n-parameter function
4.5B VND
constant

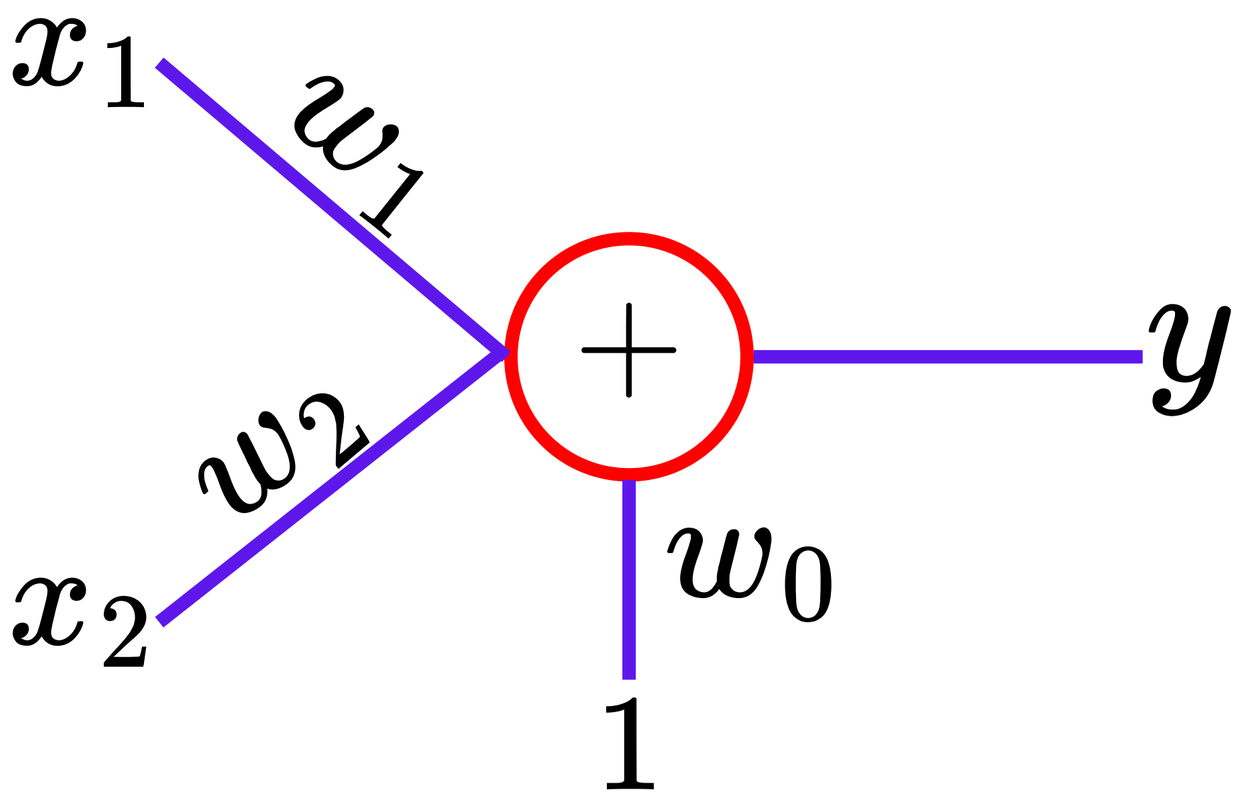
1-parameter function

$$y = ax + b$$



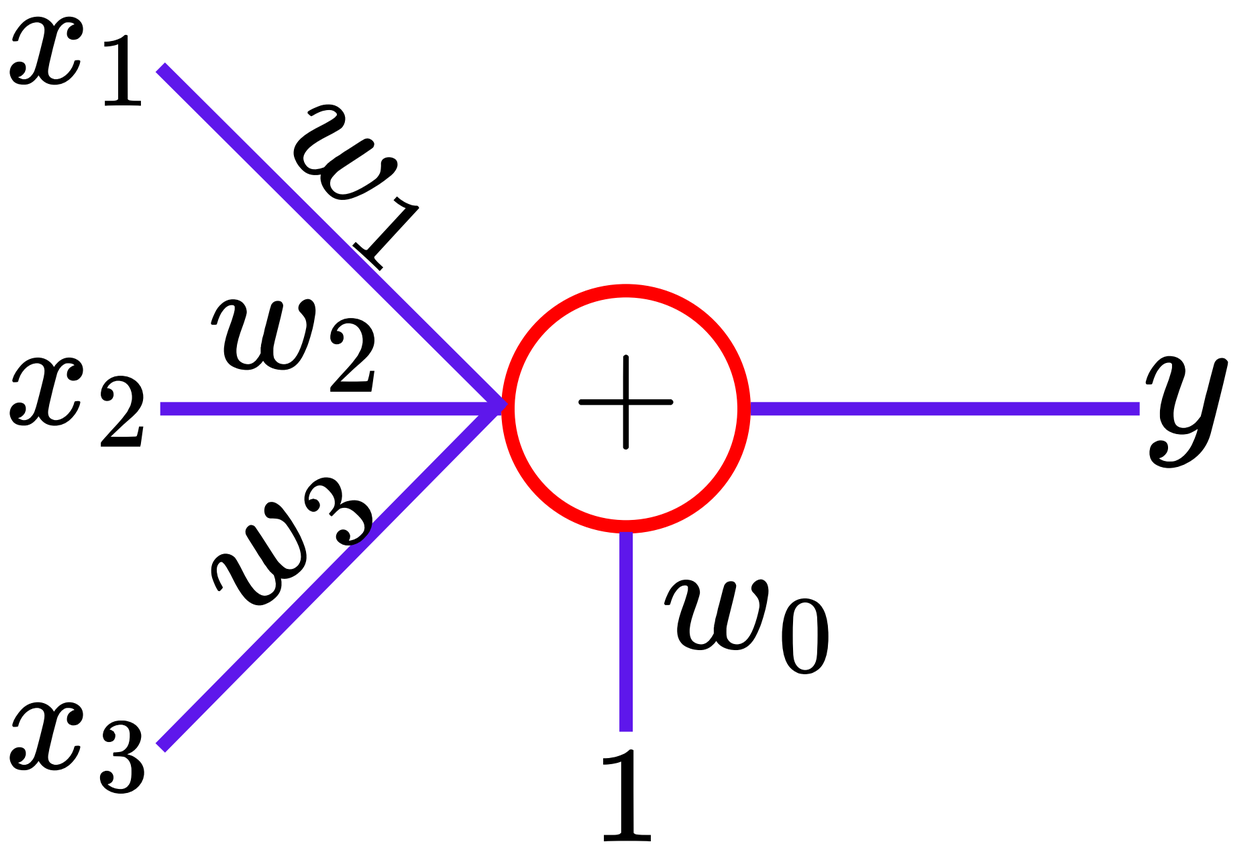
2-parameter function

$$y = w_1 x_1 + w_2 x_2 + w_0$$

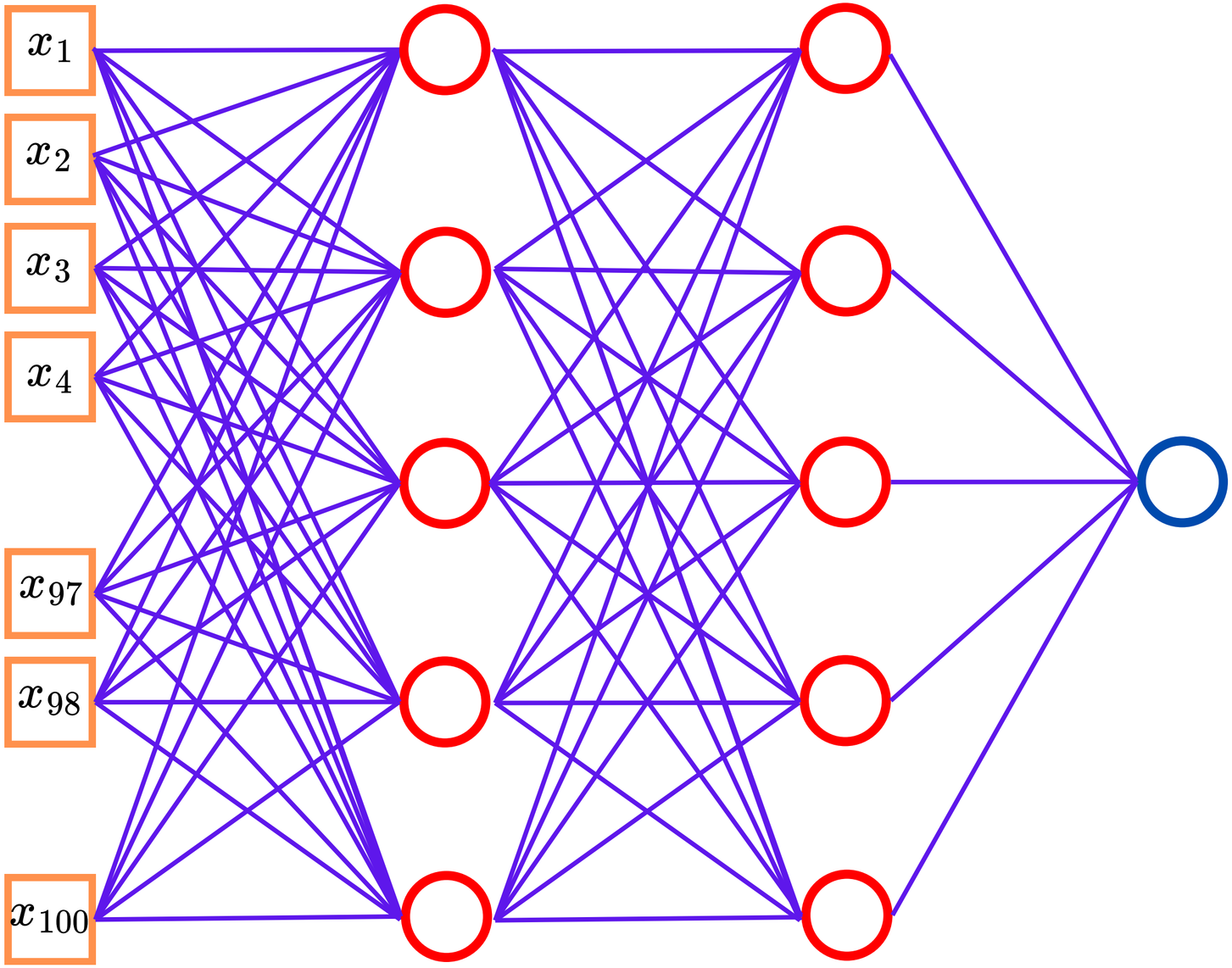


3-parameter function

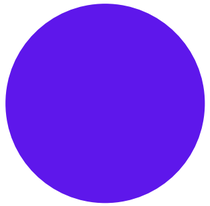
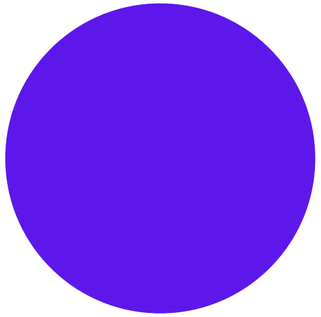
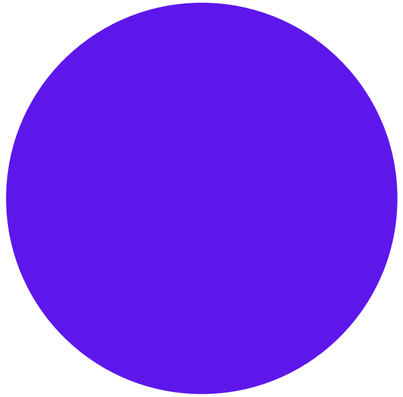
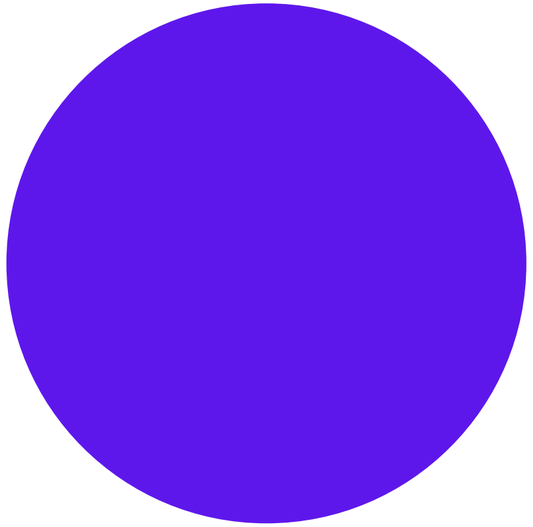
$$y = w_1 x_1 + w_2 x_2 + w_3 x_3 + w_0$$



n-parameter function



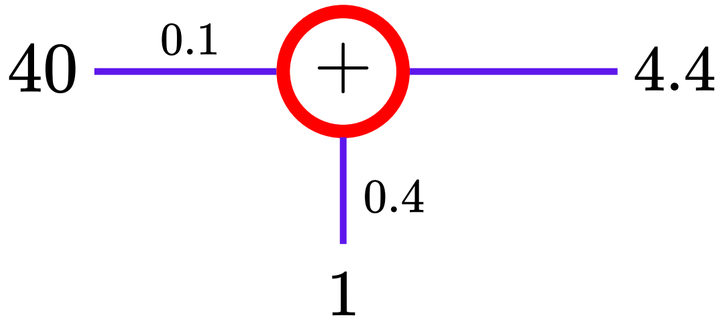
GPT versions

GPT-1	GPT-2	GPT-3	GPT-4	GPT-5
				
117 M	1.5 B	175 B	1.8 T	?

Mô hình dự đoán giá nhà

$$y = 0.1x + 0.4$$

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(\hat{Y}_i - Y_i)^2$$

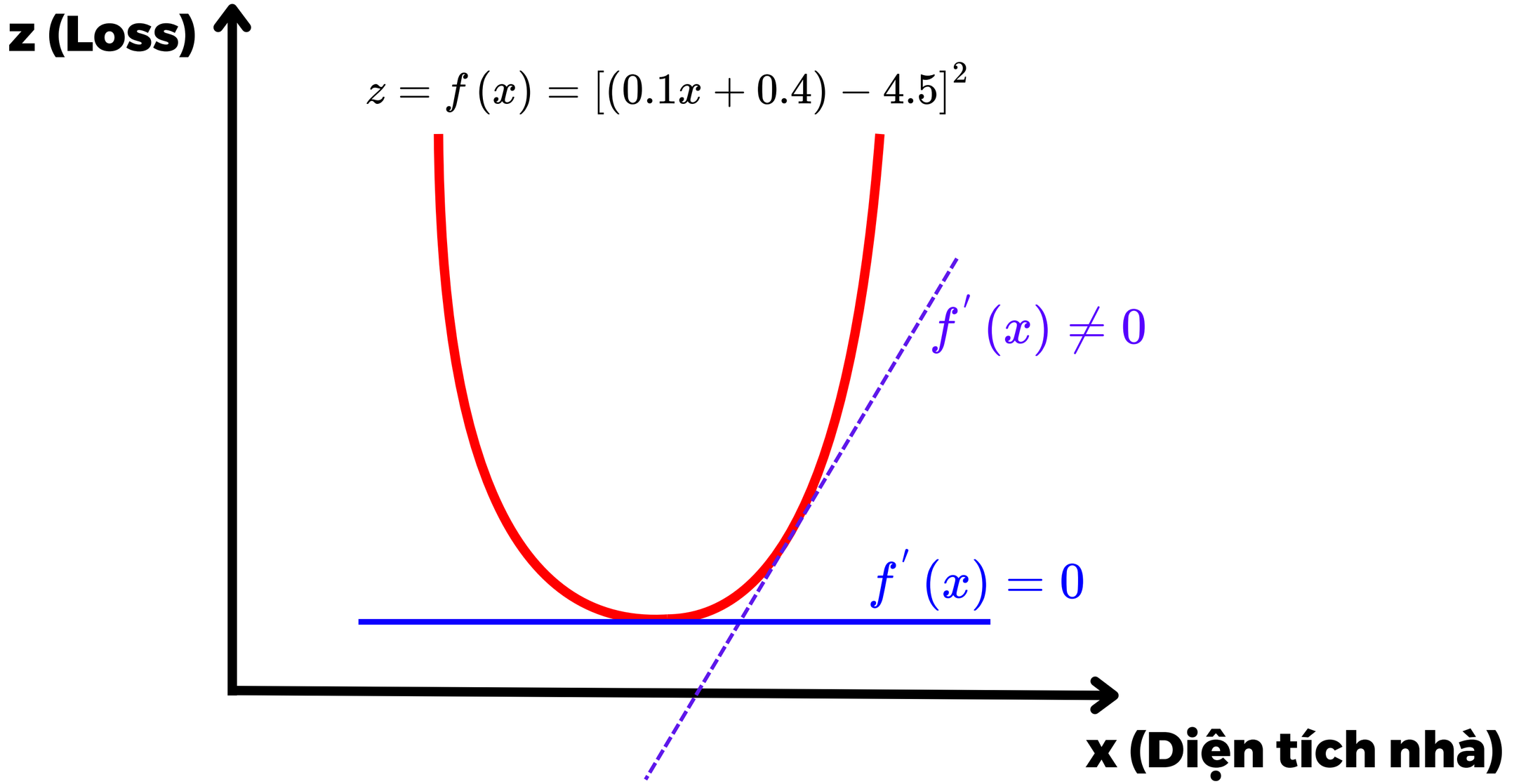


$$\left[(0.1x + 0.4) - 4.5\right]^2$$

4.5B VND

Ta cần tối thiểu hóa giá trị này

Tối thiểu hóa loss

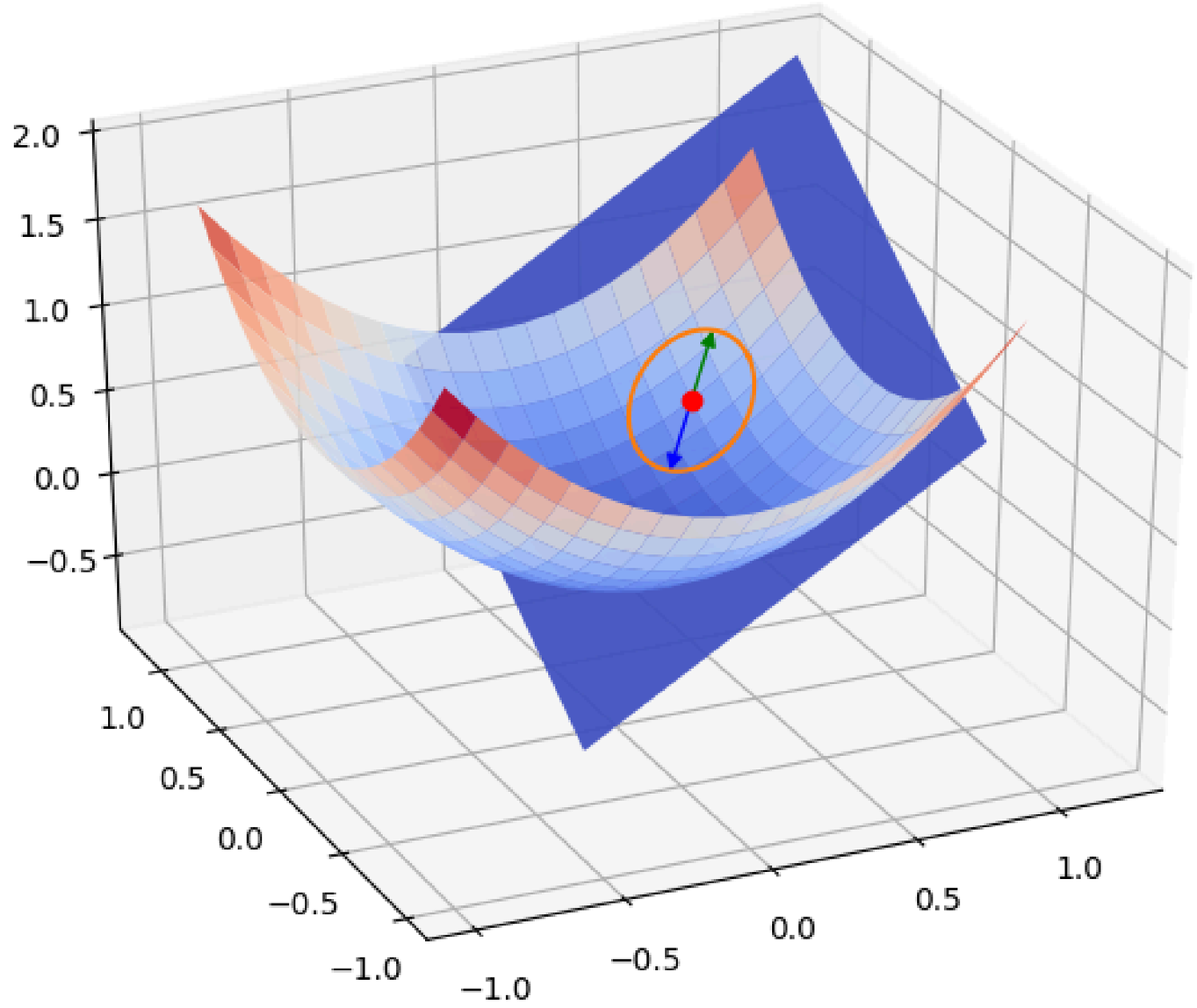


Gradient

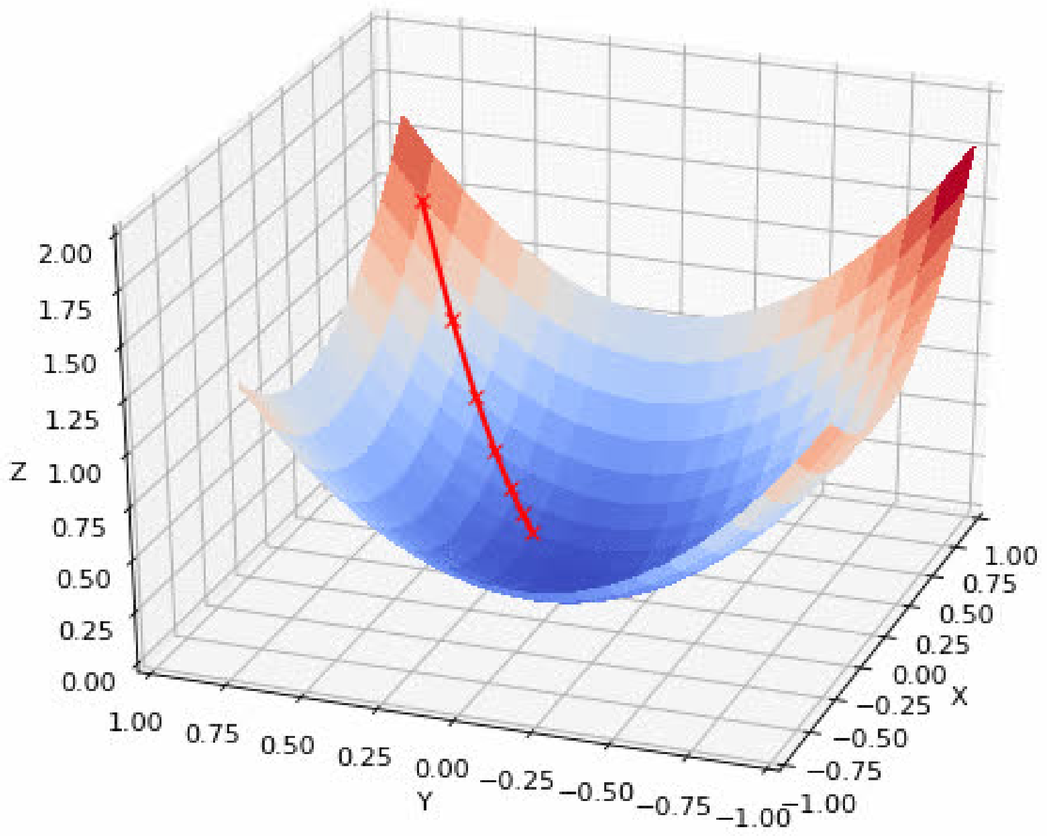
$$\bigtriangledown f\left(x,y\right)=\begin{bmatrix}\frac{\Delta f}{\Delta x}\\ \frac{\Delta f}{\Delta y}\end{bmatrix}=gradient$$

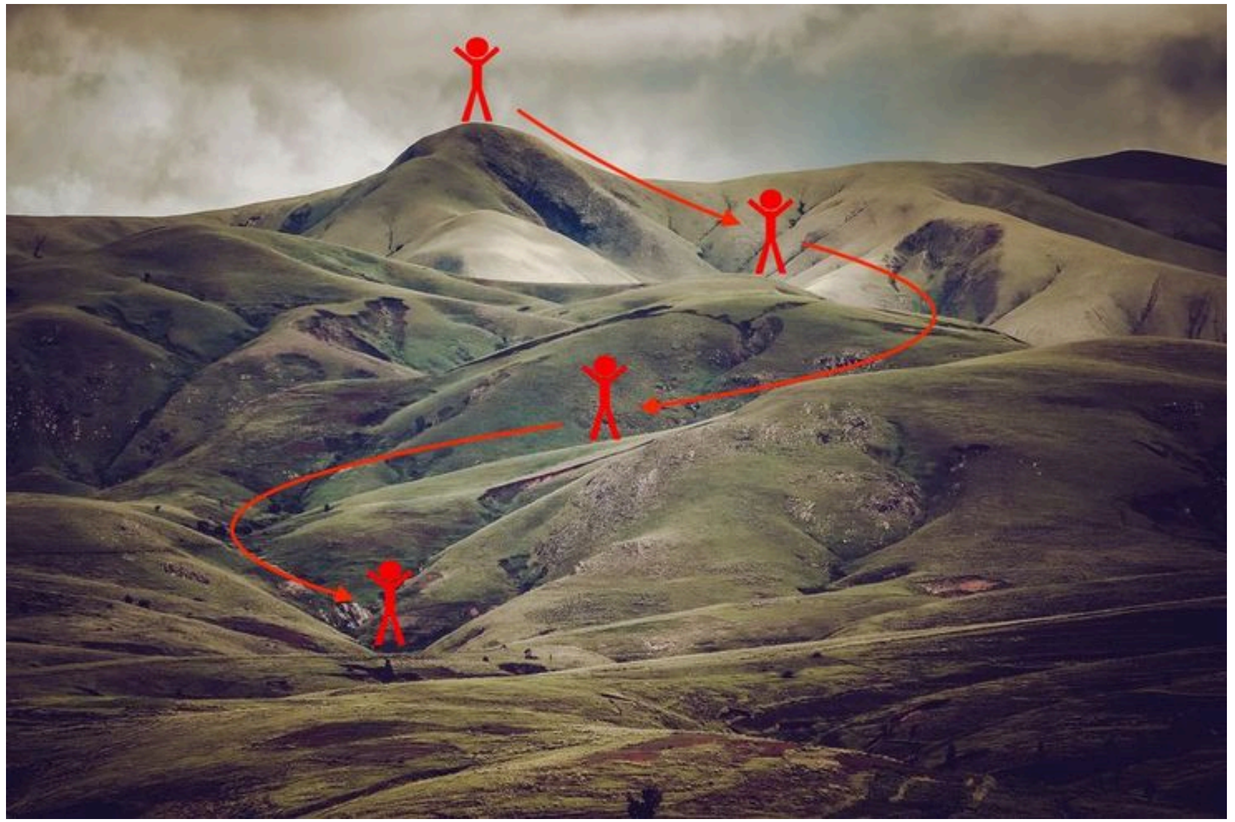
$$\bigtriangledown f\left(x,y,z\right)=\begin{bmatrix}\frac{\Delta f}{\Delta x}\\ \frac{\Delta f}{\Delta y}\\ \frac{\Delta f}{\Delta z}\end{bmatrix}=gradient$$

2D Gradient

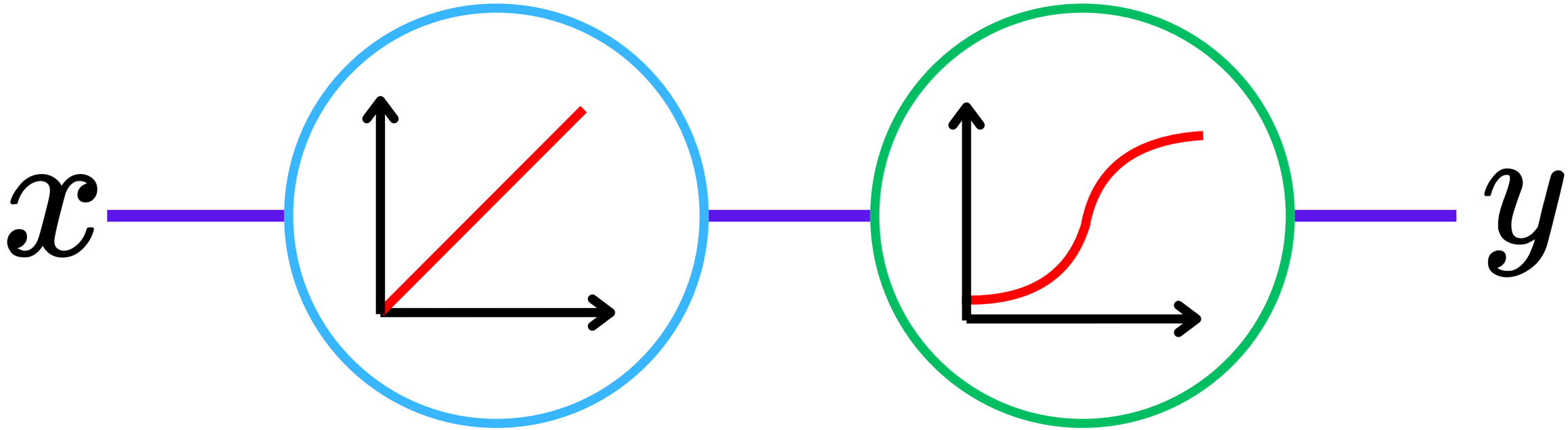


2D Gradient Descent



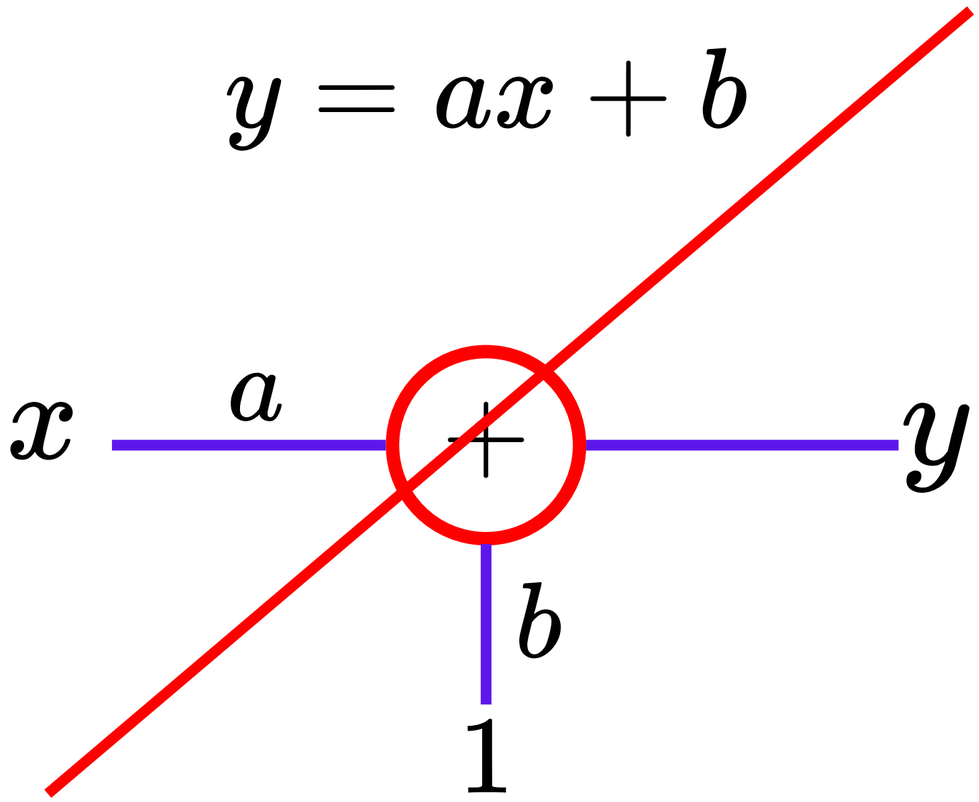


Neuron

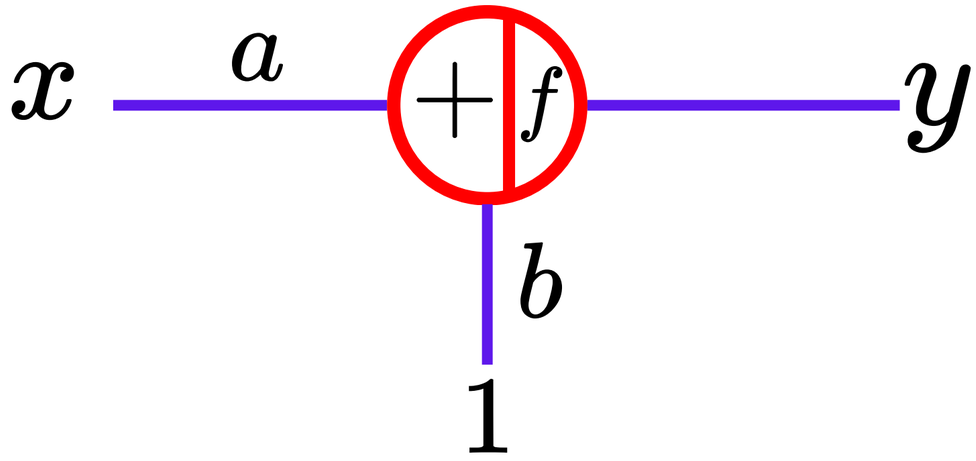


Neuron

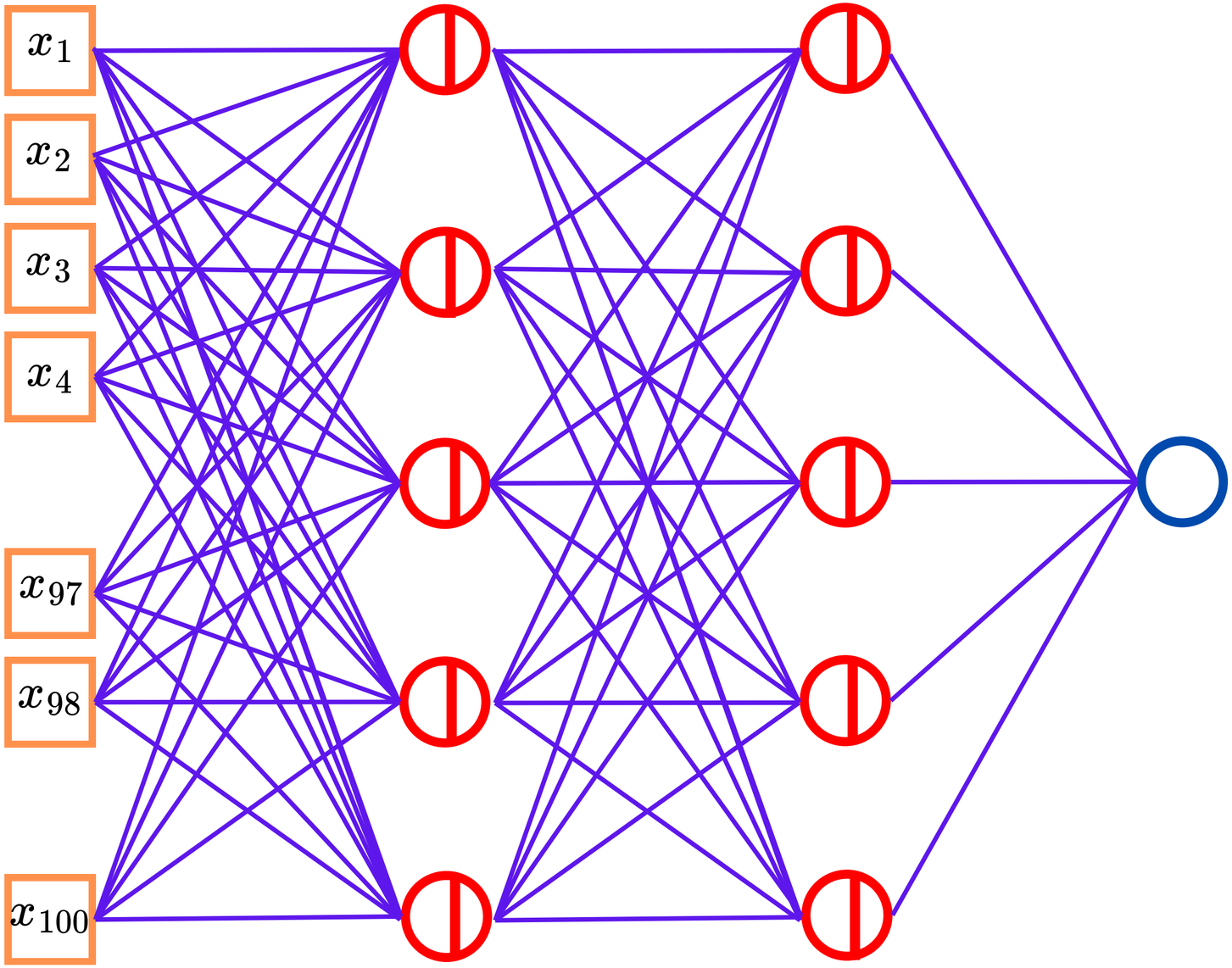
$$y = ax + b$$



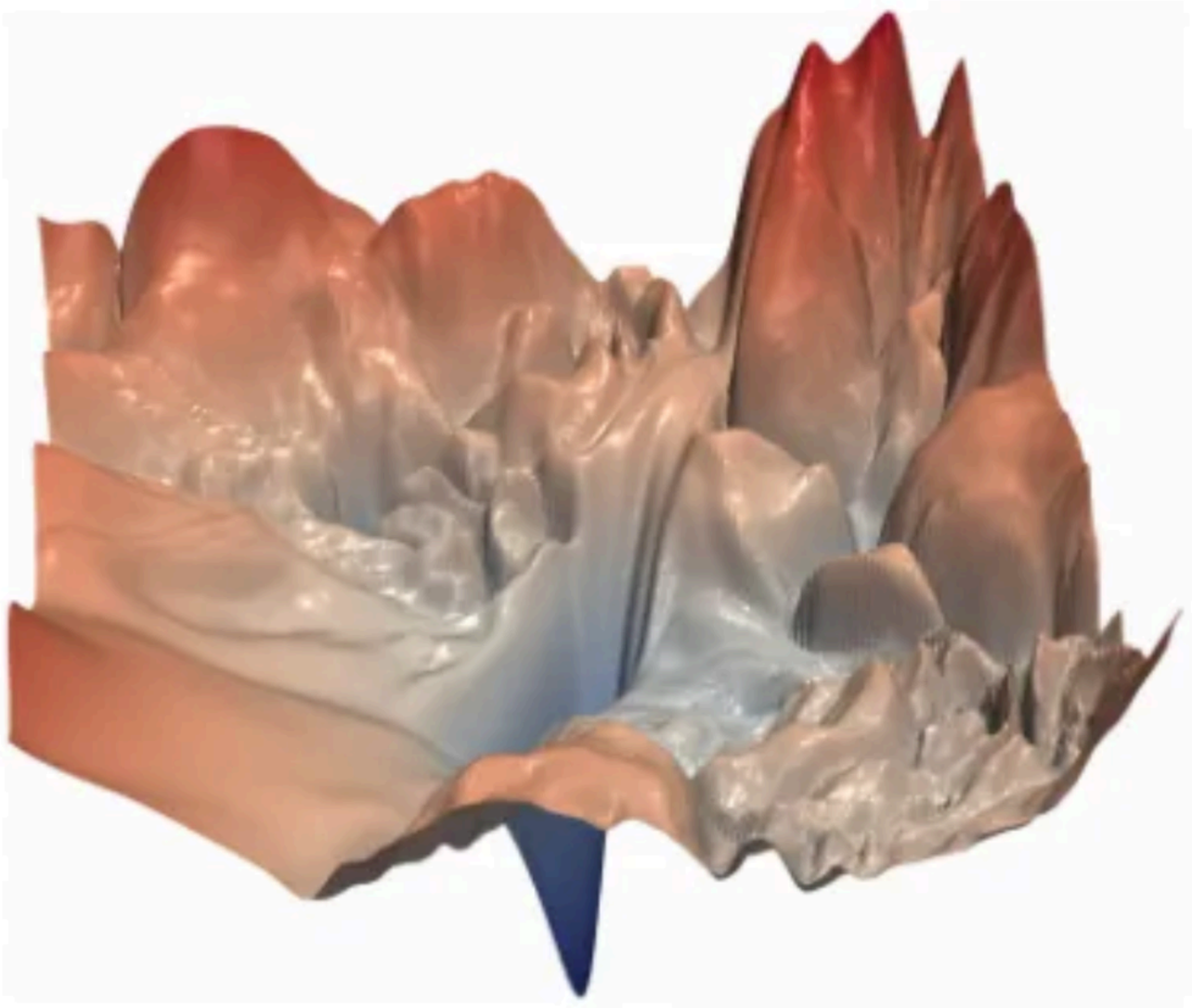
$$y = f\,(ax + b)$$



Neural Network



Gradient Descent in reality



Gradient Descent update rule

Tham số trước khi được cập nhật

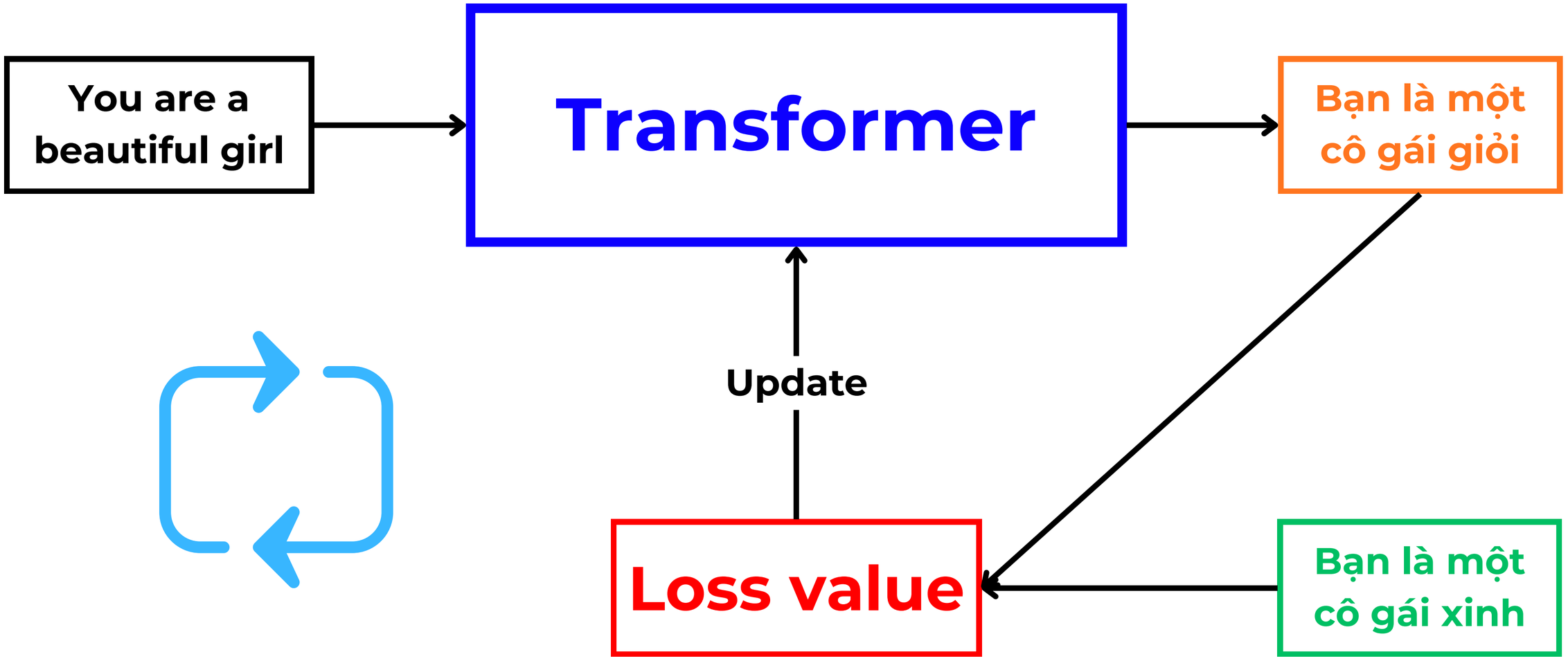
Đạo hàm riêng của hàm loss đối với tham số

$$p = p - \eta \frac{\Delta L}{\Delta p}$$

Tham số sau khi được cập nhật

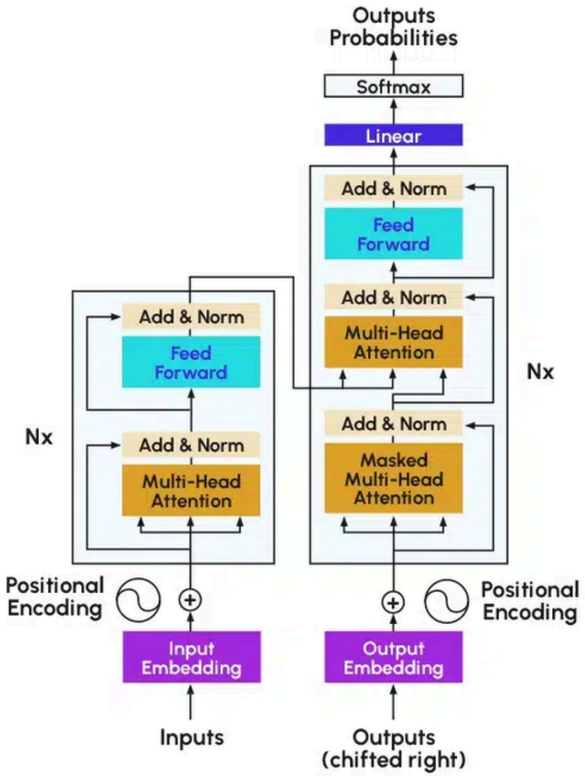
learning rate

Quá trình huấn luyện Transformer



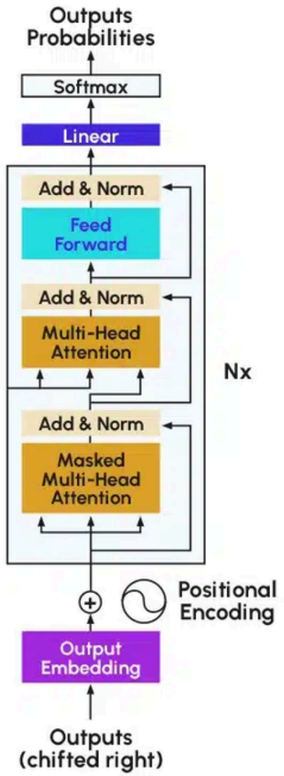
Các biến thể của Transformer

Transformer



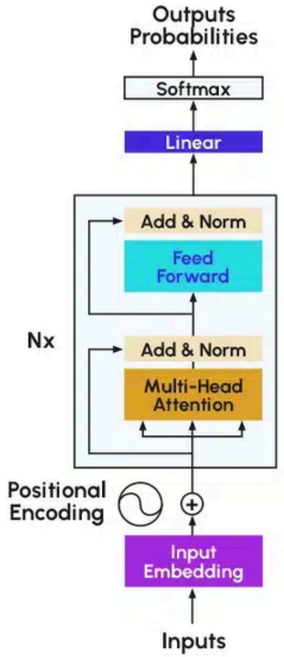
Encoder Decoder

GPT*

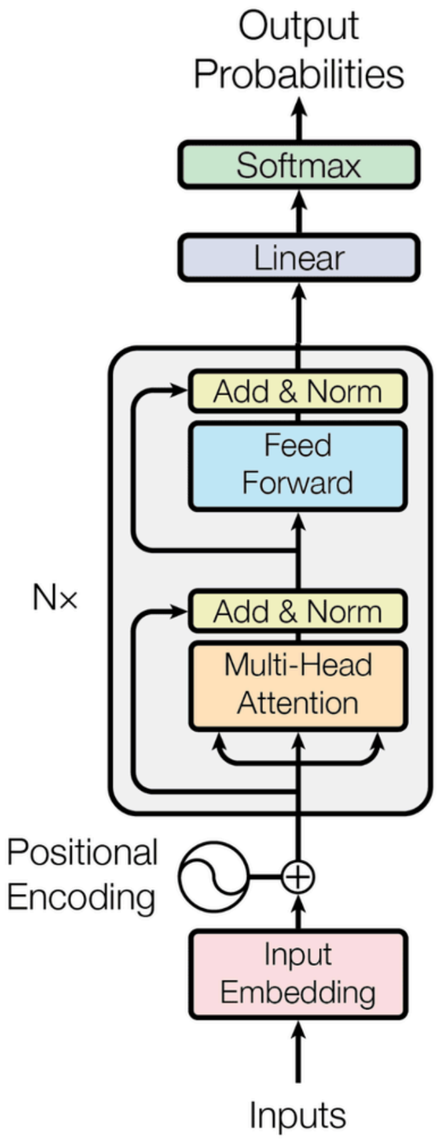


Decoder-only

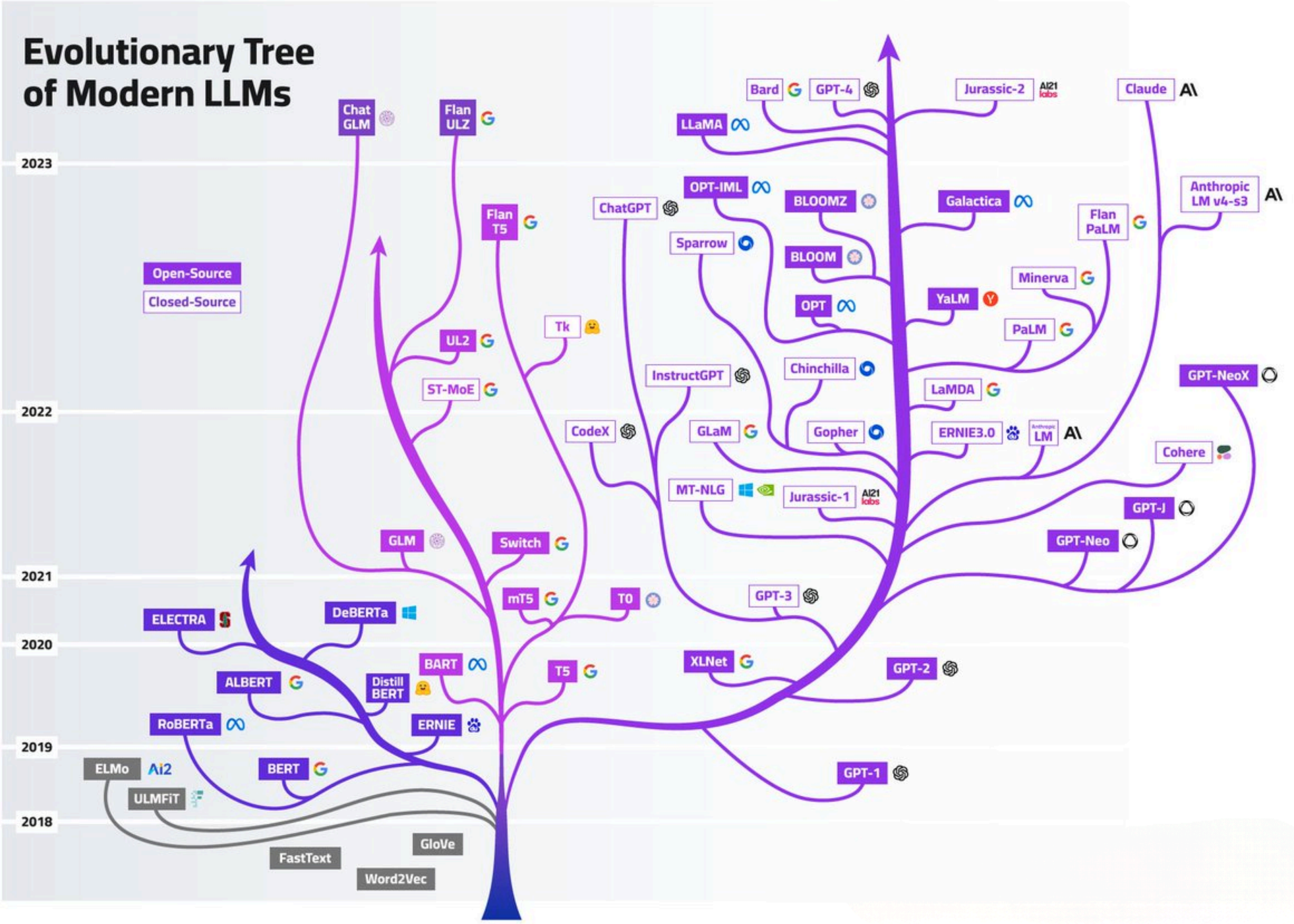
BERT*



Encoder-only

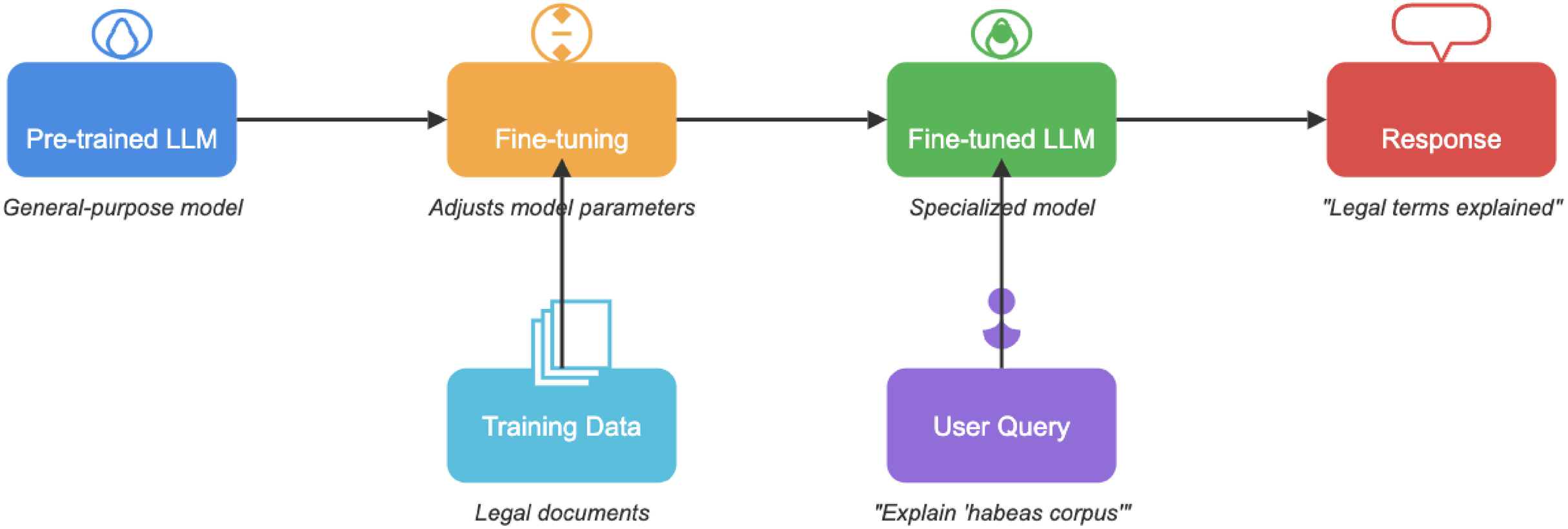


	BERT Base	BERT Large
Number of embedding dimensions, d_model	768	1024
Number of encoder blocks, N:	12	24
Number of attention heads per encoder block:	12	16
Size of hidden layer in feedforward network:	3072	4096
Total parameters:	110 million	340 million


Evolutionary Tree of Modern LLMs
2023
2022
2021
2020
2019
2018
Open-Source
Closed-Source
Chat GLM
Flan ULZ
Flan T5
Tk
UL2
ST-MoE
GLM
Switch
mT5
T0
DeBERTa
ELECTRA
BART
ALBERT
Distill BERT
T5
RoBERTa
ERNIE
ELMo
BERT
ULMFiT
GloVe
FastText
Word2Vec
Bard
GPT-4
LLaMA
Jurassic-2
Claude
OPT-IML
ChatGPT
BLOOMZ
Galactica
Flan PaLM
Anthropic LM v4-s3
Sparrow
BLOOM
Minerva
YaLM
OPT
PaLM
InstructGPT
Chinchilla
LaMDA
GPT-NeoX
CodeX
GLaM
Gopher
ERNIE3.0
Anthropic LM
Cohere
MT-NLG
Jurassic-1
GPT-J
GPT-Neo
GPT-3
XLNet
GPT-2
GPT-1

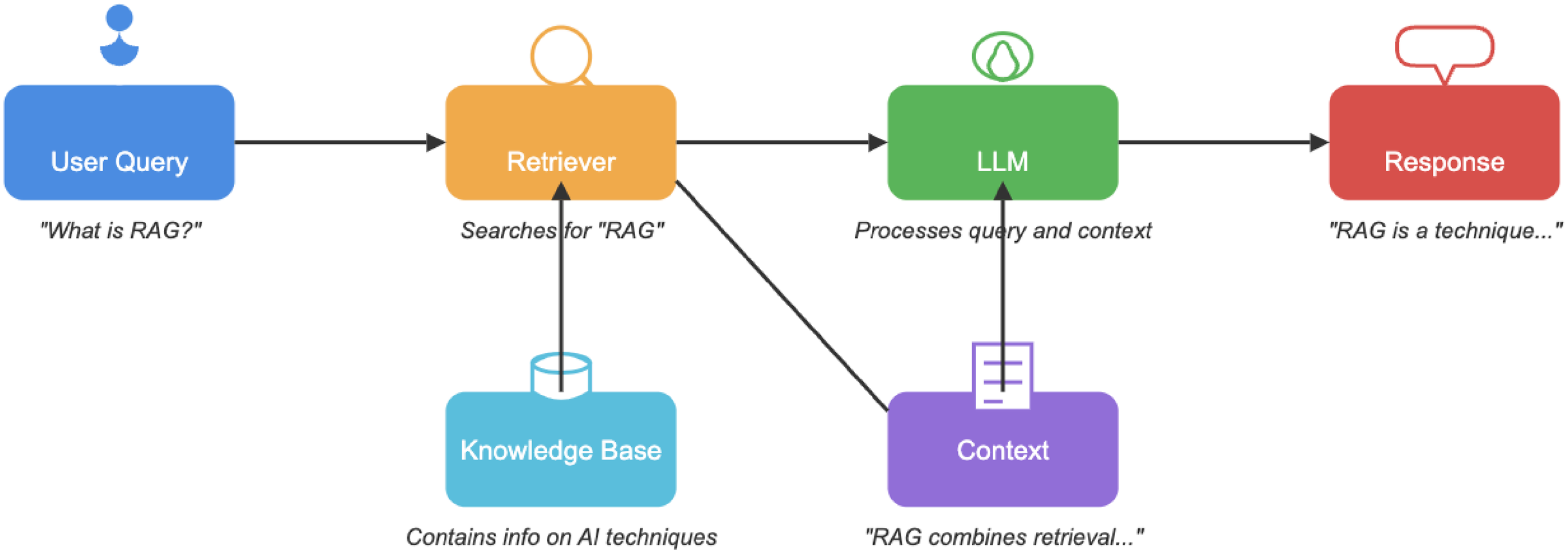
Fine-Tuning Process for LLMs

Example: Specializing in Legal Terminology



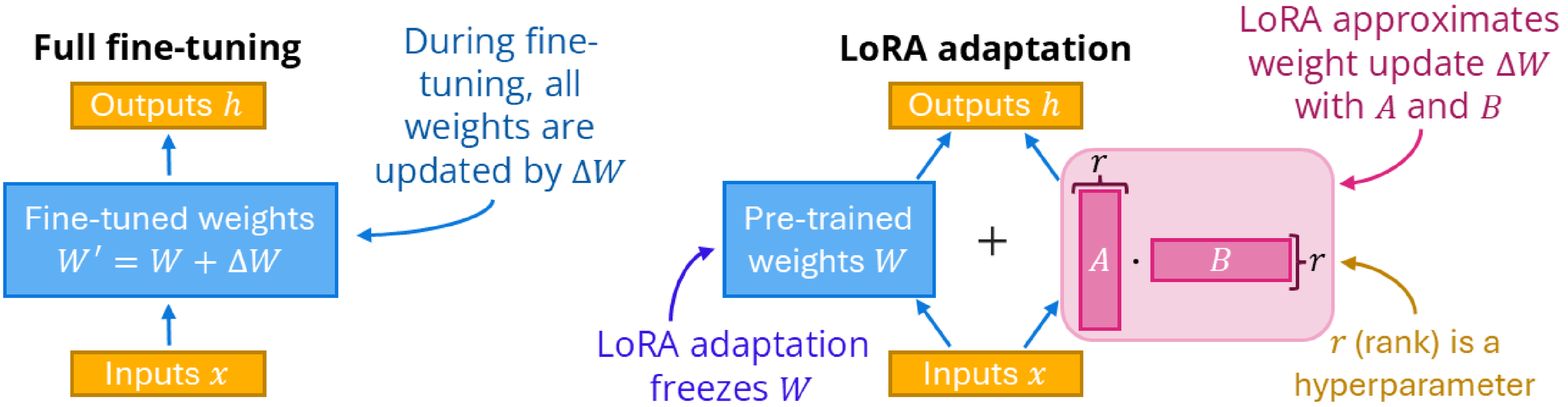
Retrieval-Augmented Generation (RAG) Process

Example: Answering "What is RAG?"



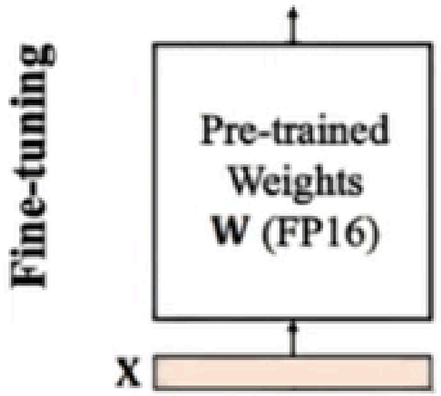
Aspect	RAG	Fine-Tuning
Use Case	General tasks, need for up-to-date info	Specialized tasks, domain-specific
Data Handling	Retrieves from external database	Incorporates data into model parameters
Flexibility	High - can update knowledge easily	Low - requires retraining for updates
Resource Use	Lower training resources, higher inference time	Higher training resources, faster inference
Privacy	Better - sensitive data stays in separate DB	Potential privacy concerns with training data

Low-Rank Adaptation



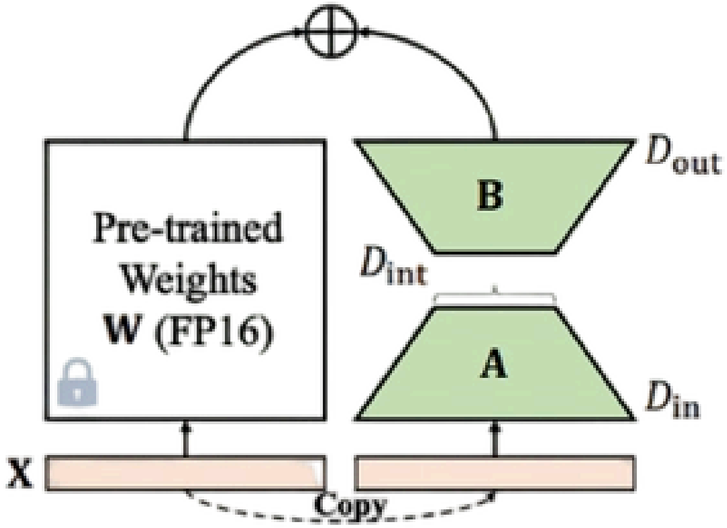
Fine-tuning

Full Fine-Tuning
16-bit precision



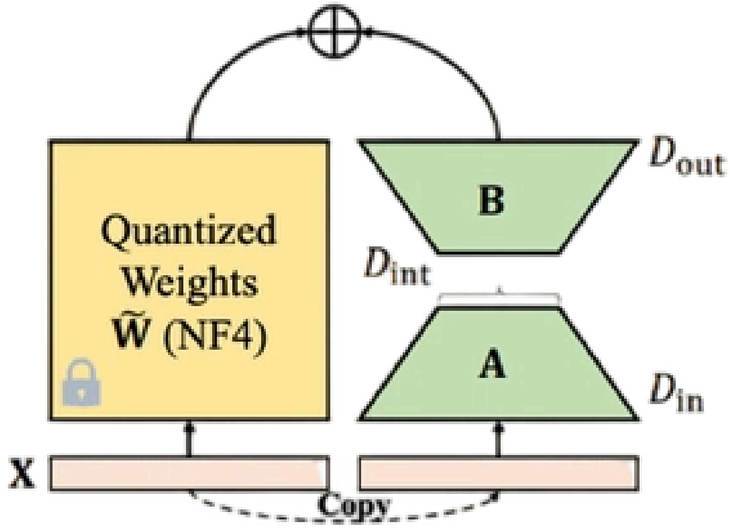
✅ Best performance
❌ Very high VRAM usage

LoRA
16-bit precision



✅ Quick training
❌ Still costly

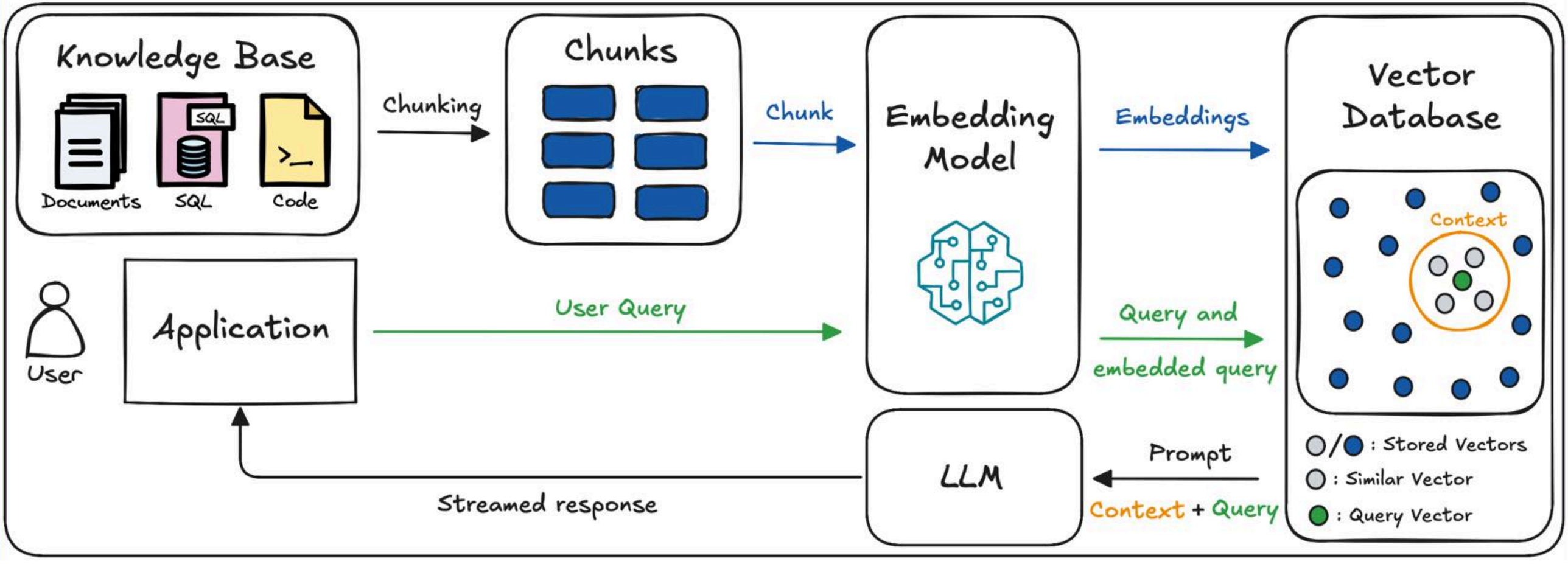
QLoRA
4-bit precision



✅ Low VRAM usage
❌ Degrades performance

RAG Workflow

1. The user enters a query.
2. Retriever fetches information relevant to the query.
3. Relevant information is augmented to the prompt as context.
4. LLM generates a response to the prompt
5. Response to the user



Data Loading

Ví dụ: Load tất cả các file pdf

Example

```python
from langchain_community.document_loaders import (
    DirectoryLoader,
    UnstructuredFileLoader,
)

loader = DirectoryLoader(
    path="./data",
    glob="**/*.pdf",
    loader_cls=UnstructuredFileLoader,
    show_progress=True,
    use_multithreading=True,
)
docs = loader.load()
```

Một vài kiểu loader

Loader	Công dụng
UnstructuredFileLoader	Loader đa năng, dùng để đọc nhiều định dạng file khác nhau. Hỗ trợ các loại dữ liệu có cấu trúc phức tạp (ảnh, đồ thị, công thức,...). Làm việc được với cả file scan. Chạy chậm so với các loader phía dưới
TextLoader	Dùng để load file text thuần như .txt, .md, .log, .py, .json dạng text.
PyPDFLoader/PyMuPDFLoader	Dùng để load file .pdf
Docx2txtLoader	Dùng để load file .docx có cấu trúc đơn giản
UnstructuredWordDocumentLoader	Dùng để load file .docx có cấu trúc phức tạp (heading, bảng, layout)
CSVLoader	Dùng để load file .csv
JSONLoader	Dùng để load file .json
PythonLoader	Dùng để load file Python source code

Ví dụ: Load nhiều loại file

Example

```python
from langchain_community.document_loaders import (
    DirectoryLoader,
    PyMuPDFLoader,
    Docx2txtLoader,
)

pdf_loader = DirectoryLoader(
    path="./data",
    glob="**/*.pdf",
    loader_cls=PyMuPDFLoader,
    show_progress=True,
    use_multithreading=True,
)

docx_loader = DirectoryLoader(
    path="./data",
    glob="**/*.docx",
    loader_cls=Docx2txtLoader,
    show_progress=True,
    use_multithreading=True,
)

documents = []
documents.extend(pdf_loader.load())
documents.extend(docx_loader.load())
```

Chunking strategies

Fixed-size chunking

Sample code

```python
from langchain_text_splitters import CharacterTextSplitter

text = """
Artificial Intelligence is a field of computer science focused on building systems
that can perform tasks requiring human intelligence. These tasks include learning,
reasoning, problem solving, understanding language, recognizing images, and making
decisions.

Machine Learning is a subfield of AI where systems learn patterns from data.
Deep Learning is a subfield of Machine Learning that uses neural networks with many layers.

"""

text_splitter = CharacterTextSplitter(
    chunk_size=100,
    chunk_overlap=20,
    separator=""
)

chunks = text_splitter.split_text(text)

for i, chunk in enumerate(chunks):
    print(f"Chunk {i + 1}:")
    print(chunk)
    print("-" * 50)
```

Result

```
Chunk 1:
Artificial Intelligence is a field of computer science focused on building systems
that can perform
--------------------------------------------------
Chunk 2:
ems
that can perform tasks requiring human intelligence. These tasks include learning,
reasoning, pr
--------------------------------------------------
Chunk 3:
rning,
reasoning, problem solving, understanding language, recognizing images, and making
decisions.
--------------------------------------------------
Chunk 4:
nd making decisions.

Machine Learning is a subfield of AI where systems learn patterns from data.
D
--------------------------------------------------
Chunk 5:
atterns from data.
Deep Learning is a subfield of Machine Learning that uses neural networks with ma
--------------------------------------------------
Chunk 6:
ral networks with many layers.
--------------------------------------------------
```

Recursive chunking

Sample code

```python
from langchain.text_splitter import RecursiveCharacterTextSplitter

text = """
Artificial Intelligence is a field of computer science focused on building systems
that can perform tasks requiring human intelligence. These tasks include learning,
reasoning, problem solving, understanding language, recognizing images, and making
decisions.

Machine Learning is a subfield of AI where systems learn patterns from data.
Deep Learning is a subfield of Machine Learning that uses neural networks with many layers.

"""

text_splitter = RecursiveCharacterTextSplitter(
    chunk_size=100,
    chunk_overlap=20,
    separators=["\\n\\n", "\\n", ". ", " ", ""]
)

chunks = text_splitter.split_text(text)

for i, chunk in enumerate(chunks):
    print(f"Chunk {i + 1}:")
    print(chunk)
    print("-" * 50)
```

Result

```
Chunk 1:
Artificial Intelligence is a field of computer science focused on building systems
that can perform
--------------------------------------------------
Chunk 2:
can perform tasks requiring human intelligence
--------------------------------------------------
Chunk 3:
. These tasks include learning,
reasoning, problem solving, understanding language, recognizing
--------------------------------------------------
Chunk 4:
recognizing images, and making decisions.

Machine Learning is a subfield of AI where systems learn
--------------------------------------------------
Chunk 5:
where systems learn patterns from data.
Deep Learning is a subfield of Machine Learning that uses
--------------------------------------------------
Chunk 6:
Learning that uses neural networks with many layers.
--------------------------------------------------
```

Semantic chunking

Sample code

```python
from langchain_experimental.text_splitter import SemanticChunker
from langchain_openai import OpenAIEmbeddings
from dotenv import load_dotenv

text = """
Artificial Intelligence is a field of computer science focused on building systems
that can perform tasks requiring human intelligence. These tasks include learning,
reasoning, problem solving, understanding language, recognizing images, and making
decisions.

Machine Learning is a subfield of AI where systems learn patterns from data.
Deep Learning is a subfield of Machine Learning that uses neural networks with many layers.

"""

load_dotenv()
embeddings = OpenAIEmbeddings()
text_splitter = SemanticChunker(
    embeddings,
    breakpoint_threshold_type="percentile",
    breakpoint_threshold_amount=50,
)

chunks = text_splitter.split_text(text.strip())

for i, chunk in enumerate(chunks):
    print(f"Chunk {i + 1}:")
    print(chunk)
    print(len(chunk))
    print("-" * 50)
```

Result

```
Chunk 1:
Artificial Intelligence is a field of computer science focused on building systems
that can perform tasks requiring human intelligence. These tasks include learning,
reasoning, problem solving, understanding language, recognizing images, and making
decisions.
259
--------------------------------------------------
Chunk 2:
Machine Learning is a subfield of AI where systems learn patterns from data. Deep Learning
is a subfield of Machine Learning that uses neural networks with many layers.
168
--------------------------------------------------
```

Document-based chunking

Sample code

```python
from langchain.text_splitter import MarkdownHeaderTextSplitter

text = """
# Data Science Overview
Data science combines statistics, programming, and domain expertise.
## Machine Learning
Machine learning automates analytical model building.
### Supervised Learning
Supervised learning uses labeled training data.
### Unsupervised Learning
Unsupervised learning finds hidden patterns.
## Deep Learning
Deep learning uses neural networks with multiple layers.
"""

text_splitter = MarkdownHeaderTextSplitter(
    headers_to_split_on=[
        ("#", "Header 1"),
        ("##", "Header 2"),
        ("###", "Header 3"),
    ]
)

chunks = text_splitter.split_text(text.strip())

for i, chunk in enumerate(chunks):
    print(f"Chunk {i + 1}:")
    print(chunk)
    print("-" * 50)
```

Result

```
Chunk 1:
page_content='Data science combines statistics, programming, and domain expertise.'
metadata={'Header 1': 'Data Science Overview'}
--------------------------------------------------
Chunk 2:
page_content='Machine learning automates analytical model building.' metadata={'Header 1':
'Data Science Overview', 'Header 2': 'Machine Learning'}
--------------------------------------------------
Chunk 3:
page_content='Supervised learning uses labeled training data.' metadata={'Header 1': 'Data
Science Overview', 'Header 2': 'Machine Learning', 'Header 3': 'Supervised Learning'}
--------------------------------------------------
Chunk 4:
page_content='Unsupervised learning finds hidden patterns.' metadata={'Header 1': 'Data
Science Overview', 'Header 2': 'Machine Learning', 'Header 3': 'Unsupervised Learning'}
--------------------------------------------------
Chunk 5:
page_content='Deep learning uses neural networks with multiple layers.' metadata={'Header
1': 'Data Science Overview', 'Header 2': 'Deep Learning'}
--------------------------------------------------
```

Hierarchical chunking

Sample code

```python
from langchain_text_splitters import RecursiveCharacterTextSplitter

text = """
Machine Learning is a branch of AI that allows computers to learn from data.
In Machine Learning, data is used to train models.
After learning from the data, the model can predict results for new data.

Deep Learning is a smaller subfield of Machine Learning.
Deep Learning uses neural networks with many layers.
Deep Learning models are commonly used in image processing, natural language processing,
speech recognition, and self-driving cars.

RAG is a technique that combines retrieval and generation.
The system first searches for relevant information from documents, then provides that
information to the LLM so it can generate a more accurate answer.
"""

parent_splitter = RecursiveCharacterTextSplitter(
    chunk_size=400,
    chunk_overlap=50
)

child_splitter = RecursiveCharacterTextSplitter(
    chunk_size=120,
    chunk_overlap=20
)

parent_chunks = parent_splitter.split_text(text)

child_chunks = []

for parent_id, parent_chunk in enumerate(parent_chunks):
    children = child_splitter.split_text(parent_chunk)

    for child_id, child_chunk in enumerate(children):
        child_chunks.append({
            "parent_id": parent_id,
            "child_id": child_id,
            "child_text": child_chunk,
            "parent_text": parent_chunk
        })

print("Number of parent chunks:", len(parent_chunks))
print("Number of child chunks:", len(child_chunks))

for item in child_chunks:
    print(f"\nParent {item['parent_id']+1} - Child {item['child_id']+1}")
    print("Child text:")
    print(item["child_text"])
    print("\nCorresponding parent text:")
    print(item["parent_text"])
    print("-" * 80)
```

Result

```
Number of parent chunks: 3
Number of child chunks: 9

Parent 1 - Child 1
Child text:
Machine Learning is a branch of AI that allows computers to learn from data.

Corresponding parent text:
Machine Learning is a branch of AI that allows computers to learn from data.
In Machine Learning, data is used to train models.
After learning from the data, the model can predict results for new data.
--------------------------------------------------------------------------------

Parent 1 - Child 2
Child text:
In Machine Learning, data is used to train models.

Corresponding parent text:
Machine Learning is a branch of AI that allows computers to learn from data.
In Machine Learning, data is used to train models.
After learning from the data, the model can predict results for new data.
--------------------------------------------------------------------------------

Parent 1 - Child 3
Child text:
After learning from the data, the model can predict results for new data.

Corresponding parent text:
Machine Learning is a branch of AI that allows computers to learn from data.
In Machine Learning, data is used to train models.
After learning from the data, the model can predict results for new data.
--------------------------------------------------------------------------------

Parent 2 - Child 1
Child text:
Deep Learning is a smaller subfield of Machine Learning.
Deep Learning uses neural networks with many layers.

Corresponding parent text:
Deep Learning is a smaller subfield of Machine Learning.
Deep Learning uses neural networks with many layers.
Deep Learning models are commonly used in image processing, natural language processing,
speech recognition, and self-driving cars.
--------------------------------------------------------------------------------

Parent 2 - Child 2
Child text:
Deep Learning models are commonly used in image processing, natural language processing,
speech recognition, and

Corresponding parent text:
Deep Learning is a smaller subfield of Machine Learning.
Deep Learning uses neural networks with many layers.
Deep Learning models are commonly used in image processing, natural language processing,
speech recognition, and self-driving cars.
--------------------------------------------------------------------------------
```

Result

```
Parent 2 - Child 3
Child text:
recognition, and self-driving cars.

Corresponding parent text:
Deep Learning is a smaller subfield of Machine Learning.
Deep Learning uses neural networks with many layers.
Deep Learning models are commonly used in image processing, natural language processing,
speech recognition, and self-driving cars.
--------------------------------------------------------------------------------

Parent 3 - Child 1
Child text:
RAG is a technique that combines retrieval and generation.

Corresponding parent text:
RAG is a technique that combines retrieval and generation.
The system first searches for relevant information from documents, then provides that
information to the LLM so it can generate a more accurate answer.
--------------------------------------------------------------------------------

Parent 3 - Child 2
Child text:
The system first searches for relevant information from documents, then provides that
information to the LLM so it can

Corresponding parent text:
RAG is a technique that combines retrieval and generation.
The system first searches for relevant information from documents, then provides that
information to the LLM so it can generate a more accurate answer.
--------------------------------------------------------------------------------

Parent 3 - Child 3
Child text:
the LLM so it can generate a more accurate answer.

Corresponding parent text:
RAG is a technique that combines retrieval and generation.
The system first searches for relevant information from documents, then provides that
information to the LLM so it can generate a more accurate answer.
--------------------------------------------------------------------------------
```

LLM-based chunking

Sample code

```python
from dotenv import load_dotenv
from langchain_openai import ChatOpenAI

load_dotenv()

text = """
Machine Learning is a branch of AI that allows computers to learn from data.
It is commonly used for prediction, classification, and recommendation systems.
Deep Learning is a subfield of Machine Learning that uses neural networks with many layers.

It is often used in image recognition, natural language processing, and speech recognition.

RAG stands for Retrieval-Augmented Generation.
It helps LLMs answer questions by first retrieving relevant information from external
documents.
"""

llm = ChatOpenAI(model="gpt-4.1-mini", temperature=0)

prompt = f"""
You are a document chunking assistant.

Split the following text into coherent chunks.
Each chunk should focus on one main idea.
Return the result as a numbered list.

Text:
{text}
"""

response = llm.invoke(prompt)
print(response.content)
```

Result

```
1. Machine Learning is a branch of AI that allows computers to learn from data. It is
commonly used for prediction, classification, and recommendation systems.

2. Deep Learning is a subfield of Machine Learning that uses neural networks with many
layers. It is often used in image recognition, natural language processing, and speech
recognition.

3. RAG stands for Retrieval-Augmented Generation. It helps LLMs answer questions by first
retrieving relevant information from external documents.
```

Agentic chunking

Sample code

```python
from dotenv import load_dotenv
from langchain_openai import ChatOpenAI
from langchain_text_splitters import RecursiveCharacterTextSplitter, CharacterTextSplitter

load_dotenv()

text = """
Machine Learning is a branch of AI that allows computers to learn from data.
It is commonly used for prediction, classification, and recommendation systems.
Deep Learning is a subfield of Machine Learning that uses neural networks with many layers.

It is often used in image recognition, natural language processing, and speech recognition.

RAG stands for Retrieval-Augmented Generation.
It helps LLMs answer questions by first retrieving relevant information from external
documents.
"""

llm = ChatOpenAI(model="gpt-4.1-mini", temperature=0)

# Step 1: Agent decides which chunking strategy to use
strategy_prompt = f"""
You are a chunking strategy agent.

Analyze the following text and choose the best chunking strategy.

Return only one word:
- recursive
- fixed_size
"""
```

Sample code

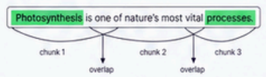
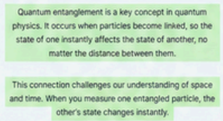
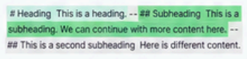
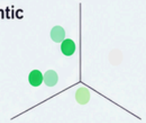
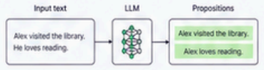
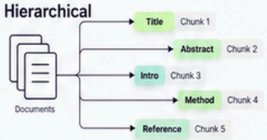
```python
"""
Rules:
- Choose recursive if the text has paragraphs, sentences, or natural structure
  that should be preserved.
- Choose fixed_size if the text has little structure or you only need simple
  chunks with equal length.

Text:
{text}
"""

strategy_response = llm.invoke(strategy_prompt)
strategy = strategy_response.content.strip().lower()

print("Selected strategy:", strategy)

if strategy == "recursive":
    text_splitter = RecursiveCharacterTextSplitter(
        chunk_size=250,
        chunk_overlap=30,
        separators=["\n\n", ". ", "\n", " ", ""]
    )
else:  # fixed_size
    text_splitter = CharacterTextSplitter(
        chunk_size=250,
        chunk_overlap=30,
        separator=""
    )

chunks = text_splitter.split_text(text.strip())

for i, chunk in enumerate(chunks, 1):
    print(f"\nChunk {i}:")
    print(chunk)
    print("-" * 50)
```

Agentic chunking

Result

```
Selected strategy: recursive

Chunk 1:
Machine Learning is a branch of AI that allows computers to learn from data.
It is commonly used for prediction, classification, and recommendation systems.
Deep Learning is a subfield of Machine Learning that uses neural networks with many layers.

--------------------------------------------------

Chunk 2:
It is often used in image recognition, natural language processing, and speech recognition.

RAG stands for Retrieval-Augmented Generation.
It helps LLMs answer questions by first retrieving relevant information from external
documents.
--------------------------------------------------
```

Tổng kết

Chunking Strategy	Cách thức hoạt động	Độ phức tạp	Phù hợp nhất cho	Ví dụ
Fixed-Size 	Chia theo số lượng token hoặc ký tự.	●○○○○	Các tài liệu nhỏ hoặc đơn giản, hoặc khi tốc độ là yếu tố quan trọng nhất.	Ghi chú cuộc họp, bài blog ngắn, email, FAQ đơn giản.
Recursive 	Chia văn bản bằng cách chia nhỏ lặp lại cho đến khi phù hợp với kích thước chunk mong muốn, thường vẫn giữ được một phần cấu trúc.	●●○○○	Các tài liệu có cấu trúc nhất định cần được giữ lại nhưng tốc độ vẫn quan trọng.	Bài nghiên cứu, hướng dẫn sản phẩm, báo cáo ngắn.
Document-Based 	Chỉ chia tại ranh giới tài liệu hoặc các yếu tố cấu trúc như tiêu đề (heading).	●○○○○	Các bộ tài liệu ngắn, độc lập hoặc các tệp có cấu trúc rõ ràng.	Bài báo tin tức, vé hỗ trợ khách hàng, tệp Markdown.
Semantic 	Chia văn bản tại các ranh giới có ý nghĩa tự nhiên (chủ đề, ý tưởng).	●●●○○	Tài liệu kỹ thuật, học thuật hoặc tường thuật, nơi chủ đề thay đổi nhưng không có dấu phân cách rõ ràng.	Bài báo khoa học, sách giáo khoa, tiểu thuyết, whitepaper.
LLM-Based 	Sử dụng mô hình ngôn ngữ để quyết định ranh giới chunk dựa trên ngữ cảnh và ý nghĩa.	●●●●○	Văn bản phức tạp, cần hiểu nghĩa tốt để chia chunk, giúp cải thiện các tác vụ hạ nguồn như Q&A.	Báo cáo dài, ý kiến pháp lý, hồ sơ y tế.
Agentic	Cho phép một AI agent quyết định cách chia dựa trên ý nghĩa và cấu trúc.	●●●●●	Các tài liệu phức tạp, tinh vi cần chiến lược tùy chỉnh.	Hồ sơ pháp lý, hợp đồng nhiều phần, chính sách doanh nghiệp.
Hierarchical 	Chia văn bản thành nhiều mức độ (phần → đoạn văn → câu).	●●●○○	Các tài liệu lớn, có cấu trúc rõ ràng, cần cả tóm tắt và chi tiết.	Sổ tay nhân viên, quy định của chính phủ, tài liệu hướng dẫn phần mềm.